ESC European Society of Cardiology

European Heart Journal (2024) **45**, 3912–4018
https://doi.org/10.1093/eurheartj/ehae178

ESC GUIDELINES

2024 ESC Guidelines for the management of elevated blood pressure and hypertension

Developed by the task force on the management of elevated blood pressure and hypertension of the European Society of Cardiology (ESC) and *endorsed by the European Society of Endocrinology (ESE) and the European Stroke Organisation (ESO)*

Authors/Task Force Members: John William McEvoy *†, (Chairperson) (Ireland), Cian P. McCarthy ‡, (Task Force Co-ordinator) (United States of America), Rosa Maria Bruno ‡, (Task Force Co-ordinator) (France), Sofie Brouwers (Belgium), Michelle D. Canavan (Ireland), Claudio Ceconi (Italy), Ruxandra Maria Christodorescu (Romania), Stella S. Daskalopoulou (Canada), Charles J. Ferro [1] (United Kingdom), Eva Gerdts (Norway), Henner Hanssen (Switzerland), Julie Harris (United Kingdom), Lucas Lauder (Switzerland/Germany), Richard J. McManus (United Kingdom), Gerard J. Molloy (Ireland), Kazem Rahimi (United Kingdom), Vera Regitz-Zagrosek (Germany), Gian Paolo Rossi [2] (Italy), Else Charlotte Sandset [3] (Norway), Bart Scheenaerts (Belgium), Jan A. Staessen (Belgium), Izabella Uchmanowicz (Poland), Maurizio Volterrani (Italy), Rhian M. Touyz *†, (Chairperson) (Canada), and ESC Scientific Document Group

* Corresponding authors: John William McEvoy, Department of Cardiology, University of Galway School of Medicine, Galway, Ireland, and National Institute for Prevention and Cardiovascular Health, Galway, Ireland. Tel: +353 91 544310, E-mail: johnwilliam.mcevoy@universityofgalway.ie; and Rhian M. Touyz, Department of Medicine, McGill University, Montreal, Canada, Department of Family Medicine, McGill University, Montreal, Canada, and the Research Institute of the McGill University Health Centre, McGill University, Montreal, Canada. Tel: +1 514 934 1934 ext 71608, E-mail: Rhian.touyz@mcgill.ca

† The two Chairpersons contributed equally to the document and are joint corresponding authors.

‡ The two Task Force Co-ordinators contributed equally to the document.

Author/Task Force Member affiliations are listed in author information.

[1]Representing the European Renal Association (ERA), [2]Representing the European Society of Endocrinology (ESE), [3]Representing the European Stroke Organisation (ESO), [4]Representing the European Geriatric Medicine Society (EuGMS).

ESC Clinical Practice Guidelines (CPG) Committee: listed in the Appendix.

ESC subspecialty communities having participated in the development of this document:

Associations: Association of Cardiovascular Nursing & Allied Professions (ACNAP), European Association of Preventive Cardiology (EAPC), European Association of Percutaneous Cardiovascular Interventions (EAPCI), Heart Failure Association (HFA).

Councils: Council for Cardiology Practice, Council on Hypertension, Council on Stroke.

Working Groups: Aorta and Peripheral Vascular Diseases, Cardiovascular Pharmacotherapy, E-Cardiology.

Patient Forum

Disclaimer. The ESC Guidelines represent the views of the ESC and were produced after careful consideration of the scientific and medical knowledge and the evidence available at the time of their publication. The ESC is not responsible in the event of any contradiction, discrepancy and/or ambiguity between the ESC Guidelines and any other official recommendations or guidelines issued by the relevant public health authorities, in particular in relation to good use of healthcare or therapeutic strategies. Health professionals are encouraged to take the ESC Guidelines fully into account when exercising their clinical judgment, as well as in the determination and the implementation of preventive, diagnostic or therapeutic medical strategies; however, the ESC Guidelines do not override, in any way whatsoever, the individual responsibility of health professionals to make appropriate and accurate decisions in consideration of each patient's health condition and in consultation with that patient and, where appropriate and/or necessary, the patient's caregiver. Nor do the ESC Guidelines exempt health professionals from taking into full and careful consideration the relevant official updated recommendations or guidelines issued by the competent public health authorities, in order to manage each patient's case in light of the scientifically accepted data pursuant to their respective ethical and professional obligations. It is also the health professional's responsibility to verify the applicable rules and regulations relating to drugs and medical devices at the time of prescription. The ESC warns readers that the technical language may be misinterpreted and declines any responsibility in this respect.

Permissions. The content of these European Society of Cardiology (ESC) Guidelines has been published for personal and educational use only. No commercial use is authorized. No part of the ESC Guidelines may be translated or reproduced in any form without written permission from the ESC. Permissions can be obtained upon submission of a written request to Oxford University Press, the publisher of the *European Heart Journal* and the party authorized to handle such permissions on behalf of the ESC (journals.permissions@oup.com).
© The European Society of Cardiology 2024. All rights reserved. For permissions, please email: journals.permissions@oup.com

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

Document Reviewers: Ana Abreu, (CPG Review Co-ordinator) (Portugal), Michael Hecht Olsen, (CPG Review Co-ordinator) (Denmark), Marco Ambrosetti (Italy), Emmanuel Androulakis (United Kingdom), Lia Evi Bang (Denmark), Jesper Nørgaard Bech (Denmark), Michael A. Borger (Germany), Pierre Boutouyrie (France), Luís Bronze (Portugal), Sergio Buccheri (Sweden), Regina Dalmau (Spain), Maria Carmen De Pablo Zarzosa (Spain), Christian Delles (United Kingdom), Maria Manuela Fiuza (Portugal), Rahima Gabulova (Azerbaijan), Bjørn Olav Haugen (Norway), Christian Heiss (United Kingdom), Borja Ibanez (Spain), Stefan James (Sweden), Vikas Kapil (United Kingdom), Meral Kayikçioglu (Turkey), Lars Køber (Denmark), Konstantinos C. Koskinas (Switzerland), Emanuela Teresa Locati (Italy), Sharon MacDonald (United Kingdom), Anastasia S. Mihailidou (Australia), Borislava Mihaylova (United Kingdom), Richard Mindham (United Kingdom), Martin Bodtker Mortensen (Denmark), Sandor Nardai (Hungary), Lis Neubeck (United Kingdom), Jens Cosedis Nielsen (Denmark), Peter M. Nilsson (Sweden), Agnes A. Pasquet (Belgium), Mónica Mendes Pedro (Portugal), Eva Prescott (Denmark), Amina Rakisheva (Kazakhstan), Ernst Rietzschel (Belgium), Bianca Rocca (Italy), Xavier Rossello (Spain), Jean-Paul Schmid (Switzerland), Eduard Shantsila (United Kingdom), Isabella Sudano (Switzerland), Ana Teresa Timóteo (Portugal), Georgios Tsivgoulis[3] (Greece), Andrea Ungar[4] (Italy), Ilonca Vaartjes (Netherlands), Frank Visseren (Netherlands), Heinz Voeller (Germany), Christiaan Vrints (Belgium), Adam Witkowski (Poland), Maria-Christina Zennaro[2] (France), and Katja Zeppenfeld (Netherlands)

All experts involved in the development of these guidelines have submitted declarations of interest, which are reported in a supplementary document to the guidelines. See the *European Heart Journal* online or www.escardio.org/Guidelines for supplementary documents, as well as evidence tables.

Click here to access the corresponding ESC CardioMed chapters.

Keywords Guidelines • Blood pressure • Hypertension • Hypertension-mediated organ damage • Blood pressure measurement • Ambulatory blood pressure monitoring • Home blood pressure monitoring • Antihypertensive medication • Hypertension treatment • Hypertension targets • Secondary hypertension • Cardiovascular disease risk estimation • Cardiovascular disease prevention • Resistant hypertension • Hypertension screening

Table of contents

1. Preamble 3918
2. Introduction 3919
2.1. What is new 3920
3. Pathophysiology of elevated blood pressure and hypertension .. 3928
4. Clinical consequences of elevated blood pressure and hypertension 3928
5. Measuring blood pressure 3929
5.1. Introduction and pertinent definitions 3929
5.2. Practical recommendations for measuring blood pressure . 3930
5.2.1. Clinical validation of equipment for measuring blood pressure 3930
5.2.2. Office blood pressure measurement 3930
5.2.3. Home blood pressure measurement 3931
5.2.4. Ambulatory blood pressure measurement 3933
5.2.5. Comparison of home and ambulatory blood pressure monitoring 3934
5.3. What is the best method for measuring blood pressure to diagnose hypertension? 3934
5.3.1. Blood pressure measurement for hypertension screening 3934
5.3.2. Blood pressure measurement for diagnosing hypertension 3934
5.4. What is the best method for measuring blood pressure for long-term management of hypertension? 3934
5.4.1. Home monitoring 3934
5.4.2. Ambulatory monitoring 3934
5.5. Measuring blood pressure in selected groups 3935
5.5.1. Pregnancy 3935
5.5.2. Atrial fibrillation 3935
5.5.3. Orthostatic hypotension 3935
5.6. Novel methods of measuring blood pressure 3935
6. Definition and classification of elevated blood pressure and hypertension, and cardiovascular disease risk assessment 3936
6.1. Definition and classification of elevated blood pressure and hypertension 3936
6.2. Principles of a risk-based approach for managing blood pressure and preventing cardiovascular disease 3936
6.2.1. Role of cardiovascular disease risk assessment 3936
6.3. Predicting cardiovascular disease risk 3937
6.3.1. 10-year cardiovascular disease risk-prediction models .. 3938
6.4. Refining cardiovascular disease risk estimation beyond risk models 3939

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

6.4.1. Sex-specific non-traditional cardiovascular disease risk modifiers 3939
6.4.2. Non-traditional cardiovascular disease risk modifiers shared by men and women 3940
6.4.3. Additional risk decision tests 3941
6.5. Summary of the cardiovascular disease risk stratification approach for allocating blood pressure treatment 3941
7. Diagnosing hypertension and investigating underlying causes 3943
7.1. Screening for hypertension 3943
7.2. Confirming the diagnosis of hypertension 3943
7.3. Communicating the diagnosis 3944
7.4. Baseline assessment and diagnostic approach 3944
7.4.1. Medical history, medication history, and physical examination 3944
7.4.2. Drug adherence and persistence with treatment 3945
7.4.3. Routine and optional tests 3945
7.4.3.1. The kidneys 3947
7.4.3.2. The heart 3947
7.4.3.3. The arteries 3948
7.4.4. Genetic testing 3948
7.5. Resistant hypertension: definition and diagnosis 3948
7.6. Secondary hypertension: when to screen/further investigations 3950
7.6.1. General considerations 3950
7.6.2. Primary aldosteronism 3950
7.6.3. Renovascular hypertension 3951
7.6.4. Obstructive sleep apnoea syndrome 3951
7.6.5. Phaeochromocytoma/paraganglioma 3954
8. Preventing and treating elevated blood pressure and hypertension 3954
8.1. Prevention strategies in early life 3954
8.2. Non-pharmacological interventions 3954
8.2.1. Dietary sodium and potassium intake 3954
8.2.1.1. Sodium 3954
8.2.1.2. Potassium 3955
8.2.2. Physical activity and exercise 3956
8.2.3. Weight reduction and diet 3957
8.2.4. Alcohol, coffee, and soft drinks 3957
8.2.5. Smoking 3957
8.3. Pharmacological interventions 3958
8.3.1. Treatment strategy to reduce adverse cardiovascular disease outcomes 3958
8.3.2. Drug classes with evidence on clinical outcomes in the target population 3958
8.3.3. New therapies with blood pressure-lowering properties that await supportive evidence from cardiovascular outcomes trials prior to guideline endorsement and routine use in hypertension 3959
8.3.4. Drug combinations and up-titrating strategies 3959
8.3.5. A practical algorithm for intensive, effective, and tolerable blood pressure lowering with drug therapy, including considerations around single-pill combinations 3959
8.3.6. Timing of blood pressure-lowering drug treatment 3962
8.4. Selecting patients for pharmacological blood pressure-lowering treatment 3962
8.5. Intensity of blood pressure-lowering therapy and ideal treatment targets 3964
8.5.1. Expected degree of blood pressure reduction with approved drugs 3964
8.5.2. The ideal target of blood pressure-lowering treatment 3964
8.5.3. Personalizing treatment strategies 3965
8.5.4. Duration and monitoring of drug therapy 3966
8.6. Device-based blood pressure lowering 3966
8.6.1. Catheter-based renal denervation 3966
8.6.2. Other devices 3968
8.7. Unintended and potentially harmful consequences of blood pressure lowering and implications for treatment targets 3968
8.7.1. Adverse effects of blood pressure-lowering medications 3968
8.7.1.1. Symptomatic adverse effects 3968
8.7.1.2. Renal effects 3968
8.7.1.3. Erectile dysfunction 3968
8.7.2. Pill burden and non-adherence 3968
8.7.3. Potentially harmful consequences of blood pressure lowering for frail older people 3968
8.7.4. Clinical inertia in blood pressure lowering 3968
9. Managing specific patient groups or circumstances 3969
9.1. Young adulthood (18–40 years) 3969
9.1.1. Definition and epidemiology 3969
9.1.2. Secondary hypertension in young adulthood 3969
9.1.3. Measurement and management of blood pressure in young adults 3969
9.2. Pregnancy 3969
9.2.1. Definition and epidemiology 3969
9.2.2. Classifying hypertension in pregnancy 3969
9.2.3. Measuring blood pressure in pregnancy 3970
9.2.4. Investigating hypertension in pregnancy 3970
9.2.5. Preventing hypertension and pre-eclampsia 3970
9.2.6. Treatment initiation and blood pressure targets 3970
9.2.7. Managing mild hypertension in pregnancy (office blood pressure 140–159/90–109 mmHg) 3970
9.2.8. Managing severe hypertension in pregnancy (>160/110 mmHg) 3971
9.2.9. Managing blood pressure post-partum 3971
9.2.10. Risk of recurrence of hypertensive disorders in a subsequent pregnancy 3971
9.3. Very old age (≥85 years), frailty, multimorbidity, and polypharmacy 3971
9.3.1. Definition of frailty 3971
9.3.2. Randomized controlled trials of blood pressure lowering in frail older patients 3972
9.3.3. Starting blood pressure-lowering treatment in very old or frail patients 3973
9.3.4. Maintaining blood pressure lowering in very old or frail patients 3973
9.4. Isolated systolic and diastolic hypertension 3973
9.4.1. Definition of isolated systolic hypertension 3973
9.4.2. Isolated systolic hypertension, risk factors, and ageing 3973
9.4.3. Isolated systolic hypertension in young adults 3974
9.4.4. Isolated diastolic hypertension 3974
9.5. Orthostatic hypotension with supine hypertension 3974

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

9.6. Diabetes ... 3975
9.6.1. Diabetes and elevated blood pressure/hypertension ... 3975
9.6.2. J-shaped curve of blood pressure and risk of cardiovascular disease in patients with diabetes ... 3975
9.6.3. Managing blood pressure in diabetes ... 3975
9.7. Chronic kidney disease ... 3975
9.7.1. Relationship between hypertension and chronic kidney disease ... 3975
9.7.2. Blood pressure lowering in chronic kidney disease ... 3976
9.7.3. Managing blood pressure in chronic kidney disease ... 3976
9.7.4. Blood pressure targets in chronic kidney disease ... 3976
9.8. Cardiac disease ... 3976
9.8.1. Blood pressure thresholds and targets in patients with cardiac disease ... 3976
9.8.2. Coronary artery disease with particular reference to the blood pressure J-curve ... 3977
9.8.3. Valvular heart disease ... 3977
9.8.4. Heart failure ... 3977
9.8.5. Heart rhythm disease (including AF) ... 3978
9.9. Chronic cerebrovascular disease and/or cognitive impairment ... 3978
9.9.1. Role of hypertension in chronic cerebrovascular disease 3978
9.9.2. Treatment in patients with history of prior stroke or transient ischaemic attack ... 3978
9.9.3. Treatment in patients with chronic cerebrovascular disease and cognitive impairment ... 3978
9.10. Aortopathy ... 3979
9.10.1. Coarctation of the aorta ... 3979
9.10.2. Bicuspid aortic valve-related aortopathy ... 3979
9.10.3. Preventing aortic dilation and dissection in high-risk patients ... 3979
9.11. Different ethnic groups ... 3979
9.12. Nocturnal hypertension ... 3979
9.12.1. Definition ... 3979
9.12.2. Epidemiology ... 3980
9.12.3. Night-time blood pressure as a cardiovascular disease risk factor ... 3980
9.12.4. Treatment of nocturnal hypertension ... 3980
9.13. Resistant hypertension ... 3980
9.13.1. Definition of resistant hypertension ... 3980
9.13.2. Non-pharmacological interventions ... 3980
9.13.3. Pharmacological interventions ... 3980
9.13.4. Devices for blood pressure lowering ... 3981
9.14. Management of specific causes of secondary hypertension 3982
9.14.1. General considerations ... 3982
9.14.2. Primary aldosteronism ... 3982
9.14.3. Renovascular hypertension ... 3982
9.14.4. Phaeochromocytoma/paraganglioma ... 3983
9.14.5. Obstructive sleep apnoea syndrome ... 3983
9.14.6. Drug-induced hypertension ... 3983
9.14.6.1. Anticancer drug-induced hypertension ... 3983
9.14.7. Other forms of secondary hypertension ... 3983
10. Acute and short-term lowering of blood pressure ... 3983
10.1. Acute blood pressure management in hypertensive emergencies ... 3983
10.1.1. Definition and characteristics of hypertensive emergencies ... 3983
10.1.2. Acute management of hypertensive emergencies ... 3984
10.1.3. Prognosis and follow-up ... 3984
10.2. Acute blood pressure management in acute intracerebral haemorrhage ... 3984
10.3. Acute blood pressure management in acute ischaemic stroke ... 3984
10.4. Acute blood pressure management in pre-eclampsia and severe hypertension in pregnancy ... 3985
10.4.1. Pre-eclampsia ... 3985
10.4.2. Severe acute hypertension in pregnancy ... 3985
10.5. Peri-operative acute management of elevated blood pressure ... 3985
10.5.1. Blood pressure-lowering drugs in the peri-operative phase ... 3985
11. Patient-centred care in hypertension ... 3986
11.1. Definition ... 3986
11.2. Communicating consequences of treatment ... 3986
11.3. Self-measuring and monitoring ... 3987
11.4. Facilitating medication adherence and persistence ... 3987
11.5. Multidisciplinary management ... 3987
12. Key messages ... 3988
13. Gaps in the evidence ... 3989
14. 'What to do' and 'what not to do' messages from the guidelines 3989
15. Evidence tables ... 3993
16. Data availability statement ... 3993
17. Author information ... 3993
18. Appendix ... 3993
19. References ... 3994

Tables of Recommendations

Recommendation Table 1 — Recommendations for measuring blood pressure (see Evidence Tables 1–8) ... 3935
Recommendation Table 2 — Recommendations for categorizing blood pressure (see Evidence Table 9) ... 3936
Recommendation Table 3 — Recommendations for assessing cardiovascular disease risk among individuals with elevated blood pressure (office systolic blood pressure 120–139 mmHg or diastolic blood pressure 70–89 mmHg) (see Evidence Tables 10 and 11) ... 3939
Recommendation Table 4 — Recommendations for refining cardiovascular disease risk (see Evidence Tables 12–14) ... 3941
Recommendation Table 5 — Recommendations for blood pressure screening (see Evidence Table 15) ... 3943
Recommendation Table 6 — Recommendations for confirming hypertension diagnosis ... 3943
Recommendation Table 7 — Recommendations for assessing adherence and persistence with treatment (see Evidence Table 16) 3945
Recommendation Table 8 — Recommendations for assessing renal hypertension-mediated organ damage ... 3947
Recommendation Table 9 — Recommendations for assessing cardiac hypertension-mediated organ damage ... 3947
Recommendation Table 10 — Recommendations for assessing vascular hypertension-mediated organ damage (see Evidence Table 17) ... 3948
Recommendation Table 11 — Recommendations for genetic testing in hypertension management ... 3948
Recommendation Table 12 — Recommendations for resistant hypertension work-up (see Evidence Table 18) ... 3950

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

Recommendation Table 13 — Recommendations for screening for secondary hypertension (see Evidence Tables 19 and 20) 3954
Recommendation Table 14 — Recommendations for screening for hypertension in children and adolescents (see Evidence Table 21) 3954
Recommendation Table 15 — Recommendations for non-pharmacological treatment of blood pressure and cardiovascular risk reduction (see Evidence Tables 22–26) 3958
Recommendation Table 16 — Recommendations for pharmacological treatment of hypertension (see Evidence Tables 27, 28, and 29) 3962
Recommendation Table 17 — Recommendations for initiating blood pressure-lowering treatment (see Evidence Tables 30–32) . 3963
Recommendation Table 18 — Recommendations for blood pressure targets with treatment (see Evidence Table 34) 3966
Recommendation Table 19 — Recommendations for follow-up in patients with treated hypertension (see Evidence Table 33) 3966
Recommendation Table 20 — Recommendations for device-based treatment of hypertension (see Evidence Table 35) 3967
Recommendation Table 21 — Recommendations for managing hypertension in young adults (see Evidence Tables 36 and 37) 3969
Recommendation Table 22 — Recommendations for managing hypertension in pregnancy (see Evidence Tables 38–40) 3971
Recommendation Table 23 — Recommendations for managing hypertension in patients who are very old or frail (see Evidence Table 41) 3973
Recommendation Table 24 — Recommendations for managing hypertension in patients with orthostatic hypotension 3975
Recommendation Table 25 — Recommendations for managing hypertension in patients with diabetes 3975
Recommendation Table 26 — Recommendations for managing hypertension in patients with chronic kidney disease 3976
Recommendation Table 27 — Recommendations for managing hypertension in patients with cardiac disease 3978
Recommendation Table 28 — Recommendations for managing hypertension in patients with chronic cerebrovascular disease and cognitive impairment 3979
Recommendation Table 29 — Recommendations for managing hypertension in different ethnic groups 3979
Recommendation Table 30 — Recommendations for treating resistant hypertension (see Evidence Tables 42 and 43) 3981
Recommendation Table 31 — Recommendations for managing hypertension in patients with renovascular hypertension (see Evidence Tables 44 and 45) 3982
Recommendation Table 32 — Recommendations for acutely managing blood pressure in patients with intracerebral haemorrhage or acute ischaemic stroke 3984
Recommendation Table 33 — Recommendations for acutely managing blood pressure in patients with severe hypertension in pregnancy and pre-eclampsia (see Evidence Table 46) 3985
Recommendation Table 34 — Recommendations for communicating consequences of treatment (see Evidence Table 47) 3987
Recommendation Table 35 — Recommendations for self-measuring and monitoring blood pressure (see Evidence Table 48) 3987
Recommendation Table 36 — Recommendations for multi/interdisciplinary blood pressure management (see Evidence Table 49) 3987

List of tables

Table 1 Classes of recommendations 3918
Table 2 Levels of evidence 3919
Table 3 New recommendations 3920
Table 4 Revised recommendations 3923
Table 5 Comparison of office, home, and ambulatory blood pressure measurement thresholds for elevated blood pressure and hypertension 3933
Table 6 Comparison of ambulatory and home blood pressure monitoring 3934
Table 7 Key illness representations and treatment beliefs: how these apply to communicating a hypertension diagnosis to the patient (note that gender influences these representations) 3945
Table 8 Routine tests recommended in the initial work-up of a patient with elevated blood pressure or hypertension 3946
Table 9 Optional tests that may be used as clinically indicated in the initial work-up of a patient with elevated blood pressure or hypertension to assess hypertension-mediated organ damage or established cardiovascular disease 3947
Table 10 Current definition of resistant hypertension 3950
Table 11 Conditions found to cause pseudo-resistance or resistance to blood pressure-lowering treatment 3950
Table 12 Drugs and conditions that affect aldosterone, renin, and aldosterone-to-renin ratio 3951
Table 13 Optional tests that should be used to screen for secondary hypertension in the presence of suggestive signs, symptoms, or medical history 3952
Table 14 Initiation of blood pressure-lowering treatment based on confirmed blood pressure category and cardiovascular disease risk 3963
Table 15 What to do and what not to do 3989

List of figures

Figure 1 Pathophysiology of elevated blood pressure and hypertension 3928
Figure 2 Persistently elevated blood pressure and hypertension lead to hypertension-mediated organ damage and cardiovascular disease 3929
Figure 3 Summary of office blood pressure measurement 3931
Figure 4 Summary of home blood pressure measurement 3932
Figure 5 Summary of ambulatory blood pressure measurement 3933
Figure 6 Blood pressure categories 3937
Figure 7 Sufficiently high cardiovascular risk conditions that warrant blood pressure-lowering treatment among adults with elevated blood pressure 3938
Figure 8 Cardiovascular disease risk modifiers to consider for up-classification of risk 3940
Figure 9 Summary of cardiovascular disease risk-stratification approach for blood pressure treatment in adults with elevated blood pressure 3942
Figure 10 Protocol for confirming hypertension diagnosis 3944
Figure 11 Definitions, assessments, and potential interventions for the three phases of adherence to BP-lowering medications 3946
Figure 12 Tests and criteria for defining hypertension-mediated organ damage and considerations for their use in clinical practice 3949
Figure 13 Summary of primary aldosteronism as a common form of secondary hypertension 3952

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

Figure 14 Summary of renovascular disease as a common form of secondary hypertension 3953
Figure 15 Summary of obstructive sleep apnoea as a common form of secondary hypertension 3953
Figure 16 Physical activity according to different types of exercise and reduction of blood pressure and overall cardiovascular disease risk 3955
Figure 17 Effects of main lifestyle factors on blood pressure and cardiovascular risk reduction 3956
Figure 18 Practical algorithm for pharmacological blood pressure lowering 3960
Figure 19 Central Illustration 3961
Figure 20 Systolic blood pressure categories and treatment target range 3965
Figure 21 Frailty assessment in the management of blood pressure 3972
Figure 22 Management of resistant hypertension 3981
Figure 23 Patient-centred care 3986
Figure 24 The five dimensions of adherence (WHO, 2003) applied to hypertension 3988

Abbreviations and acronyms

ABI	Ankle–brachial index
ABPM	Ambulatory blood pressure monitoring
ACCORD	Action to Control Cardiovascular Risk in Diabetes
ACE	Angiotensin-converting enzyme
ACR	Albumin:creatinine ratio
AF	Atrial fibrillation
AHI	Apnoea–hypopnoea index
ALARA	As low as reasonably achievable
ALLHAT	Antihypertensive and Lipid-Lowering Treatment to Prevent Heart Attack
AOBP	Automated office blood pressure (measurement)
ARB	Angiotensin receptor blocker
ARNi	Angiotensin receptor-neprilysin inhibitor
ARR	Aldosterone-to-renin ratio
ASCVD	Atherosclerotic cardiovascular disease
BMI	Body mass index
BP	Blood pressure
BSA	Body surface area
CAC	Coronary artery calcium
CAD	Coronary artery disease
CCB	Calcium channel blocker
CHAP	Chronic Hypertension and Pregnancy
CI	Confidence interval
CKD	Chronic kidney disease
CKD-EPI	Chronic Kidney Disease Epidemiology Collaboration
COVID-19	Coronavirus disease 2019
CPAP	Continuous positive airway pressure
CPG	Clinical Practice Guidelines
CT	Computed tomography
CVD	Cardiovascular disease
DASH	Dietary Approaches to Stop Hypertension
DBP	Diastolic blood pressure
DECIDE-Salt	Diet, ExerCIse and carDiovascular hEalth–Salt
EACTS	European Association for Cardio-Thoracic Surgery
ECG	Electrocardiogram
eGFR	Estimated glomerular filtration rate
EPIC	European Prospective Investigation into Cancer and Nutrition
ESC	European Society of Cardiology
ESH	European Society of Hypertension
ESPRIT	Effects of intensive Systolic blood Pressure lowering treatment in reducing RIsk of vascular evenTs
FMD-RVH	Fibromuscular dysplasia-induced renovascular hypertension
GFR	Glomerular filtration rate
GLP-1	Glucagon-like peptide-1
GP	General practitioner
HbA1c	Glycated haemoglobin
HBPM	Home blood pressure monitoring
HDL	High-density lipoprotein
HFpEF	Heart failure with preserved ejection fraction
HF(m)rEF	Heart failure with (mildly) reduced ejection fraction
HIV	Human immunodeficiency virus
HMOD	Hypertension-mediated organ damage
i.m.	Intramuscular
i.v.	Intravenous
KDIGO	Kidney Disease: Improving Global Outcomes
LA	Left atrial
LDL	Low-density lipoprotein
LV	Left ventricular
LVH	Left ventricular hypertrophy
MRA	Mineralocorticoid receptor antagonist
MRI	Magnetic resonance imaging
NNT	Number needed to treat
NT-proBNP	N-terminal pro-brain natriuretic peptide
OSAS	Obstructive sleep apnoea syndrome
PPGL	Phaeochromocytoma/paraganglioma
PREOP-ACEI	Prospective Randomized Evaluation of Preoperative Angiotensin-Converting Enzyme Inhibition
PREMs	Patient-Reported Experience Measures
PROMS	Patient-Reported Outcome Measures
PTRA	Percutaneous transluminal renal angioplasty
PWV	Pulse wave velocity
RAAS	Renin-angiotensin-aldosterone system
RADIANCE-HTN	A Study of the Recor Medical Paradise System in Clinical Hypertension
RAS	Renin–angiotensin system
RCT	Randomized controlled trial
RVH	Renovascular hypertension
RWT	Relative wall thickness
SBP	Systolic blood pressure
SCORE2	Systematic COronary Risk Evaluation 2
SCORE2-OP	Systematic COronary Risk Evaluation 2–Older Persons
SGLT2	Sodium–glucose co-transporter 2
SNP	Single-nucleotide polymorphism
SNS	Sympathetic nervous system
SPC	Single-pill combination
SPRINT	Systolic Blood Pressure Intervention Trial
SSaSS	Salt Substitute and Stroke Study
STEP	Strategy of Blood Pressure Intervention in Elderly Hypertensive Patients
STEP-1	Semaglutide Treatment Effect in People with Obesity

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

TIA	Transient ischaemic attack
TRIUMPH	Treating Resistant Hypertension Using Lifestyle Modification to Promote Health
TSH	Thyroid-stimulating hormone
WHO	World Health Organization
WML	White matter lesion

1. Preamble

Guidelines evaluate and summarize available evidence with the aim of assisting health professionals in proposing the best diagnostic or therapeutic approach for an individual patient with a given condition. Guidelines are intended for use by health professionals and the European Society of Cardiology (ESC) makes its guidelines freely available.

ESC Guidelines do not override the individual responsibility of health professionals to make appropriate and accurate decisions in consideration of each patient's health condition and in consultation with that patient or the patient's caregiver where appropriate and/or necessary. It is also the health professional's responsibility to verify the rules and regulations applicable in each country to drugs and devices at the time of prescription and to respect the ethical rules of their profession.

ESC Guidelines represent the official position of the ESC on a given topic and are regularly updated when warranted by new evidence. ESC Policies and Procedures for formulating and issuing ESC Guidelines can be found on the ESC website (https://www.escardio.org/Guidelines/Clinical-Practice-Guidelines/Guidelines-development/Writing-ESC-Guidelines). This guideline version updates and replaces the previous version from 2018.

The Members of this task force were selected by the ESC to include professionals involved in the medical care of patients with this pathology, as well as patient representatives and methodologists. The selection procedure included an open call for authors and aimed to include members from across the whole of the ESC region and from relevant ESC Subspecialty Communities. Consideration was given to diversity and inclusion, notably with respect to gender and country of origin. The task force performed a critical review and evaluation of the published literature on diagnostic and therapeutic approaches including assessment of the risk-benefit ratio. The strength of every recommendation and the level of evidence supporting them were weighed and scored according to predefined scales as outlined in *Tables 1* and *2* below. Patient-Reported Outcome Measures (PROMs) and Patient-Reported Experience Measures (PREMs) were also evaluated as the basis for recommendations and/or discussion in these guidelines. The task force followed ESC voting procedures and all approved recommendations were subject to a vote and achieved at least 75% agreement among voting members. Members of the task force with declared interests on specific topics were asked to abstain from voting on related recommendations.

The experts of the writing and reviewing panels provided declaration of interest forms for all relationships that might be perceived as real or potential sources of conflicts of interest. Their declarations of interest were reviewed according to the ESC declaration of interest rules, which can be found on the ESC website (http://www.escardio.org/guidelines) and have been compiled in a report published in a supplementary document with the guidelines. Funding for the development of ESC Guidelines is derived entirely from the ESC with no involvement of the healthcare industry.

Table 1 Classes of recommendations

Classes of recommendations	Definition	Wording to use
Class I	Evidence and/or general agreement that a given treatment or procedure is beneficial, useful, effective.	Is recommended or is indicated
Class II	Conflicting evidence and/or a divergence of opinion about the usefulness/efficacy of the given treatment or procedure.	
Class IIa	Weight of evidence/opinion is in favour of usefulness/efficacy.	Should be considered
Class IIb	Usefulness/efficacy is less well established by evidence/opinion.	May be considered
Class III	Evidence or general agreement that the given treatment or procedure is not useful/effective, and in some cases may be harmful.	Is not recommended

©ESC 2024

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

Table 2 **Levels of evidence**

Level of evidence A	Data derived from multiple randomized clinical trials or meta-analyses.
Level of evidence B	Data derived from a single randomized clinical trial or large non-randomized studies.
Level of evidence C	Consensus of opinion of the experts and/or small studies, retrospective studies, registries.

©ESC 2024

The ESC Clinical Practice Guidelines (CPG) Committee supervises and co-ordinates the preparation of new guidelines and is responsible for the approval process. In addition to review by the CPG Committee, ESC Guidelines undergo multiple rounds of double-blind peer review by external experts, including members from across the whole of the ESC region, all National Cardiac Societies of the ESC and from relevant ESC Subspecialty Communities. After appropriate revisions, the guidelines are signed off by all the experts in the task force. The finalized document is signed off by the CPG Committee for publication in the *European Heart Journal*.

ESC Guidelines are based on analyses of published evidence, chiefly on clinical trials and meta-analyses of trials, but potentially including other types of studies. Evidence tables summarizing key information from relevant studies are generated early in the guideline development process to facilitate the formulation of recommendations, to enhance comprehension of recommendations after publication, and reinforce transparency in the guidelines development process. The tables are published in their own section of ESC Guidelines and reference specific recommendation tables.

Off-label use of medication may be presented in these guidelines if a sufficient level of evidence shows that it can be considered medically appropriate for a given condition. However, the final decisions concerning an individual patient must be made by the responsible health professional giving special consideration to:

- The specific situation of the patient. Unless otherwise provided for by national regulations, off-label use of medication should be limited to situations where it is in the patient's interest with regard to the quality, safety, and efficacy of care, and only after the patient has been informed and has provided consent;
- Country-specific health regulations, indications by governmental drug regulatory agencies, and the ethical rules to which health professionals are subject, where applicable.

2. Introduction

This 2024 document updates the 2018 ESC/European Society of Hypertension (ESH) Guidelines on the management of arterial hypertension.[1] While the current document builds on prior guidelines, it also incorporates important updates and new recommendations based on current evidence. For example:

(1) The title has changed from 'Guidelines on the management of arterial hypertension' to 'Guidelines on the management of elevated blood pressure and hypertension'. This is based on evidence that the risk for cardiovascular disease (CVD) attributable to blood pressure (BP) is on a continuous exposure scale, not a binary scale of normotension vs. hypertension.[2,3] Updated evidence also increasingly demonstrates the benefit on CVD outcomes of BP-lowering medications among persons with high CVD risk and BP levels that are elevated but that do not meet traditional thresholds used to define hypertension. The term 'arterial' is removed from the title of the 2024 Guidelines, as arterial hypertension can also occur in the pulmonary arteries, which is not a focus here.

(2) The 2024 Guidelines continue to define hypertension as office systolic BP of ≥140 mmHg or diastolic BP of ≥90 mmHg. However, a new BP category called 'Elevated BP' is introduced. Elevated BP is defined as an office systolic BP of 120–139 mmHg or diastolic BP of 70–89 mmHg.

(3) A major, evidence-based change in the 2024 Guidelines is the recommendation to pursue a target systolic BP of 120–129 mmHg among adults receiving BP-lowering medications. There are several important caveats to this recommendation, including: (i) the requirement that treatment to this BP target is well tolerated by the patient, (ii) the fact that more lenient BP targets can be considered in persons with symptomatic orthostatic hypotension, those aged 85 years or over, or those with moderate-to-severe frailty

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

or limited life expectancy, and (iii) a strong emphasis on out-of-office BP measurement to confirm the systolic BP target of 120–129 mmHg is achieved. For those selected individual cases where a target systolic BP of 120–129 mmHg is not pursued, either due to intolerance or the existence of conditions that favour a more lenient BP target, we recommend targeting a BP that is as low as reasonably achievable. Personalized clinical decision-making and shared decisions with the patient are also emphasized.

(4) Another important change in the 2024 Guidelines compared with earlier versions is the increased focus on evidence related to fatal and non-fatal CVD outcomes rather than surrogate outcomes such as BP lowering alone. Except for lifestyle interventions and low-risk non-pharmacological interventions aimed at implementation or care delivery, the current guidelines require that, for a Class I recommendation to be made for a drug or procedural intervention, the evidence must show benefit on CVD outcomes and not only BP lowering.

(5) The task force comprised of a balanced representation of males and females.

(6) The present guidelines consider sex and gender as an integral component throughout the document, rather than in a separate section at the end. In this document, sex is the biological condition of being female or male from conception, based on genes, and gender is the socio-cultural dimension of being a woman or a man in a given society, based on gender roles, gender norms, gender identity, and gender relations valid in the respective society at a given timepoint.[4,5]

(7) The 2024 Guidelines are written to make them more 'user friendly'. Input from general practitioners (GPs) was obtained in this regard, and one task force member is a GP. Given the ageing population in Europe, there was also a focus on tailoring treatment with respect to frailty and into older age, which is addressed in multiple sections. Moreover, patient input and their lived experiences are considered throughout. We also now include evidence tables in the Supplementary section to provide improved transparency regarding our recommendations. As appropriate, readers who wish to seek additional details and information are referred to the Supplementary data online and to the ESC CardioMed.[6]

(8) The task force recognized that a major challenge in guideline usage is poor implementation. This likely contributes to suboptimal control of hypertension.[7–9] To address this, a dedicated section on implementation is included in the Supplementary data online. Moreover, through a new initiative, we include information from national societies following a survey on guideline implementation completed during the national society peer review of the guidelines document. It is hoped this information may help inform national societies about potential barriers to implementation.

2.1. What is new

These 2024 Guidelines contain a number of new and revised recommendations, which are summarized in *Tables 3* and *4*, respectively.

Table 3 New recommendations

Recommendations	Class[a]	Level[b]
5. Measuring blood pressure		
It is recommended to measure BP using a validated and calibrated device, to enforce the correct measurement technique, and to apply a consistent approach to BP measurement for each patient.	I	B
Out-of-office BP measurement is recommended for diagnostic purposes, particularly because it can detect both white-coat hypertension and masked hypertension. Where out-of-office measurements are not logistically and/or economically feasible, then it is recommended that the diagnosis be confirmed with a repeat office BP measurement using the correct standardized measurement technique.	I	B
Most automated oscillometric monitors have not been validated for BP measurement in AF; BP measurement should be considered using a manual auscultatory method in these circumstances, where possible.	IIa	C
An assessment for orthostatic hypotension (≥20 systolic BP and/or ≥10 diastolic BP mmHg drop at 1 and/or 3 min after standing) should be considered at least at the initial diagnosis of elevated BP or hypertension and thereafter if suggestive symptoms arise. This should be performed after the patient is first lying or sitting for 5 min.	IIa	C
6. Definition and classification of elevated blood pressure and hypertension, and cardiovascular disease risk assessment		
It is recommended to use a risk-based approach in the treatment of elevated BP, and individuals with moderate or severe CKD, established CVD, HMOD, diabetes mellitus, or familial hypercholesterolaemia are considered at increased risk for CVD events.	I	B
It is recommended that, irrespective of age, individuals with elevated BP and a SCORE2 or SCORE2-OP CVD risk of ≥10% be considered at increased risk for CVD for the purposes of risk-based management of their elevated BP.	I	B
SCORE2-Diabetes should be considered to estimate CVD risk among type 2 diabetes mellitus patients with elevated BP, particularly if they are <60 years of age.	IIa	B
History of pregnancy complications (gestational diabetes, gestational hypertension, pre-term delivery, pre-eclampsia, one or more stillbirths, and recurrent miscarriage) are sex-specific risk modifiers that should be considered to up-classify individuals with elevated BP and borderline increased 10-year CVD risk (5% to <10% risk).	IIa	B
High-risk ethnicity (e.g. South Asian), family history of premature onset atherosclerotic CVD, socio-economic deprivation, auto-immune inflammatory disorders, HIV, and severe mental illness are risk modifiers shared by both sexes that should be considered to up-classify individuals with elevated BP and borderline increased 10-year CVD risk (5% to <10% risk).	IIa	B
After assessing 10-year predicted CVD risk and non-traditional CVD risk modifiers, if a risk-based BP-lowering treatment decision remains uncertain for individuals with elevated BP, measuring CAC score, carotid or femoral plaque using ultrasound, high-sensitivity cardiac troponin or B-type natriuretic peptide biomarkers, or arterial stiffness using pulse wave velocity, may be considered to improve risk stratification among patients with borderline increased 10-year CVD risk (5% to <10% risk) after shared decision-making and considering costs.	IIb	B

Continued

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

7. Diagnosing hypertension and investigating underlying causes		
Opportunistic screening for elevated BP and hypertension should be considered: • At least every 3 years for adults aged <40 years. • At least annually for adults aged ≥40 years.	IIa	C
In individuals with elevated BP who do not currently meet risk thresholds for BP-lowering treatment, a repeat BP measurement and risk assessment within 1 year should be considered.	IIa	C
Other forms of screening for hypertension (i.e. systematic screening, self-screening, and non-physician screening) may be considered, depending on their feasibility in different countries and healthcare systems.	IIb	B
In individuals with increased CVD risk where their screening office BP is 120–139/70–89 mmHg, it is recommended to measure BP out of office, using ABPM and/or HBPM or, if not logistically feasible, make repeated office BP measurements on more than one visit.	I	B
Objective evaluation of adherence (either directly observed treatment or detecting prescribed drugs in blood or urine samples) should be considered in the clinical work-up of patients with apparent resistant hypertension, if resources allow.	IIa	B
If moderate-to-severe CKD is diagnosed, it is recommended to repeat measurements of serum creatinine, eGFR, and urine ACR at least annually.	I	C
Coronary artery calcium scoring may be considered in patients with elevated BP or hypertension when it is likely to change patient management.	IIb	B
Patients with resistant hypertension should be considered for referral to clinical centres with expertise in hypertension management for further testing.	IIa	B
It is recommended that patients with hypertension presenting with suggestive signs, symptoms, or medical history of secondary hypertension are appropriately screened for secondary hypertension.	I	B
Screening for primary aldosteronism by renin and aldosterone measurements should be considered in all adults with confirmed hypertension (BP ≥ 140/90 mmHg).	IIa	B
8. Preventing and treating elevated blood pressure		
Opportunistic screening with office BP measurements to monitor development of BP during late childhood and adolescence, especially if one or both parents have hypertension, should be considered to better predict development of adult hypertension and associated CVD risk.	IIa	B
It is recommended to restrict free sugar consumption, in particular sugar-sweetened beverages, to a maximum of 10% of energy intake. It is also recommended to discourage consumption of sugar-sweetened beverages, such as soft drinks and fruit juices, starting at young age.	I	B
In patients with hypertension without moderate-to-advanced CKD and with high daily sodium intake, an increase of potassium intake by 0.5–1.0 g/day—for example through sodium substitution with potassium-enriched salt (comprising 75% sodium chloride and 25% potassium chloride) or through diets rich in fruits and vegetables—should be considered.	IIa	A
In patients with CKD or taking potassium-sparing medication, such as some diuretics, ACE inhibitors, ARBs, or spironolactone, monitoring serum levels of potassium should be considered if dietary potassium is being increased.	IIa	C
It is recommended to take medications at the most convenient time of day for the patient, to establish a habitual pattern of medication taking to improve adherence.	I	B
In adults with elevated BP and low/medium CVD risk (<10% over 10 years), BP lowering with lifestyle measures is recommended and can reduce the risk of CVD.	I	B
In adults with elevated BP and sufficiently high CVD risk, after 3 months of lifestyle intervention, BP lowering with pharmacological treatment is recommended for those with confirmed BP ≥130/80 mmHg to reduce CVD risk.	I	A
It is recommended that in hypertensive patients with confirmed BP ≥140/90 mmHg, irrespective of CVD risk, lifestyle measures and pharmacological BP-lowering treatment is initiated promptly to reduce CVD risk.	I	A
It is recommended to maintain BP-lowering drug treatment lifelong, even beyond the age of 85 years, if well tolerated.	I	A
Because the benefit in reducing CVD outcomes is uncertain in these settings, and noting that close monitoring of treatment tolerance is advised, BP-lowering treatment should only be considered from ≥140/90 mmHg (office) among persons meeting the following criteria: • pre-treatment symptomatic orthostatic hypotension; • age ≥85 years; • clinically significant moderate-to-severe frailty; • and/or limited predicted lifespan (<3 years).	IIa	B
In cases where BP-lowering treatment is poorly tolerated and achieving a target systolic of 120–129 mmHg is not possible, it is recommended to target a systolic BP level that is 'as low as reasonably achievable' (ALARA principle).	I	A
Once BP is controlled and stable under BP-lowering therapy, at least a yearly follow-up for BP and other CVD risk factors should be considered.	IIa	C

Continued

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

9. Managing specific patient groups or circumstances		
Young adults		
Comprehensive screening for the main causes of secondary hypertension is recommended in adults diagnosed with hypertension before the age of 40 years, except for obese young adults where it is recommended to start with an obstructive sleep apnoea evaluation.	I	B
Since SCORE2 has not been validated for individuals <40 years, screening for HMOD may be considered in such young individuals with elevated BP without other increased CVD risk conditions to identify additional individuals for possible medical treatment.	IIb	B
Hypertension in pregnancy		
In consultation with an obstetrician, low- to moderate-intensity exercise is recommended in all pregnant women without contraindications to reduce the risk of gestational hypertension and pre-eclampsia.	I	B
HBPM and ABPM should be considered to exclude white-coat and masked hypertension, which are more common in pregnancy.	IIa	C
Older and frail patients		
It is recommended that treatment of elevated BP and hypertension among older patients aged <85 years who are not moderately to severely frail follows the same guidelines as for younger people, provided BP-lowering treatment is well tolerated.	I	A
When initiating BP-lowering treatment for patients aged ≥85 years, and/or with moderate-to-severe frailty (at any age), long-acting dihydropyridine CCBs or RAS inhibitors should be considered, followed, if necessary, by a low-dose diuretic if tolerated, but preferably not a beta-blocker (unless compelling indications exist) or an alpha-blocker.	IIa	B
As the safety and efficacy of BP treatment is less certain in individuals with moderate or severe frailty, clinicians should consider screening older adults for frailty using validated clinical tests; frail patients' health priorities and a shared-decision approach should be considered when deciding on BP treatments and targets.	IIa	C
If BP drops with progressing frailty, deprescription of BP-lowering medications (and other drugs that can reduce BP, such as sedatives and prostate-specific alpha-blockers) may be considered.	IIb	C
Hypertension and orthostatic hypotension		
Before starting or intensifying BP-lowering medication, it is recommended to test for orthostatic hypotension, by first having the patient sit or lie for 5 min and then measuring BP 1 and/or 3 min after standing.	I	B
It is recommended to pursue non-pharmacological approaches as the first-line treatment of orthostatic hypotension among persons with supine hypertension. For such patients, it is also recommended to switch BP-lowering medications that worsen orthostatic hypotension to an alternative BP-lowering therapy and not to simply de-intensify therapy.	I	A
Chronic kidney disease		
In hypertensive patients with CKD and eGFR >20 mL/min/1.73 m^2, SGLT2 inhibitors are recommended to improve outcomes in the context of their modest BP-lowering properties.	I	A
Other conditions		
BP-lowering drug treatment is recommended for people with pre-diabetes or obesity when confirmed office BP is ≥140/90 mmHg or when office BP is 130–139/80–89 mmHg and the patient is at predicted 10-year risk of CVD ≥10% or with high-risk conditions, despite a maximum of 3 months of lifestyle therapy.	I	A
In patients with a history of aortic valve stenosis and/or regurgitation who require BP-lowering treatment, RAS blockers should be considered as part of that treatment.	IIa	C
In patients with a history of moderate-to-severe mitral valve regurgitation who require BP-lowering treatment, RAS blockers should be considered as part of that treatment.	IIa	C
Renovascular hypertension		
Renal artery angioplasty without stenting should be considered for patients with hypertension and haemodynamically significant renal artery stenosis due to fibromuscular dysplasia.	IIa	C
Renal artery angioplasty and stenting may be considered in patients with haemodynamically significant, atherosclerotic, renal artery stenosis (stenosis of 70%–99%, or 50%–69% with post-stenotic dilatation and/or significant trans-stenotic pressure gradient) with: • Recurrent heart failure, unstable angina, or sudden-onset flash pulmonary oedema despite maximally tolerated medical therapy; • Resistant hypertension; • Hypertension with unexplained unilaterally small kidney or CKD; • Bilateral renal artery stenosis or unilateral renal artery stenosis in a solitary viable kidney.	IIb	C
Renal artery angioplasty is not recommended in patients without confirmed haemodynamically significant renal artery stenosis.	III	A
10. Acute and short-term lowering of blood pressure		
In patients with intracerebral haemorrhage presenting with systolic BP ≥220 mmHg, acute reduction in systolic BP >70 mmHg from initial levels within 1 h of commencing treatment is not recommended.	III	B

Continued

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

11. Patient-centred care in hypertension		
An informed discussion about CVD risk and treatment benefits tailored to the needs of a patient is recommended as part of hypertension management.	I	C
Motivational interviewing should be considered for patients with hypertension at hospitals and community health centres to assist patients in controlling their BP and to enhance treatment adherence.	IIa	B
Physician–patient web communications are an effective tool that should be considered in primary care, including reporting on home BP readings.	IIa	C
Home BP measurement for managing hypertension by using self-monitored BP is recommended to achieve better BP control.	I	B
Self-measurement, when properly performed, is recommended due to positive effects on the acceptance of a diagnosis of hypertension, patient empowerment, and adherence to treatment.	I	C
Enhanced self-monitoring of BP using a device paired with a connected smartphone application may be considered, though evidence to date suggests that this may be no more effective than standard self-monitoring.	IIb	B
Multidisciplinary approaches in the management of patients with elevated BP and hypertension, including appropriate and safe task-shifting away from physicians are recommended to improve BP control.	I	A

ABPM, ambulatory blood pressure monitoring; ACE, angiotensin-converting enzyme; ACR, albumin-to-creatinine ratio (urine); AF, atrial fibrillation; ALARA, as low as reasonably achievable; ARB, angiotensin receptor blocker; BP, blood pressure; CAC, coronary artery calcium; CCB, calcium channel blocker; CKD, chronic kidney disease; CVD, cardiovascular disease; eGFR, estimated glomerular filtration rate; HBPM, home blood pressure monitoring; HFpEF, heart failure with preserved ejection fraction; HIV, human immunodeficiency virus; HMOD, hypertension-mediated organ damage; MRA, mineralocorticoid receptor antagonist; RAS, renin–angiotensin system; SCORE2, Systematic COronary Risk Evaluation 2; SCORE2-OP, Systematic COronary Risk Evaluation 2–Older Persons; SGLT2, sodium–glucose co-transporter 2.
[a]Class of recommendation.
[b]Level of evidence.

Table 4 Revised recommendations

Recommendations in 2018 version	Class[a]	Level[b]	Recommendations in 2024 version	Class[a]	Level[b]
6. Definition and classification of elevated blood pressure and hypertension					
It is recommended that BP be classified as optimal, normal, high–normal, or grades 1–3 hypertension, according to office BP.	I	C	It is recommended that BP be categorized as non-elevated BP, elevated BP, and hypertension to aid treatment decisions.	I	B
CV risk assessment with the SCORE system is recommended for hypertensive patients who are not already at high or very high risk due to established CVD, renal disease, or diabetes, a markedly elevated single risk factor (e.g. cholesterol), or hypertensive LVH.	I	B	SCORE2 is recommended for assessing 10-year risk of fatal and non-fatal CVD among individuals aged 40–69 years with elevated BP who are not already considered at increased risk due to moderate or severe CKD, established CVD, HMOD, diabetes mellitus, or familial hypercholesterolaemia.	I	B
			SCORE2-OP is recommended for assessing the 10-year risk of fatal and non-fatal CVD among individuals aged ≥70 years with elevated BP who are not already considered at increased risk due to moderate or severe CKD, established CVD, HMOD, diabetes mellitus, or familial hypercholesterolaemia.	I	B
7. Diagnosing hypertension and investigating underlying causes					
It is recommended that the diagnosis of hypertension should be based on: • Repeated office BP measurements on more than one visit, except when hypertension is severe (e.g. grade 3 and especially in high-risk patients). At each visit, three BP measurements should be recorded, 1–2 min apart, and additional measurements should be performed if the first two readings differ by >10 mmHg. The patient's BP is the average of the last two BP readings. Or • Out-of-office BP measurement with ABPM and/or HBPM, provided that these measurements are logistically and economically feasible.	I	C	Where screening office BP is 140–159/90–99 mmHg, it is recommended that the diagnosis of hypertension should be based on out-of-office BP measurement with ABPM and/or HBPM. If these measurements are not logistically or economically feasible, then diagnosis can be made on repeated office BP measurements on more than one visit.	I	B
			Where screening office BP is ≥160/100 mmHg: • It is recommended that BP 160–179/100–109 mmHg be confirmed as soon as possible (e.g. within 1 month) preferably by either home or ambulatory BP measurements. • It is recommended when BP ≥180/110 mmHg that hypertensive emergency be excluded.	I	C

Continued

© ESC 2024
Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

Echocardiography is recommended in hypertensive patients when there are ECG abnormalities or signs or symptoms of LV dysfunction.	I	B	Echocardiography is recommended in patients with hypertension and ECG abnormalities, or signs or symptoms of cardiac disease.	I	B
Echocardiography may be considered when the detection of LVH may influence treatment decisions.	IIb	B	Echocardiography may be considered in patients with elevated BP, particularly when it is likely to change patient management.	IIb	B
Ultrasound examination of the carotid arteries may be considered for the detection of asymptomatic atherosclerotic plaques or carotid stenosis in patients with documented vascular disease elsewhere.	IIb	B	Ultrasound examination of the carotid or femoral arteries for detecting plaque may be considered in patients with elevated BP or hypertension when it is likely to change patient management.	IIb	B
Measurement of PWV may be considered for measuring arterial stiffness.	IIb	B	Measurement of PWV may be considered in patients with elevated BP or hypertension when it is likely to change patient management.	IIb	B
8. Preventing and treating elevated blood pressure					
Regular aerobic exercise (e.g. at least 30 min of moderate dynamic exercise on 5–7 days/week) is recommended.	I	A	Moderate intensity aerobic exercise of ≥150 min/week (≥30 min, 5–7 days/week) or alternatively 75 min of vigorous intensity aerobic exercise per week over 3 days are recommended and should be complemented with low- or moderate-intensity dynamic or isometric resistance training (2–3 times/week) to reduce BP and CVD risk.	I	A
Body-weight control is indicated to avoid obesity (BMI >30 kg/m^2 or waist circumference >102 cm in men and >88 cm in women), as is aiming at healthy BMI (about 20–25 kg/m^2) and waist circumference values (<94 cm in men and <80 cm in women) to reduce BP and CV risk.	I	A	It is recommended to aim for a stable and healthy BMI (20–25 kg/m^2) and waist circumference values (<94 cm in men and <80 cm in women) to reduce BP and CVD risk.	I	A
Increased consumption of vegetables, fresh fruits, fish, nuts, and unsaturated fatty acids (olive oil); low consumption of red meat; and consumption of low-fat dairy products are recommended.	I	A	Adopting a healthy and balanced diet, such as the Mediterranean or DASH diets, is recommended to help reduce BP and CVD risk.	I	A
It is recommended to restrict alcohol consumption to: • Less than 14 units/week for men. • Less than 8 units/week for women.	I	A	Men and women are recommended to drink less alcohol than the upper limit, which is about 100 g/week of pure alcohol. How this translates into number of drinks depends on portion size (the standards of which differ per country), but most drinks contain 8–14 g of alcohol per drink. Preferably, it is recommended to avoid alcohol to achieve the best health outcomes.	I	B
Among all anti-hypertensive drugs, ACE inhibitors, ARBs, beta-blockers, CCBs, and diuretics (thiazides and thiazide-like drugs, such as chlorthalidone and indapamide) have demonstrated effective reduction of BP and CV events in RCTs, and thus are indicated as the basis of anti-hypertensive treatment strategies.	I	A	Among all BP-lowering drugs, ACE inhibitors, ARBs, dihydropyridine CCBs, and diuretics (thiazides and thiazide-like drugs such as chlorthalidone and indapamide) have demonstrated the most effective reduction of BP and CVD events, and are therefore recommended as first-line treatments to lower BP.	I	A
It is recommended that if BP is not controlled with a three-drug combination, treatment should be increased by the addition of spironolactone or, if not tolerated, other diuretics such as amiloride or higher doses of other diuretics, a beta-blocker, or an alpha-blocker.	I	B	If BP is not controlled with a three-drug combination and in whom spironolactone is not effective or tolerated, treatment with eplerenone instead of spironolactone, or the addition of a beta-blocker if not already indicated and, next, a centrally acting BP-lowering medication, an alpha-blocker, hydralazine, or a potassium-sparing diuretic should be considered.	IIa	B
8. Preventing and treating elevated blood pressure (blood pressure targets)					
It is recommended that the first objective of treatment should be to lower BP to <140/90 mmHg in all patients and, provided that the treatment is well tolerated, treated BP values should be targeted to 130/80 mmHg or lower in most patients.	I	A	To reduce CVD risk, it is recommended that treated systolic BP values in most adults be targeted to 120–129 mmHg, provided the treatment is well tolerated.	I	A

Continued

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

A diastolic BP target of <80 mmHg should be considered for all hypertensive patients, independent of the level of risk and comorbidities.	IIa	B	In cases where on-treatment systolic BP is at or below target (120–129 mmHg) but diastolic BP is not at target (≥80 mmHg), intensifying BP-lowering treatment to achieve an on-treatment diastolic BP of 70–79 mmHg may be considered to reduce CVD risk.	IIb	C
In older patients (aged ≥65 years) receiving BP-lowering drugs: • It is recommended that systolic BP should be targeted to a BP range of 130–139 mmHg.	I	A	Because the CVD benefit of an on-treatment systolic BP target of 120–129 mmHg may not generalize to the following specific settings, personalized and more lenient systolic BP targets (e.g. <140 mmHg): should be considered among patients meeting the following criteria: • pre-treatment, symptomatic, orthostatic hypotension; • and/or age ≥85 years.	IIa	C
			Because the CVD benefit of an on-treatment systolic BP target of 120–129 mmHg may not generalize to the following specific settings, personalized and more lenient BP targets (e.g. <140/90 mmHg) may be considered among patients meeting the following criteria: • clinically significant, moderate to severe frailty at any age; • and/or limited predicted lifespan (<3 years).	IIb	C
8. Preventing and treating elevated blood pressure (renal denervation)					
Use of device-based therapies is not recommended for the routine treatment of hypertension, unless in the context of clinical studies and RCTs, until further evidence regarding their safety and efficacy becomes available.	III	B	To reduce BP, and if performed at a medium-to-high volume centre, catheter-based renal denervation may be considered for resistant hypertension patients who have BP that is uncontrolled despite a three BP-lowering drug combination (including a thiazide or thiazide-like diuretic), and who express a preference to undergo renal denervation after a shared risk-benefit discussion and multidisciplinary assessment.	IIb	B
			To reduce BP, and if performed at a medium-to-high volume centre, catheter-based renal denervation may be considered for patients with both increased CVD risk and uncontrolled hypertension on fewer than three drugs, if they express a preference to undergo renal denervation after a shared risk-benefit discussion and multidisciplinary assessment.	IIb	A
			Due to a lack of adequately powered outcomes trials demonstrating its safety and CVD benefits, renal denervation is not recommended as a first-line BP-lowering intervention for hypertension.	III	C
			Renal denervation is not recommended for treating hypertension in patients with moderately to severely impaired renal function (eGFR <40 mL/min/1.73 m^2) or secondary causes of hypertension, until further evidence becomes available.	III	C
9.1. Managing specific patient groups or circumstances					
Hypertension in pregnancy					
In women with gestational hypertension, pre-existing hypertension superimposed by gestational hypertension, or with hypertension and subclinical organ damage or symptoms, initiation of drug treatment is recommended when systolic BP is ≥140 mmHg or diastolic BP ≥90 mmHg.	I	C	In women with gestational hypertension, starting drug treatment is recommended for those with confirmed systolic BP ≥140 mmHg or diastolic BP ≥90 mmHg.	I	B

Continued

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

In all other cases, initiation of drug treatment is recommended when systolic BP is ≥150 mmHg or diastolic BP is ≥95 mmHg.	I	C	In pregnant women with chronic hypertension, starting drug treatment is recommended for those with confirmed office systolic BP ≥140 mmHg or diastolic BP ≥90 mmHg.	I	B
			In women with chronic and gestational hypertension, it is recommended to lower BP below 140/90 mmHg but not below 80 mmHg for diastolic BP.	I	C
Systolic BP ≥170 mmHg or diastolic BP ≥110 mmHg in a pregnant woman is an emergency, and admission to hospital is recommended.	I	C	Systolic BP ≥160 mmHg or diastolic BP ≥110 mmHg in pregnancy can indicate an emergency, and immediate hospitalization should be considered.	IIa	C
Diabetes					
Antihypertensive drug treatment is recommended for people with diabetes when office BP is ≥140/90 mmHg.	I	A	In most adults with elevated BP and diabetes, after a maximum of 3 months of lifestyle intervention, BP lowering with pharmacological treatment is recommended for those with confirmed office BP ≥130/80 mmHg to reduce CVD risk.	I	A
In people with diabetes receiving BP-lowering drugs it is recommended: • To target SBP to 130 mmHg and <130 mmHg if tolerated, but not <120 mmHg. • In older people (aged ≥65 years aged), to target to an SBP range of 130–139 mmHg.	I	A	In persons with diabetes who are receiving BP-lowering drugs, it is recommended to target systolic BP to 120–129 mmHg, if tolerated.	I	A
Chronic kidney disease					
In patients with diabetic or non-diabetic CKD, it is recommended that an office BP ≥140/90 mmHg be treated with lifestyle advice and BP-lowering medication.	I	A	In patients with diabetic or non-diabetic moderate-to-severe CKD and confirmed BP ≥130/80 mmHg, lifestyle optimization and BP-lowering medication are recommended to reduce CVD risk, provided such treatment is well tolerated.	I	A
In patients with diabetic or non-diabetic CKD:			In adults with moderate-to-severe CKD who are receiving BP-lowering drugs and who have eGFR >30 mL/min/1.73 m^2, it is recommended to target systolic BP to 120–129 mmHg, if tolerated. Individualized BP targets are recommended for those with lower eGFR or renal transplantation.	I	A
• It is recommended to lower systolic BP to a range of 130–139 mmHg.	I	A			
• Individualized treatment should be considered according to its tolerability and impact on renal function and electrolytes.	IIa	C			
RAS blockers are more effective at reducing albuminuria than other antihypertensive agents, and are recommended as part of the treatment strategy in hypertensive patients in the presence of microalbuminuria or proteinuria.	I	A	ACE inhibitors or ARBs are more effective at reducing albuminuria than other BP-lowering agents and should be considered as part of the treatment strategy for patients with hypertension and microalbuminuria or proteinuria.	IIa	B
Heart failure					
In patients with HFrEF, it is recommended that BP-lowering treatment comprises an ACE inhibitor or ARB, a beta-blocker and diuretic and/or MRA if required.	I	A	In patients with symptomatic HFrEF/HFmrEF, the following treatments with BP-lowering effects are recommended to improve outcomes: ACE inhibitors (or ARBs if ACE inhibitors are not tolerated) or ARNi, beta-blocker, MRA, and SGLT2 inhibitors.	I	A
In patients with HFpEF, because no specific drug has proven its superiority, all major agents can be used.	I	C	In hypertensive patients with symptomatic HFpEF, SGLT2 inhibitors are recommended to improve outcomes in the context of their modest BP-lowering properties.	I	A
			In patients with symptomatic HFpEF who have BP above target, ARBs and/or MRAs may be considered to reduce heart failure hospitalizations and reduce BP.	IIb	B

Continued

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

Stroke					
In all hypertensive patients with ischaemic stroke or TIA, an SBP target range of 120–130 mmHg should be considered.	IIa	B	In patients with confirmed BP ≥130/80 mmHg with a history of TIA or stroke a systolic BP target of 120–129 mmHg is recommended to reduce CVD outcomes, provided treatment is tolerated.	I	A
Different ethnic groups					
In black patients, initial antihypertensive treatment should include a diuretic or a CCB, either in combination or with a RAS blocker.	I	B	In black patients from Sub-Saharan Africa who require BP-lowering treatment, combination therapy including a CCB combined with either a thiazide diuretic or a RAS blocker should be considered.	IIa	B
Resistant hypertension					
Recommended treatment of resistant hypertension is: • Reinforcement of lifestyle measures, especially sodium restriction. • Addition of low-dose spironolactone to existing treatment. • Or the addition of further diuretic therapy if intolerant to spironolactone, with either eplerenone, amiloride, a higher dose thiazide/thiazide-like diuretic, or a loop diuretic. • Or the addition of bisoprolol or doxazosin.	I	B	In patients with resistant hypertension and uncontrolled BP despite use of first-line BP-lowering therapies, the addition of spironolactone to existing treatment should be considered.	IIa	B
			In patients with resistant hypertension in whom spironolactone is not effective or tolerated, treatment with eplerenone instead of spironolactone, or the addition of a beta-blocker if not already indicated, and, next, a centrally acting BP-lowering medication, an alpha-blocker, or hydralazine, or a potassium-sparing diuretic should be considered.	IIa	B
			To reduce BP, and if performed at a medium-to-high volume centre, catheter-based renal denervation may be considered for resistant hypertension patients who have BP that is uncontrolled despite a three BP-lowering drug combination, and who express a preference to undergo renal denervation after a shared risk-benefit discussion and multidisciplinary assessment.	IIb	B
10. Acute and short-term management of blood pressure					
In patients with acute intracerebral haemorrhage:			In patients with intracerebral haemorrhage, immediate BP lowering (within 6 h of symptom onset) should be considered to a systolic target 140–160 mmHg to prevent haematoma expansion and improve functional outcome.	IIa	A
• Immediate BP lowering is not recommended for patients with systolic BP <220 mmHg.	III	A			
• In patients with systolic BP ≥220 mmHg, careful acute BP lowering with i.v. therapy to <180 mmHg should be considered.	IIa	B			
In hypertensive patients with an acute cerebrovascular event, anti-hypertensive treatment is recommended:			For patients with ischaemic stroke or TIA and an indication for blood pressure lowering, it is recommended that BP lowering therapy be commenced before hospital discharge.	I	B
• Immediately for TIA.	I	A			
• After several days in ischaemic stroke.	I	A			
In severe hypertension, drug treatment with i.v. labetalol, oral methyldopa, or nifedipine is recommended.	I	C	In severe hypertension in pregnancy, drug treatment with i.v. labetalol, oral methyldopa, or oral nifedipine is recommended. Intravenous hydralazine is a second-line option.	I	C

ABPM, ambulatory blood pressure monitoring; ACE, angiotensin-converting enzyme; ACR, albumin-to-creatinine ratio (urine); ARB, angiotensin receptor blocker; ARNi, angiotensin receptor–neprilysin inhibitor; BMI, body mass index; BP, blood pressure; CCB, calcium channel blocker; CKD, chronic kidney disease; CV, cardiovascular; CVD, cardiovascular disease; DASH, Dietary Approaches to Stop Hypertension; eGFR, estimated glomerular filtration rate; HBPM, home blood pressure monitoring; HFpEF, heart failure with preserved ejection fraction; HF(m)rEF, heart failure with (mildly) reduced ejection fraction; HFrEF, heart failure with reduced ejection fraction; HMOD, hypertension-mediated organ damage; i.v., intravenous; LVH, left ventricular hypertrophy; MRA, mineralocorticoid receptor antagonist; RAS, renin–angiotensin system; RCT, randomized controlled trial; SBP, systolic blood pressure; SCORE2, Systematic COronary Risk Evaluation 2; SCORE2-OP, Systematic COronary Risk Evaluation 2–Older Persons; SGLT2, sodium–glucose co-transporter 2; TIA, transient ischaemic attack.

[a]Class of recommendation.

[b]Level of evidence.

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

© ESC 2024

3. Pathophysiology of elevated blood pressure and hypertension

Persistently high BP in systemic arteries is the hallmark of hypertension, which is the most important modifiable risk factor for all-cause and CVD morbidity and mortality globally.[2] Most patients with hypertension have essential or primary hypertension, where the exact cause remains unknown, while an estimated 10% have secondary hypertension, with an identifiable cause (notably some studies indicate that the prevalence of secondary hypertension may be substantially higher, with modern systematic screening).[10]

The pathophysiology of hypertension involves complex interactions between environmental and behavioural factors, genes, hormonal networks, and multiple organ systems (renal, cardiovascular, and central nervous system[11]) (*Figure 1*). In addition, vascular and immune mechanisms are involved.[12] Dysregulation of these processes leads to hypertension, which if uncontrolled, can lead to hypertension-mediated organ damage (HMOD) and adverse CVD outcomes.

Details on the pathophysiological processes, molecular mechanisms, and environmental and psychosocial elements that underlie hypertension are provided in the Supplementary text (Supplementary data online).

4. Clinical consequences of elevated blood pressure and hypertension

Longstanding hypertension causes organ damage and ultimately leads to cardiovascular, cerebrovascular, and clinical renal disease, which are all major contributors to the global burden of chronic disease

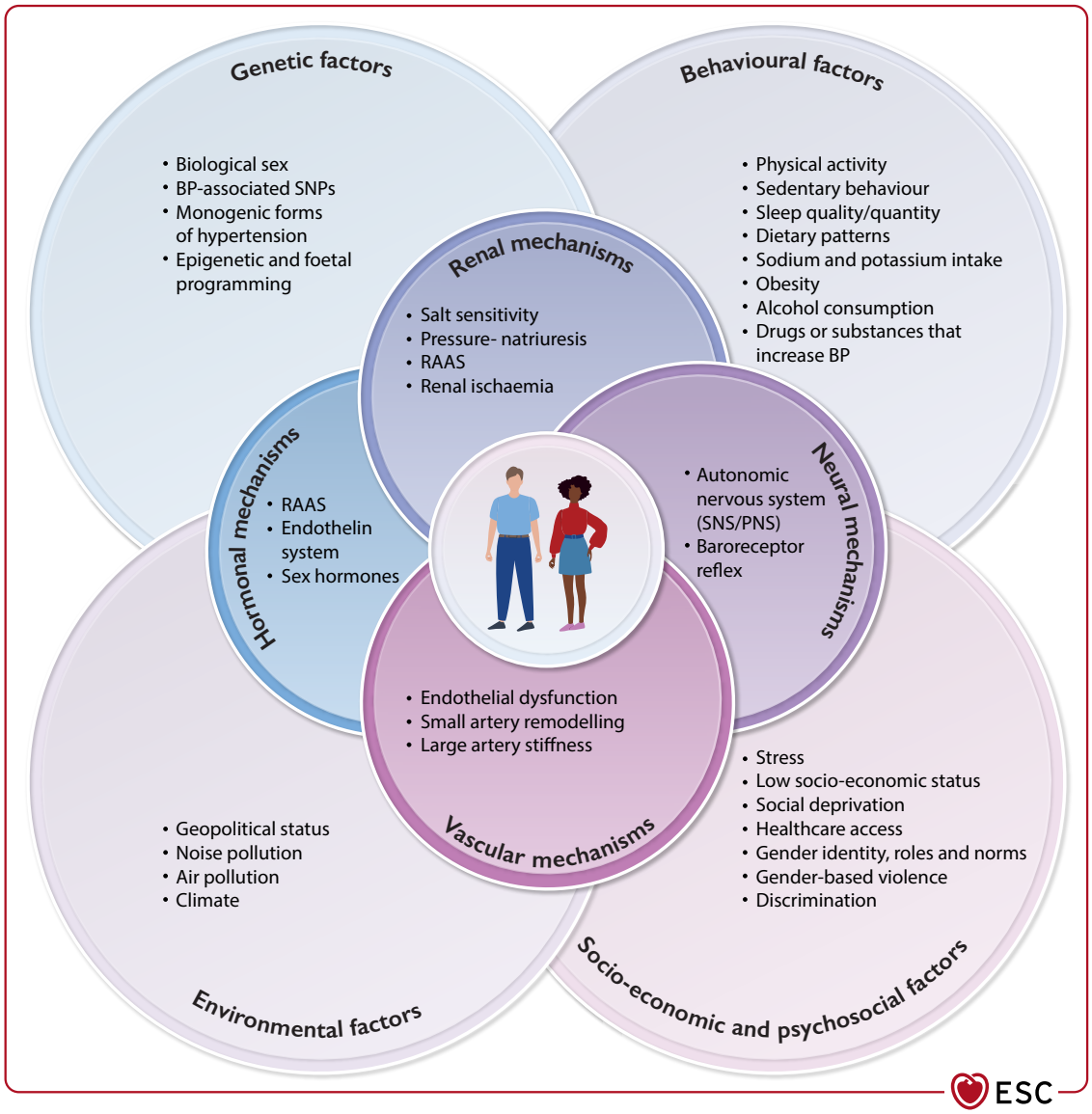


Figure 1 Pathophysiology of elevated blood pressure and hypertension. BP, blood pressure; PNS, parasympathetic nervous system; RAAS, renin-angiotensin-aldosterone system; SNP, single-nucleotide polymorphism; SNS, sympathetic nervous system. Complex interplay between genes, environmental, and behavioural factors, organs, physiological systems, and neurohumoral processes contribute to BP regulation. Dysfunction of these processes leads to hypertension. The contribution of these factors to elevated BP and hypertension may differ among males and females.

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

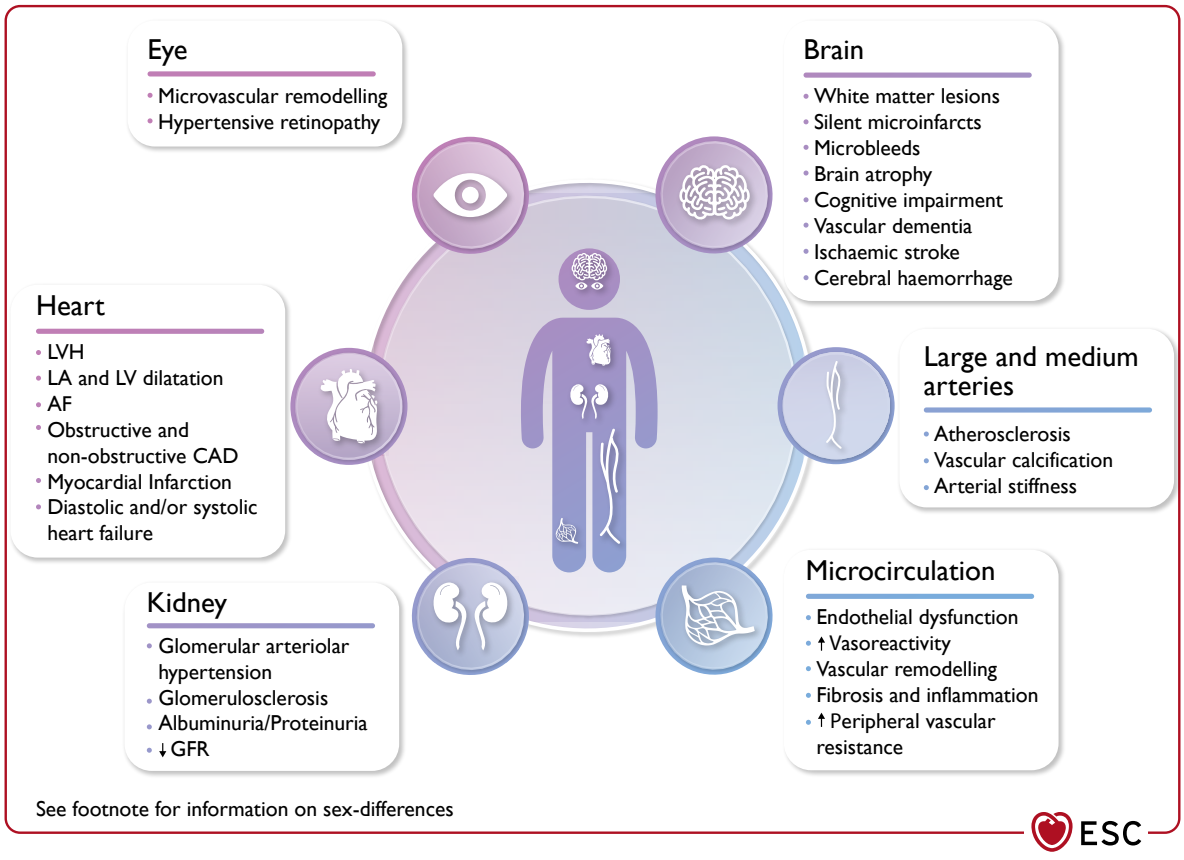


Figure 2 Persistently elevated blood pressure and hypertension lead to hypertension-mediated organ damage and cardiovascular disease. AF, atrial fibrillation; CAD, coronary artery disease; GFR, glomerular filtration rate; LA, left atrial; LV, left ventricular; LVH, left ventricular hypertrophy. See the supplementary data online for detailed information on sex differences.

(*Figure 2*).[2,13–22] Organs adversely affected by elevated BP and hypertension include the heart, brain, kidneys, eyes, and vessels (macrocirculation and microcirculation in organs with low resistance, such as the brain or kidney[23]), which undergo structural and functional changes. Although factors besides BP can contribute to these changes (i.e. dyslipidaemia, hyperglycaemia), we use the term 'hypertension-mediated organ damage' to indicate the presence of subclinical complications of hypertension that indicate high risk for subsequent clinical events. HMOD may have different profiles in men and women; for instance, left ventricular hypertrophy (LVH) and left atrial dilatation are more frequent in women.[24–28]

Evidence of HMOD usually indicates long-standing elevated BP and/or hypertension and confers incremental prognostic information regarding CVD risk in all BP categories.[29–31] Unless treated, HMOD can progress from asymptomatic to symptomatic, ultimately resulting in overt CVD events.[31]

The pathophysiological mechanisms underlying HMOD in the heart, brain, kidneys, vessels, and eyes are detailed in the Supplementary text (Supplementary data online). The clinical consequences of HMOD, especially cerebrovascular disease (stroke and cognitive decline), kidney disease (acute and chronic), and heart disease [heart failure, atrial fibrillation (AF), ischaemic heart disease, and valvular disease] are also discussed in the Supplementary text (Supplementary data online). In addition, the Supplement highlights the impact of different measures of BP on CVD risk, including systolic BP, diastolic BP, pulse pressure, and BP variability.[22,32–36]

5. Measuring blood pressure

5.1. Introduction and pertinent definitions

This section reviews practical aspects of BP measurement, including technique and clinical validation of devices. It also reviews the evidence for the most appropriate BP measurement methods when screening populations for hypertension, diagnosing hypertension, and managing patients receiving BP-lowering interventions. The current guidelines promote use of out-of-office measurement for diagnosis and ongoing management of hypertension, reflecting increasing evidence for the stronger relationship of home and ambulatory monitoring with outcomes, the ability to detect white-coat and masked hypertension, new BP treatment targets as low as 120–129 mmHg systolic (*Table 5*), and evidence supporting enabling patient involvement and shared decision-making.

Definitions:

Systolic BP: arterial BP during systole (maximum arterial pulsatile pressure). This is measured using an auscultatory device at the onset of the first Korotkoff sound. Oscillometric devices estimate systole using an algorithm that imputes from mean arterial pressure.[37]

Diastolic BP: arterial BP during diastole (minimum arterial pulsatile pressure). This is measured using an auscultatory device at the time of complete disappearance of the Korotkoff sounds (fifth sound). If there is no disappearance of sounds (no fifth sound) then the fourth

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

Korotkoff sound (muffling) is used to estimate diastolic BP. Oscillometric devices estimate diastole using an algorithm that imputes from mean arterial pressure.[37]

Inter-arm difference: systolic BP difference of >10 mmHg when BP is measured sequentially in each arm.[38]

Postural/orthostatic hypotension: decrement of ≥20 mmHg in systolic BP and/or ≥10 mmHg in diastolic BP when BP is measured in the standing position at 1 and/or 3 min after standing following a 5-min period in the sitting or lying position.

White-coat hypertension: BP that is above the threshold for diagnosing hypertension in the office but below the threshold in home/ambulatory settings, e.g. ≥140/90 mmHg in office but <135/85 mmHg at home/ambulatory daytime (or 24-h BP < 130/80 mmHg).

Masked hypertension: BP that is below the hypertension diagnostic threshold in the office but above the hypertension diagnostic threshold in home/ambulatory settings, e.g. <140/90 mmHg in clinic but ≥135/85 mmHg at home/ambulatory daytime (or 24-h BP ≥130/80 mmHg).

Office BP: also known as clinic BP. The two terms are interchangeable. This guidelines document uses 'office BP' preferentially. Of note, office BP can be measured manually or using an automated device. In addition, automated office BP (AOBP) can be conducted in a setting attended by a healthcare professional or in an unattended fashion. Finally, not all office BP measurements are equal, with some facilities using a standardized method (which is recommended and outlined below) and others unfortunately using suboptimal approaches to office BP measurement.

Home BP measurement (HBPM): an out-of-office approach to measuring BP when the patient measures their own BP at home using a validated monitor (usually an upper-arm oscillometric cuff device).

Ambulatory BP measurement (ABPM): an out-of-office BP measurement that uses a fully automated oscillometric device, usually for a 24-h period, and measures BP at set intervals.

5.2. Practical recommendations for measuring blood pressure

5.2.1. Clinical validation of equipment for measuring blood pressure

A prerequisite of BP measurement is that it must be undertaken using a device that has been clinically validated and confirmed to be accurate. Of the commercially available oscillometric BP measurement devices, as few as 6% have been adequately tested.[39–41] National and international organizations provide lists of validated monitors (e.g. www.stridebp.org, www.validatebp.org).

Since the 2018 ESC/ESH Guidelines on the management of arterial hypertension, three arbiters of device accuracy (the Association for the Advancement of Medical Instrumentation, the ESH, and the International Organization for Standardization) have published a universal standard for validating devices for measuring BP.[42] This standard will likely become widely adopted. Validation standards and methodology need to be developed and implemented for novel BP measurement devices that are non-occlusive and 'cuffless'.[43,44]

5.2.2. Office blood pressure measurement

All BP measurements can be influenced by circumstances of measurement, including position, ambient temperature, the technique of measurement, accuracy of equipment, and physical condition of the patient.[45] For BP measurements in the office, we recommend following a standardized method (*Figure 3*).

Patient preparation: BP should be measured with the patient seated comfortably after 5 min of rest. Patients should avoid exercising and stimulants (caffeine, tobacco) for at least 30 min before measurement. The patient's bladder should be emptied if needed.[46] Patients should be seated with their legs unfolded and their back supported at the time of measurement. The arm should be supported (to avoid isometric exercise-induced increases in BP). Clothing at the location of the cuff placement should be removed; rolling up of shirt sleeves should be avoided as this can result in a tourniquet effect.

BP measurement technique: auscultatory or oscillometric techniques can be used to measure BP non-invasively. The manual auscultatory approach is the traditional method of measuring systolic and diastolic BP at the brachial artery site using a stethoscope. In contrast, oscillometric devices compute mean arterial BP using the oscillation amplitude with cuff deflation (or inflation) and then estimate systolic and diastolic BP. Oscillometric devices can be semi-automated (taking one reading per activation) or fully automated (obtaining multiple readings before averaging them). Oscillometric devices are not typically validated for use in AF, and the manual auscultatory method is preferred in these circumstances when feasible.[47–49]

BP cuff selection and positioning: an appropriately sized cuff should be used, as an under-sized or over-sized cuff will artificially elevate or reduce BP, respectively.[50] The bladder length should be 75%–100% and the width 35%–50% of the arm circumference. The arm circumference can be measured at the mid-point of the olecranon and the acromion but many cuffs include sizing indicators. The cuff should be positioned on the patient's upper arm at the level of the heart with the lower edge of the cuff a few centimetres above the antecubital fossa. The stethoscope should not be placed under the cuff. In those with significant obesity where a correctly fitting upper arm cuff is not available, measurement at the lower arm or wrist can be considered as an alternative.[51]

BP measurement by manual auscultation: three BP measurements should be taken, each 1–2 min apart, and additional measurements only if the readings differ by >10 mmHg (e.g. this may occur with arrhythmias or white-coat effects). The BP recorded should be the average of the last two BP readings.

BP measurement using AOBP measurement: as noted above, AOBP using oscillometric devices may be obtained with (attended) or without (unattended) clinicians or staff present. Clear evidence regarding superiority of unattended vs. attended AOBP in managing BP to reduce rates of CVD is lacking; however, because BP readings may differ for unattended vs. attended measurements,[52] we advise that a consistent approach be used depending on local resource and preference. AOBP monitors typically make three or six readings at 1-min intervals and provide an average. AOBP correlates more closely with mean ABPM than with the manual auscultatory technique and may reduce measurement error and white-coat effects.[53]

Inter-arm BP difference: at the initial visit, BP should be measured in both arms to detect an inter-arm difference. Though devices allowing simultaneous measurement in both arms exist, sequential arm measurement is considered sufficiently reliable.[54,55] Measurement in the contralateral arm should be undertaken once the three measurements in the index arm have been taken, and if a difference is detected, further measurement in the original arm is indicated to ensure the difference is consistent. If systolic BP differs by >10 mmHg between arms, subsequent measurements are obtained using the arm with the higher BP value. Significant inter-arm BP differences may reflect arterial stenosis or

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

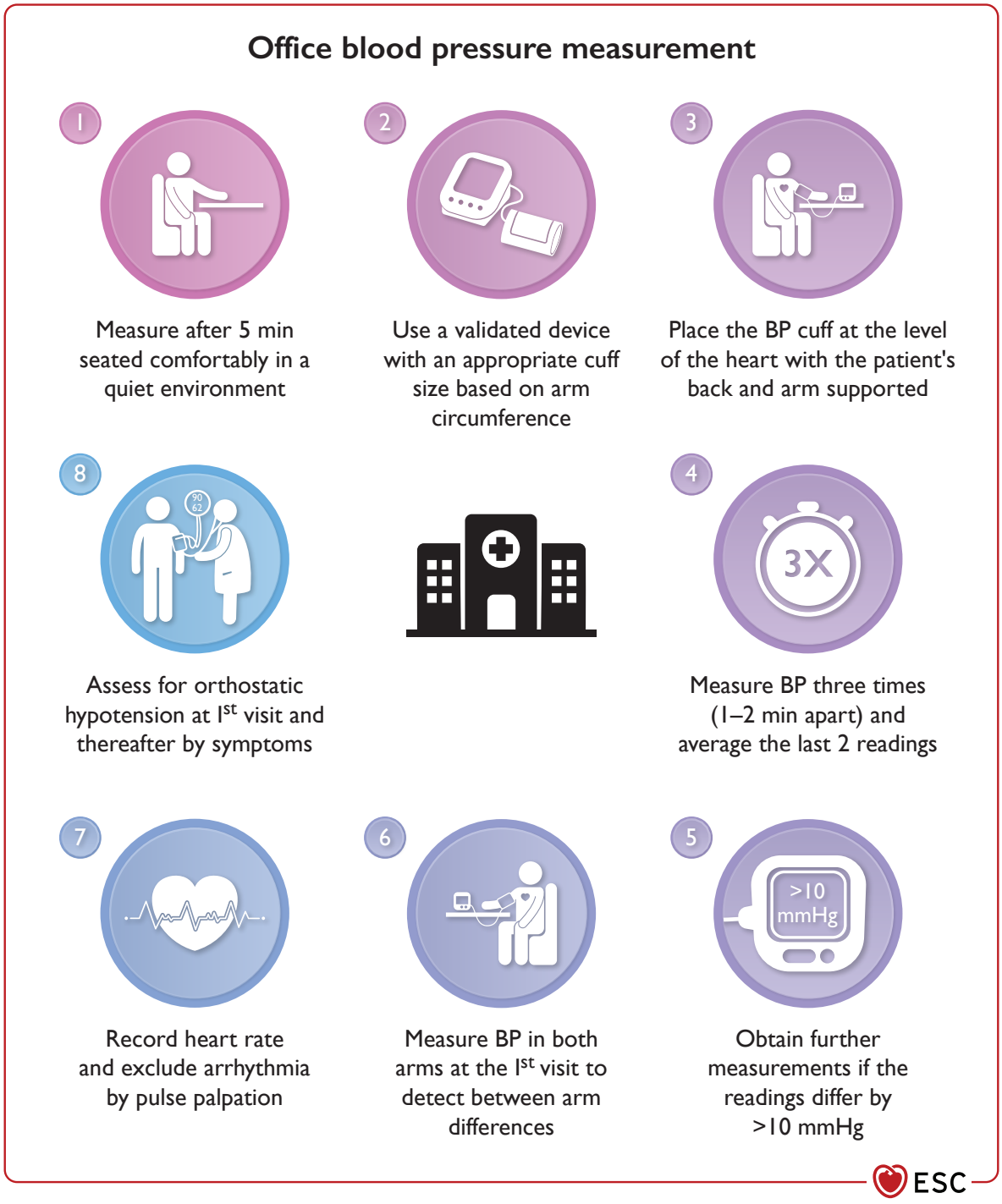


Figure 3 Summary of office blood pressure measurement. BP, blood pressure.

coarctation of the aorta, which may require investigation. Also, of note, in some patients one arm is preferred to the other for routine BP measurement (e.g. to avoid measurement of BP in an arm with an arteriovenous fistula or an arm where axillary lymph node dissection has occurred).

Postural/orthostatic hypotension: patients should be assessed for orthostatic hypotension at the initial visit and if concerning symptoms arise. After 5 min of rest in the sitting or lying position, BP should be measured at 1 min and/or 3 min after standing, with a threshold for orthostatic hypotension of ≥20/10 mmHg (systolic BP/diastolic BP) drop. Measurement after lying may be more sensitive for detecting orthostatic hypotension and may better predict falls but may be less feasible than measurement after sitting in clinical practice.[56]

Pulse assessment: heart rate should be recorded at the initial visit and arrhythmia excluded.

5.2.3. Home blood pressure measurement

HBPM is an out-of-office approach to measuring BP when the patient measures their own BP at home using a validated monitor (usually an upper-arm oscillometric cuff device).[57,58] A consistent approach to

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

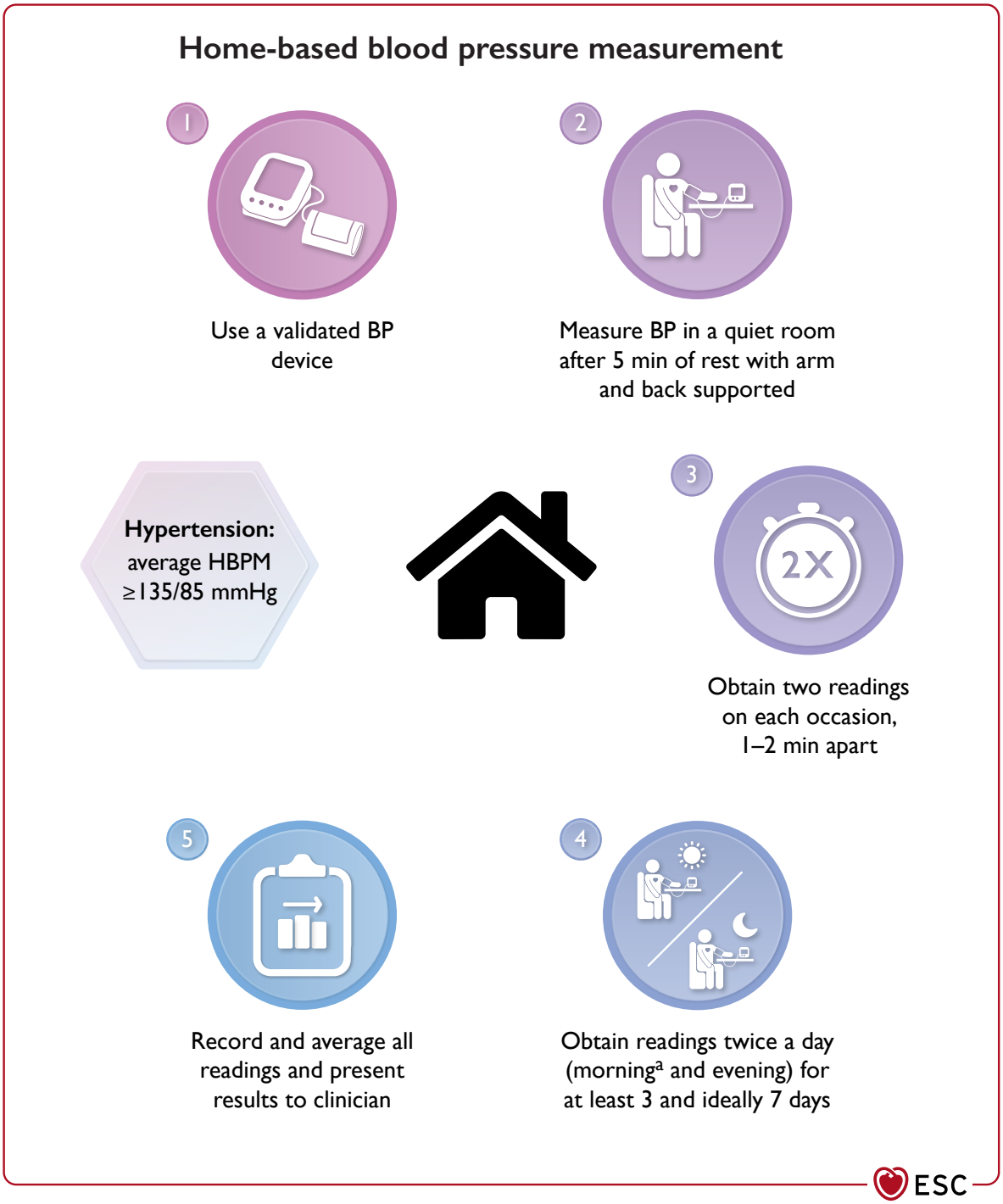


Figure 4 Summary of home blood pressure measurement. BP, blood pressure; HBPM, home blood pressure measurement. [a]Morning HBPM readings should be obtained before breakfast and before intake of medication but not immediately after awakening.

HBPM should be used (*Figure 4*). Patients should be counselled to follow the same preparation steps as used in clinics, which are outlined in *Section 5.2.2*. Two measurements should be taken at each measurement session, performed 1–2 min apart. Measurements should be made twice a day (morning and evening) at the same time for a minimum of 3 days and up to 7 days.[59] At the end of the measurement period, all readings are averaged. If the average after 3 days is close to the treatment threshold, then measurement should continue for the full 7 days. Patients should be informed to keep a record of their home BP values and to ask their healthcare provider that the device accuracy be intermittently checked. Devices older than 4 years may be inaccurate and, if inaccurate, should be replaced.[60]

An average HBPM of ≥135/85 mmHg (equivalent to an office BP of ≥140/90 mmHg) should be used to diagnose hypertension and an average systolic BP of 120–134 mmHg or diastolic BP of 70–84 mmHg should be used to diagnose elevated BP. Of note, we use the same lower BP threshold (120/70 mmHg) for both office and HBPM in defining elevated BP.[61]

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

5.2.4. Ambulatory blood pressure measurement

ABPM (summarized in *Figure 5*) is an out-of-office BP measurement that uses a fully automated device, usually for a 24-h period. The devices measure BP by the oscillometric method and are programmed to measure BP at set intervals. Readings are usually obtained at 15–30 min intervals during the day (typically 7 a.m. to 11 p.m.) and 30–60 min intervals at night (typically 11 p.m. to 7 a.m.). The software usually provides average BP measurements for daytime, night-time, and 24 h. A minimum of 70% useable BP recordings is required for a valid measurement session, typically numbering ≥27 measurements over 24 h. Preferably, seven nocturnal readings should also be obtained.[62] However, emerging data indicate that ≥8/≥4 wake/sleep readings may be adequate if more cannot be obtained.[63] Prior to using mean ABPM values (either 24 h, daytime, or night-time) the raw BP values at each measurement should be reviewed for possible outlier or erroneous values. A diary should record activities (e.g. meals and exercise) and sleep time to assist interpretation. The diagnostic thresholds for elevated BP and hypertension using ABPM, and comparison with office BP and HBPM, are provided in *Table 5*.[61,64]

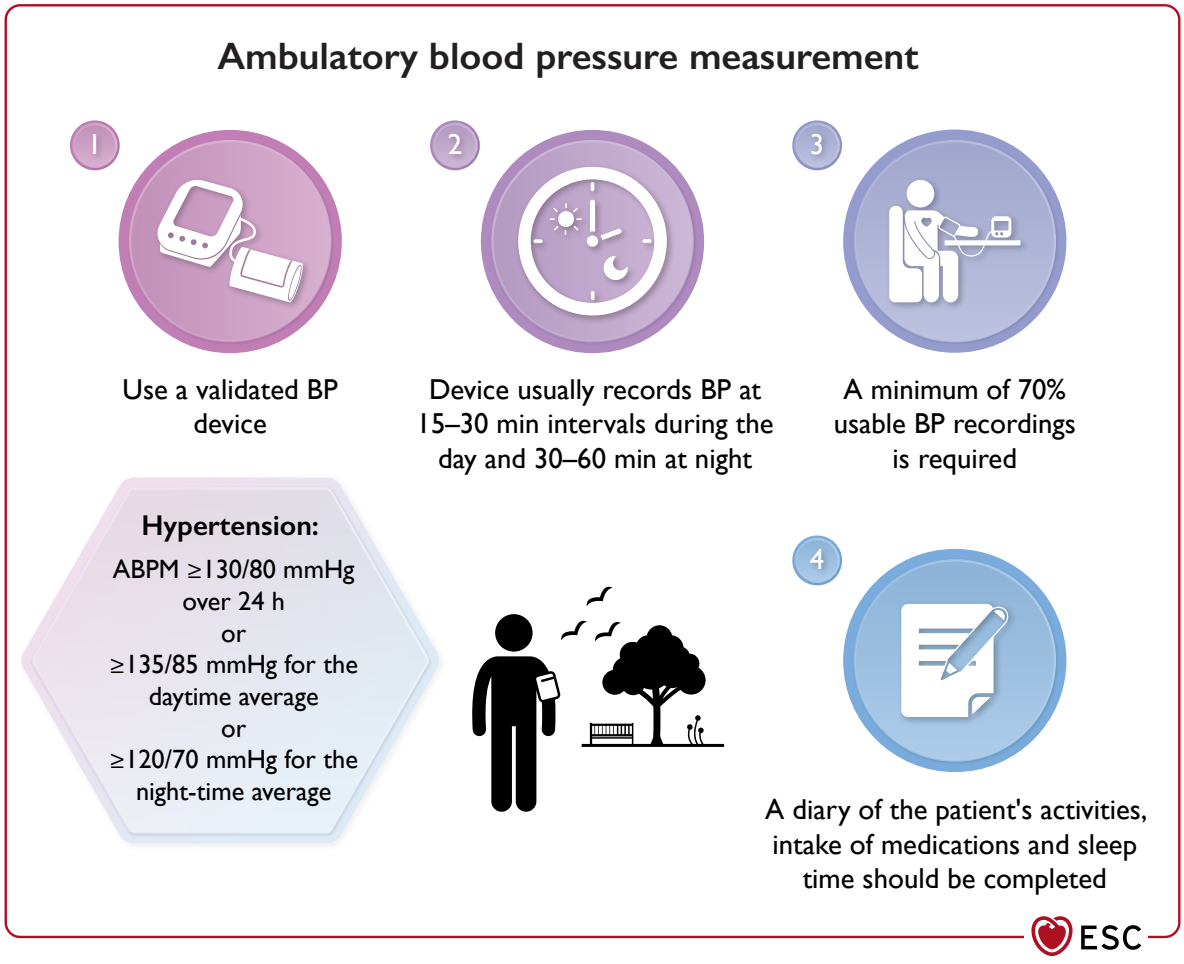


Figure 5 Summary of ambulatory blood pressure measurement. ABPM, ambulatory blood pressure measurement; BP, blood pressure.

Table 5 Comparison of office, home, and ambulatory blood pressure measurement thresholds for elevated blood pressure and hypertension

	Office BP (mmHg)[a]	Home BP (mmHg)	Daytime ABPM (mmHg)	24 h ABPM (mmHg)	Night-time ABPM (mmHg)
Reference					
Non-elevated BP	<120/70	<120/70	<120/70	<115/65	<110/60
Elevated BP	120/70–<140/90	120/70–<135/85	120/70–<135/85	115/65–<130/80	110/60–<120/70
Hypertension	≥140/90	≥135/85	≥135/85	≥130/80	≥120/70

ABPM, ambulatory blood pressure monitoring; BP, blood pressure.
[a]The BP thresholds provided assume that a standardized approach to office BP measurement is performed (*Figure 3*). However, evidence indicates that office BP measurement in routine clinical settings is often not done using a standardized approach and, in this case, the routine office BP value may be 5–10 mmHg higher than it would have been if measured using the recommended standardized approach.[65,66]

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

© ESC 2024

5.2.5. Comparison of home and ambulatory blood pressure monitoring

There is overlap between home and ambulatory monitoring in terms of differentiating between hypertensive phenotypes. However, around 15% of people will have diagnostic disagreement, of whom approximately 50% will represent clinically significant differences of >5 mmHg.[67] The advantages and disadvantages of home and ambulatory monitoring are outlined in *Table 6*.

5.3. What is the best method for measuring blood pressure to diagnose hypertension?

5.3.1. Blood pressure measurement for hypertension screening

Opportunistic screening is typically performed using office BP measurement and is key in detecting possible hypertension. However, a single screening office BP alone does not typically have sufficient diagnostic test performance to establish a diagnosis, especially for BP values close to diagnostic thresholds. Therefore, a single screening office BP requires some form of repeat BP assessment to confirm a diagnosis (preferably out-of-office or repeat office if out-of-office is not available). Relatedly, the BP threshold for acting on a screening office BP by conducting repeat BP assessments should also be lower than the office BP threshold used for diagnosing hypertension. This latter consideration is particularly relevant in the presence of increased CVD risk or markers of HMOD. Also of note, populations where masked hypertension is more prevalent include men, those who smoke, those with excessive alcohol intake, or those with diabetes or obesity.[68,69] While a screening office BP of >160/100 mmHg is almost always consistent with a diagnosis of hypertension, a small proportion of patients will have extreme white-coat effects that motivate prompt repeat BP assessment.[68] Hypertension screening approaches are discussed further in *Section 7.1*.

Table 6 Comparison of ambulatory and home blood pressure monitoring

Ambulatory monitoring
Advantages
• Can identify white-coat and masked hypertension
• Measurement in real-life settings and during usual activities
• Stronger prognostic evidence
• Night-time readings
• Abundant information from a single investigation, including short-term diurnal BP variability
• Additional BP phenotyping (e.g. nocturnal dipping status)
Disadvantages
• Relatively expensive and sometimes limited availability
• Can be uncomfortable and affect sleep
Home monitoring
Advantages
• Identify white-coat and masked hypertension
• Cheap and widely available
• Measurement at home, which may be more relaxed than at doctor's office
• Patient engagement in BP measurement and telemedicine potential
• Easily repeated and used over longer periods to assess day-to-day BP variability
Disadvantages
• Only static BP at rest is typically available
• Potential for measurement error due to improper measurement technique or unvalidated or poorly calibrated device
• Nocturnal readings not usually possible

BP, blood pressure.

© ESC 2024

5.3.2. Blood pressure measurement for diagnosing hypertension

After detecting high BP in the office, subsequent BP measurement for diagnosing hypertension depends on the clinical circumstances. Office BP has lower specificity than ABPM for detecting hypertension, so diagnosis based on office BP alone is less desirable unless resources do not allow out-of-office measurements.[70] For screening BP of 160–179 mmHg systolic or 100–109 mmHg diastolic, prompt confirmation (within 1 month) using either office or out-of-office methods is recommended, as delays in treatment are associated with increased CVD event rates.[71] For BP of ≥180/110 mmHg, assessment for hypertensive emergency is recommended. In the setting of hypertensive emergency, immediately commencing BP-lowering treatment is recommended, otherwise, prompt confirmation (preferably within a week) can be considered prior to commencing treatment (*Sections 7 and 10*).

For screening BP of 140–159/90–99 mmHg, out-of-office BP should be measured to confirm the diagnosis.[72] When treatment of elevated BP is being considered (e.g. 120–139/70–89 mmHg) for individuals with high risk CVD conditions or sufficiently high 10-year predicted CVD risk, out-of-office BP measurement is recommended, both to confirm BP and to assess for masked hypertension. Out-of-office measurements may also be helpful for individuals with office BP of 130–139/85–89 mmHg to diagnose masked hypertension. Further details on the diagnostic evaluation of hypertension are provided in *Section 7.2*.

5.4. What is the best method for measuring blood pressure for long-term management of hypertension?

While repeat office measurement of BP remains the commonest approach to long-term management of hypertension, several lines of investigation support augmenting office BP measurements with out-of-office assessment.

5.4.1. Home monitoring

There are over 50 trials of different self-monitoring-based interventions.[73] Self-monitoring is associated with lower mean systolic BP at 12 months [−3.2 mmHg; 95% confidence interval (CI) −4.9 to −1.6 mmHg].[74] Furthermore, there are known benefits of telemonitoring, digital interventions, and mobile health in managing BP.[75–78] Self-monitoring is also likely to be cost-effective.[79] Unfortunately, in clinical practice, some patients may not provide reliable information on their home BP, and both their device and measurement technique need to be checked.

5.4.2. Ambulatory monitoring

ABPM provides a reference BP measurement but repeat ABPM testing is sometimes not practical due to resource constraints and, uncommonly, low patient acceptability.[80] There is a paucity of data on treatment guided by ABPM vs. that of office or HBPM measurements. A trial of treatment guided by HBPM vs. clinic and ambulatory monitoring found equivalence in BP control and HMOD.[81] Other studies reported a non-significant trend to worse BP control with ambulatory vs. office BP monitoring, though the ambulatory group also received fewer

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

medications.[82] Potential advantages of ABPM over HBPM include diagnosing nocturnal hypertension or symptomatic transient hypotension or hypertension with exertion. As such, ABPM and HBPM should be considered complementary and additive, rather than competing approaches to long-term BP management.[83]

5.5. Measuring blood pressure in selected groups

5.5.1. Pregnancy

Monitoring BP during pregnancy is typically done at antenatal visits, which vary dependent on trimester (with increasing frequency towards term). BP tends to reach a nadir at 20–30 weeks of pregnancy before increasing towards term at 40 weeks.[84] Only a small number of automated oscillometric BP monitors have been adequately validated in pregnancy and several have failed, usually due to providing BP values that are erroneously high.[85] Auscultatory measurement with sphygmomanometry is consequently the clinical standard in pregnancy.[85] Self-monitoring at home is not yet proven to be effective in gestational hypertension.[86,87] While norms for BP during pregnancy remain unclear, the 2022 Chronic Hypertension and Pregnancy (CHAP) trial indicated benefit of targeting clinic BP below 140/90 mmHg.[88] Consideration of secondary causes of hypertension is important in young women with gestational hypertension. Further details are provided in *Section* 9.2 and the 2018 ESC Guideline for the management of cardiovascular disease during pregnancy.[89]

5.5.2. Atrial fibrillation

Hypertension is a risk factor for AF.[90,91] Oscillometric BP monitors are not always accurate in the presence of AF, due to the greater variability of BP beat to beat, so multiple auscultatory measurements are recommended.[48,92,93] Some oscillometric BP monitors include an algorithm for detecting AF, but an electrocardiogram (ECG) is still required to confirm the diagnosis.[49,94]

5.5.3. Orthostatic hypotension

Postural or orthostatic hypotension is common,[95,96] present in approximately 10% of all hypertensive adults and up to 50% of older institutionalized adults.[97,98] Orthostatic hypotension is defined as a BP drop of ≥20/10 mmHg 1 and/or 3 min after standing following a 5-min period in the seated or lying position.[99–101] Diagnosis is made in the office. Routine ABPM is not currently suitable for formally assessing orthostatic hypotension,[102] though it may help in some cases, particularly when accompanied by a patient symptom diary.[103]

5.6. Novel methods of measuring blood pressure

New methods to measure BP are under development. Continuous office and out-of-office BP recordings and ABPM and HBPM have been developed that derive beat-to-beat, reading-to-reading, and day-to-day BP variability. However, there is no agreement on the optimum approach to measuring variability, and there is no trial evidence that reducing BP variability specifically can reduce CVD events.[104] Other emerging technologies include wearable, wrist-based BP measurement devices, devices evaluating central BP, and cuffless devices implementing plethysmographic or other technologies.[105,106] However, there is at present insufficient scientific consensus on the accuracy standards and validation procedures that these cuffless devices must comply with prior to commercialization.[43,44,107–109]

In view of these issues, none of these cuffless measurement modalities are currently recommended for routine clinical use.

Recommendation Table 1 — Recommendations for measuring blood pressure (see Evidence Tables 1–8)

Recommendations	Class[a]	Level[b]
It is recommended to measure BP using a validated and calibrated device, to enforce the correct measurement technique, and to apply a consistent approach to BP measurement for each patient.[41,42]	I	B
All adult patients (≥18 years) are recommended to have their office and/or out-of-office BP measured on an opportunistic basis and recorded in their medical file, and be told what their current BP is.	I	C
Out-of-office BP measurement is recommended for diagnostic purposes, particularly because it can detect both white-coat hypertension and masked hypertension. Where out-of-office measurements are not logistically and/or economically feasible, then it is recommended that the diagnosis be confirmed with a repeat office BP measurement using the correct standardized measurement technique.[70]	I	B
It is recommended that office BP should be measured in both arms at least at the first visit, because a between-arm systolic BP difference of >10 mmHg is associated with an increased CVD risk and may indicate arterial stenosis.[55,110]	I	B
If a between-arm difference of >10 mmHg in systolic BP is recorded, then it is recommended that all subsequent BP readings use the arm with the higher BP reading.[110]	I	B
Out-of-office BP measurement is recommended for ongoing management to quantify the effects of treatment and guide BP-lowering medication titration, and/or identify possible causes of side effects (e.g. symptomatic hypotension). Where out-of-office measurements are not logistically and/or economically feasible, then ongoing management is recommended to be based on repeated office BP measurements using the correct standardized measurement technique.[74,111,112]	I	B
It is recommended that all patients undergoing BP measurement also undergo pulse palpation at rest to determine heart rate and arrhythmias such as AF.[113]	I	C
Most automated oscillometric monitors have not been validated for BP measurement in AF; BP measurement should be considered using a manual auscultatory method in these circumstances where possible.[47–49]	IIa	C

Continued

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

An assessment for orthostatic hypotension (≥20 systolic BP and/or ≥10 diastolic BP mmHg drop at 1 and/or 3 min after standing) should be considered at least at the initial diagnosis of elevated BP or hypertension and thereafter if suggestive symptoms arise. This should be performed after the patient is first lying or sitting for 5 min.	IIa	C
Other BP measures and indices (pulse pressure, BP variability, exercise BP) may be considered to provide additional clinical information on CVD risk in some circumstances.	IIb	C

© ESC 2024

AF, atrial fibrillation; BP, blood pressure; CVD, cardiovascular disease.
[a]Class of recommendation.
[b]Level of evidence.

6. Definition and classification of elevated blood pressure and hypertension, and cardiovascular disease risk assessment

6.1. Definition and classification of elevated blood pressure and hypertension

Epidemiological studies demonstrate a continuous and log-linear association between BP and adverse CVD outcomes.[22,32,33,114,115] Starting at levels as low as 90 mmHg systolic, the higher the BP the higher the relative risk of CVD including atherosclerosis.[32,114] These observational data are complemented by randomized clinical trials (RCTs),[116] which have provided experimental evidence regarding the BP range for which BP lowering with treatment is proven to reduce CVD events. Of note, some studies suggest a stronger relative risk for CVD for a given BP among females compared with males.[117,118]

A healthy lifestyle should be encouraged for all adults to prevent an increase in BP and development of hypertension.[119,120] To aid pharmacological treatment decisions, the 2024 ESC Guidelines recommend a simplified categorization of adults according to their BP (*Figure 6*). In compiling this categorization, priority was given to evidence from randomized trials over observational data. However, it is important to reiterate that the risk of CVD attributable to BP is continuous and that interpreting randomized trial data is an iterative process involving an element of subjectivity. As such, no categorization of BP can be considered immutable or flawless.

The 2024 Guidelines define hypertension as a confirmed office systolic BP of ≥140 mmHg or diastolic BP of ≥90 mmHg. For this diagnosis to be made, confirmation is recommended with out-of-office measurements (HBPM or ABPM) or at least one repeat office measurement at a subsequent visit, as detailed in *Section 5* and *Section 7.2*. This definition is based on several factors. First, meta-analyses of randomized trials provide evidence among all adults and across various settings for the benefit of BP-lowering therapy among patients with BP above this threshold.[116,121,122] Second, most adults with BP above this threshold are at increased CVD risk, typically with 10-year risk estimates of ≥10% for fatal and non-fatal CVD events.[123–125] The higher the patient's baseline absolute risk for CVD, the greater the net benefit from BP-lowering treatment and, at the population level, the lower the estimated number needed to treat (NNT).[126–128] Third, this more traditional BP threshold for hypertension is already widely used by policymakers to define a disease state, and maintaining this BP threshold to define hypertension (vs. lowering it) does not require most adults to be labelled with what is widely considered a disease.[129]

Here, we introduce a new BP category called 'elevated BP', which is defined as an office systolic BP of 120–139 mmHg or diastolic BP of 70–89 mmHg. Within this BP range, the efficacy of BP-lowering therapy has been established in meta-analyses of RCTs,[116] but average CVD risk in the elevated BP group is not sufficiently high to merit drug treatment in all patients.[123,124,130] Pharmacological treatment initiation is, however, suggested for a subgroup of patients within this BP range who are at increased global risk of CVD as identified by the risk stratification approach outlined in *Sections 6.3*, *6.4*, and 8.

Non-elevated BP is defined as a systolic BP of <120 mmHg and a diastolic BP of <70 mmHg. Fewer individuals within this BP range are at increased risk of CVD,[124] and evidence for CVD benefit with BP-lowering pharmacological treatment is lacking due to an absence of trials. We use the term 'non-elevated BP' to define this BP category in recognition that these are treatment categories and not prognostic categories. Because the relative risk for CVD starts to increase at BP below this threshold (even as low as 90 mmHg systolic BP), particularly among women,[117,118] we avoid terms like 'normal BP', 'optimal BP', or 'normotension' in defining this category.

Recommendation Table 2 — Recommendations for categorizing blood pressure (see Evidence Table 9)

Recommendation	Class[a]	Level[b]
It is recommended that BP be categorized as non-elevated BP, elevated BP, and hypertension to aid treatment decisions.[116,121,122,131–138]	I	B

© ESC 2024

BP, blood pressure.
[a]Class of recommendation.
[b]Level of evidence.

6.2. Principles of a risk-based approach for managing blood pressure and preventing cardiovascular disease

In the context of BP-lowering interventions, randomized trials demonstrate a consistent relative risk reduction in adverse CVD outcomes per unit reduction in BP.[131,139] However, many medical interventions incur costs and have side effects. Therefore, guidance is needed on selecting patients most likely to benefit from BP-lowering treatment. This is especially true among adults with elevated BP (office systolic BP of 120–139 mmHg and/or diastolic BP of 70–89 mmHg). Practical aspects for implementing a risk-based approach are further discussed in *Section 8*.

6.2.1. Role of cardiovascular disease risk assessment

The risk of adverse CVD outcomes increases log-linearly with constant increments in systolic BP and diastolic BP.[22,32,33,114,140] Concurrently, at higher BP, there is clustering of additional CVD risk factors.[141,142] Consequently, many patients with hypertension will have an estimated 10-year risk for CVD events of ≥10%,[116,121,122] which, for the purposes of these guidelines, is considered sufficiently high risk to merit consideration of BP-lowering treatment in the setting of elevated BP.[143]

Using BP thresholds for hypertension alone for allocating treatment would lead to under-treatment of many high-risk patients.[144,145,115] A substantial proportion of excess CVD events attributable to BP occur in patients with BP levels below the traditional threshold for

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

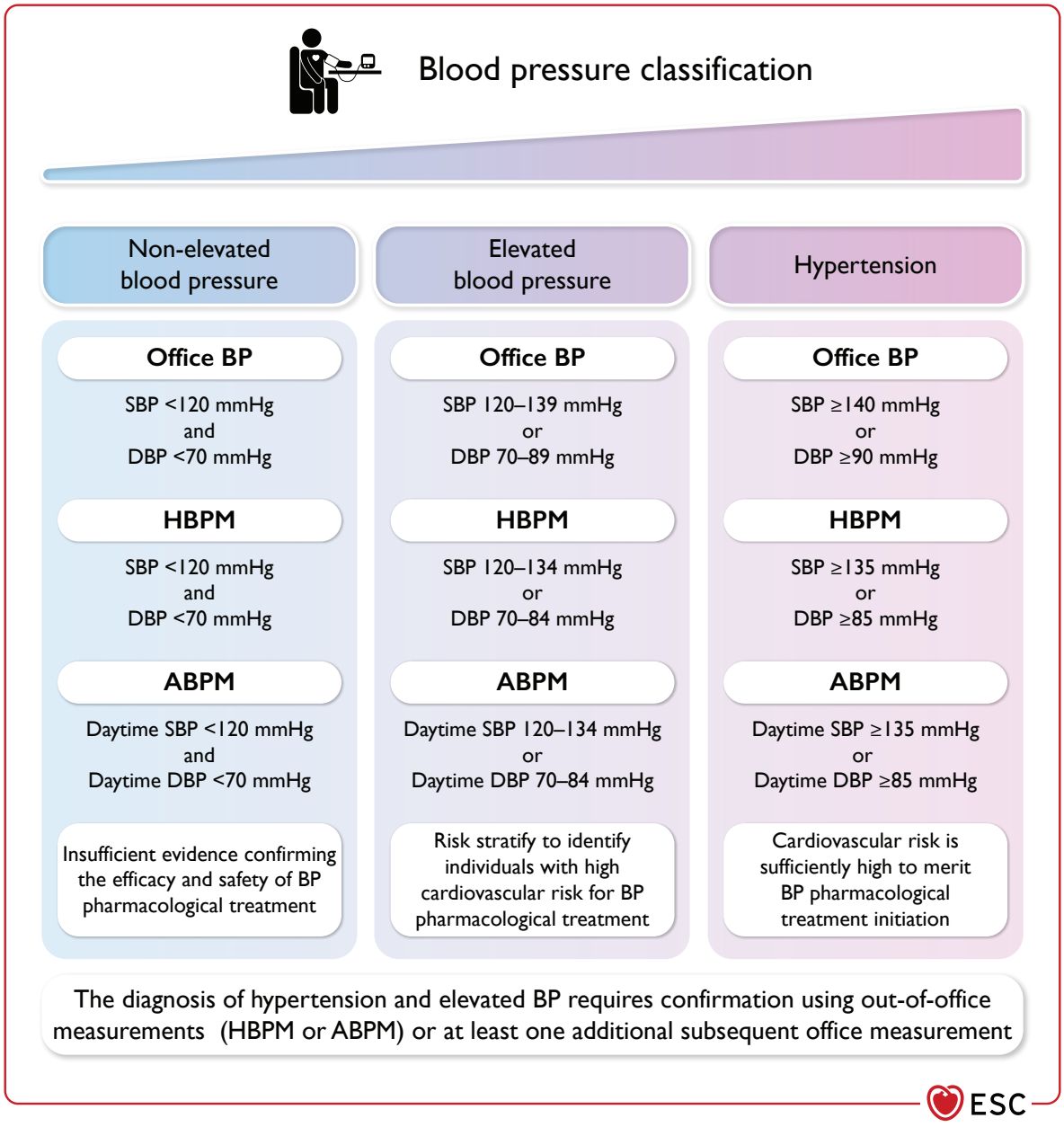


Figure 6 Blood pressure categories. ABPM, ambulatory blood pressure monitoring; BP, blood pressure; DBP, diastolic blood pressure; HBPM, home blood pressure monitoring; SBP, systolic blood pressure. We note that the respective non-daytime ABPM thresholds for elevated BP and hypertension diagnosis are listed in *Section 5* (*Table 5*).

hypertension diagnosis. As the efficacy of BP lowering on preventing CVD events extends down to a systolic BP of 120 mmHg and a diastolic BP of 70 mmHg,[116,135,136,146] patients with elevated BP and increased CVD risk can also derive benefit from BP-lowering treatment.[124,145]

The heterogeneity in CVD risk among adults with elevated BP is larger than in those with hypertension, as such patients tend to be younger, and their absolute CVD risk depends more on the prevalence of concomitant CVD risk factors.[123,147] Consequently, formally estimating the patient's CVD risk, encapsulating demographics and other CVD risk factors, is recommended to guide BP-lowering treatment decisions among patients with elevated BP.[148–151]

6.3. Predicting cardiovascular disease risk

Certain conditions on their own are associated with sufficient CVD risk such that patients with elevated BP alongside these conditions can be considered for BP-lowering therapy (*Figure 7*). These include moderate or severe chronic kidney disease (CKD),[152] established clinical CVD (coronary heart disease, cerebrovascular disease, peripheral arterial disease, or heart failure)[153–158] concomitant HMOD (see *Figure 7*; *Section 7*; Supplementary data online, *Table S1*),[31,159] diabetes mellitus, and familial hypercholesterolaemia (probable or definite).[160–163] Regarding diabetes, some adults aged <60 years with type 2 diabetes and elevated BP have 10-year CVD risk of <10%. Accordingly, the diabetes-specific

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

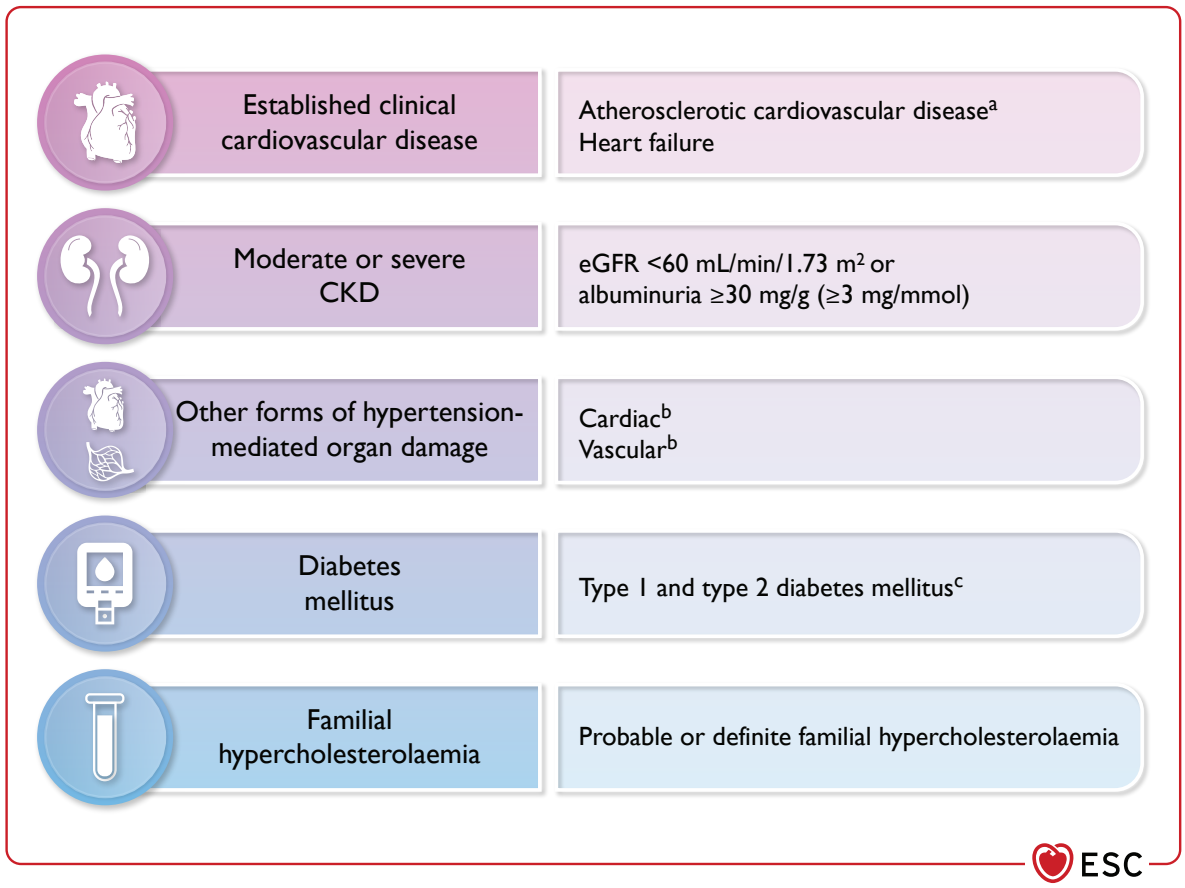


Figure 7 Sufficiently high cardiovascular risk conditions that warrant blood pressure-lowering treatment among adults with elevated blood pressure. CKD, chronic kidney disease; eGFR, estimated glomerular filtration rate. [a]Coronary heart disease, cerebrovascular disease, peripheral arterial disease. [b]See *Section 7*. [c]SCORE2-Diabetes should be considered to identify lower-risk individuals (<10% 10-year CVD risk), who may not require BP-lowering medication, particularly in individuals <60 years.

Systematic COronary Risk Evaluation 2 (SCORE2)-Diabetes risk-prediction model should be considered to confirm CVD risk is sufficiently high (≥10%) among individuals with type 2 diabetes mellitus who are aged <60 years.[164]

In the absence of these sufficiently high-risk conditions, risk-prediction models (SCORE2 and SCORE-OP) have been developed in the general population to predict 10-year risk of CVD.[165,166] In adults with elevated BP without the above sufficiently high-risk conditions, risk-prediction models are recommended to inform BP-lowering treatment decisions. Risk-prediction models are more accurate than clinical judgment or tallying of individual risk factors.[167–169]

6.3.1. 10-year cardiovascular disease risk-prediction models

Prediction models differ in their input variables, predicted endpoints (outputs), and populations in which they were derived and validated. We endorse the use of SCORE2 for individuals aged 40–69 years and SCORE2–Older Persons (SCORE2-OP) for individuals aged ≥70 years for predicting 10-year global risk of fatal and non-fatal CVD events (stroke or myocardial infarction).[165,166] The management of adults aged <40 years is discussed in *Section 9.1*. The SCORE2 and SCORE2-OP models are preferred over other 10-year risk-prediction models, as they predict both fatal and non-fatal CVD events, have been validated and recalibrated to European populations, and because SCORE2-OP is adjusted for the competing risk of non-cardiovascular mortality. Calculating SCORE2 or SCORE2-OP is recommended for individuals with elevated BP who are not already at sufficiently high CVD risk due to established CVD, moderate or severe CKD, probable or definite familial hypercholesterolaemia, diabetes mellitus, or HMOD.[165,166,170]

For the purpose of BP-lowering treatment decisions, individuals with elevated BP and a predicted 10-year CVD risk of ≥10% by SCORE2 or SCORE2-OP are considered in these guidelines to be sufficiently high risk, with details on the choice of lifestyle or drugs to facilitate BP-lowering treatment in this setting provided in *Section 8*.[171,172] A number of considerations influenced our choice to recommend a single risk threshold of ≥10%, vs. the alternative option of using age-specific risk thresholds, such as those provided in the 2021 ESC Guidelines on cardiovascular disease prevention in clinical practice.[170] For example, contemporary data indicate the heightened importance of BP control in older adults due to their higher absolute CVD risk (resulting in a lower NNT) and concomitantly to reduce age-dependent adverse outcomes attributable to increased BP, such as dementia. Recent treat-to-target trials (testing systolic BP targets of approximately 120 mmHg) used a single CVD risk inclusion threshold and were also

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

enriched with older adults.[135,136,146] In addition, the average CVD event rate in the control arm of a landmark meta-analysis showing the benefits of more intensive BP-lowering treatment was approximately equivalent to a 10% 10-year risk.[116] Finally, the task force, which included patient members, felt that age-specific risk thresholds could result in BP treatment decisions being made solely based on age, which is difficult to support scientifically or otherwise. To try to avoid any confusion with the 2021 ESC Guidelines on cardiovascular disease prevention in clinical practice, we use the terms 'sufficiently high risk' or 'increased risk' to describe a person with 10-year CVD risk of ≥10% (rather than the terms 'high risk' or 'very high risk').

Recommendation Table 3 — Recommendations for assessing cardiovascular disease risk among individuals with elevated blood pressure (office systolic blood pressure 120–139 mmHg or diastolic blood pressure 70–89 mmHg) (see Evidence Tables 10 and 11)

Recommendations	Class[a]	Level[b]
It is recommended to use a risk-based approach in the treatment of elevated BP, and individuals with moderate or severe CKD, established CVD, HMOD, diabetes mellitus, or familial hypercholesterolaemia are considered at increased risk for CVD events.[31,153–159,161–163,172]	I	B
SCORE2 is recommended for assessing 10-year risk of fatal and non-fatal CVD among individuals aged 40–69 years with elevated BP who are not already considered at increased risk due to moderate or severe CKD, established CVD, HMOD, diabetes mellitus, or familial hypercholesterolaemia.[143,165,172]	I	B
SCORE2-OP is recommended for assessing the 10-year risk of fatal and non-fatal CVD among individuals aged ≥70 years with elevated BP who are not already considered at increased risk due to moderate or severe CKD, established CVD, HMOD, diabetes mellitus, or familial hypercholesterolaemia.[143,166,172]	I	B
It is recommended that, irrespective of age, individuals with elevated BP and a SCORE2 or SCORE2-OP CVD risk of ≥10% be considered at increased risk for CVD for the purposes of risk-based management of their elevated BP.[143,165,166,172]	I	B
SCORE2-Diabetes should be considered to estimate CVD risk among type 2 diabetes mellitus patients with elevated BP, particularly if they are <60 years of age.[164]	IIa	B

© ESC 2024

BP, blood pressure; CKD, chronic kidney disease; CVD, cardiovascular disease; HMOD, hypertension-mediated organ damage; SCORE2, Systematic COronary Risk Evaluation 2; SCORE2-OP, Systematic COronary Risk Evaluation 2–Older Persons.
Established CVD: coronary artery disease, cerebrovascular disease, peripheral arterial disease, or heart failure. For details on HMOD see *Section 7*.
[a]Class of recommendation.
[b]Level of evidence.

6.4. Refining cardiovascular disease risk estimation beyond risk models

The SCORE2 and SCORE2-OP risk-prediction models incorporate traditional risk factors such as age, sex, systolic BP, cholesterol values, and smoking status to predict 10-year risk of CVD.[165,166] However, they do not include 'non-traditional' CVD risk factors (detailed below and hereafter termed 'risk modifiers'). Non-traditional CVD risk modifiers can improve the predictive performance (i.e. discrimination) of other CVD risk-prediction models, and may also apply to SCORE2 or SCORE2-OP.[173] For example, among individuals with elevated BP and borderline increased 10-year predicted CVD risk by SCORE2 or SCORE2-OP (estimates of 5% to <10%), these non-traditional CVD risk modifiers may help up-classify the patient's risk and thereby prompt BP-lowering treatment (*Figure 8*).

6.4.1. Sex-specific non-traditional cardiovascular disease risk modifiers

Sex differences in the distribution of traditional and non-traditional CVD risk factors have been documented among patients with hypertension.[174] Although sex itself is included as an input variable in the SCORE2 and SCORE2-OP, and though these models were derived separately in men and women, some sex-specific, non-traditional risk modifiers were not included, and their associated impact on CVD risk may not be fully captured by SCORE2, SCORE2-OP, or SCORE2-Diabetes.

The relationship between BP and overall CVD risk is similar in both sexes, though some studies even suggest a stronger relative risk for CVD for a given BP level among females compared with males.[117] Female-specific, non-traditional CVD risk modifiers often arise at specific times throughout the life course, especially during pregnancy and the peri-partum period. Women with a history of hypertensive disorders of pregnancy, including gestational hypertension and pre-eclampsia, have a two-fold higher long-term risk of CVD vs. women without these pregnancy conditions.[175–177] The relative long-term CVD risk associated with hypertensive disorders of pregnancy may also be higher in younger vs. older pregnant women.[178,179] Most, but not all, of the excess CVD risk associated with hypertensive disorders of pregnancy is captured by conventional CVD risk factors.[176,178] Gestational diabetes is independently associated with an approximately two-fold increase in the long-term relative risk of CVD events.[180] Other complications such as pre-term delivery, recurrent miscarriage, and one or more stillbirths are associated with a 40% relative increase in long-term CVD risk.[181–185] Accordingly, a history of specific pregnancy complications, including gestational hypertension, pre-eclampsia, gestational diabetes, pre-term delivery, one or more stillbirths, and recurrent miscarriage, can be considered as non-traditional CVD risk modifiers to up-classify women with elevated BP and borderline increased 10-year predicted CVD risk (5% to <10%) to sufficiently high risk, thereby influencing the risk-based management of their elevated BP.

Evidence whether other female-specific conditions (infertility, polycystic ovary syndrome, and premature menopause) and male-specific conditions (androgenic alopecia and erectile dysfunction) improve prediction of CVD sufficiently to inform risk-based BP-lowering treatment decisions is inconclusive at present.

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

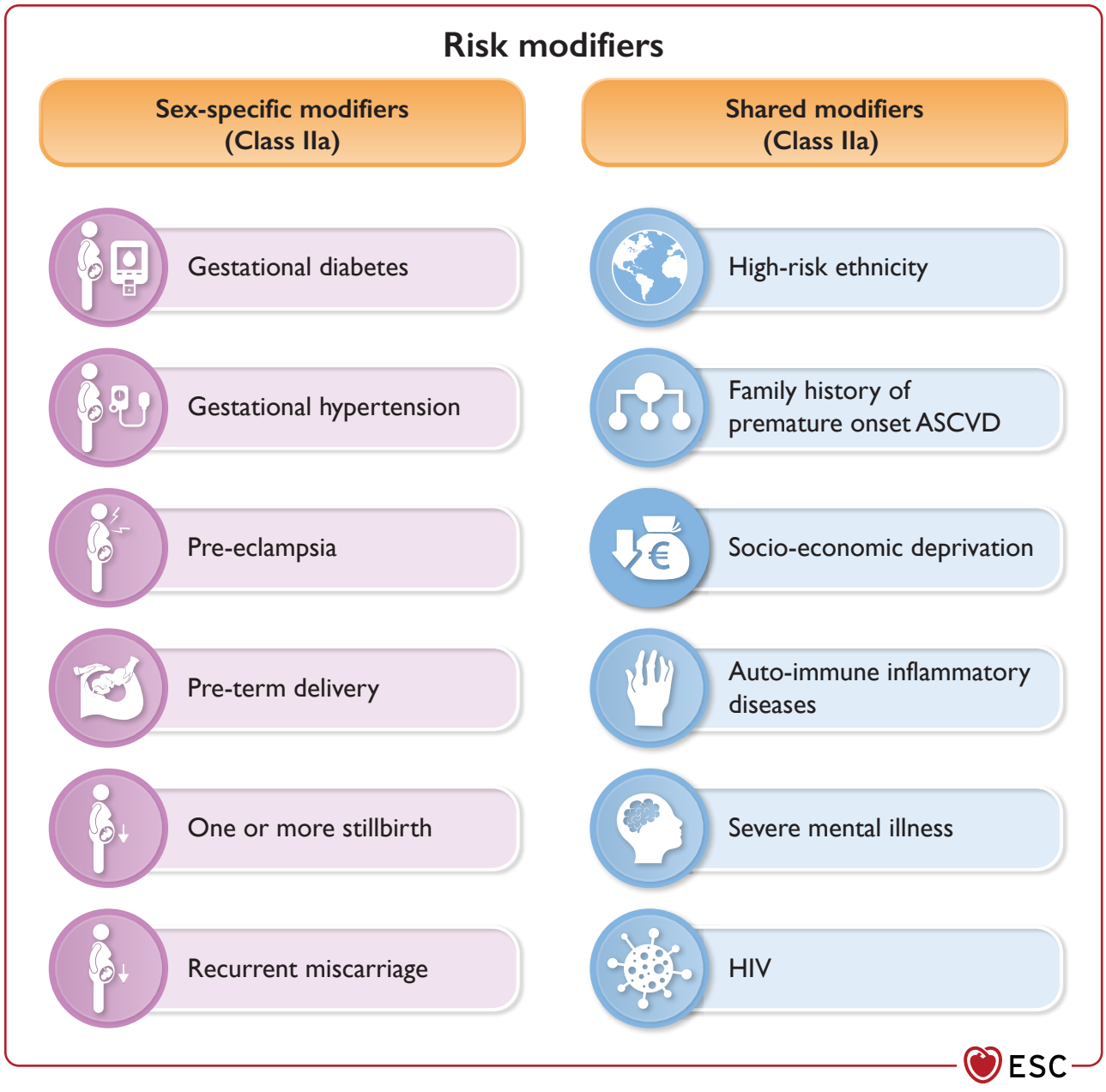


Figure 8 Cardiovascular disease risk modifiers to consider for up-classification of risk. ASCVD, atherosclerotic cardiovascular disease; HIV, human immunodeficiency virus.

6.4.2. Non-traditional cardiovascular disease risk modifiers shared by men and women

In addition to sex-specific risk modifiers, several other non-traditional risk factors are associated with an increased risk of CVD, but few have been shown to improve risk prediction or discrimination beyond traditional CVD risk factors.

We advise considering high-CVD-risk race/ethnicity (e.g. South Asian),[186–188] family history of premature onset atherosclerotic CVD (CVD event in males aged <55 years and/or females <65 years),[189,190] socio-economic deprivation,[191] inflammatory conditions (e.g. systemic lupus erythematosus, rheumatoid arthritis, and psoriasis affecting 10% or more of body surface area or requiring systemic therapy),[192–202] HIV,[203–205] and severe mental illness (major depressive disorder, bipolar disorder, and schizophrenia)[206–208] as shared non-traditional risk modifiers to up-classify the risk of individuals with a borderline increased 10-year predicted risk using SCORE2/SCORE2-OP (5% to <10%) to sufficiently high CVD risk.

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

6.4.3. Additional risk decision tests

Coronary artery calcium (CAC) scoring improves CVD risk prediction and reclassifies risk when added to conventional CVD risk factor-based estimation models.[209,210] A CAC score of >100 Agatston units or ≥75th percentile for age, sex, and ethnicity favours up-classification of CVD risk.[127] Internal or external carotid plaque may also improve CVD risk prediction.[211] Similarly, femoral artery plaque detection may improve CVD risk prediction.[212–214] Arterial stiffness, as assessed by pulse wave velocity (PWV), is associated with increased risk of CVD events and improves CVD risk stratification.[215–218] Common arterial stiffness thresholds for increased risk include carotid–femoral PWV of >10 m/s and brachial–ankle PWV of >14 m/s. After assessing 10-year predicted CVD risk and non-traditional risk factors, if a risk-based treatment decision remains uncertain for patients with elevated BP, it is reasonable to measure a CAC score or, alternatively, carotid or femoral plaque, or arterial stiffness; most especially after shared decision-making with the patient and after considering cost (see *Section 7* for more details on these tests). There is also evidence that elevated cardiac biomarker levels (specifically high-sensitivity cardiac troponin and B-type natriuretic peptide/N-terminus B-type natriuretic peptide) are significant and effective risk modifiers,[219,220] with further supportive data from hypertensive participants.[159,221,222] Of note, these cardiac biomarkers can be considered markers of HMOD (*Section 7*); however, we focus on them in this risk modifier section because they may be elevated due to other reasons besides high BP (such as atherosclerosis or heart rhythm disease).

Recommendation Table 4 — Recommendations for refining cardiovascular disease risk (see Evidence Tables 12–14)

Recommendation	Class[a]	Level[b]
History of pregnancy complications (gestational diabetes, gestational hypertension, pre-term delivery, pre-eclampsia, one or more stillbirths, and recurrent miscarriage) are sex-specific risk modifiers that should be considered to up-classify individuals with elevated BP and borderline increased 10-year CVD risk (5% to <10% risk).[183,184,223,224]	IIa	B
High-risk ethnicity (e.g. South Asian), family history of premature onset atherosclerotic CVD, socio-economic deprivation, auto-immune inflammatory disorders, HIV, and severe mental illness are risk modifiers shared by both sexes that should be considered to up-classify individuals with elevated BP and borderline increased 10-year CVD risk (5% to <10% risk).[186–191,193,198,202,204,208]	IIa	B
After assessing 10-year predicted CVD risk and non-traditional CVD risk modifiers, if a risk-based BP-lowering treatment decision remains uncertain for individuals with elevated BP, measuring CAC score, carotid or femoral plaque using ultrasound, high-sensitivity cardiac troponin or B-type natriuretic peptide biomarkers, or arterial stiffness using pulse wave velocity, may be considered to improve risk stratification among patients with borderline increased 10-year CVD risk (5% to <10% risk) after shared decision-making and considering costs.[209–211,215,218,225,226]	IIb	B

BP, blood pressure; CAC, coronary artery calcium; CVD, cardiovascular disease; HIV, human immunodeficiency virus.
[a]Class of recommendation.
[b]Level of evidence.

6.5. Summary of the cardiovascular disease risk stratification approach for allocating blood pressure treatment

Measured BP combined with 10-year CVD risk-prediction models and non-traditional risk modifiers should be used for stratifying risk when allocating BP-lowering treatment for persons with elevated BP (*Figure 9*). It is important to stress here that patients with confirmed hypertension are recommended to receive BP-lowering treatment and no further risk stratification is needed.

For patients with elevated BP, the presence of diabetes, familial hypercholesterolaemia, established CVD (defined as prior acute or chronic coronary syndrome, cerebrovascular disease, symptomatic peripheral arterial disease, or heart failure), moderate or severe CKD, or HMOD confers increased CVD risk. One caveat is that, specifically for individuals with elevated BP and type 2 diabetes mellitus only aged <60 years, SCORE2-Diabetes should be considered to identify lower CVD risk individuals (<10% over 10 years).

Otherwise, for patients without these high-risk conditions, 10-year risk of CVD should be calculated using SCORE2 (if aged 40–69 years) and SCORE2-OP (if aged ≥70 years). Patients with elevated BP and a 10-year predicted risk of CVD events ≥10% are considered sufficiently high risk to warrant BP-lowering treatment (either by lifestyle or drug treatment, see *Section 8*). For patients with elevated BP and borderline increased predicted CVD risk by SCORE2/SCORE2-OP (5% to <10% over 10 years), up-classification of risk may be considered in the presence of sex-specific or shared non-traditional risk modifiers. After considering sex-specific and shared non-traditional risk modifiers, if a risk-based BP-lowering treatment decision remains uncertain, it may be reasonable to measure CAC score, carotid or femoral plaque, high-sensitivity cardiac troponin or B-type natriuretic peptide biomarkers, or arterial stiffness.

Risk stratification for patients with non-elevated BP (systolic BP of <120 mmHg and diastolic BP of <70 mmHg) is not required for the purpose of allocating BP-lowering treatment, as the safety and efficacy of commencing BP-lowering treatment below this threshold is uncertain. Risk assessment may nonetheless be needed in this setting when considering other prevention therapies (e.g. lipid lowering).

© ESC 2024

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

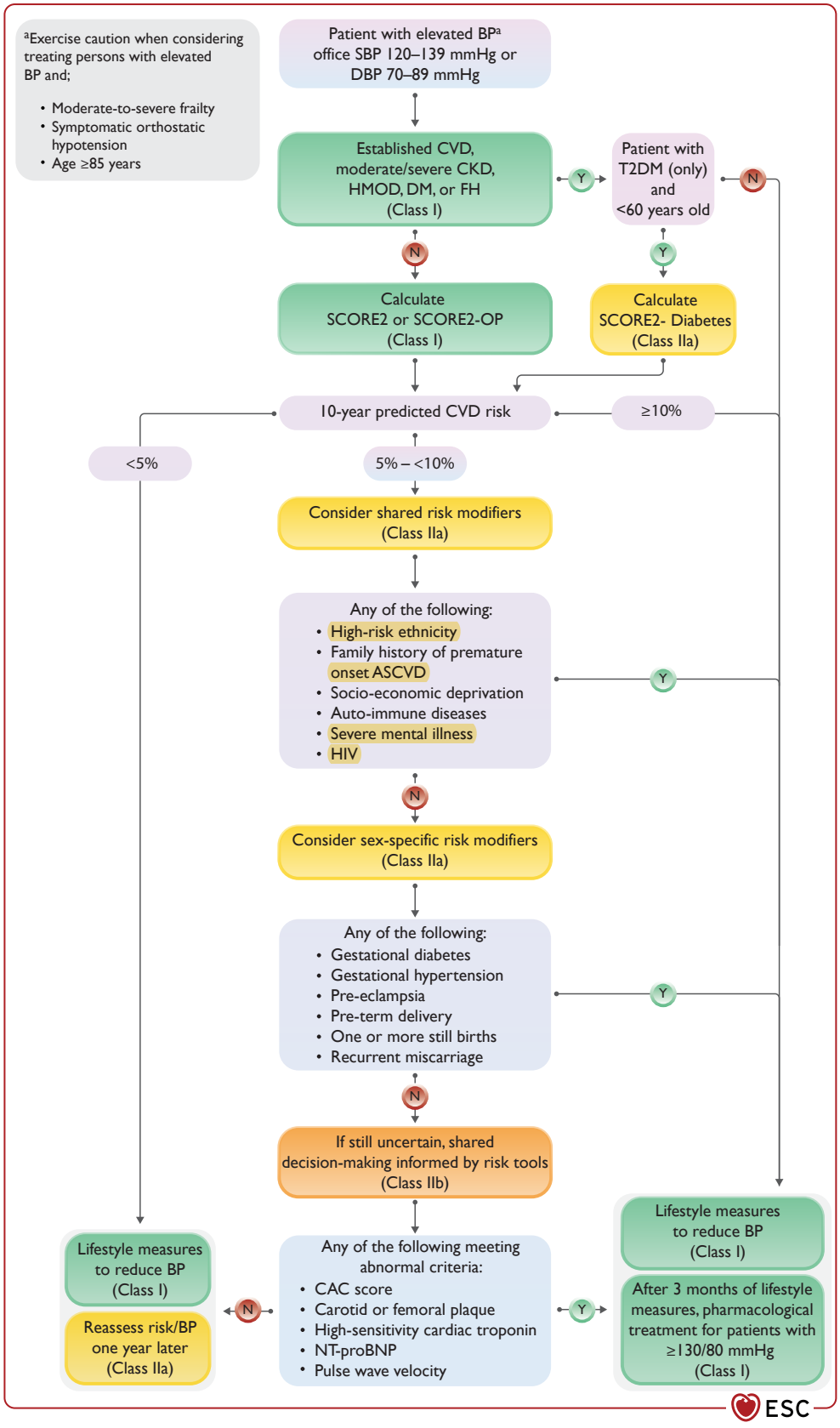


Figure 9 Summary of cardiovascular disease risk-stratification approach for blood pressure treatment in adults with elevated blood pressure. ASCVD, atherosclerotic cardiovascular disease; BP, blood pressure; CAC, coronary artery calcium; CKD, chronic kidney disease; CVD, cardiovascular disease; DBP, diastolic blood pressure; DM, diabetes mellitus; FH, familial hypercholesterolaemia; HMOD, hypertension-mediated organ damage; NT-proBNP, N-terminal pro-brain natriuretic peptide; SBP, systolic blood pressure; SCORE2, Systematic COronary Risk Evaluation 2; SCORE2-OP, Systematic COronary Risk Evaluation 2–Older Persons.

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

7. Diagnosing hypertension and investigating underlying causes

7.1. Screening for hypertension

Hypertension is predominantly an asymptomatic condition that is typically detected by systematic or opportunistic screening in a healthcare setting. Systematic screening refers to any process where individuals are identified and invited to a healthcare setting solely to measure their BP and CVD risk profile. Opportunistic screening refers to BP being measured when the patient presents to a healthcare setting for any reason, such as a routine check-up or the treatment of an acute or chronic condition. Self-screening and non-physician screening are also increasingly used.[227–230]

Few data are available on the effectiveness of different hypertension screening strategies to reduce the morbidity and mortality associated with hypertension.[231–233] More evidence is needed before systematic screening programmes with BP measurement can be recommended in all adults to reduce CVD events.[231]

Opportunistic BP screening in a primary care setting appears effective, with an estimated 90% of all adults aged >40 years in the UK having a BP check within a 5-year time period,[234] though these findings may not extrapolate to other countries. When patients provide HBPM recordings, these can also be used as part of an opportunistic screening programme (see *Section 5.2*).[235]

Despite ongoing uncertainty about the effect of hypertension screening programmes on CVD outcomes, many studies have demonstrated that screening (mostly opportunistic screening) increases hypertension detection, and that the benefits of screening likely outweigh harms.[70] Global initiatives to raise BP awareness, such as the May Measurement Month,[228] or targeted initiatives, such as the barbershop health outreach programmes,[229] are successful examples of BP screening campaigns.

Screening for hypertension, like for global CVD risk assessment, should be intermittently repeated, e.g. every 3 years. Considering the rate of progression to hypertension in European population samples,[236] it is reasonable to measure BP at least every 3 years in the case of non-elevated BP and low–moderate CVD risk (i.e. individuals aged <40 years). More frequent BP checks (i.e. yearly) should be considered in individuals 40 years or older and individuals with elevated BP not currently meeting indications for treatment[170] (*Figure 10*).

Recommendation Table 5 — Recommendations for blood pressure screening (see Evidence Table 15)

Recommendation	Class[a]	Level[b]
Opportunistic screening for elevated BP and hypertension should be considered at least every 3 years for adults aged <40 years.[236,237]	IIa	C
Opportunistic screening for elevated BP and hypertension should be considered at least annually for adults aged ≥40 years.[231,237]	IIa	C
In individuals with elevated BP who do not currently meet risk thresholds for BP-lowering treatment, a repeat BP measurement and risk assessment within 1 year should be considered.	IIa	C
Other forms of screening for hypertension (i.e. systematic screening, self-screening, and non-physician screening) may be considered, depending on their feasibility in different countries and healthcare systems.[231–233]	IIb	B

© ESC 2024

BP, blood pressure.
[a]Class of recommendation.
[b]Level of evidence.

7.2. Confirming the diagnosis of hypertension

As noted in *Section 5*, assessment at a single visit by office BP has lower specificity compared with ABPM for diagnosing hypertension.[70,238–241] Accordingly, a protocol for confirming the diagnosis of hypertension is proposed (*Figure 10*), with out-of-office BP measurement as the preferred method for confirming cases of elevated BP or hypertension. For initial screening systolic BP of >160 mmHg and/or diastolic BP of >100 mmHg, a prompt re-evaluation (within days to weeks but not >1 month) preferably with ABPM or HBPM is advisable.[71] BP of >180/110 mmHg at screening requires exclusion of hypertensive emergencies, which should be managed as appropriate (see [242] and *Section 10*) with prompt treatment. For individuals with BP of >180/110 mmHg at screening but without hypertensive emergency, prompt confirmation (preferably within a week) can be considered prior to commencing treatment.

Recommendation Table 6 — Recommendations for confirming hypertension diagnosis

Recommendations	Class[a]	Level[b]
In individuals with increased CVD risk where their screening office BP is 120–139/70–89 mmHg, it is recommended to measure BP out of office, using ABPM and/or HBPM or, if not logistically feasible, by making repeated office BP measurements on more than one visit.[70,238–241]	I	B
Where screening office BP is 140–159/90–99 mmHg, it is recommended that the diagnosis of hypertension should be based on out-of-office BP measurement with ABPM and/or HBPM. If these measurements are not logistically or economically feasible, then diagnosis can be made on repeated office BP measurements on more than one visit.[70,238–241]	I	B
Where screening office BP is ≥160/100 mmHg: • It is recommended that BP 160–179/100–109 mmHg be confirmed as soon as possible (e.g. within 1 month) preferably by either home or ambulatory BP measurements; • It is recommended when BP ≥180/110 mmHg that hypertensive emergency be excluded.	I	C

© ESC 2024

ABPM, ambulatory blood pressure measurement; BP, blood pressure; CVD, cardiovascular disease; HBPM, home blood pressure measurement.
[a]Class of recommendation.
[b]Level of evidence.

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

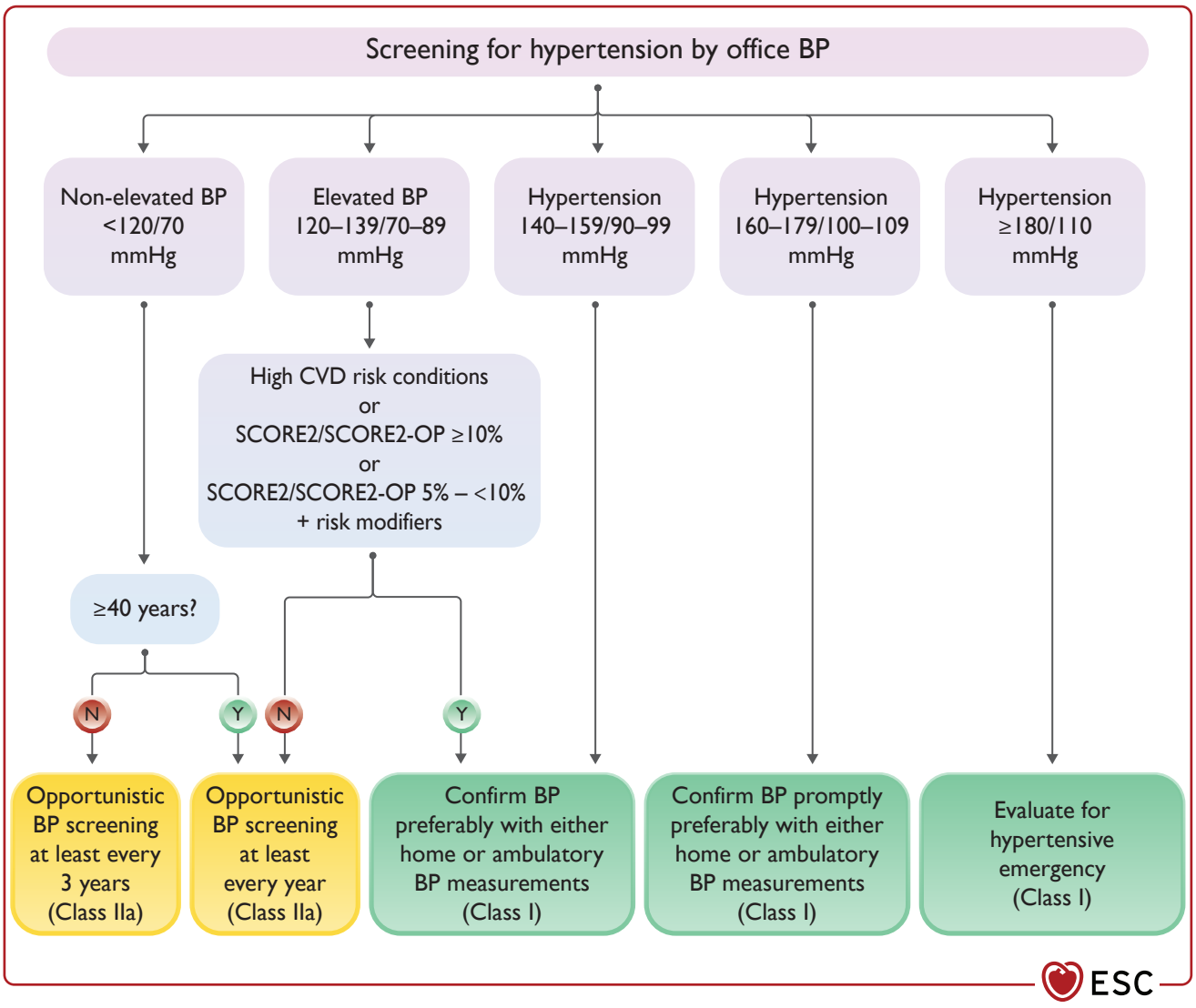


Figure 10 Protocol for confirming hypertension diagnosis. BP, blood pressure; CVD, cardiovascular disease; SCORE2, Systematic COronary Risk Evaluation 2; SCORE2-OP, Systematic COronary Risk Evaluation 2–Older Persons.

7.3. Communicating the diagnosis

Behavioural responses to health-related threats are strongly influenced by five core themes (termed 'illness representations'), which are identity, timeline, cause, consequences, and control/cure.[243,244] These illness representations form the basis of how patients understand a diagnosis, and can influence their responses after being diagnosed with hypertension.[243] This conceptual framework can help guide the clinical communication of a diagnosis of hypertension. For example, patients' understanding of the chronic nature of hypertension (i.e. timeline theme) is key for ensuring long-term engagement with medical treatment.[245] Prior to commencing treatment, it is helpful to understand the extent to which patients believe that medications are necessary and ascertain if they have concerns.[246] The core illness representations and beliefs about medicines for clinicians to consider are included in *Table 7*.

The 2021 ESC Guidelines on cardiovascular disease prevention in clinical practice recommend "an informed discussion about CVD risk and treatment benefits—tailored to the needs of a patient" as part of a diagnosis communication in hypertension.[170] This can be facilitated using an interdisciplinary healthcare-provided approach (see *Section 11*) and by visual information or other more accessible material that might optimally communicate hypertension-related risk.[128] Visualizing risk by medical imaging to motivate risk-reducing behaviour changes may also be beneficial.[247]

7.4. Baseline assessment and diagnostic approach

7.4.1. Medical history, medication history, and physical examination

The purpose of clinical evaluation is to diagnose hypertension, delineate factors potentially contributing to hypertension, identify other CVD risk factors, define relevant comorbidities, screen for potential secondary causes of hypertension (where indicated), and establish whether

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

Table 7 Key illness representations and treatment beliefs: how these apply to communicating a hypertension diagnosis to the patient (note that gender influences these representations)

Illness representation	Example patient question	Application to a hypertension diagnosis conversation
Identity	What is the disease/illness label and the related symptoms?	The condition where your systolic BP is ≥140 and/or diastolic BP ≥90 mmHg is called hypertension. We classify systolic BP 120–139 or diastolic BP 70–89 mmHg as elevated BP. For most people, this has no noticeable signs or symptoms, therefore, we need to monitor your BP to assess how medications and behavioural changes are working.
Control	Is the illness controllable through medical intervention or behavioural change?	Hypertension can usually be controlled with medication and behavioural changes such as dietary changes and regular physical activity. For some people we need to try a few different options before we get BP under control.
Timeline	Is this an acute or chronic problem?	This is a serious long-term or chronic condition condition that will require long-term management. This means that it may need to be managed throughout life.
Consequences	What are the physical and psychosocial consequences?	If hypertension is not controlled, then there is a risk of a serious acute cardiovascular disease event such as a stroke or heart attack; however, if it is managed through the right medical intervention and behavioural changes, then this risk can be reduced and the condition will have less consequences for your life.
Causes	What caused the condition?	Multiple factors contribute to someone developing hypertension. These include both non-modifiable factors (e.g. genetics and age) and modifiable factors (e.g. diet, weight, and physical activity). We are best focusing on those things that we can control to reduce your BP.
Treatment beliefs	**Example patient question**	**Application to a hypertension diagnosis communication**
Necessity	To what extent is treatment necessary?	Taking BP-lowering medication every day is necessary to keep your BP under control and to help prevent a more serious health problem developing. Do you think that these medicines will help you?
Concerns	To what extent does treatment cause concern?	Some patients have concerns about taking daily medications throughout their life, e.g. about side effects. Do you have any concerns about taking your BP medications every day?

BP, blood pressure.

there is evidence of HMOD or existing cardiac, cerebrovascular, or renal disease.

Details on medical history and physical examination steps are summarized in the supplement (see Supplementary data online, *Tables S2 and S3*), as well as drugs or substances that may increase BP (see Supplementary data online, *Table S4*).

7.4.2. Drug adherence and persistence with treatment

Adherence is defined as the extent to which a patient's behaviour, e.g. with respect to taking medication, coincides with agreed recommendations from a healthcare provider. Persistence represents the amount of time from initiation to discontinuation of therapy.[248] Adherence to medical therapies is especially suboptimal in asymptomatic conditions such as hypertension.[249–254] Non-adherence to BP-lowering therapy correlates with a higher risk of CVD events.[255,256] Objective methods to assess adherence, such as detecting prescribed drugs in blood or urine samples and directly observed treatment (witnessed pill intake during ABPM), have demonstrated their potential usefulness, particularly in the setting of apparently resistant hypertension.[257] However, all methods for testing drug adherence have limitations.

Non-adherence to BP-lowering therapy depends on many factors (*Figure 11*).[253] Effective patient–physician communication is crucial to improve adherence.[258,259] Single-pill combinations improve persistence in BP-lowering treatment and are associated with lower all-cause mortality.[260]

Recommendation Table 7 — Recommendations for assessing adherence and persistence with treatment (see Evidence Table 16)

Recommendation	Class[a]	Level[b]
Objective evaluation of adherence (either directly observed treatment or detecting prescribed drugs in blood or urine samples) should be considered in the clinical work-up of patients with apparent resistant hypertension, if resources allow.[261–263]	IIa	B

[a]Class of recommendation.
[b]Level of evidence.

7.4.3. Routine and optional tests

Routine tests include laboratory and clinical tests to detect increased CVD risk and relevant comorbidities (e.g. hyperlipidaemia and diabetes) (*Table 8*).

Optional tests should be considered in the initial assessment if they are likely to change patient management, with the main rationale being to improve CVD risk stratification.[170] As highlighted in *Section 6*, for adults with elevated BP who also have a 10-year estimated CVD risk of 5% to <10%, optional tests including those for HMOD may be considered if up-classification of risk on the basis of an abnormal test result could prompt initiation of BP-lowering therapy.[31,170] Evidence of subclinical microvascular neurodegeneration and/or lacunar brain disease due to small-vessel pathology may also indicate HMOD.[264]

© ESC 2024
Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

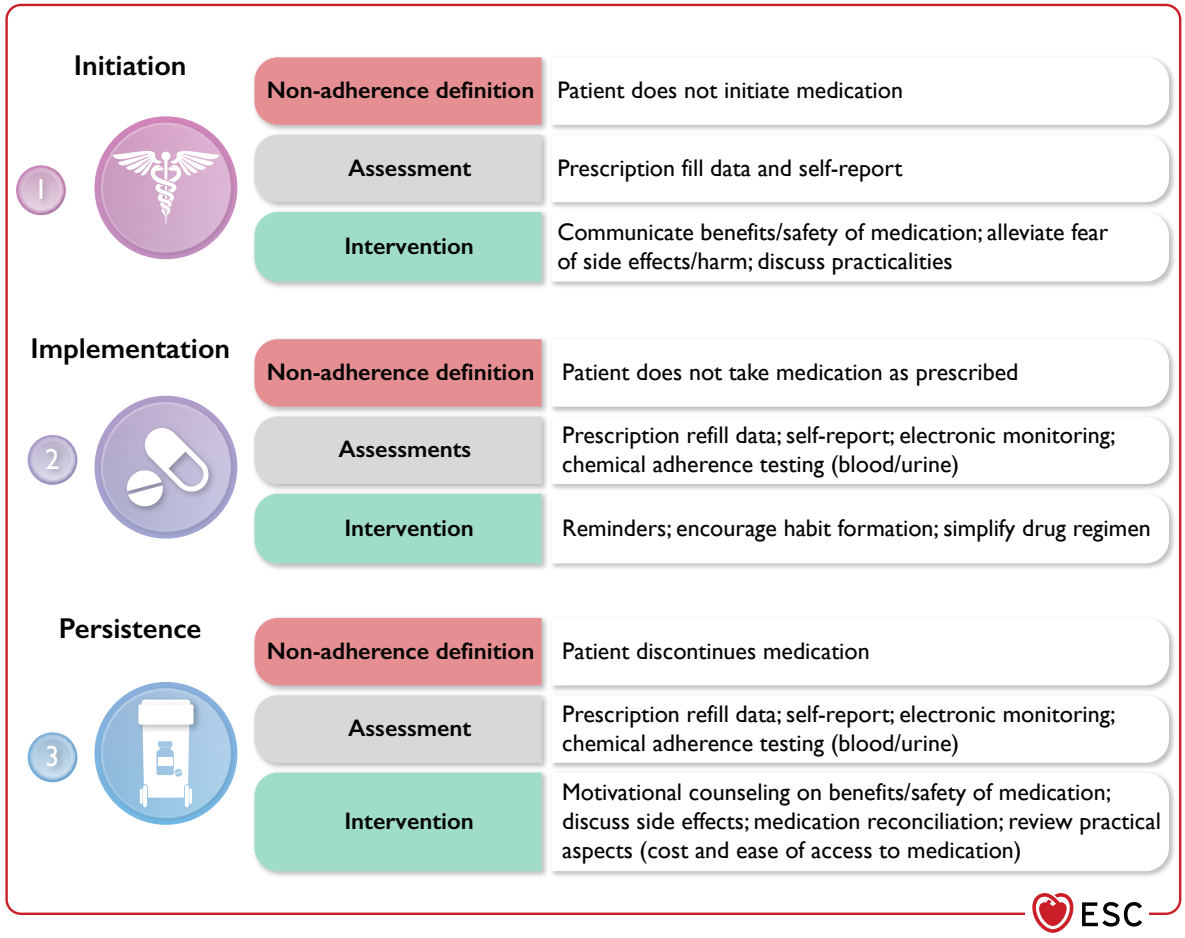


Figure 11 Definitions, assessments, and potential interventions for the three phases of adherence to BP-lowering medications.

Table 8 Routine tests recommended in the initial work-up of a patient with elevated blood pressure or hypertension

Routine test	Clinical utility
Fasting blood glucose (and HbA1c if fasting blood glucose is elevated)	Assessing CVD risk and comorbidities
Serum lipids: total cholesterol, LDL cholesterol, HDL and non-HDL cholesterol, triglycerides	Assessing CVD risk
Blood sodium and potassium, haemoglobin and/or haematocrit, calcium, and TSH	Screening secondary hypertension (primary aldosteronism, Cushing's disease, polycythaemia, hyperparathyroidism, and hyperthyroidism)
Blood creatinine and eGFR; urinalysis and urinary albumin-to-creatinine ratio	Assessing CVD risk and HMOD Guiding treatment choice Screening secondary hypertension (renoparenchymal and renovascular)
12-lead ECG	Assessing HMOD (left atrial enlargement, left ventricular hypertrophy) Assessing irregular pulse and other comorbidities (AF, previous acute myocardial infarction)

AF, atrial fibrillation; CVD, cardiovascular disease; ECG, electrocardiogram; eGFR, estimated glomerular filtration rate; HbA1c, glycated haemoglobin; HDL, high-density lipoprotein; HMOD, hypertension-mediated organ damage; LDL, low-density lipoprotein; TSH, thyroid-stimulating hormone.

While the role of optional tests for HMOD (*Table 9*) in the management of elevated BP is emphasized in these guidelines, we also note that these tests may help to optimize treatment in hypertensive adults with BP of >140/90 mmHg who are prescribed BP-lowering therapy (e.g. by facilitating patient adherence and overcoming clinician inertia in achieving an intensive BP treatment target of as low as 120 mmHg systolic). The role of visualizing HMOD in helping motivate risk-reducing changes in patients and overcome physician inertia has been tested in interventional trials (*Section 7.3*).[247,265–267]

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

© ESC 2024

Table 9 Optional tests that may be used as clinically indicated in the initial work-up of a patient with elevated blood pressure or hypertension to assess hypertension-mediated organ damage or established cardiovascular disease

Optional test	Clinical utility
Echocardiography	Assessing HMOD (hypertensive heart disease) Assessing established CVD (previous acute myocardial infarction, heart failure) Assessing thoracic aorta dilation
CAC by cardiac CT or carotid or femoral artery ultrasound imaging	Assessing HMOD (atherosclerotic plaque)
Large artery stiffness (carotid–femoral or brachial–ankle PWV)	Assessing HMOD (arterial stiffness)
High-sensitivity cardiac troponin and/or NT-proBNP	Assessing HMOD
Ankle–brachial index	Assessing established CVD (lower-extremity arterial disease)
Abdominal ultrasound	Assessing established CVD (abdominal aneurysm)
Fundoscopy	Assessing HMOD (hypertensive retinopathy) Diagnosing hypertensive emergency/malignant hypertension (haemorrhages and exudates, papilloedema)

© ESC 2024

CAC, coronary artery calcium; CT, computed tomography; CVD, cardiovascular disease; HMOD, hypertension-mediated organ damage; NT-proBNP, N-terminal pro-brain natriuretic peptide; PWV, pulse wave velocity.

HMOD assessment is also an important way to identify young adults <40 years old who have increased CVD risk, since 10-year estimated CVD risk by SCORE2 cannot be calculated in this age group (see *Section 8.1*). More details on diagnostic thresholds for HMOD by the various assessment options, including important sex differences, are provided in Supplementary data online, *Tables S1 and S5* and *Figure 12*.

Finally, some individuals may be at heightened risk for CVD events when cardiac and vascular HMOD measurements like LVH and increased PWV do not regress over time with appropriate treatment and BP control.[14,268–271]

Investigations aimed at screening for secondary hypertension are additional optional tests and are detailed in *Section 7.6*. Of note, patients with an incidental adrenal nodule or nodules (typically detected on imaging of the abdomen done for other clinical reasons) warrant screening for elevated BP and hypertension. Those with adrenal incidentalomas and hypertension warrant a basic work-up for secondary hypertension to include screening for primary aldosteronism, Cushing's syndrome and phaeochromocytoma.

7.4.3.1. The kidneys

CKD is defined as abnormalities of kidney structure or function, present for at least 3 months with implications for health.[272] Renal function is evaluated initially using serum creatinine and an estimated glomerular filtration rate (eGFR) equation (preferably race-free CKD-EPI) and typically for proteinuria.[273] Our definition of moderate-to-severe CKD requires an eGFR of <60 mL/min/1.73 m^2 or albuminuria of ≥30 mg/g (≥3 mg/mmol). Intensive BP control in patients with CKD reduces rates of CVD events.[274,275] CKD can influence the choice of BP-lowering treatment (*Sections 8* and *9*), as well as newer drugs for cardiovascular prevention, such as sodium–glucose co-transporter 2 (SGLT2) inhibitors and finerenone.

We recommend repeat measurement of eGFR and urine albumin: creatinine ratio (ACR) at least annually if clinically significant CKD is diagnosed. Renal ultrasound and Doppler examination to evaluate causes of CKD and to exclude renoparenchymal and renovascular hypertension (RVH) should also be considered.[276,277]

Recommendation Table 8 — Recommendations for assessing renal hypertension-mediated organ damage

Recommendation	Class[a]	Level[b]
It is recommended to measure serum creatinine, eGFR, and urine ACR in all patients with hypertension.[170,273]	I	A
If moderate-to-severe CKD is diagnosed, it is recommended to repeat measurements of serum creatinine, eGFR, and urine ACR at least annually.[276]	I	C
Renal ultrasound and Doppler examination should be considered in hypertensive patients with CKD to assess kidney structure and determine causes of CKD and to exclude renoparenchymal and renovascular hypertension.[276,277] CT or magnetic resonance renal angiography are alternative testing options.	IIa	C

© ESC 2024

ACR, albumin:creatinine ratio; CKD, chronic kidney disease; CT, computed tomography; eGFR, estimated glomerular filtration rate.
[a]Class of recommendation.
[b]Level of evidence.

7.4.3.2. The heart

A 12-lead ECG is a part of the initial routine work-up for all patients with hypertension and should be repeated whenever patients present with an irregular pulse or cardiac symptoms. The ECG should be analysed for LVH (Supplementary data online, *Table S1*) and AF.[31,278–282]

Echocardiography is recommended in patients with hypertension when the ECG is abnormal, murmurs are detected, or there are cardiac symptoms. A full, standardized, two-dimensional echocardiogram should be performed, preferably with tissue Doppler and strain assessment. Echocardiography can be considered for all patients with newly diagnosed hypertension, if local resources and reimbursement policies allow. Over 5 years of follow-up, subclinical left ventricular diastolic dysfunction predicts the incidence of CVD.[283–285] In addition, LVH detected by echocardiography predicts total and cardiovascular mortality and CVD events in the general population,[286,287] including in young adults.[26,268,288,289] Data on the associations with CVD of other metrics for detecting HMOD in the heart are also available.[26,268,290] Since cardiac size and function differ by sex, sex-specific thresholds for detecting HMOD in the heart are used to avoid under-diagnosis in women.[25,174,291]

Recommendation Table 9 — Recommendations for assessing cardiac hypertension-mediated organ damage

Recommendation	Class[a]	Level[b]
A 12-lead ECG is recommended for all patients with hypertension.[31,281]	I	B

Continued

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

Echocardiography is recommended in patients with hypertension and ECG abnormalities, or signs or symptoms of cardiac disease.[14,31,292]	I	B
Echocardiography may be considered in patients with elevated BP, particularly when it is likely to change patient management.[31,291]	IIb	B

© ESC 2024

BP, blood pressure; ECG, electrocardiogram.
[a]Class of recommendation.
[b]Level of evidence.

7.4.3.3. The arteries

Cardiac computed tomography (CT) may be used to measure CAC and, if intravascular contrast is administered, fully visualize coronary artery disease to improve risk stratification.[211,293] As noted in *Section 6*, CAC scoring can reclassify CVD risk upwards or downwards in addition to conventional risk factors.[127,170,211,233,294] Carotid ultrasound detects presence or absence of carotid plaque (wall thickness ≥ 1.5 mm) and stenosis. Presence of plaque in the carotid or femoral arteries improves risk prediction for CVD events in asymptomatic patients on top of conventional risk-factor assessment.[211,247,265,267,295,296] Systematic use of intima media thickness does not appear to consistently improve prediction of future CVD events.[297] Arterial stiffness is measured as carotid–femoral PWV or brachial–ankle PWV, and can contribute to predictive value and risk reclassification.[28,31,215,216] PWV is currently used mostly for research purposes or in specialist referral centres. Checking for inter-arm BP difference may identify a subclavian stenosis as vascular HMOD.[38] Other tests assessing the vasculature [such as abdominal ultrasound or ankle–brachial index (ABI)] should also be considered in patients with hypertension, when specific cardiovascular complications (abdominal aneurysm, peripheral artery disease) are clinically suspected. Finally, microvascular HMOD can be assessed by fundoscopy. A simplified classification has been proposed and validated.[298] In hypertensive individuals, the presence of mild or moderate hypertensive retinopathy is associated with an increased risk of CVD events.[299] Fundoscopy is recommended also in hypertensive diabetic patients and in the work-up of malignant hypertension and hypertensive emergencies.

Recommendation Table 10 — Recommendations for assessing vascular hypertension-mediated organ damage (see Evidence Table 17)

Recommendations	Class[a]	Level[b]
Fundoscopy is recommended if BP >180/110 mmHg in the work-up of hypertensive emergency and malignant hypertension, as well as in hypertensive patients with diabetes.	I	C
Fundoscopy for detecting hypertensive retinopathy may be considered in patients with elevated BP or hypertension.[299]	IIb	B
Ultrasound examination of the carotid or femoral arteries for detecting plaque may be considered in patients with elevated BP or hypertension when it is likely to change patient management.[211]	IIb	B
Coronary artery calcium scoring may be considered in patients with elevated BP or hypertension when it is likely to change patient management.[127,211]	IIb	B
Measurement of PWV may be considered in patients with elevated BP or hypertension when it is likely to change patient management.[28,31,215,216]	IIb	B

Continued

© ESC 2024

BP, blood pressure; PWV, pulse wave velocity.
[a]Class of recommendation.
[b]Level of recommendation.

7.4.4. Genetic testing

Hypertension is considered a complex polygenic disorder, because many genes or gene combinations influence BP.[300,301] However, some well-defined phenotypes relating to single-gene mutations (i.e. monogenic forms of hypertension) have been identified (see Supplementary data online, *Table S6*). These are rare, but knowledge of the genetic defect may allow targeted treatment of the proband and also proper management of the patient's siblings.[302,303] As such, genetic testing should be considered only for those with a high prior probability of having a monogenic condition and such patients should be referred to specialized centres. In most patients with elevated BP or hypertension, routine genetic testing is not recommended. Family history and a pedigree analysis can help to find a heritable pattern of hypertension or hypotension.[304]

Recommendation Table 11 — Recommendations for genetic testing in hypertension management

Recommendations	Class[a]	Level[b]
Genetic testing should be considered in specialist centres for patients suspected to have rare monogenic causes of secondary hypertension or for those with phaeochromocytoma/paraganglioma.[302,305]	IIa	B
Routine genetic testing for hypertension is not recommended.	III	C

© ESC 2024

[a]Class of recommendation.
[b]Level of evidence.

7.5. Resistant hypertension: definition and diagnosis

Despite availability and use of multiple BP-lowering medications, many patients worldwide have uncontrolled hypertension.[306–308] Considering this, societies have introduced the term 'drug-resistant hypertension', or 'treatment-resistant hypertension', or 'resistant hypertension',[309] which has been reported in 10%–20% of patients with hypertension.[310,311]

Resistant hypertension is not a disease *per se*. Compared with treated patients who achieve BP control, patients with resistant hypertension (by any definition) have a worse prognosis: risk of myocardial infarction, stroke, end-stage renal disease, and death in these adults may be two- to six-fold higher.[309] Secondary causes of hypertension are also more likely in the presence of resistant hypertension.[312]

All resistant hypertension definitions require a diuretic in the prescribed multiple-drug regimen, because excess salt intake and salt and water retention are key players in resistance to BP-lowering treatments (*Table 10*).[309]

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

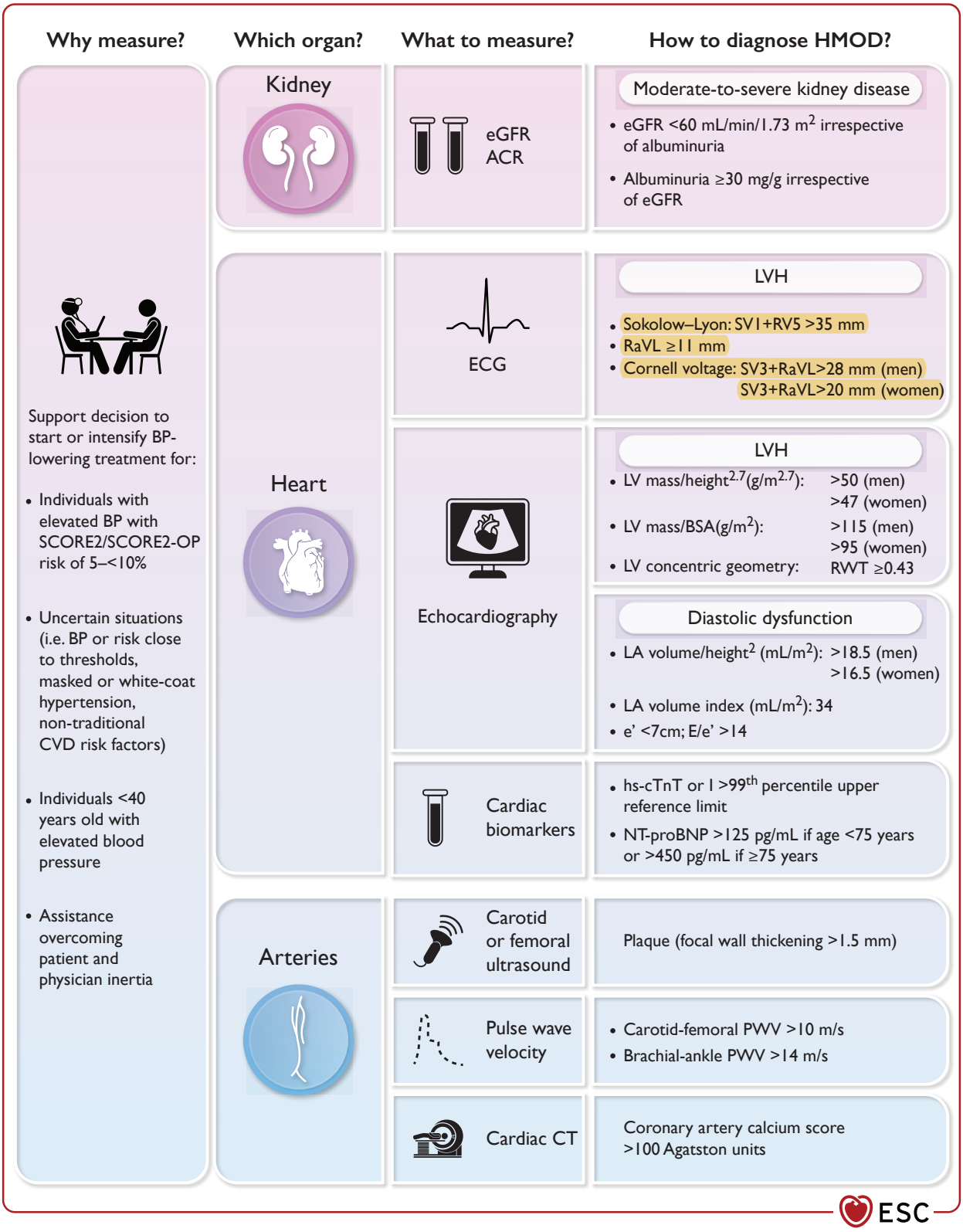


Figure 12 Tests and criteria for defining hypertension-mediated organ damage and considerations for their use in clinical practice. ACR, albumin: creatinine ratio; BP, blood pressure; BSA, body surface area; CT, computed tomography; CVD, cardiovascular disease; ECG, electrocardiogram; eGFR, estimated glomerular filtration rate; HMOD, hypertension-mediated organ damage; hs-cTnT, high-sensitivity cardiac troponin T; LA, left atrial; LV, left ventricular; LVH, left ventricular hypertrophy; NT-proBNP, N-terminal pro-brain natriuretic peptide; PWV, pulse wave velocity; RWT, relative wall thickness; SCORE2, Systematic COronary Risk Evaluation 2; SCORE2-OP, Systematic COronary Risk Evaluation 2–Older Persons. More details and references can be found in the Supplementary data online, *Tables S1 and S5*.

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

Table 10 Current definition of resistant hypertension

Definition of resistant hypertension
Hypertension is defined as resistant when a treatment strategy including appropriate lifestyle measures and treatment with maximum or maximally tolerated doses of a diuretic (thiazide or thiazide-like), a RAS blocker, and a calcium channel blocker fail to lower office systolic and diastolic BP values to <140 mmHg and/or <90 mmHg, respectively. These uncontrolled BP values must be confirmed by out-of-office BP measurements (HBPM or ABPM—*Section 5.1* for relevant BP thresholds).
Key considerations
• Resistant hypertension is not a disease, but an indicator that should be used to identify patients at high risk for CVD, in which secondary hypertension is also frequent; • Pseudo-resistant hypertension must be excluded, including that caused by non-adherence to treatment; • In patients with decreased eGFR (i.e. <30 mL/min/1.73 m^2) an adequately up-titrated loop diuretic is necessary to define resistant hypertension; • Patients with suspected resistant hypertension should be referred to specialized centres; • These ESC Guidelines do not include the terms 'controlled resistant hypertension' (BP at target but requiring ≥4 medications) or 'refractory hypertension' (BP not at target despite ≥5 medications).

ABPM, ambulatory blood pressure monitoring; BP, blood pressure; CVD, cardiovascular disease; eGFR, estimated glomerular filtration rate; HBPM, home blood pressure monitoring; RAS, renin–angiotensin system.

© ESC 2024

In addition, excluding pseudo-resistance is a prerequisite. Specifically, pseudo-resistance indicates poor adherence to BP-lowering treatment, which should be verified by careful questioning of the patient in the first instance (*Section 7.4.3*).[257] In addition, white-coat hypertension must be excluded.[263] Contributors to pseudo-resistance are listed in *Table 11*. Objective evaluation of adherence (either directly observed treatment or detecting prescribed drugs in blood or urine samples) should also be considered, if resources allow.

The work-up of patients presumed to have resistant hypertension is complex and often requires technologies that are not available to GPs.[257,309] Accordingly, we recommend these patients are referred to specialized centres.

Recommendation Table 12 — Recommendations for resistant hypertension work-up (see Evidence Table 18)

Recommendation	Class[a]	Level[b]
Patients with resistant hypertension should be considered for referral to clinical centres with expertise in hypertension management for further testing.[309,312]	IIa	B

[a]Class of recommendation.
[b]Level of evidence.

© ESC 2024

7.6. Secondary hypertension: when to screen/further investigations

7.6.1. General considerations

Secondary hypertension is more prevalent than previously thought (*Figures 13–15*).[312–317] Depending on the definition used and the cohort studied, the prevalence of secondary hypertension is 10%–35% in all hypertensive patients[318,319] and up to 50% of patients with resistant hypertension (though the latter prevalence estimate included persons with eGFR < 40 mL/min/1.73 m^2).[312] Primary aldosteronism is a common cause,[315,320] with, e.g. a high prevalence of hyperaldosteronism (up to 12%) observed in patients with BP of >180/110 mmHg.[316] Despite these numbers, screening rates for primary aldosteronism, even in high-risk groups such as those with resistant hypertension[321] and hypokalaemia,[322] are low (around 2% and 4% of eligible patients, respectively). In most healthcare systems, GPs are typically the 'gatekeeper' of access to specialized care and should be involved in screening patients for common causes of secondary hypertension, especially sleep apnoea and primary aldosteronism (Supplementary data online, *Tables S2 and S3*). Primary aldosteronism is associated with an increased risk of CVD events, which may be partly independent of BP.[323,324]

Table 11 Conditions found to cause pseudo-resistance or resistance to blood pressure-lowering treatment

Causes of pseudo-resistant hypertension
Poor adherence to and persistence with treatment
White-coat phenomenon
Poor BP measurement method
Marked brachial artery calcification (Osler phenomenon)
Clinician inertia (inadequate doses, inappropriate combinations of BP-lowering drugs)
Munchausen syndrome (rare)
Causes of resistant hypertension
Behavioural factors
Overweight/obesity
Physical inactivity
Excess daily dietary sodium
Excess habitual alcohol consumption
Use of drugs or substances that may increase BP
See Supplementary data online, *Table S4*
Undetected secondary hypertension
See *Table 13*

BP, blood pressure.

© ESC 2024

7.6.2. Primary aldosteronism

Though spontaneous or diuretic-induced hypokalaemia are strongly suggestive of primary aldosteronism, a history of hypokalaemia is not present in most patients diagnosed with this condition. The aldosterone-to-renin ratio (ARR) is thus recommended for primary aldosteronism screening (see *Figure 13*).[325] This test can easily be done in treatment-naïve patients, though it is far more common for the ARR test to be considered when patients are already being treated for elevated BP or hypertension. This is relevant because ARR can be influenced by the drugs being taken at the time of testing. Accordingly, there are 2 approaches to screen for aldosteronism among patients who are already undergoing treatment for elevated BP or hypertension:

- The first is to conduct ARR testing in treated patients with an indication for aldosteronism screening as efficiently as possible and without changing or stopping their baseline BP-lowering medications, simply to facilitate such testing. The ARR result then needs to be interpreted in the context of the specific medication(s) the patient is taking. Advantages of this approach include reducing barriers to screening and no change in medication in these patients, many of whom do not have BP controlled and in whom further deterioration in their BP control by stopping or changing medication may increase risk of CVD. Disadvantages include the interpretation of the ARR result, which depends on the specific medications taken at the time of testing.[326] Input from a hypertension specialist or endocrinologist may be necessary.

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

- To reliably estimate renin and aldosterone status (and therefore ARR), and to facilitate a 'clean' screen for aldosteronism, a second approach is to discontinue drugs that affect these variables whenever feasible before ARR testing (*Table 12*). Such interfering drugs include, beta-blockers, centrally acting drugs (e.g., clonidine and alpha-methyldopa) renin-angiotensin system (RAS) blockers and diuretics.[326] Long-acting calcium channel blockers (CCBs), either dihydropyridine or non-dihydropyridine, and alpha-receptor antagonists do not interfere with the ARR and can be used instead of interfering medications before ARR testing. Should drugs that do not interfere with the ARR be contraindicated or insufficient to control BP, centrally acting sympatholytic drugs can also then be used, but at the risk of slightly more false positives (by renin suppression). Furthermore, when mineralocorticoid receptor antagonists (MRAs) cannot be stopped for safety reasons (i.e. severe hypokalaemia or severe hypertension among patients with severe hyperaldosteronism), recent evidence suggests that the accuracy of ARR testing under this treatment is only marginally impacted, particularly in the presence of florid primary aldosteronism.[327]

Assessing sodium intake (preferably 24 h urinary sodium, or sodium-to-creatinine ratio in the morning urine sample) is also important for interpreting the ARR, as is time in menstrual cycle for females. ARR cut-offs vary depending on unit of measurement and by local laboratory. For detailed information, readers are referred to the latest primary aldosteronism guidelines.[328,329]

7.6.3. Renovascular hypertension

Renovascular hypertension (RVH) defines a condition where renal artery occlusion or stenosis decreases renal perfusion pressure to a level that activates the renin–angiotensin–aldosterone system (RAAS), thereby raising BP. Major causes are atherosclerosis and fibromuscular dysplasia (*Figure 14* and Supplementary data online, *Tables S1* and *S2*). Atherosclerosis is the most common form of RVH, especially in older adults.[318] Fibromuscular dysplasia is a systemic non-atherosclerotic vascular disease involving medium-sized muscular arteries. When renal arteries are involved, fibromuscular dysplasia may induce RVH (FMD-RVH), especially in children and younger women.[330–332]

Though not highly sensitive, very elevated renin levels raise the suspicion for RVH. The work-up of RVH (*Table 13*) is based on imaging tests, such as renal artery Doppler ultrasound, with bilateral assessment of renal arterial resistive index, or abdominal CT angiography, or magnetic resonance imaging (MRI), in line with current ESC Guidelines on the diagnosis and treatment of peripheral arterial diseases,[277] which will be updated in 2024. Of note, bystander renal artery stenosis may be present in patients with essential hypertension, without causing secondary hypertension due to RVH. Since fibromuscular dysplasia is a systemic disease, CT or MRI angiography from head to pelvis is recommended in patients with FMD-RVH.[277,332]

7.6.4. Obstructive sleep apnoea syndrome

Obstructive sleep apnoea syndrome (OSAS) is prevalent in hypertension and particularly in resistant hypertension, with studies indicating that up to 60% of patients with resistant hypertension have features of OSAS.[314] OSAS should be suspected in patients with hypertension and suggestive symptoms (see Supplementary data online, *Table S2*), in all patients with resistant hypertension, and in patients with non-dipping or reverse-dipping pattern at 24 h BP monitoring, especially if obese (*Figure 15*). Using validated questionnaires may help identify patients at high risk of OSAS.[333] Lack of suggestive symptoms does not rule out OSAS. A simplified polysomnogram confirms the diagnosis [apnoea–hypopnoea index (AHI) > 5] and can quantify the severity of OSAS (mild: AHI < 15; moderate: AHI of 15–30; severe: AHI > 30).[334]

Table 12 Drugs and conditions that affect aldosterone, renin, and aldosterone-to-renin ratio

Factor	Effect on plasma aldosterone levels	Effect on renin levels	Effect on ARR
Serum potassium status			
Hypokalaemia	↓	→↑	↓ **(FN)**
Potassium loading	↑	→↓	↑
Sodium restriction	↑	↑↑	↓ **(FN)**
Sodium loading	↓	↓↓	↑ **(FP)**
Drugs			
Beta-adrenergic blockers	↓	↓↓	↑ **(FP)**
Calcium channel blockers (DHPs)	→↓	→↑	→↓ **(FN** with short-acting DHPs**)**
ACE inhibitors	↓	↑↑	↓ **(FN)**
ARBs	↓	↑↑	↓ **(FN)**
Potassium-sparing diuretics	↑	↑↑	↓ **(FN)**
Potassium-wasting diuretics	→↑	↑↑	↓ **(FN)**
Alpha-2 agonists (clonidine, methyldopa)	↓	↓↓	↑ **(FP)**
NSAIDs	↓	↓↓	↑ **(FP)**
Steroids	↓	→↓	↑ **(FP)**
Contraceptive agents (drospirenone)	↑	↑	↑ **(FP)**

↑, raised; ↓, lowered; →, no effect; ACE, angiotensin-converting enzyme; ARB, angiotensin receptor blocker; ARR, aldosterone-to-renin ratio; DHPs, dihydropyridines; FN, false negative; FP, false positive; NSAID, non-steroidal anti-inflammatory drug.

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

© ESC 2024

Table 13 Optional tests that should be used to screen for secondary hypertension in the presence of suggestive signs, symptoms, or medical history

Cause of secondary hypertension	Screening test
Primary aldosteronism	Aldosterone-to-renin ratio Helpful information can also be provided by reviewing prior potassium levels (hypokalaemia increases the likelihood of coexistent primary hyperaldosteronism)
Renovascular hypertension	Renal doppler ultrasound Abdominal CT angiogram or MRI
Phaeochromocytoma/paraganglioma	24 h urinary and/or plasma metanephrine and normetanephrine
Obstructive sleep apnoea syndrome	Overnight ambulatory polysomnography
Renal parenchymal disease	Plasma creatinine, sodium, and potassium eGFR Urine dipstick for blood and protein Urinary albumin-to-creatinine ratio Renal ultrasound
Cushing's syndrome	24 h urinary free cortisol Low-dose dexamethasone suppression test
Thyroid disease (hyper- or hypothyroidism)	TSH
Hyperparathyroidism	Parathyroid hormone Calcium and phosphate
Coarctation of the aorta	Echocardiogram Aortic CT angiogram

CT, computed tomography; eGFR, estimated glomerular filtration rate; MRI, magnetic resonance imaging; TSH, thyroid-stimulating hormone.

© ESC 2024

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

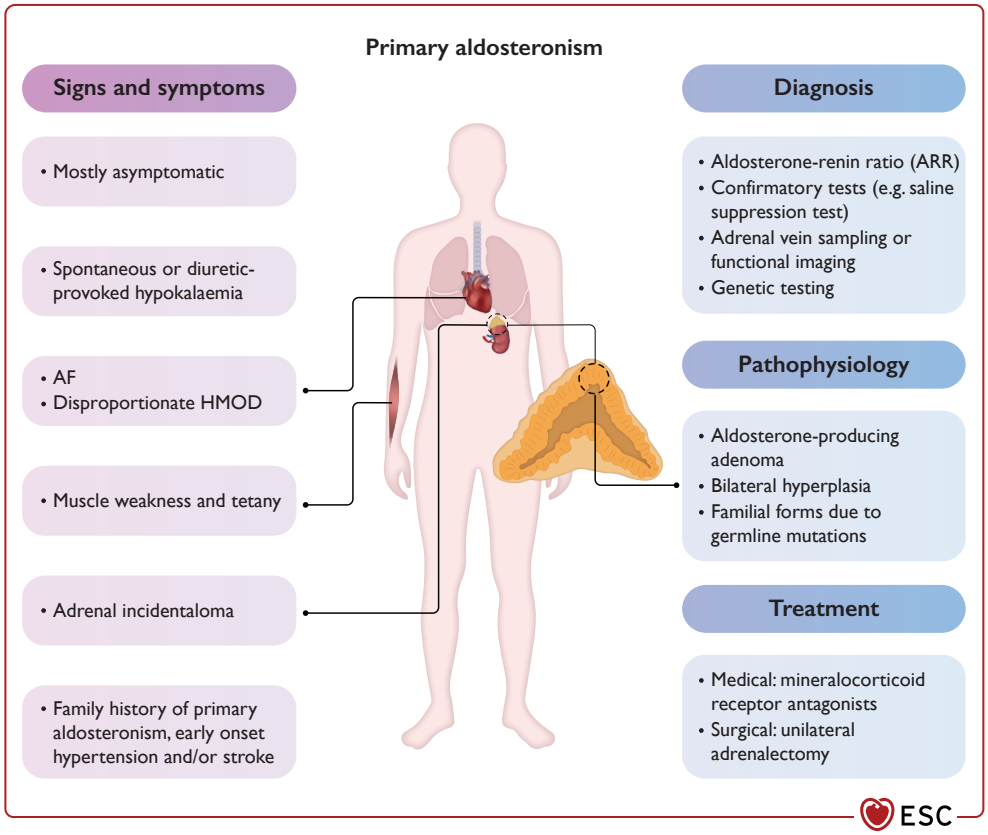


Figure 13 Summary of primary aldosteronism as a common form of secondary hypertension. AF, atrial fibrillation; HMOD, hypertension-mediated organ damage.

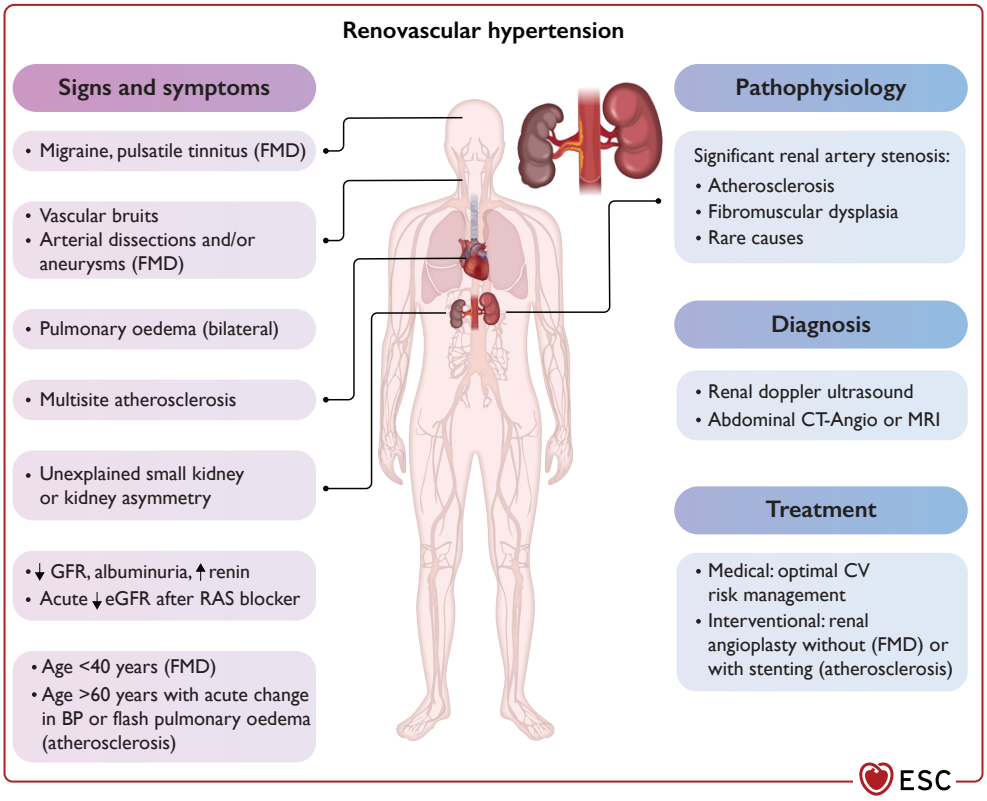


Figure 14 Summary of renovascular disease as a common form of secondary hypertension. CT-Angio, computed tomography angiography; CV, cardiovascular; FMD, fibromuscular dysplasia; GFR, glomerular filtration rate; MRI, magnetic resonance imaging; RAS, renin–angiotensin system.

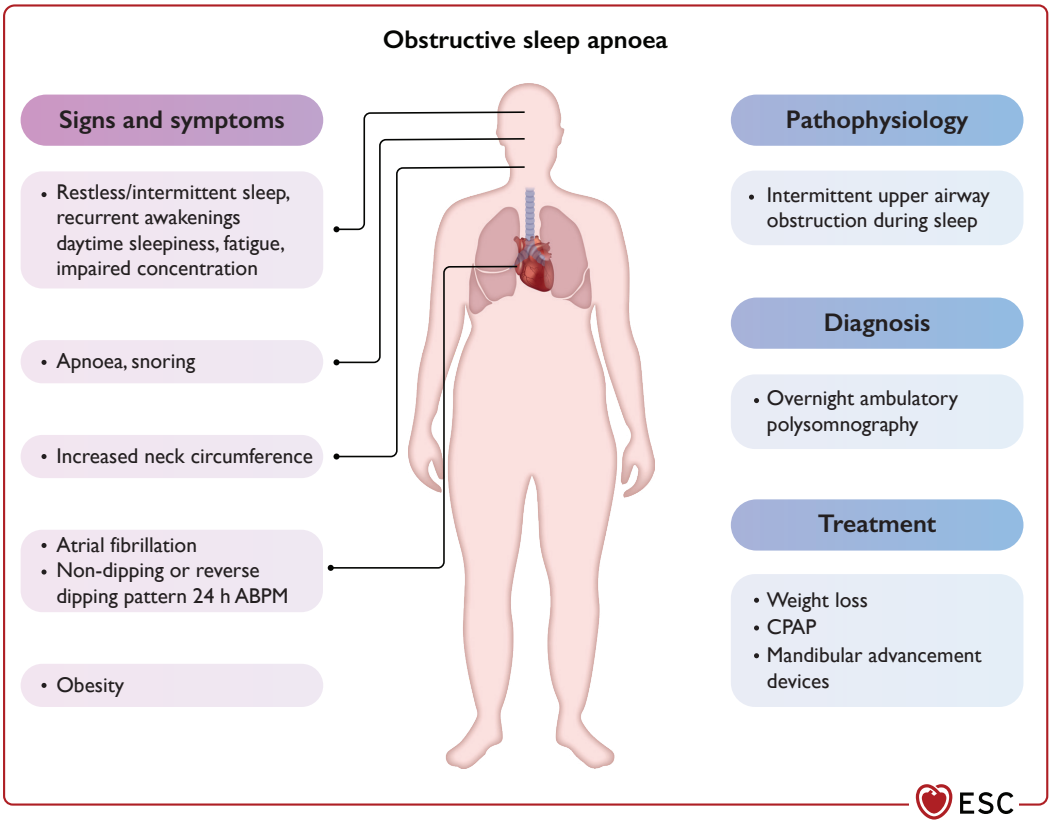


Figure 15 Summary of obstructive sleep apnoea as a common form of secondary hypertension. AF, atrial fibrillation; ABPM, ambulatory blood pressure monitor; CPAP, continuous positive airway pressure.

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

7.6.5. Phaeochromocytoma/paraganglioma

Phaeochromocytomas/paragangliomas (PPGLs) are a rare form of secondary hypertension characterized by a highly heterogeneous clinical presentation.[335,336] PPGLs are usually discovered incidentally.[337]

A PPGL should be suspected in the presence of signs and symptoms of catecholamine excess or in syndromic PPGL, in patients with a family history of PPGL, and in carriers of a germline mutation in one of the PPGL-causing genes.[338] Since normetanephrine and metanephrine are secreted constitutively, as opposed to the highly variable nature of catecholamine secretion, they are preferred as screening tests for PPGL (*Table 13*).

Recommendation Table 13 — Recommendations for screening for secondary hypertension (see Evidence Tables 19 and 20)

Recommendations	Class[a]	Level[b]
It is recommended that patients with hypertension presenting with suggestive signs, symptoms or medical history of secondary hypertension are appropriately screened for secondary hypertension.[312,314,315,323,339]	I	B
Screening for primary aldosteronism by renin and aldosterone measurements should be considered in all adults with confirmed hypertension (BP ≥140/90 mmHg).[313,316,323,339]	IIa	B

© ESC 2024

BP, blood pressure.
[a]Class of recommendation.
[b]Level of evidence.

8. Preventing and treating elevated blood pressure and hypertension

The ultimate goal of preventing and treating elevated BP and hypertension is to reduce CVD, to improve quality of life, and to prevent premature death. Crucially, besides BP, other CVD risk factors need to be comprehensively addressed (e.g. smoking, glucose, dyslipidaemia) as detailed in the 2021 ESC Guidelines on cardiovascular disease prevention in clinical practice.[170] Indeed, it is worth emphasizing that, when combined, these CVD risk factors have multiplicative (not additive) effects on CVD risk.[340]

8.1. Prevention strategies in early life

Detailed information on this topic is provided in the Supplementary data online. High BP tracks from childhood to adulthood.[341,342] Hypertension in childhood was redefined in a 2022 ESC Consensus Document.[343]

Recommendation Table 14 — Recommendations for screening for hypertension in children and adolescents (see Evidence Table 21)

Recommendation	Class[a]	Level[b]
Opportunistic screening with office BP measurements to monitor development of BP during late childhood and adolescence, especially if one or both parents have hypertension, should be considered to better predict development of adult hypertension and associated CVD risk.[344–346]	IIa	B

© ESC 2024

BP, blood pressure; CVD, cardiovascular disease.
[a]Class of recommendation.
[b]Level of evidence.

8.2. Non-pharmacological interventions

A major underlying contributor to elevated BP and hypertension in the general adult population is unhealthy lifestyle, with severe consequences for all-cause and CVD mortality. As such, we give lifestyle interventions to reduce BP a special status in our recommendations throughout these guidelines (*Figures 16* and *17*). This is reflected by a modified approach to the class of recommendations for lifestyle interventions compared with medical interventions (pharmacological or procedural). Given the salutary benefits of healthy lifestyle on a range of outcomes that extend well beyond BP-lowering effects,[347] including broad mental and physical health benefits, we do not require lifestyle interventions to have RCT evidence for efficacy in reducing CVD events through BP lowering to achieve a Class I recommendation. In deciding to give lifestyle interventions this status, the task force also recognizes that: (i) lifestyle interventions are less likely to be subjected to clinical outcomes trials (e.g. due to funding limitations and lack of interest from industry), and (ii) the risks of adverse effects and toxicity relating to healthy lifestyle interventions are low. In contrast, in these guidelines, to achieve a Class I recommendation (irrespective of level of evidence) there needs to be evidence that medical interventions that reduce BP also decrease CVD events by BP lowering.

8.2.1. Dietary sodium and potassium intake

8.2.1.1. Sodium

Reducing dietary salt (sodium chloride) intake in individuals with high baseline intake lowers CVD event rates.[348] Extensive observational studies have reported dose–response associations between high dietary sodium intake and CVD events.[349–351] The potential impact of salt reduction on population health is significant, particularly in countries where the population's average salt intake is high. Pooled data from long-term follow-up salt-reduction trials demonstrate that reducing salt by 2.5 g/day is associated with an approximately 20% reduction in CVD events at the population level.[349]

The health benefits of salt reduction are likely mediated, largely, by BP-lowering effects.[352–354] An almost linear relationship has been described in a dose–response meta-analysis between sodium intake ranging from 0.4 to 7.6 g/day and reduction of systolic and diastolic BP is independent of baseline BP.[355,356] Women appear to be, on average, more sodium sensitive than men,[357] and may have greater outcome benefits when receiving comparable sodium-restricted diets.[358] Trial evidence for the BP-lowering benefits of salt reduction extend down to daily sodium intakes of <1.5 g/day.[356,358–361]

The task force acknowledges that the observational data linking sodium intake to CVD outcomes are mixed and that some studies have not found a link between salt intake and CVD.[362,363] In addition, a potential J-curve exists between sodium intake and CVD events (whereby some analyses suggest that sodium reduction to very low levels could be harmful).[363,364] While there are differences of opinion, the task force agreed that, on balance, (i) observational J-curve data are often due to reverse causality or confounding,[114,348,365,366] (ii) the relationship between dietary sodium and stroke is typically linear in shape, without any J-curve, (iii) if the J-curve were causal, the adverse effect of very low sodium on CVD would have to be mediated by some harmful mechanism that overcomes the expected benefit mediated by BP lowering (which is unlikely), and (iv) estimation of sodium intake using spot-urine sodium testing (which was commonly done in studies reporting a J-curve) may not be as valid as other methods.[367] For example, most (but not all)[362] reports measuring 24 h urine sodium excretion (a surrogate measure of sodium intake) have not reported a J-curve association with CVD.[350,364] Furthermore the causal evidence demonstrating reduced CVD with sodium restriction (using potassium-enriched salt substitutes) in the Salt Substitute and Stroke Study (SSaSS)

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

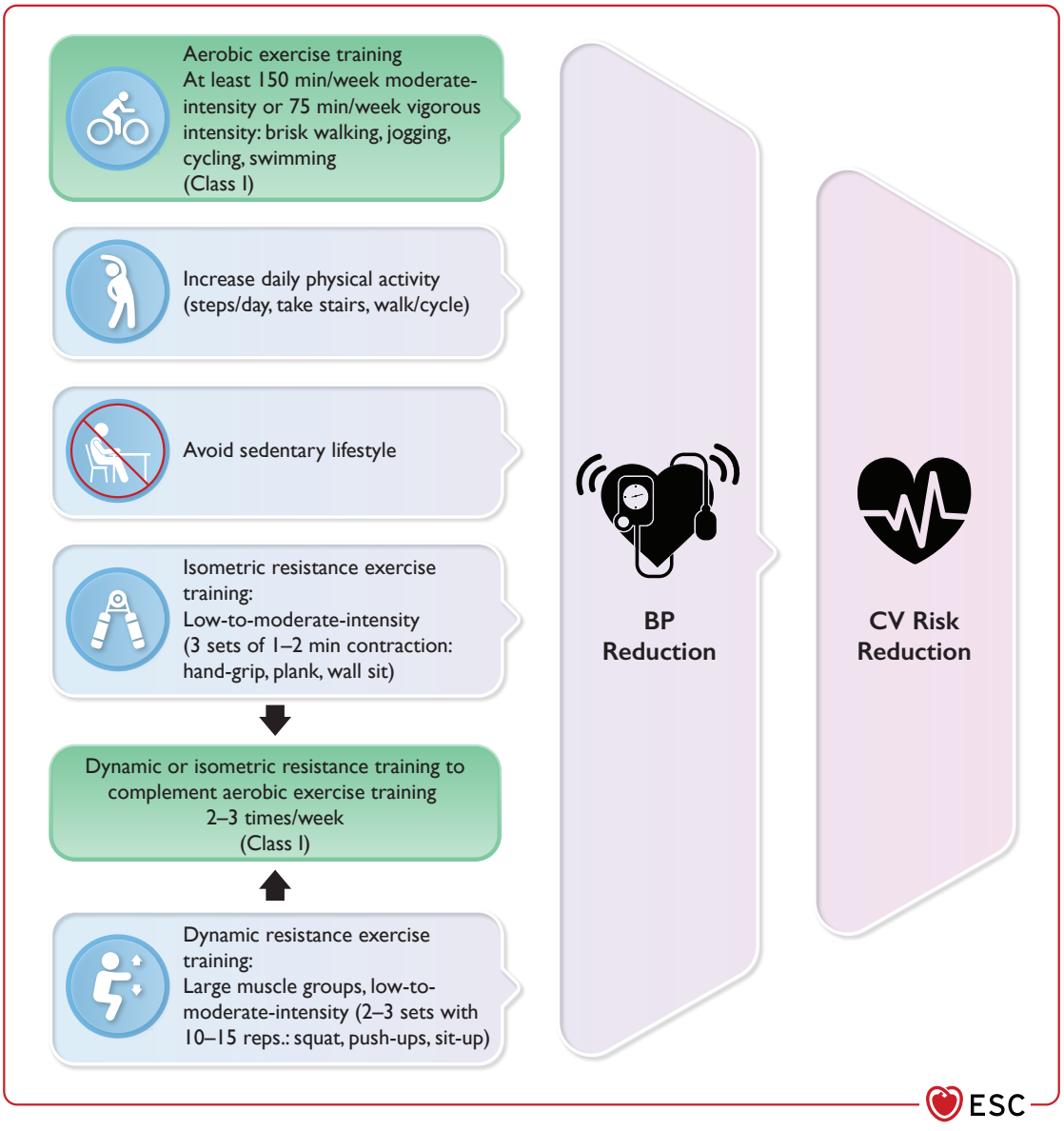


Figure 16 Physical activity according to different types of exercise and reduction of blood pressure and overall cardiovascular disease risk. Priority is given to aerobic exercise training (green). BP, blood pressure; CVD, cardiovascular disease.

and Diet, ExerCIse and carDiovascular hEalth-Salt (DECIDE-Salt) trials was compelling,[348,368] even though sodium restriction in these trials was not below 2 g/day.

It is recommended to restrict total dietary sodium intake to approximately 2 g/day or less (equivalent to approximately 5 g or about a teaspoon of salt per day). This includes added salt and salt already contained in food. While the feasibility of this sodium target can be debated, the evidence for the benefits of this sodium target among patients with elevated BP or hypertension is sufficient, particularly in terms of lowering BP. The optimal sodium intake in the general population with non-elevated BP is less clear (noting also that the BP-lowering effect of salt reduction among patients with non-elevated BP appears lower).[353,354] A more feasible compromise in the general population might be a target sodium intake range of 2–4 g/day.[369,370] It needs to be emphasized that large parts of daily sodium intake occur by means of sodium consumption contained in processed foods.

8.2.1.2. Potassium

Optimal dietary potassium intake, e.g. by consuming diets rich in fruits and vegetables, has BP-lowering effects and may be associated with lower CVD risk.[348,364,368,371–373] The association between potassium intake, systolic BP, and CVD events may be sex-specific, being stronger in women.[374] The World Health Organization (WHO) recommends over 3.5 g/day (~90 mmol/day) of dietary potassium.[375] Excessive potassium supplementation should, however, be avoided[374] and CKD guidelines recommend dietary potassium restriction to <2.4 g/day in persons with advanced CKD (see Supplementary data online).[376]

A lower urinary sodium-to-potassium ratio (Na^+/K^+ ratio; a surrogate for reduced dietary sodium intake complemented by increased potassium intake) has been associated with a greater reduction in systolic and diastolic BP than with a higher ratio.[348,377]

In patients with hypertension and high dietary sodium, increased dietary intake of potassium (in addition to lower dietary sodium) should be considered.[348,350,378] In patients with persistently high sodium intake (>5 g/day)

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

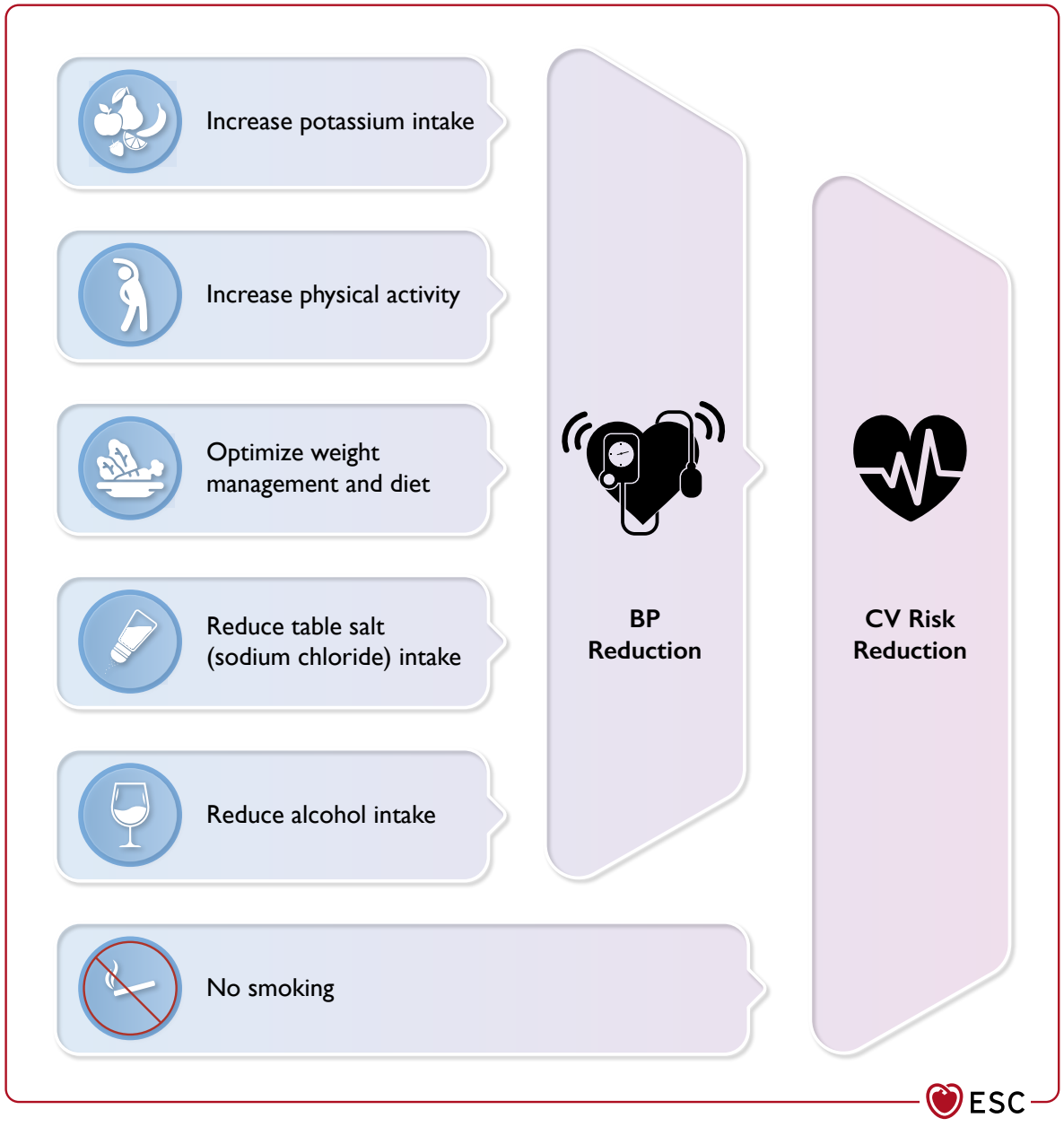


Figure 17 Effects of main lifestyle factors on blood pressure and cardiovascular risk reduction. BP, blood pressure; CV, cardiovascular. Smoking cessation reduces overall cardiovascular risk but not BP (long arrow). Salt reduction reduces BP and (for persons with high baseline intake) reduces cardiovascular risk. Increased potassium intake and higher physical activity, as well as optimized weight management, reduce BP and are associated with lower overall cardiovascular risk (short arrows).

and without moderate-to-advanced CKD, particularly women, an increase in potassium intake by 0.5–1.0 g/day may be considered to achieve a favourable Na^+/K^+ ratio of 1.5–2.0 and to reduce CVD risk. Potassium supplementation can be achieved by substituting sodium using potassium enriched salts (75% sodium chloride and 25% potassium chloride)[368,379,380] or by increasing dietary potassium intake [e.g. a 125 g (medium) banana contains about 450 mg potassium, or unsalted boiled spinach (840 mg/cup) or mashed avocado (710 mg/cup)]. In patients with CKD and/or those taking potassium-sparing medication, such as some diuretics, angiotensin-converting enzyme (ACE) inhibitors, angiotensin receptor blockers (ARBs) or spironolactone, serum levels of potassium should be monitored (noting that phlebotomy recommendations to avoid spuriously high potassium results should be followed).

8.2.2. Physical activity and exercise

In a systematic review and meta-analyses, aerobic (endurance) exercise was suggested as the first-line exercise therapy for reducing BP in patients with elevated BP and hypertension vs. alternative forms of exercise, such as dynamic or isometric resistance training.[381] In patients with hypertension, regular aerobic exercise substantially lowers systolic BP by up to 7–8 mmHg and diastolic BP by up to 4–5 mmHg.[381,382] For non-white patients with hypertension, dynamic resistance training elicits BP reductions that appear comparable to aerobic exercise.[383] Isometric resistance training also achieves clinically relevant BP reductions in patients with hypertension, but results are inconsistent and more data from more high-quality intervention trials are required (see Supplementary data online).[381,384,385] With respect to mode and intensity of aerobic exercise,

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

high-intensity interval training elicits comparable BP reductions to moderate continuous exercise, with high-intensity interval training achieving greater improvement in physical fitness.[386]

In patients with known hypertension, engaging in physical activity is associated with reduced CVD mortality risk vs. sedentary patients with hypertension.[387]

An exaggerated BP response to exercise may yield diagnostic merits for predicting incident hypertension and CVD. In a meta-analysis, an exaggerated BP response to exercise was associated with an increased risk for masked hypertension.[388] The risk of coronary heart disease also increases with higher systolic BP during exercise, independent of systolic BP at rest.[389]

Prior recommendations for at least 150 min/week of moderate intensity aerobic exercise (≥30 min, 5–7 days/week) can be maintained.[1,390] Alternatively, 75 min of vigorous-intensity exercise per week over 3 days may be performed, with additional benefits derived by achieving 300 min of moderate-intensity or 150 min of vigorous-intensity aerobic physical activity per week.[390,391] As acute aerobic exercise induces intensity-dependent short-term reductions in ambulatory BP after exercise, patients with elevated BP and hypertension may benefit from daily exercise to improve their 24 h BP profile and avoid BP peaks on sedentary days.[392] Aerobic exercise should be complemented by low- or moderate-intensity resistance training (2–3 times per week), e.g. dynamic resistance, starting at 2–3 sets of 10–15 repetitions at 40%–60% of one-repetition maximum[393] or isometric resistance training with three sets of 1–2 min contractions, such as hand-grip, plank, or wall sit (*Figure 16*).[381,394]

In uncontrolled hypertension at rest, high-intensity exercise should be applied with caution, with resting systolic BP of >200 mmHg and diastolic BP of >110 mmHg indicating relative contraindications.[395] Age, sex, gender,[396] ethnicity, and comorbidities, as well as individual preferences, should be considered for individual exercise prescription. Detailed information on exercise prescription in terms of frequency, intensity, time (duration) and type and progression are available in the 2020 ESC Guidelines on sports cardiology and exercise in patients with cardiovascular disease,[390] which include recommendations for pre-participation screening and cardiopulmonary exercise testing.[390]

8.2.3. Weight reduction and diet

Visceral obesity is common and associated with incident hypertension.[397,398] An average weight loss of 5 kg has been associated with an average systolic and diastolic BP reduction of 4.4 and 3.6 mmHg, respectively.[399] Data show that, starting at an index body mass index (BMI) of 40 kg/m^2, a median weight loss of 13% is associated with a 22% lower risk for hypertension.[400,401] Maintaining even moderate weight loss of 5%–10% of initial body weight can improve not only BP, but also glucose and lipid metabolism, and potentially reduce premature all-cause mortality.[402–404] However, achieving long-term effects in patients with hypertension via weight loss is challenging and the magnitude of these effects remains unclear.[405,406] Weight stabilization during middle-age appears to be an important and attainable goal to prevent obesity-related increase in BP later in life.[407]

Evidence-based diets, such as the Mediterranean diet and the Dietary Approaches to Stop Hypertension (DASH) diet, are established interventions in patients with hypertension to reduce their BP and CVD risk.[408,409] Additional information on healthy dietary patterns is provided in the 2021 ESC Guidelines on cardiovascular disease prevention in clinical practice and Supplementary data online.[170]

In combination with weight-loss and exercise interventions[410] and low sodium intake,[411] the DASH diet has added effect on BP reduction.[412] Pharmacological treatment of obesity with orlistat achieved a slight reduction of 2.6 mmHg in systolic BP.[413] The greatest BP-lowering effects of weight-loss medications may be achieved with the glucagon-like peptide 1 (GLP-1) receptor agonists.[414–416] For example, in the Semaglutide Treatment Effect in People with Obesity (STEP-1) trial, the GLP-1 analogue semaglutide resulted in a mean weight reduction of 12.4% and a 5.1 mmHg reduction in systolic BP.[415]

8.2.4. Alcohol, coffee, and soft drinks

In a 2020 Cochrane review, the short-term effects of alcohol on BP were dose dependent; low-dose alcohol (<14 g) did not affect BP within 6 h, medium-dose (14–28 g) decreased both systolic and diastolic BP, and high-dose alcohol (>30 g) first decreased BP up to 12 h and then increased BP following >13 h of consumption by 3.7 mmHg systolic and 2.4 mmHg diastolic.[417] The trials in this Cochrane review included small numbers of women. In the longer term, no evidence has been found for a protective effect of chronic alcohol consumption on hypertension, for either sex. In contrast, even low-dose alcohol consumption (10 g/day) increases chronic risk of hypertension by 14% in men, but not in women.[418] As per the 2021 ESC Guidelines on cardiovascular disease prevention in clinical practice, men and women are recommended to stay within the upper limit of drinking alcoholic beverages (100 g/week of pure alcohol). Defining number of drinks depends on portion size, the standards of which differ per country, but translates to 8–14 g/drink.[170] Emerging data indicate it is likely healthiest to avoid all alcohol, where possible.[419]

Coffee intake is not associated with a higher risk of hypertension in the general population; in fact, higher coffee consumption may be associated with a lower risk for incident hypertension.[420] Data regarding the association between tea drinking and CVD are inconclusive, though mechanistic trials have suggested benefits on BP lowering.[421] In contrast, energy drinks with high concentrations of ingredients such as taurine and caffeine increase BP and may lead to acute or chronic cardiovascular complications in young adults.[422–424]

Consuming two or more servings per day of sugar-sweetened beverages was associated with a 35% higher risk of coronary artery disease in women in the Nurses' Health Study.[425] In the European Prospective Investigation into Cancer and Nutrition (EPIC) cohort, both sugar- and artificially sweetened soft drinks were associated with higher all-cause mortality.[426] In children and adolescents, sugar-sweetened beverages increased systolic BP and the risk for incident hypertension.[427] It is recommended to restrict free sugar consumption, in particular sugar-sweetened beverages, to a maximum of 10% of energy intake.

8.2.5. Smoking

To stop smoking is arguably the most effective measure in preventing major CVD events at the individual level, likely through improved vascular health.[428–431] Estimated health benefits will be even more substantial looking at all-cause morbidity and mortality, e.g. including smoking cessation for cancer prevention.

The effects of electronic cigarettes (e-cigarettes) on BP remain unclear and to date there are no robust outcomes data. However, growing evidence suggests that e-cigarettes can increase BP (see Supplementary data online).[432,433]

Among adults, smoking affects ambulatory BP by raising daily BP,[434] but effects of chronic smoking on office BP appear to be small.[435] Smoking cessation advice helps, but more intensive interventions are superior.[436,437] As recommended by previous ESC Guidelines, smoking cessation is recommended to reduce CVD risk and improve non-CVD health.[1,170]

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

Recommendation Table 15 — Recommendations for non-pharmacological treatment of blood pressure and cardiovascular risk reduction (see Evidence Tables 22–26)

Recommendations	Class[a]	Level[b]
Restriction of sodium to approximately 2 g per day is recommended where possible in all adults with elevated BP and hypertension [this is equivalent to about 5 g of salt (sodium chloride) per day or about a teaspoon or less].[353,354]	I	A
Moderate intensity aerobic exercise of ≥150 min/week (≥30 min, 5–7 days/week) or alternatively 75 min of vigorous intensity aerobic exercise per week over 3 days are recommended and should be complemented with low- or moderate-intensity dynamic or isometric resistance training (2–3 times/week) to reduce BP and CVD risk.[1,381,390–393]	I	A
It is recommended to aim for a stable and healthy BMI (e.g. 20–25 kg/m^2) and waist circumference values (e.g. <94 cm in men and <80 cm in women) to reduce BP and CVD risk.[399–401]	I	A
Adopting a healthy and balanced diet, such as the Mediterranean or DASH diets, is recommended to help reduce BP and CVD risk.[412,438,439]	I	A
Men and women are recommended to drink less alcohol than the upper limit, which is about 100 g/week of pure alcohol. How this translates into number of drinks depends on portion size (the standards of which differ per country), but most drinks contain 8–14 g of alcohol per drink. Preferably, it is recommended to avoid alcohol to achieve the best health outcomes.[170,419,440,441]	I	B
It is recommended to restrict free sugar consumption, in particular sugar-sweetened beverages, to a maximum of 10% of energy intake. It is also recommended to discourage consumption of sugar-sweetened beverages, such as soft drinks and fruit juices, starting at a young age.[425–427]	I	B
It is recommended to stop tobacco smoking, initiate supportive care and refer to smoking cessation programmes, as tobacco use strongly and independently causes CVD, CVD events, and all-cause mortality.[428,429,431,437]	I	A
In patients with hypertension without moderate to advanced CKD and with high daily sodium intake, an increase of potassium intake by 0.5–1.0 g/day—for example through sodium substitution with potassium-enriched salt (comprising 75% sodium chloride and 25% potassium chloride) or through diets rich in fruits and vegetables—should be considered.[348,368,373,374,442]	IIa	A
In patients with CKD or taking potassium-sparing medication, such as some diuretics, ACE inhibitors, ARBs, or spironolactone, monitoring serum levels of potassium should be considered if dietary potassium is being increased.	IIa	C

ACE, angiotensin-converting enzyme; ARB, angiotensin receptor blocker; BMI, body mass index; BP, blood pressure; CKD, chronic kidney disease; CVD, cardiovascular disease; DASH, Dietary Approaches to Stop Hypertension.
[a]Class of recommendation.
[b]Level of evidence.

© ESC 2024

8.3. Pharmacological interventions

8.3.1. Treatment strategy to reduce adverse cardiovascular disease outcomes

The main goal of reducing BP is to prevent adverse CVD outcomes. The relative risk reduction afforded by a fixed degree of BP reduction is largely independent of pre-treatment BP.[116] There is a clear relationship between the intensity of BP lowering and the relative and absolute reduction in risk of CVD events for all adults, regardless of age (at least up to 85 years), sex, prior CVD, diabetes, or AF.[116,131,443–445] With this strong evidence for the 'the lower the better, but within reason' paradigm, decision rules are required for selecting patients most likely to benefit from treatment.[172] In this section, a summary of evidence for BP-lowering drug treatment is provided, followed by strategies for their use for preventing CVD.

8.3.2. Drug classes with evidence on clinical outcomes in the target population

The major drug classes with robust evidence for BP-mediated reduction in CVD events are ACE inhibitors, ARBs, dihydropyridine CCBs, diuretics (thiazides and thiazide-like diuretics such as hydrochlorothiazide, chlorthalidone, and indapamide), and beta-blockers (see Supplementary data online, *Tables S7 and S8*).[122,446,447] The first four are recommended as first-line options for starting hypertension treatment in the general population. Beta-blockers can be added preferentially in circumstances such as in the presence of angina or heart failure, after myocardial infarction, or for controlling heart rate, where they are the cornerstone of therapy.[122,448,449] In such settings, second-generation (cardioselective) and, specifically, third-generation (vasodilating) beta-blockers are preferred.[450] However, beta-blockers are less effective than ACE inhibitors, ARBs, CCBs, or diuretics at preventing stroke, and have a higher discontinuation rate due to side effects.[451,452] Beta-blockers and diuretics, especially when combined, are associated with an increased risk of new-onset diabetes in predisposed patients.[453,454] The effect of RAS blockers and CCBs on preventing progression of HMOD also appears to be superior to beta-blockers.[455–458] Beta-blockers should also be avoided in patients with isolated systolic hypertension or more generally with arterial stiffness, as they increase stroke volume (given the lower heart rate).[218]

When therapy and adherence with the above-mentioned drug classes is optimized but insufficient to reach BP goals, other drug classes can be used for treating hypertension. Of these, spironolactone, an MRA, appears to be the most effective at further lowering BP in resistant hypertension; however, more evidence of CVD risk-lowering effects with MRAs among all hypertensive populations, especially those without resistant hypertension, is needed.[459] Specifically, while use of MRAs in patients with heart failure has provided clinical evidence on the effectiveness of MRAs for preventing CVD events, dedicated outcome trials in patients with primary hypertension without heart failure are lacking. Because the present guidelines require trial evidence for CVD outcome benefit for a BP-lowering drug or procedure to achieve a Class I recommendation, and given no outcome trials of MRAs have been conducted in general samples of patients with primary hypertension, we have given MRAs a Class IIa recommendation (see below). We acknowledge that spironolactone was provided a Class I recommendation in the 2018 ESC/ESH Guidelines on the management of arterial hypertension. However, to be consistent with our requirement for trial evidence for CVD outcomes benefit in patients with hypertension, the task force agreed to provide a Class IIa recommendation for spironolactone in these 2024 Guidelines. Importantly, it was also agreed that a Class IIa recommendation (i.e. should be considered) is an endorsement of MRAs for

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

treating resistant hypertension but one that acknowledges some uncertainty of outcomes benefit. Future outcome trials of MRAs, perhaps including finerenone,[460–462] are encouraged in patients with hypertension.

Clinical outcome evidence from trials for other BP-lowering drug classes, such as alpha-blockers, hydralazine, minoxidil, other potassium-sparing diuretics, and centrally acting agents, is less compelling and caution regarding adverse effects is warranted. However, they may be a final addition if all other therapeutic efforts are insufficient to decrease BP. Of note, in the Antihypertensive and Lipid-Lowering Treatment to Prevent Heart Attack (ALLHAT) trial, the alpha-blocker arm was stopped early due to futility of benefit for the CVD outcome.[463]

8.3.3. New therapies with blood pressure-lowering properties that await supportive evidence from cardiovascular outcomes trials prior to guideline endorsement and routine use in hypertension

A few other drug classes with indication in heart failure have emerged that also have BP-lowering properties. For example, the angiotensin receptor–neprilysin inhibitor (ARNi) sacubitril/valsartan, which was initially developed for hypertension,[464] reduces CVD mortality and morbidity in patients with heart failure, an effect that may have been mediated, in part, by superior BP lowering compared with enalapril alone.[465,466] In a *post hoc* subgroup analysis, sacubitril/valsartan lowered BP in adults with both heart failure with preserved ejection fraction (HFpEF) and resistant hypertension.[467] In the context of research studies, sacubitril/valsartan has been used in higher doses (200 mg or 400 mg once daily) for treating hypertension.[464,466,468,469]

SGLT2 inhibitors have shown favourable effects on CVD events and renal haemodynamics in patients with and without type 2 diabetes, and in heart failure trials.[470] In these trials, SGLT2 inhibitors did lower BP, though only modestly.[471] Small trials among adults with hypertension have confirmed the potential for BP lowering with this drug class.[472,473]

Other new drugs with preliminary data include GLP-1 agonists and the new non-steroidal MRAs, e.g. finerenone, in managing hypertension.[415,460,461] In addition, novel aldosterone synthase inhibitors (baxdrostat and lorundrostat) have significantly lowered BP in patients with uncontrolled hypertension in phase 2 trials.[474,475]

The dual endothelin-A and -B receptor antagonist aprocitentan also reduced office and 24 h BP compared with placebo at 4 weeks in patients with resistant hypertension in a phase 3 trial.[476] Zilebesiran, an investigational RNA interference agent administered subcutaneously, inhibits hepatic angiotensinogen synthesis and a single dose reduced 24 h BP over approximately 6 months.[477]

8.3.4. Drug combinations and up-titrating strategies

To treat hypertension, many patients will require more than one BP-lowering medication. Combining drugs from different drug classes can have additive or synergistic effects and lead to greater BP reduction than increasing the dose of one drug.[478–483] The superior BP-lowering efficacy of combination therapy is mediated, at least in part, by the potential of combination therapy to target multiple pathophysiological pathways contributing to perturbed BP in each patient.[484] A further benefit of combination therapy is the potential to use lower doses of each individual BP-lowering agent, which may reduce side effects and improve adherence and persistence,[485] though the evidence for this hypothesis has been questioned.[486]

Upfront low-dose combination therapy is therefore recommended in persons with hypertension, with the potential advantages of fewer side effects and swifter BP control being important for long-term adherence.[487–489] If combination BP-lowering therapy is pursued, single-pill combinations are preferred. For those with elevated BP who have an indication for BP-lowering treatment, monotherapy is recommended in the first instance.

One caveat to combination therapy in hypertension is that patient-level response to individual BP-lowering drug classes can be heterogeneous (suggesting some patients may benefit from more personalized treatment compared with routine combinations).[490] This is relevant also with respect to race/ethnicity (see *Section 9*). Another caveat is that the evidence for reduced CVD outcomes with BP-lowering drugs in combination therapy is based on observational studies.[491–493] There are no outcomes data from prospective trials that prove superiority of upfront combination therapy (either as single-pill combinations or as separate pills) over upfront monotherapy in the isolated treatment of hypertension.[486] Therefore, we considered giving upfront combination therapy (either as separate pills or as single-pill combinations) a Class IIa recommendation in these guidelines. However, given the totality of evidence for outcomes benefit in observational studies, randomized trial data for better BP control and adherence, and importantly, also given CVD outcomes benefit for polypills (a form of single-pill combination) in randomized trials,[494–496] we chose to provide a Class I recommendation for upfront combination therapy in adults with confirmed hypertension, in agreement with 2018 ESC recommendations.

The major four drug classes (ACE inhibitors, ARBs, dihydropyridine CCBs, and thiazide or thiazide-like diuretics) are recommended as first-line BP-lowering medications, either alone or in combination.[122,447,484,497] An exception is the combination of two RAS blockers, which is not recommended.[498–500] For most hypertensive patients, a single-pill combination, initially containing two of these major drug classes, and initially at low dose, is recommended.[489,501,502] Doses of BP-lowering drugs are presented in the Supplementary data online, *Tables S7 and S8*.

When BP is still uncontrolled under maximally tolerated triple-combination (RAS blocker, CCB, and diuretic) therapy, and after adherence is assessed, the patient should be considered resistant and referred to an expert centre for appropriate work-up (see *Section 7.5*). At the same time, the addition of spironolactone should be considered.[459] If spironolactone is not tolerated, eplerenone or other MRA, or beta-blockers (if not already indicated), should be considered. Eplerenone may need to be dosed higher (50–200 mg) for effective BP lowering. In a meta-analysis, eplerenone 25 mg did not lower BP.[503] Due to the shorter time of action than spironolactone, eplerenone may need to be administered twice daily for treating hypertension. An alternative to MRA as fourth-line treatment for BP lowering is the use of beta-blockers for persons who do not already have a compelling indication. A vasodilating beta-blocker (e.g. labetalol, carvedilol, or nebivolol) is preferred when a beta-blocker is chosen.[504] However, we note that the BP-lowering effects of beta-blockade appears to be less potent than spironolactone in the setting of resistant hypertension.[459]

Only thereafter should hydralazine, other potassium-sparing diuretics (amiloride and triamterene), centrally acting BP-lowering medications, or alpha-blockers be considered. Given multiple side-effects, minoxidil should only be considered if all other pharmacological agents prove ineffective in resistant hypertension.[505]

As noted above, polypills combining fixed doses of BP-lowering treatment, lipid-lowering therapy and, if indicated, aspirin are effective in more general CVD prevention.[496,506–509] However, the polypill is not available for routine clinical use in many European countries.

8.3.5. A practical algorithm for intensive, effective, and tolerable blood pressure lowering with drug therapy, including considerations around single-pill combinations

The aim of the algorithm in *Figure 18* is to introduce a low-dose double- and then triple-combination strategy while monitoring tolerance

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

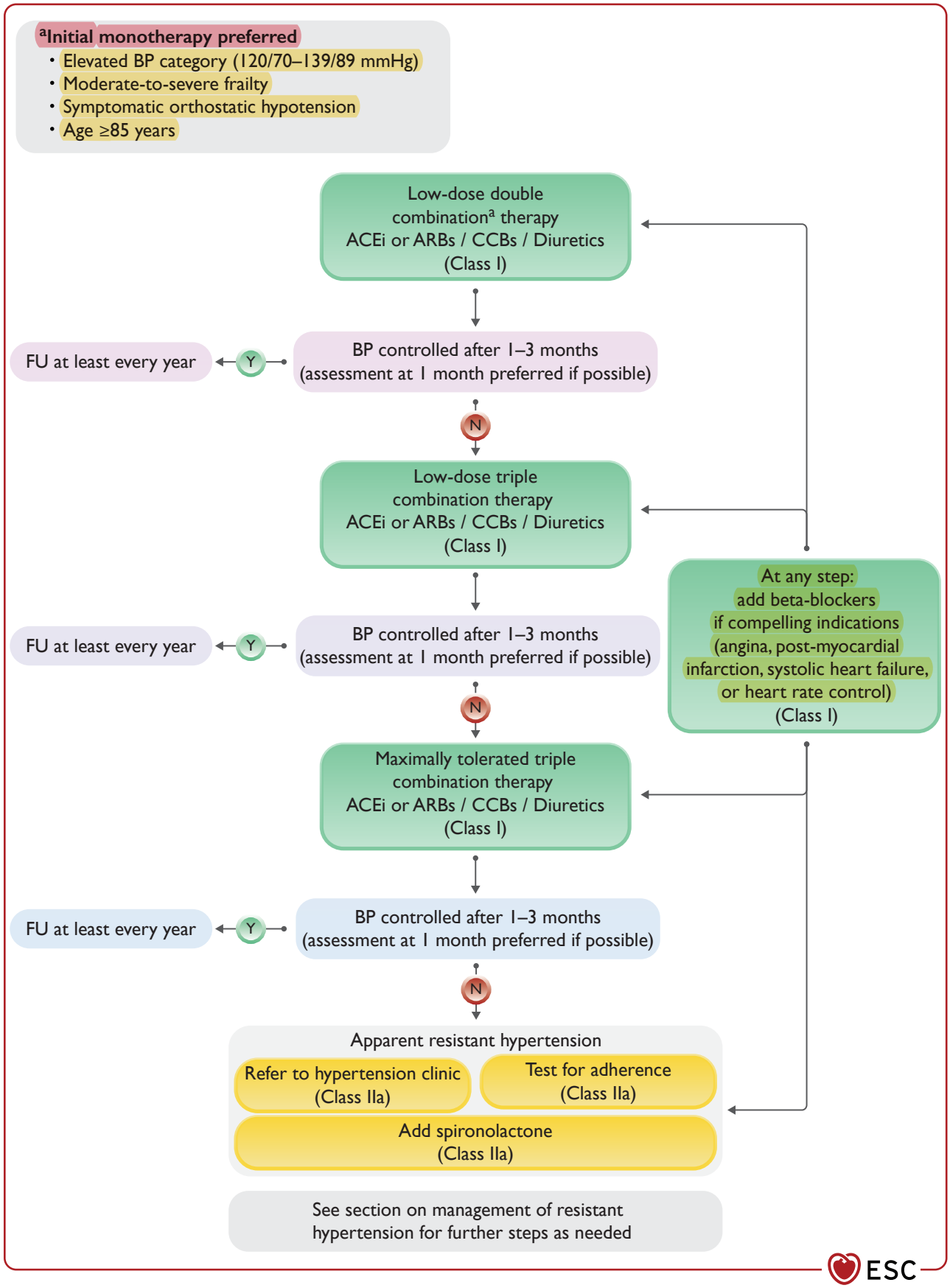


Figure 18 Practical algorithm for pharmacological blood pressure lowering. ACEi, angiotensin-converting enzyme inhibitor; ARB, angiotensin receptor blocker; BP, blood pressure; CCB, calcium channel blocker; FU, follow-up.

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

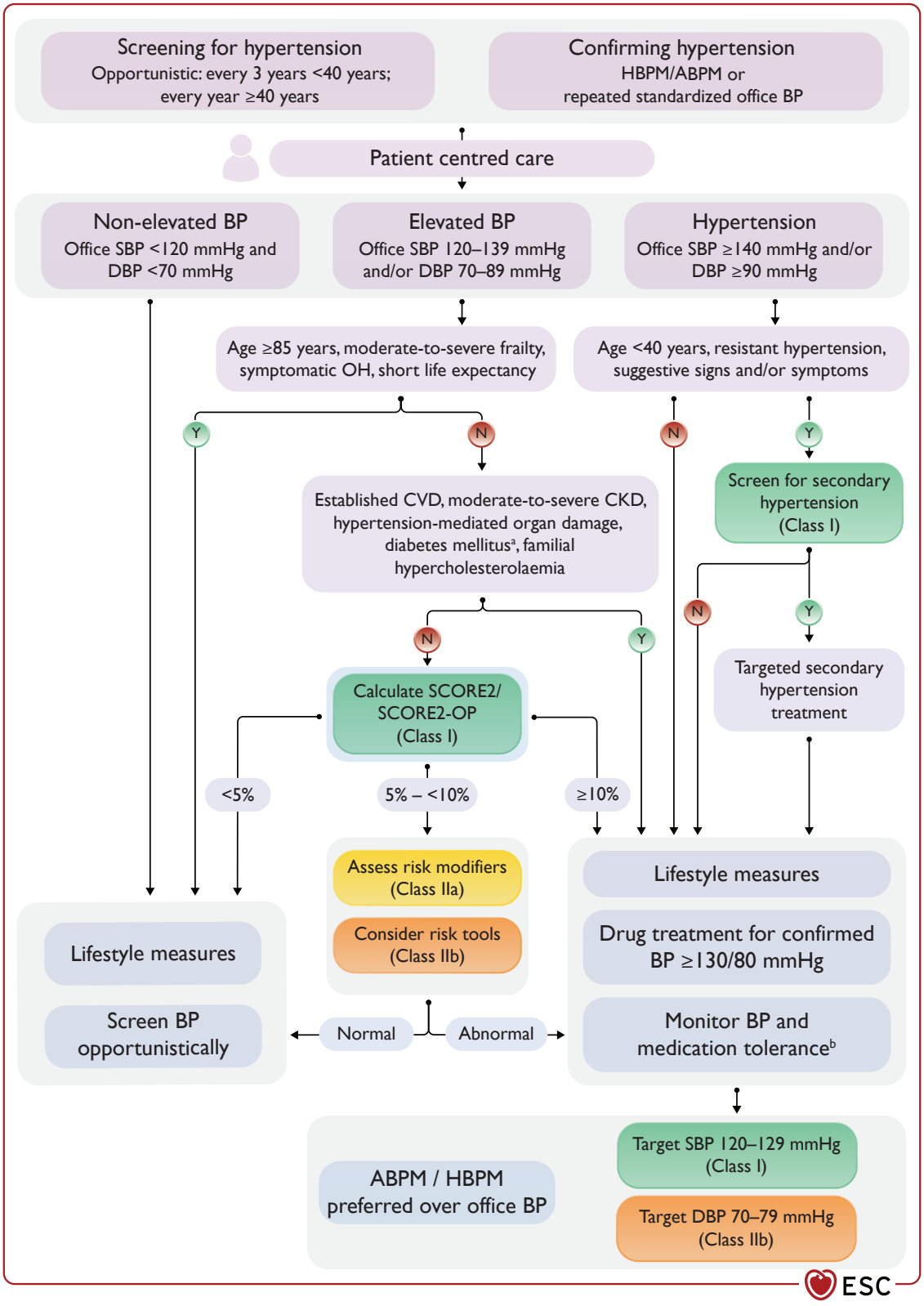


Figure 19 Central Illustration. ABPM, ambulatory blood pressure monitoring; BP, blood pressure; CVD, cardiovascular disease; CKD, chronic kidney disease; DBP, diastolic blood pressure; HBPM, home blood pressure monitoring; OH, orthostatic hypotension; SBP, systolic blood pressure; SCORE2, Systematic COronary Risk Evaluation 2; SCORE2-OP, Systematic COronary Risk Evaluation 2–Older Persons. Summary algorithm for BP classification and management. See *Section 5* for recommendations on out-of-office confirmation of the three BP categories. [a]Adults with type 2 diabetes mellitus only and younger than 60 years should be considered for SCORE2-Diabetes assessment. [b]Though scientific data demonstrate that, under research conditions, the optimal target BP is ≤120/70 mmHg, the target BP recommended by these guidelines in routine practice is 120–129/70–79 mmHg. If achieving this target is not possible, or if treatment is not well tolerated, then BP should be treated to as low as reasonably achievable. For persons with elevated BP, treatment with lifestyle measures for 3 months is first recommended, prior to considering medications.

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

among patients with hypertension, and only afterwards to start up-titrating doses to maximum amounts.

Initiation with monotherapy, slower up-titration, and lower dosing should be considered in the setting of elevated BP and increased CVD risk, or in moderate-to-severe frailty, limited life expectancy, symptomatic orthostatic hypotension, or older people (aged ≥85 years). Ideally, BP should be treated to target within 3 months to retain the confidence of the patient, to ensure long-term adherence, and to reduce CVD risk.[71]

An overview of the recommended approach to BP management in all adult patients is provided in *Figure 19* (Central Illustration). Also, of note, teleconsultation, multidisciplinary or nurse-led care, or patient self-monitoring can help with achieving BP control in certain healthcare systems.[75,510,511]

8.3.6. Timing of blood pressure-lowering drug treatment

Current evidence does not show benefit of diurnal timing of BP-lowering drug administration on major CVD outcomes.[512] It is important that medication is taken at the most convenient time of day to improve adherence. Patients should also be encouraged to take medications at the same time each day and in a consistent setting, to help ensure adherence.[246,513]

Recommendation Table 16 — Recommendations for pharmacological treatment of hypertension (see Evidence Tables 27, 28, and 29)

Recommendations	Class[a]	Level[b]
Among all BP-lowering drugs, ACE inhibitors, ARBs, dihydropyridine CCBs, and diuretics (thiazides and thiazide-like drugs such as chlorthalidone and indapamide) have demonstrated the most effective reduction of BP and CVD events, and are therefore recommended as first-line treatments to lower BP.[122,446]	I	A
It is recommended that beta-blockers are combined with any of the other major BP-lowering drug classes when there are other compelling indications for their use, e.g. angina, post-myocardial infarction, heart failure with reduced ejection fraction, or for heart rate control.[122,448–450]	I	A
It is recommended to take medications at the most convenient time of day for the patient to establish a habitual pattern of medication taking to improve adherence.[246,513]	I	B
Given trial evidence for more effective BP control vs. monotherapy, combination BP-lowering treatment is recommended for most patients with confirmed hypertension (BP ≥140/90 mmHg) as initial therapy. Preferred combinations are a RAS blocker (either an ACE inhibitor or an ARB) with a dihydropyridine CCB or diuretic. Exceptions to consider include patients aged ≥85 years, those with symptomatic orthostatic hypotension, moderate-to-severe frailty, or elevated BP (systolic BP 120–139 mmHg or diastolic BP 70–89 mmHg) with a concomitant indication for treatment.[131,480,483,484,489]	I	B
In patients receiving combination BP-lowering treatment, fixed-dose single-pill combination treatment is recommended.[484,489,501,502,514]	I	B
If BP is not controlled with a two-drug combination, increasing to a three-drug combination is recommended, usually a RAS blocker with a dihydropyridine CCB and a thiazide/thiazide-like diuretic, and preferably in a single-pill combination.[489]	I	B
If BP is not controlled with a three-drug combination, adding spironolactone should be considered.[459]	IIa	B
If BP is not controlled with a three-drug combination and in whom spironolactone is not effective or tolerated, treatment with eplerenone instead of spironolactone,[503] or the addition of a beta-blocker if not already indicated[459] and, next, a centrally acting BP-lowering medication,[515] an alpha-blocker,[515] hydralazine, or a potassium-sparing diuretic should be considered.[516]	IIa	B
Combining two RAS blockers (ACE inhibitor and an ARB) is not recommended.[498–500,517]	III	A

Continued

ACE, angiotensin-converting enzyme; ARB, angiotensin receptor blocker; BP, blood pressure; CCB, calcium channel blocker; CVD, cardiovascular disease; RAS, renin–angiotensin system.
[a]Class of recommendation.
[b]Level of evidence.

8.4. Selecting patients for pharmacological blood pressure-lowering treatment

Commencement of BP-lowering treatment is often decided based on office BP measurements but, where possible, the present guidelines strongly recommend using out-of-office BP measurement for confirming elevated BP and hypertension (see *Section 5*). As detailed in *Section 6*, an office BP of <120/70 mmHg is categorized in these guidelines as non-elevated BP.

When a patient is diagnosed with confirmed hypertension (sustained BP ≥ 140/90 mmHg), starting BP-lowering treatment is recommended irrespective of CVD risk, which should consist of a simultaneous combination of lifestyle interventions and pharmacological therapy. Lifestyle interventions are crucial as an initial treatment step, and must be strongly emphasized with the patient, but concurrent pharmacological therapy is recommended. This concurrent initiation of lifestyle and pharmacological therapy should not give patients the impression that lifestyle changes are of lesser importance, and the patient should be counselled that these lifestyle changes may allow subsequent discontinuation or down-titration of medication, which can be used as motivation to persist with lifestyle changes. After treatment initiation, the patient should be seen frequently (e.g. every 1–3 months with a GP or specialist) until BP is controlled. BP should be controlled, preferably within 3 months (see also *Section 6* and algorithm *Figure 18*). If lifestyle changes are effective in BP lowering, pharmacological treatments may subsequently be down-titrated or stopped as appropriate.

When office BP is 120–139/70–89 mmHg, the patient is considered as having elevated BP, and further CVD risk stratification is recommended to guide therapy (*Table 14*).

- In patients with elevated BP who are not at increased risk for CVD (10-year CVD risk <10%) and do not have other high-risk conditions or risk modifiers, BP-lowering lifestyle measures are recommended.

© ESC 2024

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

Table 14 Initiation of blood pressure-lowering treatment based on confirmed blood pressure category and cardiovascular disease risk

Blood pressure (mmHg)	Non-elevated BP (<120/70)	Elevated BP (120/70 to 139/89)		Hypertension (≥140/90)
Risk		(a) All adults with SBP 120–129 mmHg (b) SBP 130–139 AND 10-year estimated CVD risk <10% AND no high-risk conditions or risk modifiers or abnormal risk tool tests	(a) SBP 130–139 AND high-risk conditions (e.g. established CVD, diabetes mellitus, CKD, FH or HMOD) (b) SBP 130–139 AND 10-year estimated CVD risk ≥10% (c) SBP 130–139 AND 10-year estimated CVD risk 5% - <10% AND risk modifiers or abnormal risk tool tests	Assumed all at sufficiently high risk to benefit from pharmacological treatment
Treatment	Lifestyle measures for prevention Screen BP and CVD risk opportunistically	Lifestyle measures for treatment Monitor BP and CVD risk yearly	Lifestyle measures and pharmacological treatment (after 3-month delay). Monitor BP yearly once treatment control is established	Lifestyle measures and pharmacological treatment (immediate) Monitor BP yearly once treatment control is established
Target (mmHg)	**Maintain BP <120/70**	**Aim BP 120–129/70–79 mmHg[a]**		

BP, blood pressure; CKD, chronic kidney disease; CVD, cardiovascular disease; DBP, diastolic blood pressure; FH, familial hypercholesterolaemia; HMOD, hypertension-mediated organ damage; SBP, systolic blood pressure.
[a]Caution in adults with orthostatic hypotension, moderate-to-severe frailty, limited life expectancy, and older patients (aged ≥85 years).

While there are not enough outcomes data for a drug recommendation in lower CVD risk adults with elevated BP, there is some evidence to suggest that relative treatment effects of BP lowering are similar across a wide range of predicted risk categories, including among those with a <10% risk.[518] Therefore, while we do not provide a formal recommendation for this, if initial lifestyle measures are not successful after 6–12 months, drug treatment might be discussed on an individual basis among lower CVD-risk adults when BP is between 130/80 and <140/90 mmHg.[518]

- In patients with elevated BP (office BP of 120–139/70–89 mmHg) who are at sufficiently high risk for CVD (e.g. 10-year CVD risk ≥ 10%) or in the presence of high-risk conditions or borderline 10-year CVD risk (5% - <10%) combined with risk modifiers or abnormal risk tool tests, BP-lowering lifestyle measures should be initiated for 3 months. Following this, pharmacological therapy is recommended for persons with confirmed BP of ≥130/80 mmHg, when these lifestyle changes have not worked or are not being implemented (*Section 8.2*) Prompt addition of pharmacological therapy, if needed by 3 months, should be emphasized, to avoid therapeutic inertia.[519] For those with BP of 120–129/70–79 mmHg, ongoing and intensified lifestyle intervention is preferred.

The above recommendations apply to all individuals with elevated BP, irrespective of age. However, recognizing the lack of conclusive evidence and added risk of side effects among certain subgroups, the task force also recommends that, among patients with elevated BP, BP-lowering treatment should always be started based on individual clinical judgment and shared decision-making.

In addition, consideration of BP-lowering drug treatment should be deferred until BP is >140/90 mmHg in the following settings: pre-treatment symptomatic orthostatic hypotension, age ≥85 years, clinically significant moderate-to-severe frailty, and/or limited predicted lifespan (<3 years) due to high competing risk (including eGFR < 30 mL/min/1.73 m^2). Patients with elevated BP in these settings are less likely to obtain sufficient net benefit from BP-lowering drug therapy or to tolerate intensive drug therapy. *Section 9* contains more information on the treatment of specific subgroups, including older and frail adults.

Recommendation Table 17 — Recommendations for initiating blood pressure-lowering treatment (see Evidence Tables 30–32)

Recommendations	Class[a]	Level[b]
In adults with elevated BP and low/medium CVD risk (<10% over 10 years), BP lowering with lifestyle measures is recommended and can reduce the risk of CVD.[119,120,348,408,411,520,521]	I	B
In adults with elevated BP and sufficiently high CVD risk[c], after 3 months of lifestyle intervention, BP lowering with pharmacological treatment is recommended for those with confirmed BP ≥130/80 mmHg to reduce CVD risk.[116,522]	I	A
It is recommended that in hypertensive patients with confirmed BP ≥140/90 mmHg, irrespective of CVD risk, lifestyle measures and pharmacological BP-lowering treatment are initiated promptly to reduce CVD risk.[116,121,122]	I	A
It is recommended to maintain BP-lowering drug treatment lifelong, even beyond the age of 85 years, if well tolerated.[523–525]	I	A
Because the benefit in reducing CVD outcomes is uncertain in these settings, and noting that close monitoring of treatment tolerance is advised, BP-lowering treatment should only be considered from ≥140/90 mmHg among persons meeting the following criteria: pre-treatment symptomatic orthostatic hypotension, age ≥85 years, clinically significant moderate-to-severe frailty, and/or limited predicted lifespan (<3 years).[131,524,526,527]	IIa	B

BP, blood pressure; CVD, cardiovascular disease.
[a]Class of recommendation.
[b]Level of evidence.
[c]10-year estimated CVD risk of ≥10%; or 10-year estimated CVD risk of 5% - ≤10% plus risk modifiers or abnormal risk tool tests; or high-risk conditions (e.g. established CVD, diabetes, moderate or severe CKD, familial hypercholesterolaemia, or hypertension-mediated organ damage).

© ESC 2024
Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

8.5. Intensity of blood pressure-lowering therapy and ideal treatment targets

8.5.1. Expected degree of blood pressure reduction with approved drugs

The magnitude of BP reduction achieved with the main classes of BP-lowering medications (ACE inhibitors, ARBs, dihydropyridine CCBs, diuretics, and beta-blockers) as monotherapy is similar overall.[528,529] BP reduction with standard doses of any of these five classes can be expected to be approximately 9/5 mmHg with office BP and 5/3 mmHg with ABPM.[478] These BP-lowering effects may attenuate over time.[530] Combination therapy (e.g. with three drugs at half standard dose) over the short term can lower office BP by up to 20/11 mmHg.[478,531] The reason why beta-blockers are not considered first-line BP-lowering medications (outside of compelling indications) is not because of inferior BP-lowering properties (particularly for vasodilating beta-blockers),[532] but because of inferior efficacy in reducing CVD events (particularly stroke) among patients with hypertension, and tolerance issues.[533–536]

The BP-lowering effect of each BP medication class generally increases with the dose administered, though this relationship is not linear.[537] Effects of each medication can also vary at the individual level, sometimes requiring personalization by matching the patient with the best medication for them.[490] The magnitude of BP reduction for any BP-lowering medication may increase as a function of the pre-treatment BP, which is also known as Wilders principle.[538,539]

The BP-lowering effect of pharmacological therapy is typically evident after 1–2 weeks of treatment,[540] but the maximum effect might take longer to manifest. Therefore, the advised follow-up after 1–3 months (1 month preferred with a GP or specialist) allows for assessment of tolerance/safety, but also allows enough time to gauge the full BP-lowering effect of each drug titration (see *Section 8.3.4*).

8.5.2. The ideal target of blood pressure-lowering treatment

As discussed in *Section 6*, optimal control of BP translates into CVD risk reduction, thereby reducing morbidity and mortality in the population.[116,478,541]

BP threshold is defined as the BP at which BP-lowering treatment is initiated, while BP target is the BP goal with treatment.

The BP threshold to initiate BP-lowering therapy is not necessarily the same as the recommended BP target once therapy is commenced (in other words, treatment threshold and treatment target may not be the same for a given patient). Specifically, for hypertensive patients in whom BP-lowering treatment is recommended above a baseline BP of ≥140/90 mmHg, the recommended target of BP-lowering therapy is 120–129/70–79 mmHg, provided treatment is well tolerated (see Supplementary data online). Persons with elevated BP who receive treatment are also recommended to achieve a target of 120–129/70–79 mmHg.

Therefore, the treatment target in the 2024 Guidelines is always 120–129/70–79 mmHg (but only if treatment is tolerated and with certain exceptions where more lenient targets are advised). In contrast, the treatment threshold may differ based on CVD risk, specifically in the elevated BP category. For example, in addition to hypertensive adults with BP ≥140/90 mmHg, there are individuals with an office systolic BP of 130–139 mmHg and/or diastolic BP of 80–89 mmHg who have sufficiently high CVD risk to recommend BP-lowering drug treatment.

The BP target range of 120–129/70–79 mmHg recommended in these guidelines reflects the most current evidence from contemporaneous RCTs[135,136,146,542–545] and from meta-analyses of RCTs.[131] Of note, this treatment target reduces CVD events in older adults[136,523] with evidence for efficacy of more intensive BP-lowering treatment targets established up to age 85 years.[131] Furthermore, research data indicate that, to optimally reduce CVD risk, achieving an on-treatment BP of 120/70 mmHg is the best point on the BP target range provided in our guideline recommendations (*Figure 20*). However, while we strongly considered recommending a treatment target of exactly 120/70 mmHg with out-of-office BP confirmation, we instead chose a target range of 120–129/70–79 mmHg (preferably with out-of-office BP confirmation but also allowing for office BP) for the following reasons: providing flexibility to patients and clinicians; feedback from external peer review; feedback from patients that lifestyle is preferred to medication unless BP is in the hypertensive range; the knowledge that contemporary treat-to-target intensive BP trials included only persons with baseline systolic BP of ≥130 mmHg; and a recognition that the BP values recorded under research conditions using systematic approaches to measurement (while strongly recommended by these guidelines) are not always the same as BP values recorded under routine clinical care, which can be 5–10 mmHg higher.[65,66]

In addition, the trial data confirming efficacy for our recommended treatment target of 120–129/70–79 mmHg do not necessarily apply to moderately-to-severely frail adults who were generally excluded from trials. Furthermore, the data supporting this BP target among adults aged >85 years are inconclusive.[131] Frailty can occur at different ages and is, together with tolerability of BP-lowering treatment, an important characteristic when considering the BP target for a given patient. Accordingly, personalized BP-lowering treatment should be instituted in people aged ≥85 years and/or those with significant frailty. Recommended indicators of frailty in guiding BP-lowering treatment are given in *Section 9*.

Several important nuances are highlighted and warrant consideration prior to implementing the new BP target of 120–129/70–79 mmHg among patients receiving BP-lowering therapy:

- Evidence for a systolic BP-lowering treatment target of 120–129 mmHg is strong (Class I, level of evidence A).
- Evidence for a specific diastolic BP-lowering treatment target is less strong in those who are treated to a systolic target of 120–129 mmHg. While most adults treated to a systolic BP target of 120–129 mmHg will also achieve a diastolic BP of 70–79 mmHg, not all will.[543,546] Furthermore, adults who achieve systolic BP control are generally at low relative risk for CVD, even when diastolic BP is 70–90 mmHg.[547,548] Nonetheless, due in part to the known higher risk of isolated diastolic hypertension among younger adults,[549] the task force agreed that it is reasonable to target an on-treatment diastolic BP of 70–79 mmHg among patients with diastolic BP of ≥80 mmHg who are already at the systolic BP target of 120–129 mmHg (Class IIb, level of evidence C).
- The task force acknowledges the 2021 ESC Guidelines on cardiovascular disease prevention in clinical practice,[170] which take a stepwise approach to their recommendations for BP-lowering treatment. The 2021 Guidelines recommended an on-treatment systolic BP target of 130–139 mmHg as the first step and then—based on patient preferences, risk, and frailty—to aim for a target on-treatment systolic BP of <130 mmHg as the second step. While we recognize the potential value of this two-step approach, which many clinicians may choose to follow, the current guidelines emphasize one on-treatment BP target (120–129/70–79 mmHg, provided treatment is tolerated). This one-step approach is based on the evidence, and motivated to discourage therapeutic inertia around BP lowering. As an illustrative example of the latter concern for therapeutic inertia, an

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

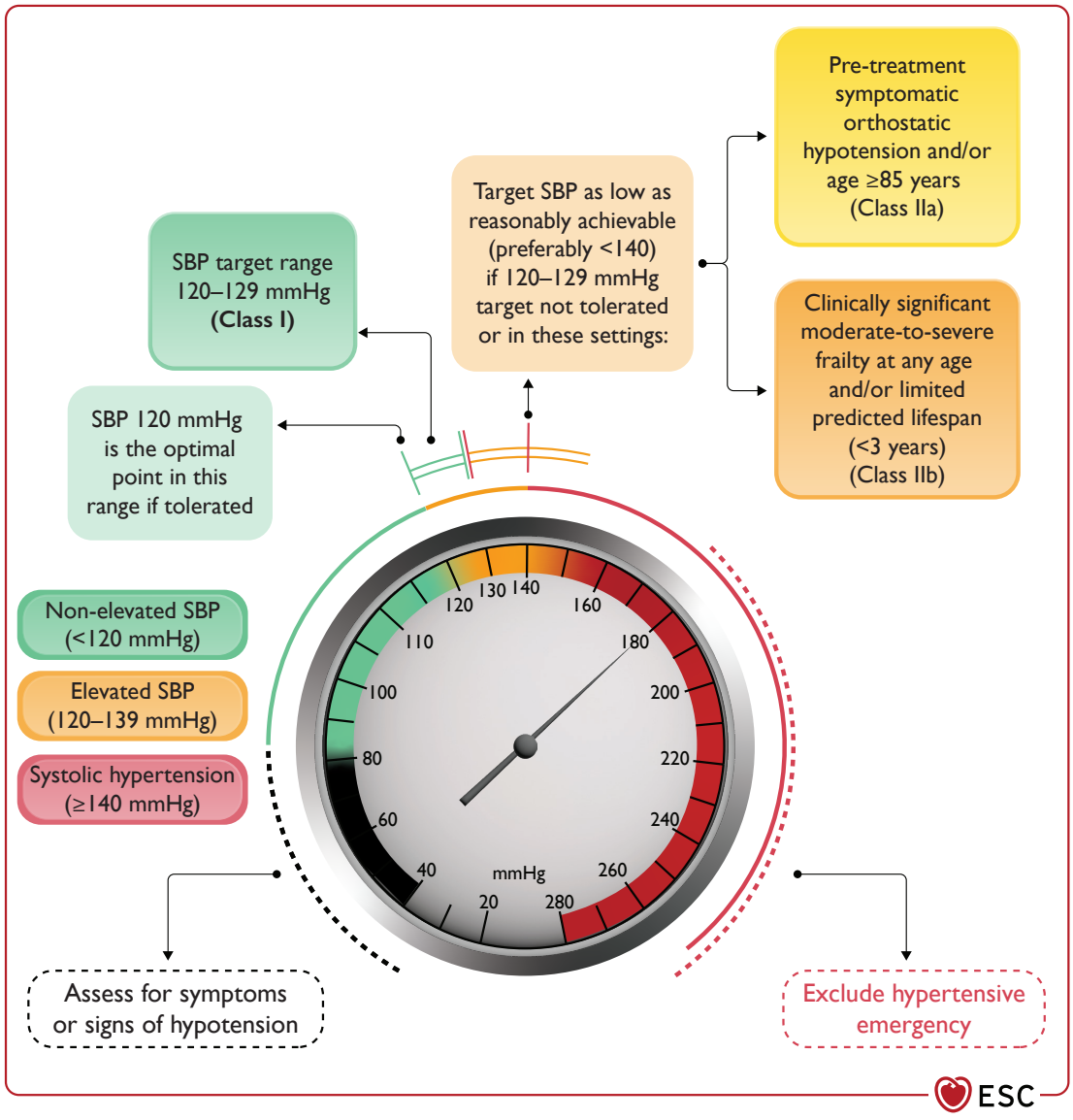


Figure 20 Systolic blood pressure categories and treatment target range. BP, blood pressure; SBP, systolic blood pressure.

on-treatment systolic BP of 135 mmHg (office) may be considered reasonable when reviewing the 2021 ESC Guidelines on cardiovascular disease prevention in clinical practice but, we hope, will not be considered reasonable to those who follow the updated 2024 ESC Guidelines presented here. Specifically, it is well established that an on-treatment systolic BP of 135 mmHg is not optimal relative to more intensive control.[116,131,135,136,445,542,543,545]

- We acknowledge that the results from RCTs cannot always be extrapolated to routine clinical care. In addition, we recognize the increased risk of side effects among patients receiving more intensive BP-lowering treatments, compared with traditional BP targets.[545,550] Accordingly, an important caveat to our treatment target of 120–129/70–79 mmHg is the recommendation to pursue this target only when treatment is well tolerated. In cases where BP-lowering treatment is not well tolerated and a target of 120–129/70–79 mmHg is not possible, it is recommended to follow the 'as low as reasonably achievable' (ALARA) principle, by targeting treatment to a BP that is as low as reasonably achievable.
- In addition to adults with significant frailty and/or who are ≥85 years of age, the evidence for a BP-lowering treatment target of 120–129/70–79 mmHg may also not generalize to patients with: (i) pre-treatment symptomatic orthostatic hypotension, (ii) limited predicted lifespan (e.g. <3 years),[527] and/or (iii) high levels of competing risk for non-CVD death including CKD with eGFR < 30 mL/min/1.73 m^2) (see *Section 9*).

Finally, as outlined in *Section 5*, these guidelines endorse a 'trust but verify' approach to office BP measurements, and, where possible, confirming BP with accurate out-of-office BP measurements (ABPM, HBPM) is recommended prior to starting treatment, to monitor the treatment effect of BP-lowering medication.

8.5.3. Personalizing treatment strategies

Though promising, there is little to no evidence to date from CVD outcome trials to use novel biomarkers for individualizing BP-lowering treatment.[551,552] Nonetheless, different patient groups

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

Recommendation Table 18 — Recommendations for blood pressure targets with treatment (see Evidence Table 34)

Recommendations	Class[a]	Level[b]
To reduce CVD risk, it is recommended that treated systolic BP values in most adults be targeted to 120–129 mmHg, provided the treatment is well tolerated.[22,122,131,523,541]	I	A
In cases where BP-lowering treatment is poorly tolerated and achieving a systolic of 120–129 mmHg is not possible, it is recommended to target a systolic BP level that is 'as low as reasonably achievable' (ALARA principle).[22,122,131,523,541]	I	A
Because the CVD benefit of an on-treatment systolic BP target of 120–129 mmHg may not generalize to the following specific settings, personalized and more lenient BP targets (e.g. <140 mmHg) should be considered among patients meeting the following criteria: pre-treatment symptomatic orthostatic hypotension, and/or age ≥85 years.[131]	IIa	C
Because the CVD benefit of an on-treatment systolic BP target of 120–129 mmHg may not generalize to the following specific settings, personalized and more lenient BP targets (e.g. <140/90 mmHg) may be considered among patients meeting the following criteria: clinically significant moderate-to-severe frailty at any age, and/or limited predicted lifespan (<3 years).	IIb	C
In cases where on-treatment systolic BP is at or below target (120–129 mmHg) but diastolic BP is not at target (≥80 mmHg), intensifying BP-lowering treatment to achieve an on-treatment diastolic BP of 70–79 mmHg may be considered to reduce CVD risk.	IIb	C

© ESC 2024

ALARA, as low as reasonably achievable; BP, blood pressure; CVD, cardiovascular disease.
[a]Class of recommendation.
[b]Level of evidence.

can be identified, and medication initiation can be tailored to pre-existing conditions, such as diabetes mellitus, CKD, AF, post-myocardial infarction, heart failure, metabolic syndrome, and proteinuria/albuminuria (*Section 9*). History of previous side effects and possible and compelling indications also need to be considered when selecting treatment (see Supplementary data online, *Tables S9 and S10*). Cardioselective beta-blockers may be used in low dose in chronic asthma,[553,554] in line with their use in patients with heart failure with chronic asthma.

For considerations of BP-lowering treatment among specific patient populations of interest, including different racial/ethnic populations, see *Section 9*.

8.5.4. Duration and monitoring of drug therapy

BP-lowering treatment is usually chronic, often lifelong. This raises the question of long-term efficiency, long-term side effects, adherence, and persistence with therapy. While BP-lowering therapies typically provide an overall durable effect, some attenuation of effect may be seen over time.[66,530] First-line BP-lowering medication classes appear to be safe for long-term use.[555–557] Once BP is controlled, at least a yearly follow-up is advised. Because of the known temporal variability in BP[558,559] and medication efficacy in the long term,[530] medication changes may be necessary over time (see Supplementary data online).

Recommendation Table 19 — Recommendations for follow-up in patients with treated hypertension (see Evidence Table 33)

Recommendation	Class[a]	Level[b]
Once BP is controlled and stable under BP-lowering therapy, at least a yearly follow-up for BP and other CVD risk factors should be considered.	IIa	C

© ESC 2024

BP, blood pressure; CVD, cardiovascular disease.
[a]Class of recommendation.
[b]Level of evidence.

8.6. Device-based blood pressure lowering

Several device-based therapies designed to lower BP have been investigated.[560,561] To date, the best evidence exists for catheter-based renal denervation.

8.6.1. Catheter-based renal denervation

Sympathetic nervous system overactivity contributes to the development and progression of hypertension.[562] Renal denervation aims to interrupt afferent and efferent sympathetic nerves in the adventitia and perivascular tissue of the renal arteries.[563] The 2018 ESC/ESH Guidelines on the management of arterial hypertension did not recommend the use of device-based therapies for routine treatment of hypertension, unless in the context of clinical studies and RCTs.[1] This was based on negative data using first-generation radiofrequency catheters (see Supplementary data online).

More recent data from sham-controlled trials investigating second-generation radiofrequency and ultrasound catheters demonstrated a BP-lowering efficacy in a broad range of patients, with and without concomitant BP-lowering medications, including those with resistant hypertension.[564–568] Long-term, non-randomized, follow-up data from the Global Symplicity Registry,[569] Symplicity HTN-3 trial,[570] Spyral HTN-ON MED pilot trial,[571] and A Study of the Recor Medical Paradise System in Clinical Hypertension (RADIANCE-HTN) SOLO trial[572] indicate a sustained BP-lowering effect for up to 3 years. A single-centre open-label study suggested sustained BP reductions up to 10 years.[573] These data also highlight a potentially important advantage of renal denervation, namely that the BP-lowering effect of this intervention might be 'always on', making this approach attractive for patients with suboptimal medication adherence.[254] Some patients may prefer a one-off procedure rather than taking daily medications chronically and may request renal denervation.

Of significance, there are no reported procedure-related serious safety signals in the first- and second-generation trials beyond the usual risk of femoral arterial access procedures (noting that most trials to date were not powered for safety outcomes and that the task force could find no published meta-analysis data on exact rates of major bleeding and major femoral artery vascular access complications after renal denervation procedures). However, the rate of major bleeding

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

and major femoral artery vascular access complications for coronary angiography using a femoral approach is typically reported as 1%–4%[574,575] but has been reported as 5%–10% in some studies.[575] Trials investigating radial access for renal denervation are currently ongoing (ClinicalTrials.gov identifier: NCT05234788). After renal denervation, there is a 0.25%–0.5% rate of renal artery stenosis/dissection requiring stenting.[576] Long-term follow-up data up to 3 years have not suggested worsening of renal function beyond the expected rates in patients with hypertension with mildly-to-moderately reduced renal function.[569,577] Of note, sham-controlled trials to date excluded patients with severely reduced kidney function at baseline.[564,566–568]

Despite the clear promise of renal denervation in reducing BP, there are some concerns that warrant consideration, as we indicate in the recommendations. First, the effect of current renal denervation catheter technologies on BP lowering is relatively modest for an invasive procedure (meta-analyses report placebo-corrected systolic BP lowering of approximately 6 mmHg on office BP assessment and 4 mmHg on 24 h ABPM).[578] As such, the average BP-lowering effect of renal denervation appears no more than for one standard BP-lowering medication. Accordingly, many adults undergoing renal denervation will likely require ongoing, post-procedural, BP-lowering drugs.

Second, the cost-effectiveness of renal denervation has not been fully established. Since effects of current renal denervation technologies are similar to that of one standard BP-lowering medication, most of which are generic, it is difficult to see a scenario where renal denervation could be proven cost-effective for most patients. An exception might be patients who are at very high risk of CVD events and who have uncontrolled BP due to resistant hypertension (with or without non-adherence).[579,580]

Third, there are no adequately powered outcomes trials demonstrating that renal denervation reduces CVD events and is safe in the long term. While observational reports have suggested associations between renal denervation and reduced risk for CVD events,[581,582] these observational data have major inferential limitations including a significant potential for confounding. While BP lowering is typically a good surrogate for CVD benefit, there is no guarantee that this is true with renal denervation and, furthermore, off-target effects independent of BP could influence CVD and other adverse-event rates after the procedure. Because of the lack of outcomes trials, renal denervation cannot reach the Class I indication threshold set by this task force. Arguments that outcomes trials will not be funded are insufficient to influence guideline recommendations. However, it is hoped that the position of these guidelines will motivate industry to sponsor the necessary renal denervation outcomes trials.

Fourth, related to the lack of outcomes data, the potentially 'always on' effect of renal denervation could backfire if late complications emerge. Medications causing complications or side effects can simply be stopped and replaced with alternative medications when such problems emerge; this is not true with renal denervation.

Fifth, the impact of scaling up renal denervation on usual cardiac department catheterization laboratory workflows is of some concern. Specifically, it is important that renal denervation procedures do not delay timely access to other elective procedures with proven efficacy in reducing CVD outcomes.[583]

Sixth, there is still no direct evidence to gauge whether renal denervation procedures are successful and that the kidneys are denervated and do not reinnervate over time.[584] Relatedly, the concept of responders and non-responders to renal denervation (and the hypothesis that predictors of response might be found to help identify patients most suitable for the procedure) is questioned. Medical interventions, including drugs, are naturally subject to inter-individual variability in response.[567] Additionally, there are few examples in medicine of consistent and clinically useful predictors of treatment response for medical conditions that have complex genetic and environmental underpinnings (i.e. conditions like hypertension).

A multidisciplinary hypertension team, including experts in hypertension and percutaneous cardiovascular interventions, is recommended to evaluate the indication and to perform the procedure.[585] Based on the available evidence, renal denervation may be considered for patients who have uncontrolled, true resistant hypertension with a three-drug combination and who express a preference to undergo renal denervation.[566,568,585] This recommendation is informed, in part, by the higher risk of CVD events in this subgroup, which represents a major unmet clinical need and which also means that cost-effectiveness considerations are likely to be optimal in this setting. In patients who are non-adherent or intolerant to multiple BP-lowering medications, particularly first-line agents, and who have high predicted CVD risk and a BP that is not at target, renal denervation may, for the same reasons, be considered if the patients express a preference to undergo renal denervation after a tailored shared decision-making process. The shared decision-making process requires that the patients are fully informed about the benefits, limitations, and risks associated with renal denervation.

Recommendation Table 20 — Recommendations for device-based treatment of hypertension (see Evidence Table 35)

Recommendations	Class[a]	Level[b]
To reduce BP, and if performed at a medium-to-high volume centre, catheter-based renal denervation may be considered for resistant hypertension patients who have BP that is uncontrolled despite a three BP-lowering drug combination (including a thiazide or thiazide-like diuretic), and who express a preference to undergo renal denervation after a shared risk-benefit discussion and multidisciplinary assessment.[564,566–568,586–590]	IIb	B
To reduce BP, and if performed at a medium-to-high volume centre, catheter-based renal denervation may be considered for patients with both increased CVD risk and uncontrolled hypertension on fewer than three drugs, if they express a preference to undergo renal denervation after a shared risk-benefit discussion and multidisciplinary assessment.[564,566–568,586–590]	IIb	A
Due to a lack of adequately powered outcomes trials demonstrating its safety and CVD benefits, renal denervation is not recommended as a first-line BP-lowering intervention for hypertension.	III	C
Renal denervation is not recommended for treating hypertension in patients with moderate-to-severely impaired renal function (eGFR <40 mL/min/1.73 m^2) or secondary causes of hypertension, until further evidence becomes available.	III	C

BP, blood pressure; CVD, cardiovascular disease; eGFR, estimated glomerular filtration rate.
[a]Class of recommendation.
[b]Level of evidence.

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

© ESC 2024

8.6.2. Other devices

Most device-based therapies investigated for BP-lowering therapy in hypertension have been aimed at modulating the autonomic nervous system activity (baroreflex activation therapy, endovascular baroreflex amplification therapy, and carotid body ablation).[560] Cardiac neuromodulation therapy aims to lower left ventricular preload by variably altering the atrioventricular interval using a dual-chamber, rate-responsive, implantable pulse generator in patients indicated for implantation or replacement of a dual-chamber pacemaker.[591,592] Some, though not all, of these devices have shown promising results in non-randomized, single-arm studies. A sham-controlled trial investigating central iliac arteriovenous stent anastomosis was terminated early after longer-term follow-up data indicated an increase in heart failure in the stent group.[561] Therefore, the use of these device-based therapies is not recommended for routinely treating hypertension until further evidence regarding their safety and efficacy becomes available (see Supplementary data online).

8.7. Unintended and potentially harmful consequences of blood pressure lowering and implications for treatment targets

8.7.1. Adverse effects of blood pressure-lowering medications

8.7.1.1. Symptomatic adverse effects

BP-lowering medications have multiple side effects, which may be more common in females.[536,593,594] Although generally well tolerated, common side effects include headaches, cough, dizziness or light-headedness, diarrhoea or constipation, fatigue, ankle swelling, and erectile problems, depending on the drug class (see Supplementary data online, *Table S9*).[536,550,593–597]

In randomized trials of adults aged >60 years, the overall rate of symptomatic BP-lowering drug withdrawal was higher than the rate of placebo withdrawal (approximately 15% vs. 5%).[593] A systematic review, which included 280 638 participants in 58 RCTs, reported no evidence for an increased relative risk of falls in those taking BP-lowering drugs.[550] There was, however, an increased relative risk of mild hyperkalaemia, acute kidney injury, hypotension, and syncope. Furthermore, very frail adults were excluded from BP-lowering trials, which is relevant because such patients are more prone to adverse effects and polypharmacy (see Supplementary data online).[596]

8.7.1.2. Renal effects

A systematic review reported an increased risk of acute kidney injury and hyperkalaemia associated with BP-lowering treatment.[550] Analyses of outcomes by specific drug class showed that drugs affecting the RAAS were more likely to be associated with acute kidney injury and hyperkalaemia.[550]

Patients with significant CKD tend to be excluded from RCTs.[137,545,598] It is important to remember these exclusion criteria, and that patients with CKD are more likely to suffer from resistant hypertension, when extrapolating the results of more intensive BP lowering to patients with moderate-to-severe CKD (see *Section 9*).[599]

8.7.1.3. Erectile dysfunction

Older classes of BP-lowering drugs (including diuretics, beta-blockers, and centrally acting medications) are associated with erectile dysfunction.[600] However, newer classes have neutral effects.[601] Angiotensin receptor antagonists may have beneficial effect on erectile function.[602]

8.7.2. Pill burden and non-adherence

More intensive treatment of elevated BP and hypertension may be associated with an increased risk of polypharmacy and pill burden, which are themselves associated with non-adherence.[603,604] Single-pill, fixed-dose drug combinations can help to reduce pill burden and are recommended to improve adherence (refer to *Section 8.3.4*).

Increased intensity of BP lowering (while ultimately cost reducing in terms of CVD reduction)[605] can also result in higher upfront direct and indirect healthcare costs, with more people requiring medication and higher demand for technology-based adherence strategies, which can be challenging to implement, especially in resource-poor settings.[604]

8.7.3. Potentially harmful consequences of blood pressure lowering for frail older people

Unintended consequences of BP lowering (hypotension, syncope, falls) can be hazardous for frail older people in particular.[606] Retrospective studies have shown that adults aged >75 years from the general population, who would have met the criteria for inclusion in the Systolic Blood Pressure Intervention Trial (SPRINT), had a rate of injurious falls and syncope that was nearly five times that of the standard care group in the trial. This suggests that healthy participant bias may have contributed to the findings of SPRINT and other similar BP-lowering trials, and that the results may not fully generalize to older adults in more routine clinical care.[607]

Patients' functional ability should be considered in addition to age to help negate any unintended consequences of BP lowering in a frailer cohort. Despite their chronological age, older patients with hypertension who are fit and can independently carry out activities of daily living will benefit from guideline-directed treatment similar to younger cohorts.[131] However, tailoring treatment targets and treatment plans for frail older patients is necessary to avoid unintended consequences. This should include assessing frailty, including cognitive status, risk of falls, propensity for symptomatic orthostatic hypotension, polypharmacy, and other comorbid conditions.[608,609] Of note, and as detailed in *Section 9*, some data indicate a benefit of more intensive BP-lowering on cognitive function.[523,610,611] For those with loss of function but preserved activities of daily living, a more detailed geriatric assessment is required to explore the risks and benefits of treatment, as well as considerations for tailoring therapeutic strategies where appropriate. For patients who are both functionally impaired and unable to carry out activities of daily living, the therapeutic goals of hypertension treatment should be personalized, and medications discontinued where appropriate (see *Section 9.3*).[596]

8.7.4. Clinical inertia in blood pressure lowering

The fear of serious adverse events with BP-lowering medications is often cited as a reason for clinical inertia, although the evidence to date from meta-analyses of RCTs suggests these side-effect concerns may be exaggerated.[550,612] However, RCTs often select populations with less frailty and multimorbidity who are more likely to tolerate treatment.[613] Consequently, fewer adverse effects might be reported than would be expected in the general population. It remains up to individual clinicians to initiate shared decision-making with each patient, especially patients in vulnerable groups and those who have experienced previous adverse events, weighing up potential benefits against risks of treatment.[614,615]

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

9. Managing specific patient groups or circumstances

9.1. Young adulthood (18–40 years)

9.1.1. Definition and epidemiology

In the present guidelines 'young adulthood' is defined as age 18–40 years. The prevalence of hypertension in young adults is increasing in men and women.[616–618] Unhealthy lifestyle, gender, obesity, and socio-economic factors contribute.[617,619–621] Hypertension-attributable CVD burden in young adults, evaluated as mortality or years of living with disability, has increased in the last decades, especially in low- and middle-income countries and in men.[622] Hypertension awareness, treatment, and control in young adults is lower than in other age categories, a result driven by worse control in young men.[623]

Systolic and diastolic hypertension and isolated diastolic hypertension are associated with increased CVD risk in the young (see Supplementary data online).[624] Isolated systolic hypertension in the young is discussed in *Section 9.4*.

9.1.2. Secondary hypertension in young adulthood

Secondary hypertension is more frequent in younger than in later-onset hypertension, with a prevalence of 15%–30% in hypertensive young adults reported from some referral centres.[625,626] Major causes of secondary hypertension include drug-induced hypertension (e.g. oestrogen-progesterone oral contraceptives; cold medication) and primary aldosteronism. The use of recreational drugs/substances, as well as supplements and energy drinks should be investigated (see *Section 7*).

Combined oestrogen-progesterone contraceptives are among the most common causes of drug-induced hypertension in young women,[627,628] and should not be used in hypertensive women unless there is no other method available or acceptable to the patient.[629] Conversely, progestin-only contraceptives are generally considered safe in women with hypertension.[89,630,631] Fibromuscular dysplasia should be considered as a cause of secondary hypertension in young women,[626,632] whereas primary aldosteronism, the most common form of secondary hypertension, is equally common in different age classes.[316] Screening for secondary hypertension is thus recommended in young adults with hypertension. However, in obese young adults, primary hypertension is more common, though OSAS should also be considered in this instance.[633]

9.1.3. Measurement and management of blood pressure in young adults

Out-of-office BP measurement is recommended in young adults for confirming diagnosis, since the white-coat phenomenon occurs in the young.[634] Because of the lower absolute CVD risk in this age category compared with older adults, hard-endpoint randomized trials of BP lowering in young adults have not been performed. However, since relative risk reduction by BP-lowering treatment is homogeneous in any age group, including those <55 years old,[131] young adults with suitable indications are also expected to benefit from BP-lowering therapy. The hypertension management algorithm based on CVD risk proposed in *Section 6* is not fully applicable in young adults, since SCORE2 has not been validated for individuals <40 years old. Even risk stratification based on lifetime risk assessment does not apply to very young adults (e.g. 20–30 years of age).[128] In the absence of established CVD, diabetes mellitus, familial hypercholesterolaemia, and moderate or severe CKD a BP-lowering treatment initiation threshold of office 140/90 mmHg is appropriate in most young adults. However, HMOD assessment may be considered in patients aged <40 years to stratify individuals with elevated BP into a higher risk category. For example, arterial stiffness better reclassifies CVD risk in individuals aged <50 years than in older individuals.[28,215] Echocardiographic left ventricular mass also maintains its added reclassification and discrimination on top of risk scores in young adults (see *Section 6* for discussion of risk modifiers).[289]

Irrespective of cardiovascular risk, all young adults with elevated BP are recommended to follow lifestyle guidance for BP lowering. A discussion about family planning should be taken with young women of childbearing potential at each visit.[635,636]

Adherence to treatment is low in young adults, <50% in some studies.[251] Therefore, communicating the importance of adherence, education, and follow-up clinics is important. (see *Sections 7* and *11*).

Recommendation Table 21 — Recommendations for managing hypertension in young adults (see Evidence Tables 36 and 37)

Recommendations	Class[a]	Level[b]
Comprehensive screening for the main causes of secondary hypertension is recommended in adults diagnosed with hypertension before the age of 40 years, except for obese young adults where it is recommended to start with an obstructive sleep apnoea evaluation.[316,626]	I	B
Since SCORE2 has not been validated for individuals <40 years, screening for HMOD may be considered in such young individuals with elevated BP without other increased CVD risk conditions to identify additional individuals for possible medical treatment.[28,215]	IIb	B

BP, blood pressure; CVD, cardiovascular disease; HMOD, hypertension-mediated organ damage; SCORE2, Systematic COronary Risk Evaluation 2.
[a]Class of recommendation.
[b]Level of evidence.

9.2. Pregnancy

9.2.1. Definition and epidemiology

Hypertension in pregnancy is typically defined as systolic BP of ≥140 mmHg and/or diastolic BP of ≥90 mmHg, measured using repeated BP readings in the office or hospital on two separate occasions or ≥15 min apart in severe hypertension (≥160/110 mmHg).[1,637,638]

Hypertension in pregnancy is the second leading cause of maternal death after maternal peri-partum haemorrhage.[639] Approximately 7% of pregnancies are complicated by hypertension, of which 3% are due to pre-eclampsia and around 1% are chronic or pre-existing hypertension.[640] Women with a history of hypertensive disorders during pregnancy are at increased risk of subsequent hypertension and CVD.[640–642]

9.2.2. Classifying hypertension in pregnancy

Hypertension in pregnancy includes:

- **Chronic hypertension:** precedes pregnancy, develops before 20 weeks of gestation, persists for >6 weeks post-partum, and may be associated with proteinuria.
- **Gestational hypertension:** develops after 20 weeks of gestation and usually resolves within 6 weeks post-partum.

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026
© ESC 2024

- **Antenatally unclassifiable hypertension:** BP is first recorded after 20 weeks of gestation, and hypertension is diagnosed but it is unclear if chronic or not; reassessment is necessary 6 weeks post-partum.
- **Pre-eclampsia:** gestational hypertension accompanied by new-onset: (i) proteinuria (>0.3 g/day or ≥30 mg/mmol ACR), (ii) other maternal organ dysfunction, including acute kidney injury (serum creatinine ≥ 1 mg/dL), liver dysfunction (elevated transaminases > 40 UI/L with or without right upper quadrant or epigastric abdominal pain), neurological complications (convulsions, altered mental status, blindness, stroke, severe headaches, and persistent visual scotomata), or haematological complications (platelet count < 150 000/μL, disseminated intravascular coagulation, haemolysis), or (iii) uteroplacental dysfunction (such as foetal growth restriction, abnormal umbilical artery Doppler waveform analysis, or stillbirth).[643] The only cure for pre-eclampsia is delivery, which is recommended at 37 weeks' gestation, or earlier in high-risk cases. Of note, proteinuria is not mandatory for diagnosing pre-eclampsia but is present in about 70% of cases.[644] Also, as proteinuria may be a late manifestation of pre-eclampsia, it should be suspected when *de novo* hypertension is accompanied by headache, visual disturbances, abdominal pain, or abnormal laboratory tests, specifically low platelets and/or abnormal liver function.

Other potential causes for high BP, including pain and anxiety, must be excluded when treating hypertension during pregnancy.

9.2.3. Measuring blood pressure in pregnancy

See *Section 5.5.1* for information on BP measurement approaches in pregnancy.[645] It is important to restate here that oscillometric devices tend to under-estimate the true BP and are unreliable in severe pre-eclampsia; only a few have been validated in pregnancy. Importantly, only the relatively few devices validated for measuring BP in pregnancy and pre-eclampsia should be used (https://stridebp.org).

9.2.4. Investigating hypertension in pregnancy

Basic laboratory investigations include urinalysis, blood count, haematocrit, liver enzymes, serum creatinine, and serum uric acid. Serum uric acid is increased in pre-eclampsia and identifies women at increased risk of adverse maternal and foetal outcomes in hypertensive pregnancies.[646]

All pregnant women should be assessed for proteinuria in early pregnancy (e.g. 11–14 weeks' gestation).[647] A dipstick test of ≥1+ should prompt further investigations, including ACR, which can be quickly determined in a single spot-urine sample.[648] An ACR of <30 mg/mmol (<0.3 mg/mg) can rule out proteinuria.[649] Higher values should prompt 24 h urine collection.

In one study, 10% of pregnant women with chronic hypertension had secondary hypertension (estimated to affect 0.24% of all pregnancies).[650] Secondary hypertension during pregnancy is associated with an increased risk of adverse outcomes.[650] The most common cause of secondary hypertension during pregnancy is CKD. The onset of hypertension during the first trimester, at the peak of human chorionic gonadotropin (HCG) secretion, should prompt consideration of primary aldosteronism.[651] Phaeochromocytoma in pregnant women is rare (0.002% of all pregnancies) but highly morbid.[652,653]

9.2.5. Preventing hypertension and pre-eclampsia

Low-to-moderate-intensity exercise, especially if supervised and initiated during the first trimester of pregnancy, decreases the incidence of developing gestational hypertension.[654] As such, after consultation with their obstetrician, all pregnant women should participate in physical activity, unless contraindicated.[655] Factors indicating risk of pre-eclampsia are discussed in the Supplementary data online.

Women at high or moderate risk of pre-eclampsia should be advised to take 100–150 mg of aspirin daily at bedtime from gestational weeks 12–36.[647,656,657]

Oral calcium supplementation of 0.5–2 g daily is recommended for preventing pre-eclampsia in women with low dietary intake of calcium (<600 mg daily).[658,659]

9.2.6. Treatment initiation and blood pressure targets

Acute management of BP in pre-eclampsia and eclampsia is detailed in *Section 10.4*.

Meta-analyses have found no evidence for an increased risk for delivering small-for-gestational-age babies in pregnant women with mild hypertension receiving BP-lowering medications.[660] Despite a historical paucity of trial data, previous European guidelines[1,89] recommended initiating BP-lowering drug treatment (i) in all women with persistently elevated office BP of ≥150/90 mmHg, and (ii) in women with gestational hypertension (with or without proteinuria), pre-existing hypertension with superimposed gestational hypertension, or hypertension with subclinical HMOD, when office BP is >140/90 mmHg.

In the CHAP trial, treating pregnant women with chronic hypertension and BP of ≥140/90 mmHg reduced the occurrence of pre-eclampsia with severe features, and reduced medically indicated pre-term birth <35 weeks, compared with only treating severe hypertension (BP ≥ 160/105 mmHg).[88] Tight BP control (target diastolic BP < 85 mmHg) compared with less-tight BP control (target diastolic BP < 100 mmHg) reduces the incidence of subsequent severe maternal hypertension (BP ≥ 160/110 mmHg), but not foetal or other maternal outcomes in women with mild hypertension at baseline (diastolic BP of 85–105 mmHg).[661]

Treatment with BP-lowering drugs in all pregnant women with confirmed BP of ≥140/90 mmHg is recommended to reduce the progression to severe hypertension and the related risks for adverse pregnancy outcomes.[660,661] In women with pre-existing and gestational hypertension with and without pre-eclampsia, we recommend lowering BP below 140 mmHg for systolic and to 80–90 mmHg for diastolic BP.[661] Evidence to support a BP target as low as 120–129/70–79 mmHg is lacking in pregnancy, though such evidence exists for non-pregnant patients receiving BP-lowering medication.

9.2.7. Managing mild hypertension in pregnancy (office blood pressure 140–159/90–109 mmHg)

RAS inhibitors are not recommended in pregnancy due to adverse foetal and neonatal outcomes. The BP-lowering drugs of choice are: beta-blockers (most data are available for labetalol, a non-selective beta-blocker that also acts as an alpha-blocker in higher doses; metoprolol and bisoprolol are also considered safe), dihydropyridine CCBs (most data are available for nifedipine, which is generally considered first choice, also felodipine, nitrendipine, amlodipine, and isradipine can be used), and methyldopa.[662,663] A meta-analysis suggests that beta-blockers and CCBs are more effective than methyldopa in preventing severe hypertension.[660] Of note, however, atenolol should be avoided, as it is associated with foetal growth restriction.[664,665] Methyldopa has been associated with an increased risk of post-partum depression and caution is therefore advised both intra-partum and post-partum.[637]

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

Hydralazine can be particularly effective for severe hypertension in pregnancy and can be administered intravenously in hypertensive admissions (*Section 10*).[666–668] While thiazide diuretics in pregnancy have limited safety data and should be used with caution, other diuretics such as furosemide are not contraindicated and may be necessary in some situations (see Supplementary data online).[669]

9.2.8. Managing severe hypertension in pregnancy (>160/110 mmHg)

Acute onset of severe hypertension persisting for more than 15 min is considered a hypertensive emergency in pregnancy and is covered in *Section 10.4.2*.

9.2.9. Managing blood pressure post-partum

For women with hypertension during pregnancy, BP should be measured within 6 h of delivery and, if possible, daily for at least a week after discharge from the hospital.[637] Post-partum hypertension is common in the first week and associated with prolonged hospitalization.[670]

Women with hypertension in pregnancy are at increased risk of chronic hypertension,[671] CKD,[672] and CVD.[177,223,640] The relative risk of chronic hypertension is highest in the first 6 months following delivery, motivating regular screening in these women.[673] Women with gestational hypertension, especially those with pre-eclampsia, have higher risk of masked hypertension.[674] BP measurements, ideally including out-of-office measurements, urine analyses, and CVD risk assessment, should at least be performed 6–12 weeks, 6 months, and 12 months post-partum and, after that, annually. Recent data indicate the potential utility of self-monitoring of BP during the busy early post-partum period.[675]

All BP-lowering drugs are excreted into breast milk.[637] Except for propranolol, atenolol, acebutolol, and nifedipine, most drugs are excreted in very low concentrations in breast milk (see Supplementary data online, *Table S11*).[637]

9.2.10. Risk of recurrence of hypertensive disorders in a subsequent pregnancy

About 20%–30% of women with hypertensive disorders in a previous pregnancy will experience recurrence in a subsequent pregnancy.[676,677] The earlier the onset of hypertension in the first pregnancy, the higher the risk of recurrence in a subsequent pregnancy.[677]

Further details on managing hypertension and other cardiovascular disorders in pregnancy are available elsewhere.[89,637]

Recommendation Table 22 — Recommendations for managing hypertension in pregnancy (see Evidence Tables 38–40)

Recommendations	Class[a]	Level[b]
In women with gestational hypertension, starting drug treatment is recommended for those with confirmed office systolic BP ≥140 mmHg or diastolic BP ≥90 mmHg.[661]	I	B
In pregnant women with chronic hypertension, starting drug treatment is recommended for those with confirmed office systolic BP ≥140 mmHg or diastolic BP ≥90 mmHg.[88,660,661,678]	I	B
In women with chronic and gestational hypertension, it is recommended to lower BP below 140/90 mmHg but not below 80 mmHg for diastolic BP.	I	C
Dihydropyridine CCBs (preferably extended-release nifedipine), labetalol, and methyldopa are recommended first-line BP-lowering medications for treating hypertension in pregnancy.	I	C
In consultation with an obstetrician, low- to moderate-intensity exercise is recommended in all pregnant women without contraindications to reduce the risk of gestational hypertension and pre-eclampsia.[654,655]	I	B
Systolic BP ≥160 mmHg or diastolic BP ≥110 mmHg in pregnancy can indicate an emergency, and immediate hospitalization should be considered.	IIa	C
HBPM and ABPM should be considered to exclude white-coat and masked hypertension, which are more common in pregnancy.[679]	IIa	C
RAS blockers are not recommended during pregnancy.[680,681]	III	B

ABPM, ambulatory blood pressure monitoring; BP, blood pressure; CCB, calcium channel blocker; HBPM, home blood pressure monitoring; RAS, renin–angiotensin system.
[a]Class of recommendation.
[b]Level of evidence.

9.3. Very old age (≥85 years), frailty, multimorbidity, and polypharmacy

9.3.1. Definition of frailty

The most common definition of frailty is an age-associated, biological syndrome characterized by decreased biological reserves, due to dysregulation of several physiological systems.[682] This puts an individual at risk when facing physiological stressors, and is associated with poor outcomes, such as disability, hospitalization, and death.[683] The estimated prevalence of frailty in people aged >65 years is 7%–16% and is greater in women than in men.[684,685] Although the main determinant of frailty is age, chronological age must be differentiated from biological age.[686] An older patient can be fit and robust while a multimorbid young patient can be frail. Using multiple drugs may have more unpredictable effects on BP in older patients, because of increased competition for underlying mechanisms responsible for their degradation and elimination, and because the ability of the baro-[687] and chemo-reflex[688] systems in maintaining a steady treated BP level can decline with ageing.

With respect to BP, two issues compound interpretation of the frailty literature. First, frailty on its own is a strong predictor of mortality and cardiovascular complications[689] and is accompanied by a decrease in systolic BP.[690] This raises the issue of the so-called BP J-curve (see *Section 9.8*) and reverse causality, with frailty rather than excessive BP lowering being the root cause of adverse health outcomes. Only properly randomized and controlled clinical trials can differentiate between the effects of frailty vs. overly intensive BP-lowering treatment, but unfortunately, few BP-lowering trials have included a substantial proportion of frail patients. Second, there is no consensus on how to grade frailty in day-to-day clinical practice.[606] Complex frailty scales exist for application in research,[523,691] but unless they are electronically generated,[692] they are typically not practical in routine clinical care. Nonetheless, the clinical frailty scale (*Figure 21*) is intuitive and easy to administer and has been validated against 5-year risk of death.[596,693]

© ESC 2024
Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

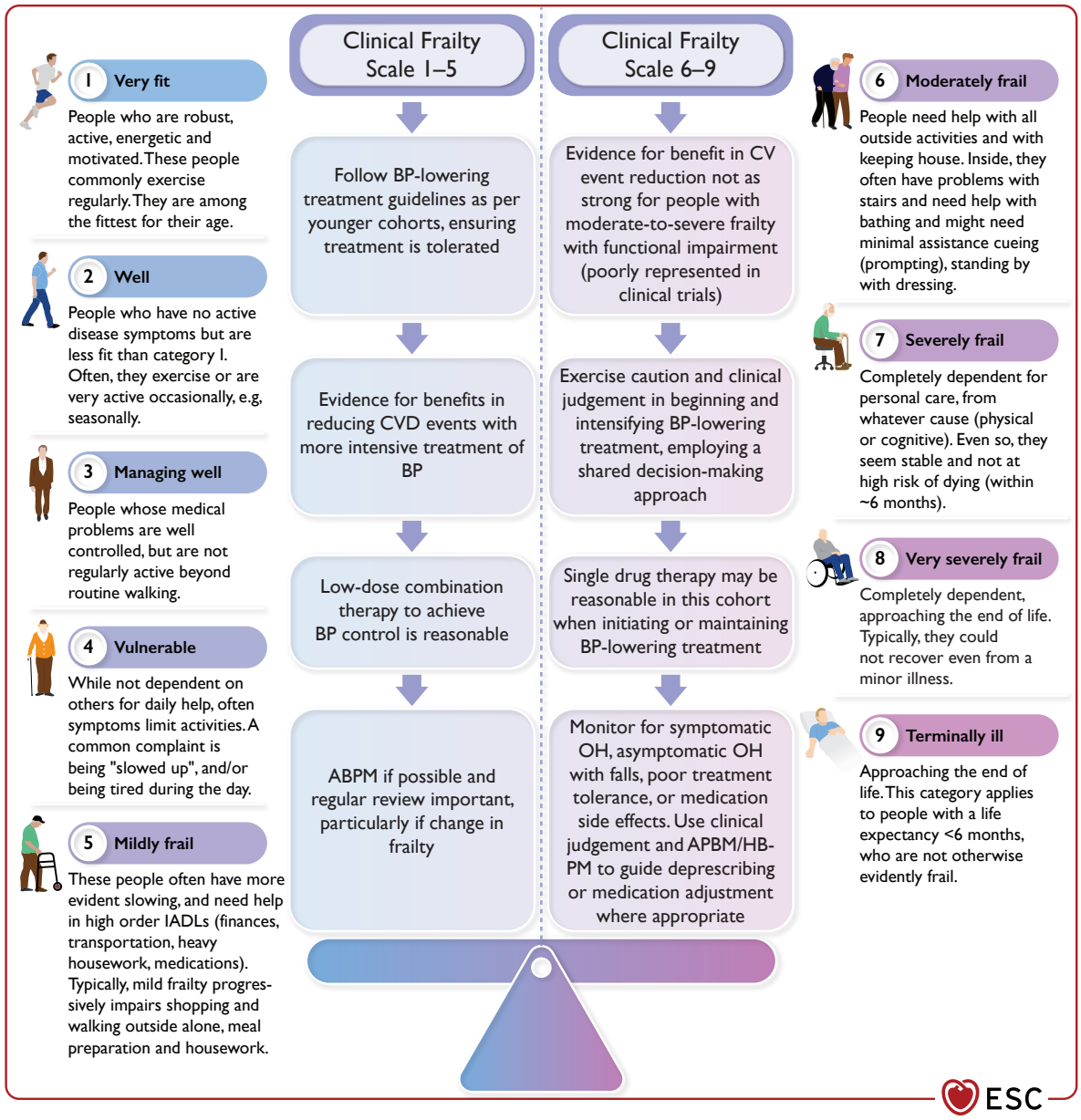


Figure 21 Frailty assessment in the management of blood pressure. ABPM, ambulatory blood pressure monitoring; BP, blood pressure; CV, cardiovascular; CVD, cardiovascular disease; HBPM, home blood pressure monitoring; IADLs, independent activities of daily living; OH, orthostatic hypotension. Adapted from Rockwood *et al.*[693]

9.3.2. Randomized controlled trials of blood pressure lowering in frail older patients

Few adults aged ≥85 years have been included in trials.[131] In addition, generalizing data from RCTs to very frail patients may not be possible.[692,694–697] However, the currently available evidence from RCTs has not demonstrated weakening of the benefits of BP-lowering treatment (i.e. no effect modification) among frailer patients enrolled in these trials, although these participants likely had no more than mild frailty (see Supplementary data online).[523,694,698]

In the absence of robust randomized evidence, several observational studies have suggested that lowering BP might not be warranted or even be harmful in patients with significant frailty or multimorbidity, particularly when BP is not very high. For instance, a systematic review and meta-analysis of non-randomized studies that investigated associations between BP and risk of mortality in older patients found evidence for interaction by frailty status, suggesting that lowering BP might be harmful in this patient group.[699] However, as noted above and in *Section 9.8*, these observational J-curve findings are unreliable when guiding clinical care, as unidentified biases potentially confound the results. For instance, in addition to reverse causality, stiffness of the large arteries is associated with both low diastolic BP and increased mortality.[697] In addition, absolute CVD risk increases with age, indicating that fewer older than younger patients with hypertension may need to be treated to prevent one adverse health outcome.[700]

Therefore, given the totality of evidence from clinical trials,[523,694,701,702] very old and frail patients with hypertension should not be denied the potential benefits of BP-lowering treatment down to a target of 120–129/70–79 mmHg. However, personalized decision-

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

making should be a priority in the very old and frail. Together with management of BP, a major consideration should also be whether reversible causes of frailty can be addressed,[609] e.g. by treating underlying comorbidities or undergoing supervised muscle-strengthening physiotherapy or supervised exercise and co-ordination and balance training.[703]

9.3.3. Starting blood pressure-lowering treatment in very old or frail patients

All patients must be fully informed about the benefits and risks of starting BP-lowering treatment, so that their preference is considered. Among 34 hypertension guidelines, 18 recommended 150 mmHg as the systolic goal in frailer and/or older patients, but four endorsed systolic targets <130 mmHg or <120 mmHg.[704] Treatment can be started with a long-acting dihydropyridine calcium channel antagonist.[596,705] To achieve BP control, an ACE inhibitor, or if contraindicated, an ARB, can also be used. Thereafter, low-dose thiazides or thiazide-like diuretics are preferred unless there is a specific contraindication, such as gout, orthostatic hypotension, or disturbed micturition (including micturition syncope).[596,705] Beta-blockers are less desirable as they reduce heart rate, cause fatigue, and increase the systolic pulse wave amplitude, which is insufficiently buffered in stiff central elastic arteries. Vasodilating beta-blockers and direct vasodilators (e.g. hydralazine and minoxidil) are associated with increased risk of orthostasis. Though data are conflicting,[706] alpha-blockers are also considered less desirable as they appear to be associated with an increased risk of orthostasis and falls in very old patients (aged ≥85 years).[707,708] Alpha-1 blockers, such as doxazosin, prazosin, terazosin (also used to treat prostate symptoms), are particularly prone to causing orthostasis.[98] Once the appropriate combination is found, a combination tablet with variable composition of two agents may optimize adherence. Starting with combination therapy is not advised in most very old and/or frail patients, unless BP is very high.

9.3.4. Maintaining blood pressure lowering in very old or frail patients

If very old and frail patients tolerate BP-lowering treatment well, there is no automatic need to deprescribe or discontinue treatment; however, this should be kept under review. In the case of progressive frailty, systolic BP tends to drop,[709] such that deprescription of a BP-lowering drug might become necessary. To identify candidate drugs for deprescribing, a patient's current medications should be reviewed to identify BP-lowering drugs that may have become contraindicated due to concomitant prescriptions or newly developed comorbidities.[705] To help guide deprescription of BP-lowering agents, ABPM can be used to detect orthostatic hypotension or a highly variable BP not buffered by autonomic nervous reflexes.[687,688]

Recommendation Table 23 — Recommendations for managing hypertension in patients who are very old or frail (see Evidence Table 41)

Recommendations	Class[a]	Level[b]
It is recommended that treatment of elevated BP and hypertension among older patients aged <85 years who are not moderately to severely frail follows the same guidelines as for younger people, provided BP-lowering treatment is well tolerated.[131,523,524]	I	A
It is recommended to maintain BP-lowering drug treatment lifelong, even beyond the age of 85 years, if well tolerated.[523–525]	I	A
Because the benefit in reducing CVD outcomes is uncertain in these settings, and noting that close monitoring of treatment tolerance is advised, BP-lowering treatment should only be considered from ≥140/90 mmHg among persons meeting the following criteria: pre-treatment symptomatic orthostatic hypotension, age ≥85 years, clinically significant moderate-to-severe frailty, and/or limited predicted lifespan (<3 years).[131,524,526,527]	IIa	B
As the safety and efficacy of BP treatment is less certain in individuals with moderate or severe frailty, clinicians should consider screening older adults for frailty using validated clinical tests; frail patients' health priorities and a shared-decision approach should be considered when deciding on BP treatments and targets.[523,524,613,710]	IIa	C
When initiating BP-lowering treatment for patients aged ≥85 years, and/or with moderate-to-severe frailty (at any age), long-acting dihydropyridine CCBs or RAS inhibitors should be considered, followed if necessary by low-dose diuretic if tolerated, but preferably not a beta-blocker (unless compelling indications exist) or an alpha-blocker.[711]	IIa	B
If BP drops with progressing frailty, deprescription of BP-lowering medications (and other drugs that can reduce BP, such as sedatives and prostate-specific alpha-blockers) may be considered.[712]	IIb	C

BP, blood pressure; CCB, calcium channel blocker; CVD, cardiovascular disease; RAS, renin–angiotensin system.
[a]Class of recommendation.
[b]Level of evidence.

9.4. Isolated systolic and diastolic hypertension

9.4.1. Definition of isolated systolic hypertension

Isolated systolic hypertension is typically defined as systolic BP of ≥140 mmHg with a diastolic BP of <90 mmHg. While isolated systolic hypertension is uncommon in younger patients,[713] it is the most common type of hypertension in older patients; >80% of untreated patients with hypertension aged >60 years have isolated systolic hypertension.[714]

9.4.2. Isolated systolic hypertension, risk factors, and ageing

Systolic BP increases with age in men and women until the eighth decade of life, while diastolic BP gradually increases up until the fifth or sixth decade of life, after which it either plateaus or decreases. As a result, the pulse pressure (the difference between the systolic and diastolic BP) gradually widens from middle age.[34] These BP changes are related to increased aortic stiffening with age.[715,716]

Since most older patients with hypertension have isolated systolic hypertension, and since with advancing age, risk of CVD events is driven by systolic rather than diastolic BP,[717] management of isolated systolic hypertension in older adults is broadly in line with that of combined systolic-diastolic hypertension seen in younger adults.[718] Early isolated

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026
© ESC 2024

systolic hypertension studies used systolic BP treatment targets of 160 or 150 mmHg.[718] However, results from the SPRINT and the Strategy of Blood Pressure Intervention in Elderly Hypertensive Patients (STEP) trials (mean BP at study entry of 140/78 mmHg and 146/82 mmHg, respectively, indicating that many of the patients had isolated systolic hypertension) confirm that lower systolic BP targets are effective in reducing CVD events in patients with isolated systolic hypertension (see Supplementary data online).[135,136]

Since relative risk reduction by BP-lowering treatment is homogeneous in any age group, whereas absolute risk reduction is larger with advancing age,[131] therapeutic inertia in older patients with isolated systolic hypertension should be avoided (see *Section 9.3*). As noted earlier, beta-blockers should be avoided in patients with isolated systolic hypertension or more generally with arterial stiffness, as they increase stroke volume (given the lower heart rate).[218]

9.4.3. Isolated systolic hypertension in young adults

In young adults (<40 years old), the presence of isolated systolic hypertension poses different pathophysiological and clinical considerations. In young patients with isolated systolic hypertension, arterial stiffness[713] and relative risk of CVD events[624] appear to be similar to those without isolated systolic hypertension and lower than young adults with combined systolic-diastolic hypertension and isolated diastolic hypertension. Indeed, younger patients with isolated systolic hypertension appear to comprise a heterogeneous group.[719] For these reasons, it might be reasonable to assess central BP and arterial stiffness in these individuals, as recommended by other scientific societies.[720,721] Out-of-office BP measurement is recommended to exclude white-coat hypertension, which is often associated with isolated systolic hypertension in the young.[634]

9.4.4. Isolated diastolic hypertension

Isolated diastolic hypertension is defined as a systolic BP of <140 mmHg with a diastolic BP of ≥90 mmHg. The isolated diastolic hypertension phenotype is more commonly seen in younger adults and, particularly, younger adults with obesity or other metabolic derangements.[722,723] In older adults with this phenotype, consideration should be given to whether the diastolic BP was accurately measured.[724]

Patients with isolated diastolic hypertension appear to have a slightly increased relative risk for CVD of 5%–30%.[548,723,725] However, because patients with isolated diastolic hypertension are younger, they tend to have few events, and very large samples are required to show this association. Furthermore, because the absolute risk for CVD among these individuals is low (typically <10% over 10 years), it is less clear if isolated diastolic hypertension should prompt initiation of BP-lowering medication, particularly among persons in whom baseline systolic BP is already at the target of 120–129 mmHg.[723] Irrespective, patients with isolated diastolic hypertension should be followed up, as they are at increased risk for systolic hypertension.[723]

Finally, it is also worth noting that when a patient achieves a target systolic BP of 120–129 mmHg with BP-lowering treatment, there is little to no high-quality trial evidence that further intensifying BP-lowering medication to achieve both systolic BP of <120 mmHg and also diastolic BP of <70 mmHg improves CVD prognosis.[547,723]

9.5. Orthostatic hypotension with supine hypertension

Patients with orthostatic hypotension need not be hypotensive and indeed, many have supine elevated BP or supine hypertension. Furthermore, many patients with orthostatic hypotension are asymptomatic. Orthostatic hypotension is present in around 10% of all adults with hypertension and is defined as a drop in BP of ≥20/10 mmHg after rising from either a sitting or lying position to a standing position (see *Section 5.5.3*).[97,99] Assessment for orthostatic hypotension should be timed to occur at 1 and/or 3 min after standing. Because seated to standing assessment can lead to under-detection of orthostatic hypotension, it is preferable, where possible, to test for orthostatic hypotension using a supine (lying) to standing assessment (see *Section 5*).[56,98,726]

Assessing for orthostatic hypotension is important in managing adults with elevated BP or hypertension for several reasons. First, findings of trials linking more intensive BP control to improved outcomes may not generalize to patients with orthostatic hypotension, particularly when it is severe in magnitude (standing systolic BP < 110 mmHg[97]) and/or symptomatic. Second, orthostatic hypotension may be associated with symptoms that may limit the patient's tolerability of more intensive BP-lowering approaches. Third, orthostatic hypotension may be associated with an increased risk of adverse effects commonly co-attributed to pharmacological BP lowering (such as hospitalizations for hypotension).[727] Fourth, orthostatic hypotension is associated with increased risk for CVD.[728]

However, the frequency of orthostatic hypotension is not increased in the more intensive BP-lowering arms of randomized trials compared with the less intensive BP-lowering arms.[726,727,729,730] As such, and in contrast to common belief, it does not appear that more intensive treatment of BP (which almost always requires more BP-lowering medication) worsens orthostatic hypotension. In contrast, there is some evidence that more intensive treatment of hypertension may actually reduce the risk of orthostatic hypotension.[730,731]

The aetiology of orthostatic hypotension may be considered as neurogenic or non-neurogenic, with the latter being far more common.[99] Patients with orthostatic hypotension may have underlying neurodegenerative diseases, diabetes, B12 deficiency,[732] renal failure, dehydration, prolonged recumbency, deconditioning, and triggering medications (like alpha-blockers, beta-blockers, diuretics, nitrates, antidepressants, and antipsychotics). Of note, ACE inhibitors, ARBs, and dihydropyridine CCBs are examples of BP-lowering medications that appear to have less impact on orthostatic hypotension, and their adverse impact, if any, on orthostatic hypotension typically occurs in the first 2 weeks or so after starting or intensifying treatment.[733]

Managing patients with supine hypertension and orthostatic hypotension is a common clinical conundrum. More detailed reviews on this topic are available elsewhere.[98,99] The approach to managing orthostatic hypotension should be non-pharmacological at first. Patients with orthostatic hypotension should be asked to change position slowly, maintain adequate hydration, and avoid alcohol and large meals. Compression stockings, crossing legs while standing, and abdominal binders may also help and should be trialled.[734,735] Abdominal heating pads and a head-up bed position can reduce supine (typically nocturnal) hypertension, which may reduce nocturnal diuresis and daytime orthostatic hypotension.[736]

The treatment of orthostatic hypotension among those with supine hypertension is not to automatically down-titrate BP-lowering medications. Rather, reversible causes should be sought and treated (including discontinuation of offending medications), and patients requiring BP-lowering medication should be switched to BP-lowering medications that are less likely to cause orthostatic hypotension. When symptoms are disabling and the above interventions do not help, particularly in neurogenic orthostatic hypotension, the best evidence exists for midodrine to reverse orthostatic hypotension, and this may be given in conjunction with ongoing BP-lowering medications when supine hypertension exists.[99] An alternative option to midodrine is droxidopa, though this is less readily available. Specialist referral is prudent when persons with supine hypertension are prescribed these orthostatic hypotension treatments, as these agents can increase supine BP more than standing BP.

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

Recommendation Table 24 — Recommendations for managing hypertension in patients with orthostatic hypotension

Recommendation	Class[a]	Level[b]
Before starting or intensifying BP-lowering medication, it is recommended to test for orthostatic hypotension, by first having the patient sit or lie for 5 min and then measuring BP 1 and/or 3 min after standing.[97,99]	I	B
It is recommended to pursue non-pharmacological approaches as the first-line treatment of orthostatic hypotension among persons with supine hypertension. For such patients, it is also recommended to switch BP-lowering medications that worsen orthostatic hypotension to an alternative BP-lowering therapy and not to simply de-intensify therapy.[726,727,729,730]	I	A

© ESC 2024

BP, blood pressure.
[a]Class of recommendation.
[b]Level of evidence.

9.6. Diabetes

9.6.1. Diabetes and elevated blood pressure/hypertension

Patients with diabetes (both type 1 and type 2) often have elevated BP or hypertension, and are about twice as likely to suffer a major CVD event over the medium to long term compared with those without diabetes.[737] Diabetes is also a major cause of microvascular events, such as retinopathy and nephropathy.[738,739] Although the risk of CVD in patients with diabetes varies by screening and diagnostic methods,[740,741] as well as with the presence of other CVD risk factors,[740,742] on average, patients with diabetes are at ≥10% 10-year risk for CVD. However, formal risk estimation with the use of SCORE2-Diabetes among type 2 diabetes mellitus patients should be considered if they are aged <60 years (see *Section 6*).[164,739]

9.6.2. J-shaped curve of blood pressure and risk of cardiovascular disease in patients with diabetes

Evidence on the BP threshold and target for treatment in patients with diabetes has been subject to debate. Reports of a J-shaped association between BP and risk of CVD in diabetes,[743] and the lack of a clear benefit of treatment on cardiac outcomes at lower BP in some meta-analyses,[744–746] has led to some cautious recommendations for intensive treatment in this patient population.

An individual patient data meta-analysis by the Blood Pressure Lowering Treatment Trialists' Collaboration tested treatment effects in 103 325 patients with diabetes and provided evidence against effect modification by categories of baseline BP down to a systolic BP of 120 mmHg.[445] BP reduction in patients with diabetes is expected to reduce the risk of diabetes-associated complications including retinopathy, vasculopathy, and nephropathy (albuminuria and end-stage renal disease), which adds weight to the importance of reducing BP in these patients (see Supplementary data online).[745–747] This task force also considered the fact that a proportion of patients with diabetes have orthostatic hypotension due to diabetic neuropathy,[748] which might affect the tolerability of BP lowering.

9.6.3. Managing blood pressure in diabetes

We recommend that all patients with diabetes are offered pharmacological BP-lowering treatment with a BP target of 120–129/70–79 mmHg, if feasible and tolerated.[136,146,445,747,749–752] The task force further sees no strong evidence for a differential BP treatment targets in patients with diabetes and those without.[136,146,445,746,747] While the Action to Control Cardiovascular Risk in Diabetes (ACCORD) trial reported a null primary endpoint for more intensive BP targets in diabetes, stroke was marginally reduced.[137] Furthermore, extended follow-up of ACCORD,[750] as well as analyses of intensive BP reduction in those randomized to the standard glycaemic arm,[749] provide evidence suggesting benefit consistent with the SPRINT, STEP, and Effects of intensive Systolic blood Pressure lowering treatment in reducing RIsk of vascular evenTs (ESPRIT) trials.[136,146,545] Overall, all major BP-lowering medication classes are effective in preventing CVD in people with or without diabetes. Of note, however, albuminuria is more common in diabetes and, for this reason, ACE inhibitors and ARBs have potential advantages that may warrant consideration for BP-lowering in patients with diabetes (see Supplementary data online, *Table S10*).[753]

Evidence for modifying BP-lowering treatment in patients with pre-diabetes is somewhat limited. Furthermore, the relative effect of BP lowering is relatively consistent across different categories of BMI as a measure of obesity.[754,755] It is also noteworthy that elevated BP itself may increase the risk of diabetes,[755] emphasizing the potential role of BP lowering in preventing diabetes in addition to preventing CVD. Among the major classes of BP-lowering drugs, ACE inhibitors and ARBs are effective in preventing new-onset diabetes and can be considered in patients at risk of diabetes and who are indicated for BP-lowering therapy.[164,755]

Recommendation Table 25 — Recommendations for managing hypertension in patients with diabetes

Recommendations	Class[a]	Level[b]
In most adults with elevated BP and diabetes, after a maximum of 3 months of lifestyle intervention, BP lowering with pharmacological treatment is recommended for those with confirmed office BP ≥130/80 mmHg to reduce CVD risk.[445,749]	I	A
BP-lowering drug treatment is recommended for people with pre-diabetes or obesity when confirmed office BP is ≥140/90 mmHg or when office BP is 130–139/80–89 mmHg and the patient is at predicted 10-year risk of CVD ≥10% or with high-risk conditions, despite a maximum of 3 months of lifestyle therapy.[445]	I	A
In persons with diabetes who are receiving BP-lowering drugs, it is recommended to target systolic BP to 120–129 mmHg, if tolerated.[136,146,445,747,749–752]	I	A

© ESC 2024

BP, blood pressure; CVD, cardiovascular disease.
[a]Class of recommendation.
[b]Level of evidence.

9.7. Chronic kidney disease

9.7.1. Relationship between hypertension and chronic kidney disease

Approximately 850 million people worldwide have CKD, with >80% of them hypertensive, and the prevalence is expected to rise to 1.56 billion

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

by 2025.[756–759] The pathogeneses of hypertension and CKD are closely entwined.[760] Resistant hypertension, masked hypertension, HMOD, and higher night-time BP are common in patients with CKD.[761] CVD is one of the largest contributors to mortality in patients with CKD, with hypertension being a major risk factor.[760,762]

For the purposes of these guidelines, adults with moderate-to-severe CKD and elevated BP are at sufficiently high risk to be considered for BP-lowering drug therapy as outlined in *Section 8* and the Central Illustration (*Figure 19*). We use Kidney Disease: Improving Global Outcomes (KDIGO) categories to define CKD-based risk, and our definition of moderate-to-severe CKD comprises persons with an eGFR of <60 mL/min/1.73 m^2 and/or albuminuria of ≥30 mg/g (≥3 mg/mmol).[19] For persons with mild CKD and elevated BP, a CVD risk assessment should be conducted before deciding on BP-lowering treatment.

9.7.2. Blood pressure lowering in chronic kidney disease

BP lowering in patients with CKD is associated with beneficial effect on CVD events and mortality.[275,763–766] BP lowering reduces progression of CKD and the incidence of end-stage renal disease, but this tends to be only in those with significant proteinuria at baseline.[766,767]

9.7.3. Managing blood pressure in chronic kidney disease

Patients with CKD should receive lifestyle advice, especially regarding reducing sodium intake. Dietary potassium supplementation recommendations are provided in *Section 8*, with caution required among persons with moderate-to-severe CKD. While exercise appears to have little effect on improving BP in patients with CKD[768] or patients on dialysis,[769] those with CKD on ACE inhibitor monotherapy have protection against adverse kidney outcomes, CVD events, cardiovascular death, and all-cause death.[770,771] Both ACE inhibitors and ARBs reduce the risk of CVD events and kidney failure compared with placebo; however, ACE inhibitors appear to do so with higher probability than ARBs.[772,773] Patients with CKD usually require combination therapy, and this should be initiated as a combination of a RAS inhibitor and a CCB or diuretic. In patients with eGFR < 30 mL/min/1.73 m^2, an adequately up-titrated loop diuretic is necessary to define resistant hypertension. Chlorthalidone, typically added to a loop diuretic, also effectively lowers BP and reduces microalbuminuria in patients with resistant hypertension with stage 4 CKD (eGFR of 15–30 mL/min/1.73 m^2).[774] The combination of an ACE inhibitor and an ARB is not recommended in CKD or any other BP-treatment scenario.

9.7.4. Blood pressure targets in chronic kidney disease

Evidence regarding BP targets in patients with CKD is complex and controversial. The 2021 KDIGO Guideline suggested that adults with elevated BP and CKD be treated to a target systolic BP of <120 mmHg, when tolerated, using standardized office BP measurement (Class of Recommendation IIb).[19] This suggestion was based, in part, on the SPRINT trial.[545] It should be noted that patients with 24 h urine protein excretion ≥ 1 g/day or eGFR < 20 mL/min/1.73 m^2 were excluded from SPRINT. In patients with CKD, after a median follow-up of 3.3 years, the hazard ratio for the primary composite cardiovascular outcome was 0.81 (95% CI 0.63–1.05) and for all-cause death it was 0.72 (95% CI 0.53–0.99). Although intensive BP lowering in SPRINT resulted in greater early decline in eGFR, there was no evidence that this reduction in eGFR attenuated the beneficial effects of the SPRINT intervention on CVD events or death.[775]

Several systematic reviews have examined the benefit of intensive BP control in patients with CKD (see Supplementary data online). Some have shown no benefit of intensive BP control on renal outcomes,[764,767] while others showed lower mortality in intensively treated vs. non-intensively treated patients.[275] Highlighting the beneficial effects of SGLT2 inhibitors in persons with CKD[776,777] and finerenone in persons with CKD and diabetes[460–462,778] is relevant, though these drugs are not currently marketed for BP-lowering effects alone.

Recommendation Table 26 — Recommendations for managing hypertension in patients with chronic kidney disease

Recommendations	Class[a]	Level[b]
In patients with diabetic or non-diabetic moderate-to-severe CKD and confirmed BP ≥130/80 mmHg, lifestyle optimization and BP-lowering medication are recommended to reduce CVD risk, provided such treatment is well tolerated.[275,766]	I	A
In adults with moderate-to-severe CKD who are receiving BP-lowering drugs and who have eGFR >30 mL/min/1.73 m^2, it is recommended to target systolic BP to 120–129 mmHg, if tolerated. Individualized BP targets are recommended for those with lower eGFR or renal transplantation.[274,779]	I	A
In hypertensive patients with CKD and eGFR >20 mL/min/1.73 m^2, SGLT2 inhibitors are recommended to improve outcomes in the context of their modest BP-lowering properties.[776,777]	I	A
ACE inhibitors or ARBs are more effective at reducing albuminuria than other BP-lowering agents and should be considered as part of the treatment strategy for patients with hypertension and microalbuminuria or proteinuria.[780–782]	IIa	B

ACE, angiotensin-converting enzyme; ARB, angiotensin receptor blocker; BP, blood pressure; CKD, chronic kidney disease, eGFR, estimated glomerular filtration rate; SGLT2, sodium–glucose co-transporter 2.
[a]Class of recommendation.
[b]Level of evidence.

9.8. Cardiac disease

9.8.1. Blood pressure thresholds and targets in patients with cardiac disease

Recommended BP thresholds for initiating BP-lowering therapy and recommended BP targets in patients receiving therapy are provided in *Sections 6* and *8*. All patients with a history of CVD (including coronary artery disease) are at increased risk of recurrent CVD. As such, these patients are recommended to be treated with BP-lowering therapy for confirmed baseline BP of ≥130/80 mmHg and the recommended treatment target BP of 120–129/70–79 mmHg, provided treatment is tolerated (see *Sections 6* and *8*). As stated in *Section 8*, it should be remembered that a systolic BP of 120 mmHg (especially by out-of-office assessment) is likely the optimal point in the target range recommended in these guidelines. In addition to considering patients with known CVD at sufficiently high risk for more intensive BP treatment targets, the

© ESC 2024

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

task force considers patients with both severe valvular heart disease and symptomatic heart failure to also be at sufficiently high risk. We also note that, whether used for angina control or BP control, a beta-blocker should not be combined with a non-dihydropyridine CCB.

9.8.2. Coronary artery disease with particular reference to the blood pressure J-curve

Important considerations in patients with coronary artery disease are: (i) the BP J-curve (an observation suggesting that over-intensive BP lowering may increase CVD risk in some patients), and (ii) compelling indications for specific classes of BP-lowering medications.

The J-curve phenomenon describes increased risk for CVD observed among patients with the lowest and highest BP in the dataset, with the best CVD outcome rates typically observed among those with BP in the normal range (e.g. systolic BP of 100–120 mmHg and diastolic BP of 60–80 mmHg). For this reason, the J-curve is sometimes also called the U-curve, with both terms typically used interchangeably.[697,783,784]

However, observational data do not consistently demonstrate a BP J-curve with CVD risk.[697] It is more commonly observed among patients with established clinical CVD, such as those with coronary artery disease, or in secondary prevention cohorts.[785,786] Furthermore, the J-curve is more commonly observed when analysing diastolic BP values vs. systolic BP values, though it has been described for both.[114,697] This stronger relationship with diastolic BP has informed the hypothesis that the J-curve may be caused by reduced perfusion of major organs at low BP, which is particularly operative for diastolic BP in the coronary vasculature when considering ischaemic heart disease events (since coronary blood flow is largely confined to diastole).[786,787]

If excessive lowering of BP causes CVD events, this needs to be addressed in treatment recommendations provided by BP management guidelines. The 2018 ESC/ESH Guidelines on the management of arterial hypertension introduced, for the first time, lower bounds of BP-lowering treatment targets, implying that treatment be de-intensified for patients with on-treatment BP below these bounds (i.e. <120 mmHg systolic or <70 mmHg diastolic).[1] The 2023 ESH document also makes this argument.[788]

However, since 2018, compelling evidence has demonstrated that the BP J-curve evident in observational datasets is highly unlikely to reflect a causal process and can instead be attributed to residual confounding and/or reverse causation.[33,114,115,546,697,789–793]

Accordingly, while low BP can indicate a high-risk state, particularly among older adults and those with comorbidities, there is no evidence that this is a causal phenomenon. Indeed, if there is another indication for BP-lowering therapy (e.g. in a patient with wide pulse pressure and a baseline systolic BP of >140 mmHg but diastolic BP of <60 mmHg), the evidence suggests that such therapy should be provided if tolerated to reduce CVD risk.

There is a limit to how low BP can be treated without potentially tipping the scales in favour of CVD harm vs. CVD benefit. However, it is not clear what that limit is and how much it differs based on comorbidities. Currently, the data do not suggest that risk for CVD can be causally increased by treating any patient to the recommended intensive BP target outlined in these guidelines of as low as 120/70 mmHg. We also do not recommend stopping or de-intensifying BP-lowering medication among asymptomatic patients with on-treatment BP of <120/70 mmHg. It should be recognized, though, that there are no robust data demonstrating that an on-treatment systolic BP of <90 mmHg or an on-treatment diastolic BP of <50 mmHg is safe from a CVD perspective and there is clear potential for harm. Furthermore, it must be emphasized that the above discussion of the BP J-curve relates solely to CVD risk and does not consider the known non-CVD side effects of BP-lowering drugs, like, e.g. orthostatic hypotension, syncope, and renal injury. We do know that patients treated to a more intensive BP target of 120/70 mmHg are at increased risk for these side effects,[550] which is why these guidelines stress that this more intensive target should only be pursued among those in whom treatment is being tolerated (*Section 8*).

The second consideration in managing BP in patients with coronary artery disease is the recommended use of BP-lowering medications with compelling indications based on outcomes trials that demonstrated CVD outcomes benefits in the setting of coronary artery disease. These recommendations are provided in the recommendation table below.

9.8.3. Valvular heart disease

Most patients with both severe heart valve disease and heart failure, defined by the 2021 ESC/European Association for Cardio-Thoracic Surgery (EACTS) Guidelines for the management of valvular heart disease,[794] can be considered at increased risk of CVD. Accordingly, it is recommended they are treated with BP-lowering therapy for confirmed baseline BP of >130/80 mmHg, and their recommended target of treatment is BP of 120–129/70–79 mmHg, provided treatment is tolerated. Persons with mild-to-moderate heart valve disease should have a CVD risk assessment prior to deciding their BP-lowering treatment threshold and target.

Vasodilating ARBs/ACE inhibitors are preferable over vasodilating dihydropyridine CCBs because of the link between valvular heart disease and subsequent heart failure and given the stronger efficacy evidence for ARBs/ACE inhibitors in the setting of heart failure once manifested.[795] In aortic valve stenosis, concomitant hypertension influences both the aortic root, the aortic valve, and the left ventricular structure and function.[796] In this subgroup, treatment preferably with ARBs/ACE inhibitors should be considered. A beta-blocker may be added if BP remains >140/90 mmHg.[796,797]

9.8.4. Heart failure

Patients with symptomatic heart failure are at increased risk of CVD. Therefore, it is recommended that these patients are treated with BP-lowering therapy for confirmed baseline BP of >130/80 mmHg and their recommended treatment target is BP of 120–129/70–79 mmHg, provided treatment is tolerated and with out-of-office confirmation of on-treatment BP. Of note, many patients with systolic heart failure on maximal heart failure therapies have BP of <120/70 mmHg, and we do not recommend de-intensifying such treatment unless indicated by symptomatic side effects. Besides referencing the new evidence for ARNi and SGLT2 inhibitor therapies,[795] our 2024 recommendations for heart failure are largely unchanged from the 2018 ESC/ESH Guidelines on the management of arterial hypertension. Non-dihydropyridine CCBs should not be used in heart failure. Frailty and hypotension risk should be assessed in older heart failure patients being considered for ARNi and SGLT2 inhibitor therapies, and older patients should be closely followed to ensure they are tolerating such treatments. For more information on the management of heart failure, we direct readers to the latest ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure.[798]

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

Recommendation Table 27 — Recommendations for managing hypertension in patients with cardiac disease

Recommendations	Class[a]	Level[b]
In patients with a history of myocardial infarction who require BP-lowering treatment, beta-blockers and RAS blockers are recommended as part of that treatment.[538]	I	A
In patients with symptomatic angina who require BP-lowering treatment, beta-blockers and/or CCBs are recommended as part of that treatment.[538]	I	A
In patients with symptomatic HFrEF/HFmrEF, the following treatments with BP-lowering effects are recommended to improve outcomes: ACE inhibitors (or ARBs if ACE inhibitors are not tolerated) or ARNi, beta-blockers, MRAs, and SGLT2 inhibitors.[795]	I	A
In hypertensive patients with symptomatic HFpEF, SGLT2 inhibitors are recommended to improve outcomes in addition to their modest BP-lowering properties.[795]	I	A
In patients with a history of aortic valve stenosis and/or regurgitation who require BP-lowering treatment, RAS blockers should be considered as part of that treatment.[794,796]	IIa	C
In patients with a history of moderate-to-severe mitral valve regurgitation who require BP-lowering treatment, RAS blockers should be considered as part of that treatment.[794]	IIa	C
In patients with symptomatic HFpEF who have BP above target, ARBs and/or MRAs may be considered to reduce heart failure hospitalizations and reduce BP.[795,799,800]	IIb	B

© ESC 2024

ACE, angiotensin-converting enzyme; ARB, angiotensin receptor blocker; ARNi, angiotensin receptor–neprilysin inhibitor; BP, blood pressure; CCB, calcium channel blocker; HFpEF, heart failure with preserved ejection fraction; HF(m)rEF, heart failure with (mildly) reduced ejection fraction; MRA, mineralocorticoid receptor antagonist; RAS, renin–angiotensin system; SGLT2, sodium–glucose co-transporter 2.
[a]Class of recommendation.
[b]Level of evidence.

9.8.5. Heart rhythm disease (including AF)

Not all patients with heart rhythm disease, including those with AF, are at increased risk for CVD and, as such, the management of BP among patients with heart rhythm disease should be the same as for the general adult population.[443] However, there is a close relationship between increased BP and AF risk, hence, ensuring good BP control is important.[801,802] Management of heart rhythm disorders should follow recommendations in guidelines specific to these conditions.[803]

9.9. Chronic cerebrovascular disease and/or cognitive impairment

9.9.1. Role of hypertension in chronic cerebrovascular disease

Hypertension is a risk factor for chronic cerebrovascular disease through its direct effects on brain structure and microvasculature. This manifests as transient ischaemic attack (TIA) and stroke in the acute setting, but chronic hypertension can lead to covert stroke and white-matter ischaemic change over time, resulting in cognitive decline and progressive vascular dementia.[804–808] Hypertension is also associated with increased risk of Alzheimer's disease,[809] and is a risk factor for developing AF, heart failure, and CKD, all of which are associated with increased risk of developing cognitive impairment and dementia.[810–813] For the purposes of these guidelines, adults with a history of stroke or TIA and elevated BP are considered at sufficiently high risk to be considered for BP-lowering drug therapy as outlined in *Section 8* and the Central Illustration (*Figure 19*).

9.9.2. Treatment in patients with history of prior stroke or transient ischaemic attack

In patients with a prior history of TIA and ischaemic stroke, BP-lowering treatment reduces the risk of any recurrent stroke by 20%.[814–817] The aetiology of stroke can affect the degree of risk reduction with pharmacological treatment, with greater reductions observed for intracerebral haemorrhage and lacunar ischaemic stroke syndromes.[818–820]

Most prior guidelines recommend an intensive BP target in patients with a prior history of stroke, typically using combination treatment (ACE inhibitor/ARB plus either a calcium channel antagonist or a thiazide/thiazide-like diuretic), with therapy commencing immediately after TIA and within a few days of ischaemic stroke (see Supplementary data online and see *Section 10.3* for acute BP management during hospitalization for stroke).[814,815,821–824]

Regimens containing an ACE inhibitor and thiazide/thiazide-like diuretic may be superior to beta-blockers in terms of stroke risk reduction.[825,826] Regarding intensive BP control after stroke, typically targeting a systolic BP of <130 mmHg, individual trials were somewhat inconclusive, but a meta-analysis showed a reduced risk of recurrent stroke of 22% in the intensive treatment group randomized to a target systolic BP as low as 120 mmHg.[543,824,827,828] Caveats to this recommendation would be for frail patients, who have a much higher rate of stroke and recurrent stroke than the general population, and who are more sensitive to adverse effects of BP-lowering agents (see *Section 9.3*).[596,606,607,829]

9.9.3. Treatment in patients with chronic cerebrovascular disease and cognitive impairment

Treatment of hypertension represents a key mechanism for reducing the global burden of dementia at the population level.[830] Epidemiological studies have reported associations between mid-life hypertension and development of cognitive decline in later life, with, e.g. mid-life hypertension increasing relative risk of lifetime dementia by 20%–54%.[831–837] In one observational meta-analysis, an increased risk for dementia emerged with systolic BP of >130 mmHg.[831]

Evidence for lowering BP to reduce the risk of dementia is limited due to heterogeneity in populations studied, cognitive testing methods used, and the varied use of dementia or cognitive impairment or both as a primary outcome.[838,839] Findings from individual studies have mixed results (see Supplementary data online).[264,839–843] Studies on effects of BP-lowering treatment on white-matter intensities concluded that patients in the intensive-control arm had less white-matter intensity accumulation than in the standard-treatment arm.[841,844] Studies in which people with stroke and TIA were included reported a reduced risk of dementia and cognitive decline for the active-treatment group, but a mixed signal for dementia alone.[841,845] However, individual studies

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

may have been under-powered and more recent meta-analyses do convincingly support efficacy in reducing dementia with BP-lowering treatment.[610,611] Indeed, these meta-analyses reported a reduced risk of incident dementia or cognitive impairment with BP lowering of 7%–13%.[610,611] While one trial suggested superiority of long-acting CCBs,[264] it is unclear if any first-line BP-lowering agent is preferable for preventing dementia and cognitive impairment.[846,847]

The role of competing risk mechanisms including orthostatic hypotension[848] and BP variability[849] may be important factors in treatment decisions for people with frailty, multimorbidity, and/or chronic cerebrovascular disease.

Recommendation Table 28 — Recommendations for managing hypertension in patients with chronic cerebrovascular disease and cognitive impairment

Recommendations	Class[a]	Level[b]
It is recommended that the BP-lowering drug treatment strategy for preventing recurrent stroke should comprise a RAS blocker plus a CCB or a thiazide-like diuretic.[820,823,825,826]	I	A
In patients with confirmed BP ≥130/80 mmHg with a history of TIA or stroke a systolic BP target of 120–129 mmHg is recommended to reduce CVD outcomes, provided treatment is tolerated.[824,827,828]	I	A

© ESC 2024

BP, blood pressure; CCB, calcium channel blocker; CVD, cardiovascular disease; RAS, renin–angiotensin system; TIA, transient ischaemic attack.
[a]Class of recommendation.
[b]Level of evidence.

9.10. Aortopathy

9.10.1. Coarctation of the aorta

Aortic coarctation is associated with CVD in the long term, even following early surgical or percutaneous treatment. The most common complications are associated with hypertension, which is common in aortic coarctation. When aortic coarctation is not treated, patients often develop severe hypertension and HMOD (especially LVH and left ventricular dysfunction, aortopathy, and cerebrovascular complications).[850–852] No formal RCTs to define optimal medical treatment of hypertension in aortic coarctation have been conducted, therefore, patients not suitable for, or having undergone, intervention should be treated for hypertension following the core algorithm for the general population.

9.10.2. Bicuspid aortic valve-related aortopathy

Bicuspid aortic valve is the most common congenital heart disease and is sometimes associated with aortopathy or aortic coarctation. Bicuspid aortic valve disease is associated with an increased risk of valve malfunction and adverse aortic events.[853,854] This risk is exacerbated by hypertension.

Beyond aortic dilation and aneurysm formation, bicuspid aortic valve disease is also a risk factor for aortic dissection and rupture.[855] Blood pressure should be carefully monitored and controlled.[856]

9.10.3. Preventing aortic dilation and dissection in high-risk patients

A modest dilatation of the ascending aorta or aortic root is often associated with chronic hypertension and HMOD. An additional cause of aortopathy (bicuspid valve, coarctation, Marfan or other syndromes) should be considered in more severe cases.[857] Patients with aortic dilatation should have their BP optimally controlled following the core algorithm for the general hypertension population.

In patients with Marfan syndrome, prophylactic use of ARBs, ACE inhibitors, or beta-blockers may reduce complications or progression of aortic dilation.[857–860] More information is available in the 2024 ESC Guidelines for the management of peripheral arterial and aortic diseases.[861]

9.11. Different ethnic groups

Influx and settlement of migrant populations in Europe have contributed to regional population growth and changes in its composition.[862] Ethnic minority populations are disproportionally affected by hypertension and hypertension-mediated complications, compared with historically native Europeans, with data suggesting migrant women are particularly vulnerable.[536,863] In particular, hypertension is more prevalent in those of African descent.[863,864] The predominant group of European black ethnicity originates from sub-Saharan Africa,[863] but specific studies on the management and control of hypertension in this population are lacking, and data are often extrapolated from studies in the African American population.[864] This assumption requires caution, as differences likely exist between these populations in terms of CVD risk, economic, and sociological status,[865,866] as well as responses to BP-lowering drugs.[867]

Black patients have a greater prevalence of low-renin, salt-sensitive hypertension and may be more predisposed to HMOD than white patients, possibly in part due to increased vascular stiffness.[864,868,869] Salt restriction, thiazide or thiazide-like diuretics, and CCBs appear particularly useful in black patients with hypertension, whereas RAS blocker monotherapy may be less effective.[870–873] If combination therapy is needed, in a recent RCT conducted in sub-Saharan African countries, amlodipine plus either hydrochlorothiazide or perindopril proved to be equally effective and superior, respectively, to hydrochlorothiazide plus perindopril.[874] When RAS blockers are used in combination therapy, ARBs may be preferable to ACE inhibitors, as angioedema appears more common with ACE inhibitors in black patients.

Despite some recent progress,[875] data on hypertension epidemiology and management in European immigrant patients are still lacking.[863,875–877]

Recommendation Table 29 — Recommendations for managing hypertension in different ethnic groups

Recommendation	Class[a]	Level[b]
In black patients from Sub-Saharan Africa who require BP-lowering treatment, combination therapy including a CCB combined with either a thiazide diuretic or a RAS blocker should be considered.[874]	IIa	B

© ESC 2024

BP, blood pressure; CCB, calcium channel blocker; RAS, renin–angiotensin system.
[a]Class of recommendation.
[b]Level of evidence.

9.12. Nocturnal hypertension

9.12.1. Definition

Nocturnal hypertension is defined as night-time BP of >120 mmHg systolic and/or >70 mmHg diastolic by 24 h ABPM. Nocturnal hypertension can occur as day–night sustained hypertension or isolated nocturnal hypertension (daytime BP < 135/85 mmHg on 24 h ABPM). Physiologically, BP is

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

expected to decrease during sleep by 10%–20% relative to daytime BP.[878] Night-time dipping patterns are classified into four groups:[879,880]

- **Inverse dipping (riser):** nocturnal increase in BP (night-to-day ratio of >1.0).
- **Non-dipper:** reduced night-time BP dip of <10% (or night-to-day ratio of >0.9 and ≤1.0).
- **Normal dipping:** fall in night-time BP of >10% and <20% (or night-to-day ratio of 0.8 to 0.9).
- **Extreme dipping:** marked fall in night-time BP of >20% (or night-to-day ratio of <0.8).

Patients with nocturnal hypertension may be dippers or non-dippers. Of note, the long-term reproducibility of dipping patterns appears to be low.[881,882]

9.12.2. Epidemiology

Nocturnal hypertension has been observed in up to half of patients with hypertension,[883–886] and is associated with increased HMOD,[883] impaired renal function, and diabetes mellitus.[887] Nocturnal hypertension appears to be more prevalent in black[888–890] and Asian[891,892] populations. Masked uncontrolled hypertension, which occurs in 30% of patients treated for hypertension, is more often due to poorly controlled nocturnal BP than daytime BP on ABPM.[893]

Environmental factors, including sleep duration and higher humidity,[894] nocturia,[895] OSAS,[896] obesity, high salt intake in salt-sensitive patients,[897] orthostatic hypotension, autonomic dysfunction, CKD,[898–900] diabetic neuropathy/diabetes,[901] and old age[62] are associated with non-dipping. Moreover, nocturnal hypertension and absent night-time dipping pattern are more common in secondary hypertension.[902,903]

9.12.3. Night-time blood pressure as a cardiovascular disease risk factor

Nocturnal hypertension is a risk factor for adverse CVD events,[904] cerebrovascular disease, including stroke,[905] and cardiovascular mortality.[891,906,907] Night-time BP may provide more prognostic information than daytime BP, perhaps as it is less dependent on physical activities. Non-dipping[908–910] and reverse dipping (nocturnal rise in BP) may also be associated with increased CVD risk.[62,910–913] A nocturnal rise in BP is associated with an increased risk of dementia and Alzheimer's disease in older men.[914] There is also some evidence that extreme dipping, particularly in untreated patients, is associated with an increased risk for CVD events.[35,886]

9.12.4. Treatment of nocturnal hypertension

There is no reliable evidence that BP-lowering medication should be routinely dosed at bedtime. The diurnal timing of drug administration is discussed in *Section 8.3.4*. In patients with secondary hypertension, the underlying cause (OSAS, primary aldosteronism) should be treated as discussed in *Section 9.14*.

9.13. Resistant hypertension

9.13.1. Definition of resistant hypertension

Resistant hypertension is defined as BP remaining above goal despite three or more BP-lowering drugs of different classes at maximally tolerated doses, of which one is a diuretic (*Table 11*).[915] Resistant hypertension should be managed at specialized centres with the expertise and resources to exclude pseudo-resistant hypertension (adherence testing) and causes of secondary hypertension.[916]

9.13.2. Non-pharmacological interventions

The Treating Resistant Hypertension Using Lifestyle Modification to Promote Health (TRIUMPH) trial demonstrated significant clinic and ambulatory BP reductions in patients with resistant hypertension participating in a 4-month lifestyle intervention comprising diet and exercise interventions delivered within a cardiac rehabilitation programme.[917]

9.13.3. Pharmacological interventions

BP-lowering treatment of resistant hypertension with single-pill combinations is recommended to reduce the pill burden, thereby increasing drug adherence and persistence.[492]

As resistant hypertension often, and especially in CKD,[918] represents a state of salt retention and volume expansion secondary to relative aldosterone excess,[516,919,920] BP control may be improved by switching hydrochlorothiazide to long-acting thiazide-like diuretics, such as chlorthalidone.[921,922] However, a recent trial of chlorthalidone vs. hydrochlorothiazide—which probably included a sizeable proportion of adults with resistant hypertension—did not demonstrate any difference in systolic BP or CVD outcomes between the two medications. In the subgroup of patients with prior CVD, there was a strong trend of benefit with chlorthalidone on CVD outcomes.[447] Of note, the risk of hypokalaemia was higher in the chlorthalidone group than in the hydrochlorothiazide group.[447] In patients with eGFR < 30 mL/min/1.73 m^2, an adequately up-titrated loop diuretic is necessary to define resistant hypertension.

Most patients with resistant hypertension require the addition of non-first-line BP-lowering drugs (*Figure 22*). Of these, low-dose spironolactone (25–50 mg daily) should be considered first.[459,515,923–925] In patients with resistant hypertension and type 2 diabetes, spironolactone (25–50 mg daily) reduced BP and albuminuria.[926] The use of spironolactone can be precluded by limited tolerability due to anti-androgenic side effects resulting in breast tenderness or gynaecomastia (in about 6%), impotence in men, and menstrual irregularities in women.[927] The efficacy and safety of spironolactone for treating resistant hypertension have not yet been established in patients with significant renal impairment. Moreover, spironolactone, especially in addition to RAS inhibitors, increases the risk of hyperkalaemia.[927,928] Therefore, spironolactone should be restricted to patients with an eGFR of ≥30 mL/min/1.73 m^2 and a plasma potassium concentration of ≤4.5 mmol/L.[459] Steroidal MRAs are contraindicated in patients with an eGFR of <30 mL/min/1.73 m^2. Serum electrolytes and kidney function should be monitored soon after initiation and frequently thereafter. In patients with resistant hypertension and CKD (eGFR of 25–45 mL/min/1.73 m^2), the oral potassium binder patiromer enabled more patients to continue treatment with spironolactone.[929]

If spironolactone is not tolerated due to anti-androgen side effects, eplerenone may be used. If eplerenone is used, higher doses (i.e. 50–200 mg daily) and twice-daily dosing may be necessary to achieve a BP-lowering effect.[503] Of note, eplerenone is not licensed for hypertension treatment in many countries.

When not already prescribed for a compelling indication, beta-blockers should be considered in the treatment of resistant hypertension, though their BP-lowering effects appear to be less potent than spironolactone in the setting of resistant hypertension.[459]

Amiloride and clonidine have data suggesting they are as effective as spironolactone for BP lowering, though they lack outcomes data. A non-exhaustive list of additional medications sometimes used for BP-lowering purposes includes other centrally acting BP-lowering medications (e.g. methyldopa), hydralazine, aliskiren, minoxidil, triamterene, and loop diuretics (*Figure 22*).[515,516] As noted earlier, minoxidil use is often limited by side effects.

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

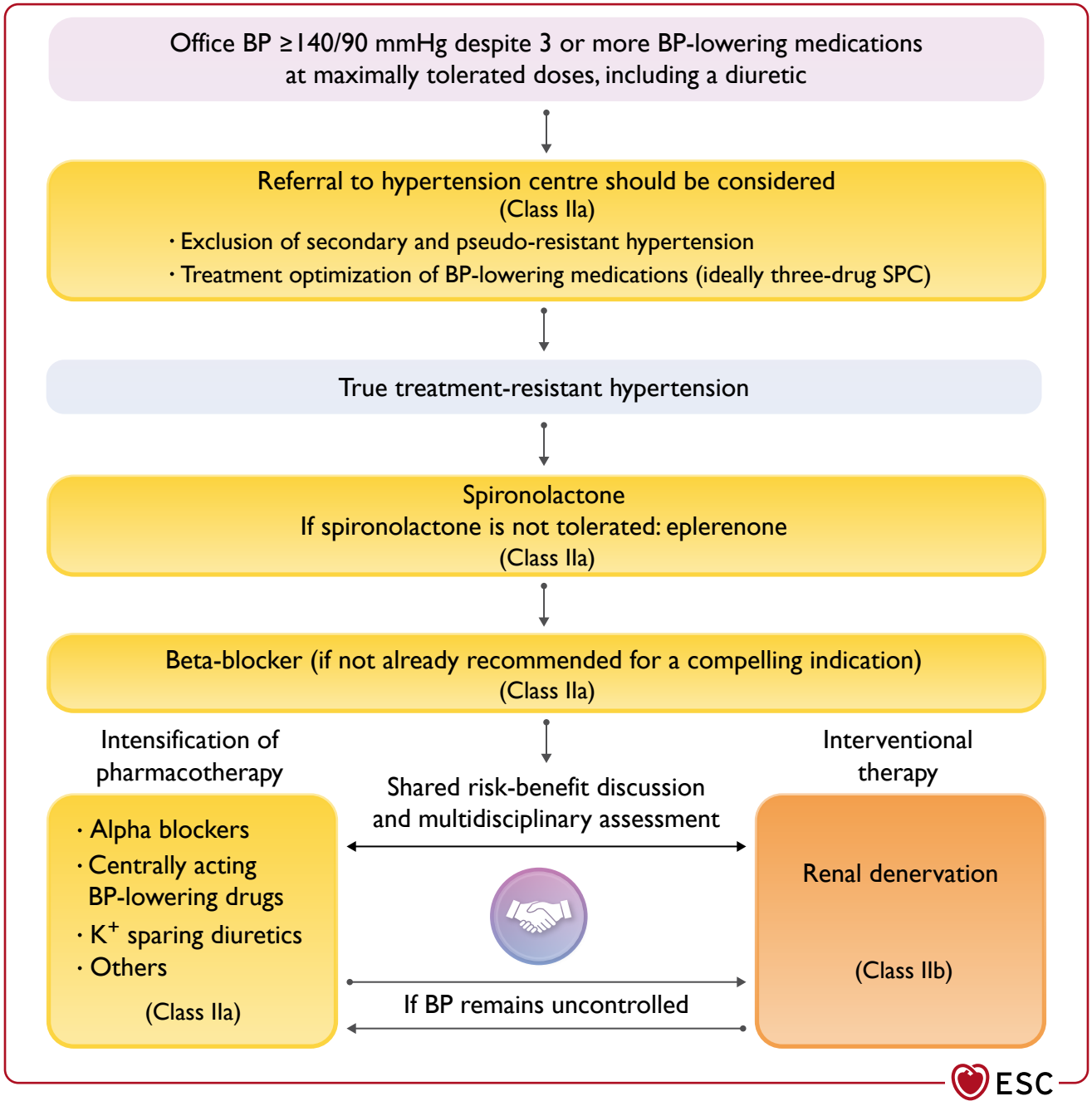


Figure 22 Management of resistant hypertension. BP, blood pressure; K^+, potassium; SPC, single-pill combination.

9.13.4. Devices for blood pressure lowering

Several devices have been investigated for treating resistant hypertension. Of these, the most evidence is available for catheter-based renal denervation. As discussed in *Section 8.6.1*, several randomized, sham-controlled trials have been published, demonstrating a BP-lowering efficacy over 24 h for radiofrequency and ultrasound renal denervation in a broad spectrum of hypertension, including resistant hypertension.[568,585] Other devices are still under investigation and are not recommended for routine use in clinical practice (*Section 8.6.2*).

Recommendation Table 30 — Recommendations for treating resistant hypertension (see Evidence Tables 42 and 43)

Recommendations	Class[a]	Level[b]
In patients with resistant hypertension and uncontrolled BP despite use of first-line BP lowering therapies, the addition of spironolactone to existing treatment should be considered.[459,515]	IIa	B

Continued

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

In patients with resistant hypertension in whom spironolactone is not effective or tolerated, treatment with eplerenone instead of spironolactone,[503] or the addition of a beta-blocker if not already indicated[459] and, next, a centrally acting BP-lowering medication,[515] an alpha-blocker,[515] hydralazine, or a potassium-sparing diuretic[516] should be considered.	IIa	B
To reduce BP, and if performed at a medium-to-high volume centre, catheter-based renal denervation may be considered for resistant hypertension patients who have BP that is uncontrolled despite a three BP-lowering drug combination, and who express a preference to undergo renal denervation after a shared risk-benefit discussion and multidisciplinary assessment.[564,566–568,586–590]	IIb	B

© ESC 2024

BP, blood pressure.
[a]Class of recommendation.
[b]Level of evidence.

9.14. Management of specific causes of secondary hypertension

9.14.1. General considerations

These guidelines will describe only the general principles of managing the most common forms of secondary hypertension. For the rarer forms of secondary hypertension, patients should be referred to specialized hypertension centres.

By definition, secondary hypertension should be, for the most part, cured when the underlying cause has been unambiguously identified and removed. However, in clinical practice, this is not always the case. Vascular remodelling, a common feature of a delayed diagnosis of secondary hypertension, affects renal function and can account for residual high BP in some patients with secondary hypertension. The rate of cure is higher when the diagnosis is made early in the course of the disease. Most common forms of secondary hypertension are listed in *Table 13*.

9.14.2. Primary aldosteronism

Primary aldosteronism (Conn syndrome) is the most common form of secondary hypertension. The management of primary aldosteronism depends on its subtype, particularly on adrenal lesions being unilateral or bilateral, because the unilateral forms are amenable to surgical treatment while the latter require lifelong medical treatment. In sporadic forms, unilateral primary aldosteronism is distinguished from bilateral primary aldosteronism by adrenal vein sampling or functional imaging with radiolabelled tracers.[930–932] In the much less common familial forms (necessitating a family history be taken), genetic testing for germline mutations is necessary.[933]

For unilateral primary aldosteronism, surgical removal of the offending adrenal gland is typically considered, unless the patient is older or has comorbidities of concern. Surgery is not an option for bilateral primary aldosteronism. Medical treatment is currently based on MRAs. Among MRAs, spironolactone is the most widely available. The effective dose, usually 50–100 mg once daily, can be titrated up to 300–400 mg once daily, if necessary. Eplerenone is also used and, despite being less potent than spironolactone and requiring twice-daily administration, it has the advantage of causing less gynaecomastia and erectile dysfunction in men.[934] Newer agents, such as the non-steroidal MRAs finerenone and exarenone, and the aldosterone synthase inhibitor baxdrostat, which lower BP in resistant hypertension,[326,474] are also being tested for treating primary aldosteronism. Of the familial forms, only glucocorticoid-remediable primary aldosteronism, now reclassified as familial hyperaldosteronism type 1, can be corrected with dexamethasone,[935] usually with low doses that are free of glucocorticoid effects and can be safely used during pregnancy.[936] For detailed information, readers are referred to the latest primary aldosteronism guidelines.[328,329]

9.14.3. Renovascular hypertension

Patients with RVH should receive medical therapy to reduce BP in the first instance. Percutaneous transluminal renal angioplasty (PTRA) without stenting is the treatment of choice for fibromuscular dysplasia, and can restore renal perfusion pressure and lower BP.[937] When this is not feasible, RAS blockers are the drugs of choice for treatment, but they require careful monitoring of renal function over time, as they can cause acute renal failure in those with tight bilateral stenoses or a stenosed solitary functioning kidney. Possible involvement of the carotid, coronary, and other major arteries, possibly leading to dissection if BP is not controlled, should also be considered, as fibromuscular dysplasia is now recognized as a systemic disease affecting multiple vascular beds.

Patients with significant atherosclerotic renal artery stenosis are at very high risk of CVD and renal events. It is recommended that PTRA and stenting are performed in experienced centres due to the high risk of restenosis. Unfortunately, though these studies did not solely recruit patients with true significant atherosclerotic RVH, publication of some null trials[938,939] have decreased the enthusiasm for investigating atherosclerotic renal artery stenosis. This could result in more uncontrolled hypertension, recurrent flash pulmonary oedema (Pickering syndrome), and worsening renal function ultimately leading to end-stage renal disease.[940]

Recommendation Table 31 — Recommendations for managing hypertension in patients with renovascular hypertension (see Evidence Tables 44 and 45)

Recommendations	Class[a]	Level[b]
Renal artery angioplasty without stenting should be considered for patients with hypertension and haemodynamically significant renal artery stenosis due to fibromuscular dysplasia.[941]	IIa	C
Renal artery angioplasty and stenting may be considered in patients with haemodynamically significant, atherosclerotic, renal artery stenosis (stenosis of 70%–99%, or 50%–69% with post-stenotic dilatation and/or significant trans-stenotic pressure gradient) with: • Recurrent heart failure, unstable angina, or sudden onset flash pulmonary oedema despite maximally tolerated medical therapy; • Resistant hypertension; • Hypertension with unexplained unilaterally small kidney or CKD; • Bilateral renal artery stenosis or unilateral renal artery stenosis in a solitary viable kidney.[942,943]	IIb	C

Continued

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

In patients with an indication to renal artery revascularization and technically unfeasible, or failed, renal artery angioplasty and stenting, open surgical revascularization may be considered.	IIb	C
Renal artery angioplasty is not recommended in patients without confirmed haemodynamically significant renal artery stenosis.[c] [938,939]	III	A

© ESC 2024

CKD, chronic kidney disease.
[a]Class of recommendation.
[b]Level of evidence.
[c]A haemodynamically relevant stenosis is usually defined by a luminal narrowing of >70% or 50%–70% with post-stenotic dilatation.

9.14.4. Phaeochromocytoma/paraganglioma

Phaeochromocytomas are rare adrenal tumours that secrete catecholamines and are present in <0.2% of patients with hypertension. A small percentage (<10%) of catecholamine-producing tumours are extra-adrenal and are derived from sympathetic and non-sympathetic nerves. More than 35% of the non-syndromic PPGLs are due to germline mutations.[338] These mutations should be screened for because, when found, they can drive management of the proband and the family and also inform the choice of functional imaging. Moreover, some germline mutations, such as those involving succinate dehydrogenase B, carry a risk of malignant adrenal tumours.[301,336]

Sympathetic PPGLs are usually secreting and present with chronic, episodic, or labile hypertension. Adrenergic crises cause hypertensive emergencies and should be treated with an intravenous (i.v.) alpha-1-blocker, such as phentolamine, doxazosin or terazosin, or labetalol. When given i.v. (1–2 mg/kg) twice weekly as a bolus over 1 min followed by a continuous infusion, labetalol also has alpha-blocker properties and has the advantage of allowing titration of the infusion based on the BP response, and avoids tachycardia via beta-blockade.

Identifying a single tumour mandates surgical excision after adequate pharmacological preparation, because secreting PPGLs can cause fatal events with no warning. Administering doxazosin or terazosin, followed by a beta-blocker, usually controls BP and adrenergic crises. As PPGLs are associated with a redistribution of volume from the periphery to the cardiopulmonary system,[944] patients with PPGLs have peripheral hypovolaemia that exposes them to the risk of profound hypotension, particularly right after tumour excision. Therefore, adequate fluid administration should be carefully managed.

9.14.5. Obstructive sleep apnoea syndrome

The management of this prevalent condition should be driven by the result of a polysomnography study, which should provide the value of the AHI (the average number of episodes per hour) and the sleep position in which apnoeic–hypopnoeic episodes occur. For mild OSAS (AHI < 15), weight loss and advice on sleep hygiene are usually sufficient. For moderate (AHI of 15–30) and severe (AHI > 30) OSAS, continuous positive airway pressure (CPAP) is indicated and usually improves BP control and helps to resolve resistant hypertension. If CPAP is not tolerated, the site of upper airway obstruction should be determined by an Ear, Nose, and Throat evaluation with drug-induced sleep endoscopy as a potential step to corrective surgery.

9.14.6. Drug-induced hypertension

Over-the-counter medications, prescribed drugs, and drug abuse (recreational substances and misuse of drugs) can cause hypertension (Supplementary data online, *Table S4*).

9.14.6.1. Anticancer drug-induced hypertension

Growing evidence indicates that, while contemporary anticancer and anti-angiogenic drugs improve cancer survival, they can also cause hypertension (Supplementary data online, *Table S4*). This is especially evident in patients treated with vascular endothelial growth factor inhibitors, in whom BP increases in 80%–90%.[945] Tyrosine kinase inhibitors and proteasome inhibitors also increase BP, as do adjuvant therapies (corticosteroids, calcineurin inhibitors, non-steroidal anti-inflammatory drugs, and anti-androgen hormone therapy). Hypertension caused by anticancer drugs is often dose limiting and may be reversible after therapy interruption or discontinuation. Evidence-based clinical trials specifically addressing patients who develop hypertension due to cancer therapy are lacking. It is recommended that management of hypertension in these patients follows that for the general population.[945,946] Managing these complex patients requires multidisciplinary healthcare involving oncologists, hypertension specialists, cardiologists, and nephrologists,[945,946] as highlighted in the 2022 ESC Guidelines on cardio-oncology.[946]

9.14.7. Other forms of secondary hypertension

Other forms of secondary hypertension, such as genetic causes of hypertension (Liddle's syndrome, glucocorticoid-remediable aldosteronism), excess liquorice, Cushing's syndrome, thyroid disease, hyperparathyroidism, aortic coarctation, and acromegaly are rare. Affected patients should be referred to specialized centres for treatment.

10. Acute and short-term lowering of blood pressure

10.1. Acute blood pressure management in hypertensive emergencies

10.1.1. Definition and characteristics of hypertensive emergencies

Hypertensive emergency is defined as BP of ≥180/110 mmHg (see *Figure 10*) associated with acute HMOD, often in the presence of symptoms. Hypertensive emergencies are potentially life-threatening and require immediate and careful intervention to reduce BP, often with i.v. therapy.

Symptoms of hypertensive emergency depend on the organs affected but may include headache, visual disturbances, chest pain, shortness of breath, dizziness, and other neurological deficits. In patients with hypertensive encephalopathy, somnolence, lethargy, tonic–clonic seizures, and cortical blindness may precede a loss of consciousness; however, focal neurological lesions are rare and should raise the suspicion of stroke.

As outlined in *Section 7*, we define HMOD among patients with chronically elevated BP or hypertension as the presence of specific cardiac, vascular, and renal alterations.[31,159] However, in the setting of hypertensive emergency, more acute manifestations of organ damage are relevant for management.

Acute manifestations of organ damage include:

- Patients with severe acute hypertension associated with other clinical conditions likely to require urgent reduction in BP, e.g. acute onset of aortic dissection, myocardial ischaemia, eclampsia, or heart failure.

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

- Malignant hypertension, defined as extreme BP elevations and acute microvascular damage (microangiopathy) affecting various organs.[947] The hallmark of this condition is small-artery fibrinoid necrosis in the kidneys, retina, and brain. The acute microangiopathy is typically characterized clinically by retinopathy (flame haemorrhages, cotton wool spots, and/or papilloedema). Other manifestations of microangiopathy include disseminated intravascular coagulation, encephalopathy (in about 15% of cases), acute heart failure, and acute deterioration in renal function.
- Patients with sudden severe hypertension due to phaeochromocytoma, which can result in severe acute organ damage.

The term 'hypertension urgency' describes severe hypertension in patients without clinical evidence of acute organ damage. While these patients require BP reduction, they do not usually require admission to hospital, and BP reduction is best achieved with oral medication according to the drug treatment algorithm presented in *Section 8*. However, these patients may require more urgent outpatient review to ensure that their BP is controlled.

Acute and severe increases in BP can sometimes be precipitated by sympathomimetics such as methamphetamine or cocaine, when caution around beta-blocker use is also needed. Many patients in an emergency department with acute pain or distress may have acutely elevated BP that will normalize when the pain and distress are relieved, rather than requiring any specific intervention to lower BP.

A diagnostic work-up is necessary for patients with a suspected hypertensive emergency (see Supplementary data online, *Table S12*).

10.1.2. Acute management of hypertensive emergencies

Key considerations in defining treatment are:

(1) Establishing the affected target organ(s) and whether they require any specific interventions other than BP lowering.
(2) Determining whether there is a precipitating cause for the acute rise in BP and/or another concomitant health condition present that might affect the treatment plan (e.g. pregnancy).
(3) The recommended timing and magnitude of BP lowering required for safe BP reduction.

These considerations will inform the type of BP-lowering treatment required. Regarding BP-lowering drugs, i.v. treatment using a short half-life drug is typically ideal to allow careful titration of the BP response to treatment. This requires a higher dependency clinical area with facilities for continuous or near-continuous haemodynamic monitoring. Recommended drug treatments for specific hypertensive emergencies are provided in the Supplementary data online, *Table S13*.

Rapid and uncontrolled or excessive BP lowering is not recommended in hypertensive emergency as this can lead to further complications. Although i.v. drug administration is recommended for most hypertensive emergencies, oral therapy with ACE inhibitors, ARBs, or beta-blockers (shorter-acting formulations like captopril or metoprolol) can also be effective. However, low initial doses should be used because these patients can be very sensitive to these agents, and treatment should take place in hospital. Further comprehensive details on the clinical management of hypertensive emergencies are available elsewhere.[242]

10.1.3. Prognosis and follow-up

The survival of patients with hypertensive emergencies has improved over the past few decades, but these patients remain at high risk and should be screened for secondary hypertension.

10.2. Acute blood pressure management in acute intracerebral haemorrhage

In acute intracerebral haemorrhage, an increased BP is common and is associated with a greater risk of haematoma expansion and death, and a worse prognosis for neurological recovery. In trials testing immediate BP lowering (within <6 h) to a systolic target of <140 mmHg, the achieved systolic BP in the intervention group was typically 140–160 mmHg and was reported to reduce the risk of haematoma expansion.[948,949] Excessive acute drops in systolic BP (>70 mmHg) may be associated with acute renal injury and early neurological deterioration and should be avoided.[950,951]

10.3. Acute blood pressure management in acute ischaemic stroke

The beneficial effects of BP reduction in acute ischaemic stroke remain unclear. In patients not receiving i.v. thrombolysis or mechanical thrombectomy, there is no evidence for actively lowering BP unless it is extremely high (e.g. >220/120 mmHg). If BP is extremely high, an initial moderate relative reduction of 10%–15% over a period of hours may be considered.[952] The reason for a more conservative approach to acute BP management is that cerebral autoregulation may be impaired in acute stroke, and maintaining cerebral perfusion relies on systemic BP.

In contrast, patients who are treated with i.v. thrombolysis or mechanical thrombectomy (or both) should have more proactive management of severe hypertension, because they have an increased risk of reperfusion injury and intracranial haemorrhage. In patients undergoing treatment with i.v. thrombolysis, BP should be lowered to <185/110 mmHg prior to thrombolysis and then maintained at <180/105 mmHg over the following 24 h.[953] In patients undergoing treatment with mechanical thrombectomy (with or without i.v. thrombolysis) there is limited evidence from clinical trials, but BP should also be lowered to <180/105 mmHg prior to thrombectomy and maintained over the next 24 h.[953,954] Therefore, patients with acute ischaemic stroke and a BP of <180/105 mmHg in the first 72 h after stroke do not seem to benefit from the introduction or reintroduction of BP-lowering medication.[955] For stable patients who remain hypertensive (≥140/90 mmHg) ≥3 days after an acute ischaemic stroke, initiation or reintroduction of BP-lowering medication is recommended.

Recommendation Table 32 — Recommendations for acutely managing blood pressure in patients with intracerebral haemorrhage or acute ischaemic stroke

Recommendations	Class[a]	Level[b]
For patients with ischaemic stroke or TIA and an indication for BP lowering, it is recommended that BP-lowering therapy be commenced before hospital discharge.[819,820,823]	I	B
In patients with acute ischaemic stroke, early BP lowering with BP-lowering therapy should be considered in the first 24 h in the following settings:		
• In patients who are eligible for re-perfusion therapy with intravenous thrombolysis or mechanical thrombectomy, BP should be carefully lowered and maintained at <180/105 mmHg for at least the first 24 h after treatment.[956–960]	IIa	B

Continued

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

• In patients with ischaemic stroke not receiving re-perfusion treatment and BP of ≥220/110 mmHg, BP should be carefully lowered by approximately 15% during the first 24 h after stroke onset.[956–960]	IIa	C
In patients with intracerebral haemorrhage, immediate BP lowering (within 6 h of symptom onset) should be considered to a systolic target 140–160 mmHg to prevent haematoma expansion and improve functional outcome.[948,949]	IIa	A
In patients with intracerebral haemorrhage presenting with systolic BP ≥220 mmHg, acute reduction in systolic BP >70 mmHg from initial levels within 1 h of commencing treatment is not recommended.[950,951,960–963]	III	B

© ESC 2024

BP, blood pressure; TIA, transient ischaemic attack.
[a]Class of recommendation.
[b]Level of evidence.

Recommendation Table 33 — Recommendations for acutely managing blood pressure in patients with severe hypertension in pregnancy and pre-eclampsia (see Evidence Table 46)

Recommendation	Class[a]	Level[b]
In pre-eclampsia or eclampsia with hypertensive crisis, drug treatment with i.v. labetalol or nicardipine and magnesium is recommended.[971]	I	C
In pre-eclampsia or eclampsia associated with pulmonary oedema, nitroglycerin given as an i.v. infusion is recommended.[242]	I	C
In severe hypertension in pregnancy: • drug treatment with i.v. labetalol, oral methyldopa, or oral nifedipine is recommended. Intravenous hydralazine is a second-line option.[666–668,969,971]	I	C

© ESC 2024

i.v., intravenous.
[a]Class of recommendation.
[b]Level of evidence.

10.4. Acute blood pressure management in pre-eclampsia and severe hypertension in pregnancy

10.4.1. Pre-eclampsia

Pre-eclampsia is discussed in *Section 9*. Here we focus on its management in the acute setting. Pre-eclampsia is cured by delivery. Most international societies, including the ESC, recommend an intensive approach to BP lowering in pre-eclampsia.[89,964,965] In women with pre-eclampsia and severe hypertension, immediately reducing systolic BP to <160 mmHg and diastolic BP to <105 mmHg using i.v. labetalol or nicardipine (with administration of magnesium sulfate if appropriate and consideration of delivery if appropriate) was recommended in the 2018 ESC/ESH Guidelines on the management of arterial hypertension and the 2022 ESC Guidelines for management of cardiovascular disease in pregnancy.[1,89] The objective of treatment is to lower BP within 150–180 min.

Magnesium sulfate [4 g i.v. over 5 min, then 1 g/h i.v.; or 5 g intramuscularly (i.m.) into each buttock, then 5 g i.m. every 4 h] is recommended for eclampsia treatment but also for women with pre-eclampsia who have severe hypertension and proteinuria or hypertension and neurological symptoms or signs.[966] There is a risk of hypotension when magnesium is given concomitantly with nifedipine.[967] If BP control is not achieved by 360 min despite two medications, consulting critical care is recommended for intensive care unit admission, stabilization, and delivery (if appropriate).[966] Since plasma volume is reduced in pre-eclampsia, diuretic therapy should be avoided.

10.4.2. Severe acute hypertension in pregnancy

Severe hypertension in pregnancy (without pre-eclampsia) may necessitate acute BP-lowering therapies. Severe hypertension in pregnancy is defined in general as systolic BP of >160 mmHg and diastolic BP of >110 mmHg and is associated with adverse maternal and peri-natal outcomes independent of pre-eclampsia and potentially of the same magnitude as eclampsia itself.[89,968]

There are differences in rate of BP control between i.v. labetalol and i.v. hydralazine in severe hypertension in pregnancy.[969] While evidence is conflicting,[667,668] hydralazine may be associated with more peri-natal adverse events than other drugs.[970] Nifedipine seems to provide lower BP with lower rates of neonatal complications than labetalol.[971]

10.5. Peri-operative acute management of elevated blood pressure

Details are provided in the ESC Guidelines on cardiovascular assessment and management of patients undergoing non-cardiac surgery.[972] Peri-operative hypertension, hypotension, and BP variability are associated with haemodynamic instability and poor clinical outcomes for patients undergoing surgery.[973] Pre-operative risk assessment for BP management, therefore, should involve assessing for underlying end-organ damage and comorbidities.[974] Postponing necessary non-cardiac surgery is not usually warranted for patients with minor or moderate elevations in BP, as they are not at higher CVD risk.[130,975]

Avoiding large fluctuations in BP in the peri-operative course is important, and planning a strategy for a patient should account for the baseline office BP.[974–977]

There is insufficient evidence for reduced or increased peri-operative BP targets compared to usual care BP targets to lower peri-operative events.[978] No specific measure of BP appears better than any other for predicting risk of peri-operative events.[975]

10.5.1. Blood pressure-lowering drugs in the peri-operative phase

Routine initiation of a beta-blocker peri-operatively is not necessary.[979]

Pre-operative initiation of beta-blockers in advance of high-risk, non-cardiac surgery may be considered in patients who have known coronary artery disease or myocardial ischaemia[980] or two or more significantly elevated clinical risk factors in order to reduce the incidence of peri-operative myocardial infarction.[979] Peri-operative continuation of beta-blockers is recommended for patients currently taking beta-blockers.[981]

Some studies suggest that continued use of ACE inhibitors is associated with a higher risk of peri-operative hypotension and subsequent end-organ damage including kidney injury, myocardial infarction, and stroke.[982] In the Prospective Randomized Evaluation of Preoperative Angiotensin-Converting Enzyme Inhibition (PREOP-ACEI) trial, transient pre-operative interruption of ACE inhibitor therapy was associated with a decreased risk of intra-operative hypotension.[983] A subsequent systematic review also showed a decreased risk of intra-operative hypotension with withholding ACE inhibitors/ARBs before surgery, but no association

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

with decreased mortality or CVD outcomes.[984] On the other hand, vigilance is needed because withholding ACE inhibitors has also been shown to increase post-operative hypertension.[985] In patients with heart failure, loop diuretics can be continued in patients prone to volume overload.[986] CCBs are generally considered safe pre-operatively.

11. Patient-centred care in hypertension

11.1. Definition

Patient-centred care is defined as an attitude of the healthcare professional that closely aligns with the patient's preferences and needs.[987] In the patient-centred approach (*Figure 23*), patients are viewed as active participants in health services, who work as partners alongside healthcare professionals. A patient-centred approach is associated with higher satisfaction rates, better adherence to recommendations and prescriptions, and better treatment, particularly in the management of chronic illness, such as hypertension.[988] While there is limited evidence for the efficacy and effectiveness of specific shared decision-making intervention strategies in hypertension care,[989] it is viewed as an ethical imperative in healthcare practice and health policy, and in clinical guidelines.[130]

11.2. Communicating consequences of treatment

In line with patient-centred care, it is important to assess whether patients understand their hypertension-related risk, the rationale for any hypertension treatment, the benefits and harms of hypertension treatment, and that the treatment plan is also centrally guided by what matters most to the patient. Risk communication is challenging, and providers need to be led by the individual's preferences when presenting more detailed numeric and visual representations of risk and the likely benefits and harms of hypertension treatment. Socio-demographic differences in healthcare need to be considered in patient–provider communication.[990,991]

Standard approaches to communicate consequences of treatment can involve 10-year risk of a CVD event with SCORE2 or SCORE2-OP. Alternatively, individual risk and risk reduction can be communicated in terms of 'risk age' or 'heart age' (*Section 7.3*).



Figure 23 Patient-centred care.

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

Recommendation Table 34 — Recommendations for communicating consequences of treatment (see Evidence Table 47)

Recommendation	Class[a]	Level[b]
An informed discussion about CVD risk and treatment benefits tailored to the needs of a patient is recommended as part of hypertension management.[992]	I	C
Motivational interviewing should be considered for patients with hypertension at hospitals and community health centres to assist patients in controlling their BP and to enhance treatment adherence.[993,994]	IIa	B
Physician–patient web communications are an effective tool that should be considered in primary care, including reporting on home BP readings.[995]	IIa	C

© ESC 2024

BP, blood pressure; CVD, cardiovascular disease.
[a]Class of recommendation.
[b]Level of evidence.

11.3. Self-measuring and monitoring

Self-care refers to individual responsibility for healthy lifestyle behaviours, as well as the actions required to cope with health conditions.[996,997] In the context of hypertension, it also includes self-management and self-measurement of BP.

Self-management includes lifestyle behaviour (diet, exercise, smoking, alcohol), co-management of medical treatments, and support for adhering to prescribed medication.[998] Self-monitoring allows high BP to be detected early,[999] and enables patients to co-manage medications with their healthcare provider.[1000,1001] Suitably validated and correctly used digital devices have the potential to support co-management,[1002,1003] and facilitate remote monitoring of BP.[76,81,1004]

Recommendation Table 35 — Recommendations for self-measuring and monitoring blood pressure (see Evidence Table 48)

Recommendations	Class[a]	Level[b]
Home BP measurement for managing hypertension by using self-monitored BP is recommended to achieve better BP control.	I	B
Self-measurement, when properly performed, is recommended due to positive effects on the acceptance of a diagnosis of hypertension, patient empowerment, and adherence to treatment.[1001]	I	C
Enhanced self-monitoring of BP using a device paired with a connected smartphone application may be considered, though evidence to date suggests that this may be no more effective than standard self-monitoring.[1005,1006]	IIb	B

© ESC 2024

BP, blood pressure.
[a]Class of recommendation.
[b]Level of evidence.

11.4. Facilitating medication adherence and persistence

Adherence (*Figure 24*) to BP-lowering drug regimens in clinical practice is almost always lower than seen in clinical trials.[1007] Most apparent treatment-resistant hypertension is accounted for by non-adherence.[1008] Adherence should always be assessed with a no-blame approach. Various methods are available to assess adherence and, along with details on barriers to adherence, are described in the Supplementary data online and *Table S14*.[1009]

Adherence may also be facilitated by an optimal therapeutic regimen, which can be achieved by medication reviews carried out at appropriate intervals. Several factors should be considered: (i) identifying drug-related adverse events and appropriate dosing levels, (ii) using long-acting drugs that require once-daily dosing (preferably drugs that are long-acting due to pharmacokinetic properties rather than galenic formulation), (iii) avoiding complex dosing schedules, (iv) using single-pill combinations whenever feasible, (v) taking into account the financial capacity of the patient to pay for a given regimen in the longer term, if relevant, or other pertinent aspects of the local or national healthcare systems, and (vi) enlisting support of a family member or other social support to facilitate medication adherence and persistence (see Supplementary data online, *Table S15*).[1010]

While there have been advancements in digital tools to support self-management of chronic illness including hypertension, there is little efficacy evidence evaluating these interventions. Therefore, it is premature to make recommendations about specific digital tools.

11.5. Multidisciplinary management

A collaborative approach to managing hypertension, using team-based care among physicians, nurses, pharmacists, dietitians, and physiotherapists, offers significant benefits over physician-only care. Multidisciplinary care is intended to be collaborative and complementary to regular medical care[1011] and is associated with lower systolic and diastolic BP[227,229,1012,1013] and improved outcomes.[230,1014] Task-shifting away from physicians is necessary to meet the huge need for the management of elevated BP and hypertension in the population.[1015] Prescribing remains a physician duty, but prescribing can be conducted under collaborative practice agreements with the multidisciplinary team in many countries.

Further details on patient-centred care in hypertension is provided in the Supplementary data online.

Recommendation Table 36 — Recommendations for multi/interdisciplinary blood pressure management (see Evidence Table 49)

Recommendation	Class[a]	Level[b]
Multidisciplinary approaches in the management of patients with elevated BP and hypertension, including appropriate and safe task-shifting away from physicians, are recommended to improve BP control.[227,229,230,1012–1014,1016]	I	A

© ESC 2024

BP, blood pressure.
[a]Class of recommendation.
[b]Level of evidence.

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

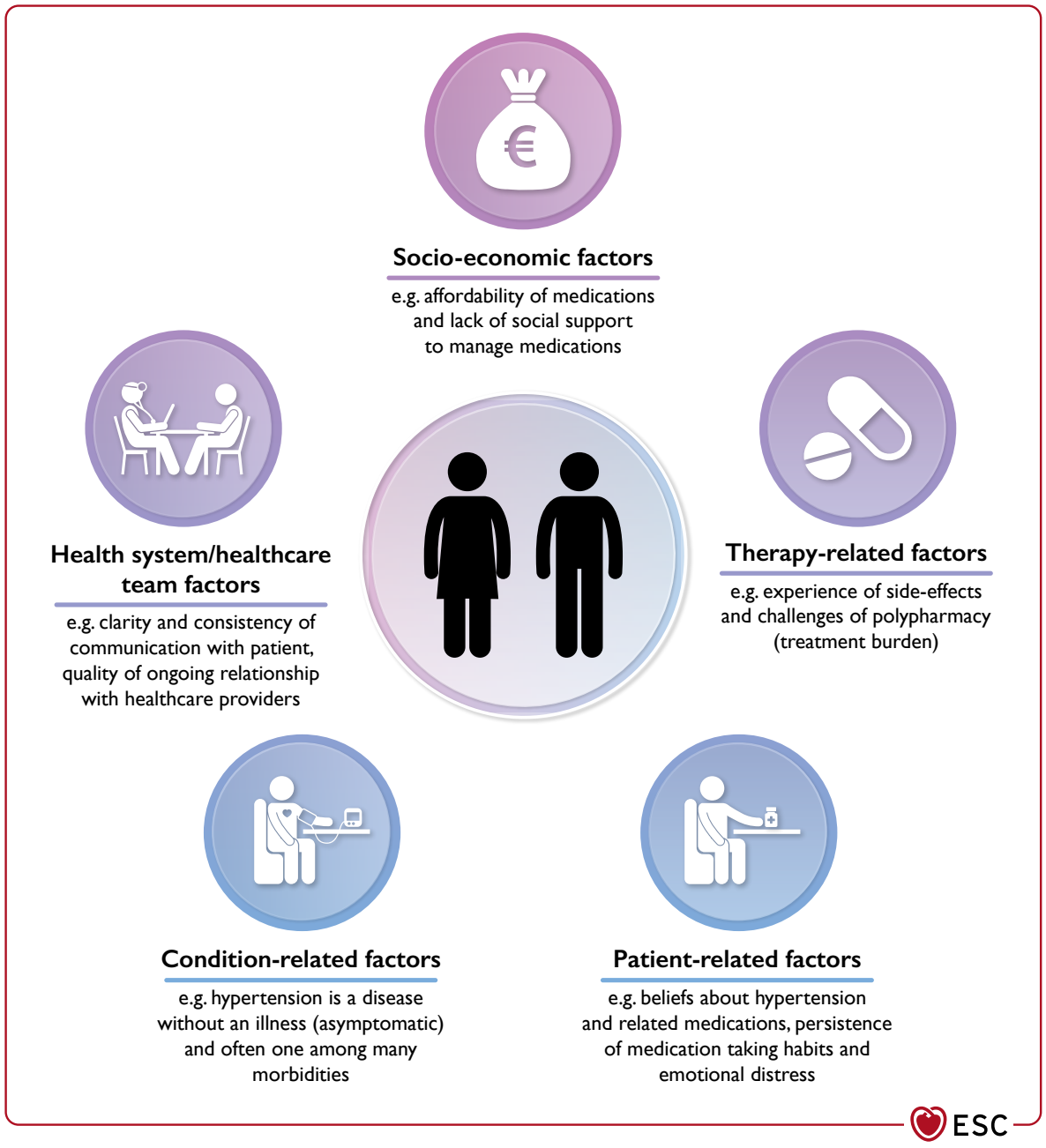


Figure 24 The five dimensions of adherence (WHO, 2003) applied to hypertension.

12. Key messages

(1) Given the demographic transition and the worldwide ageing of populations, the number of individuals with elevated BP or hypertension is increasing worldwide.
(2) The trajectory of BP control appears to be worsening in North America, in some (but not all) European countries, and elsewhere around the world.
(3) The risk for CVD attributable to BP is on a continuous log-linear exposure variable scale, not a binary scale of normotension vs. hypertension.
(4) BP-lowering drugs can reduce CVD risk even among individuals not traditionally classified as hypertensive. Accordingly, a new BP category called 'elevated BP' is introduced. Elevated BP is defined as an office systolic BP of 120–139 mmHg or diastolic BP of 70–89 mmHg. Hypertension remains defined as office BP of ≥140/90 mmHg.
(5) Hypertension in women is under-studied in basic, clinical, and population research.
(6) HMOD suggests long-standing or severe hypertension and is associated with increased CVD risk.
(7) Absolute CVD risk must be considered when assessing and managing elevated BP.
(8) Despite the growing number of hypertension guidelines, the rates of diagnosis, treatment, and control of hypertension (and elevated BP) remain suboptimal. A major factor underlying this is poor

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

implementation of evidence-based guidelines in real-world clinical practice.

(9) One of the most important changes in the 2024 Guidelines is the focus on evidence related to CVD outcomes of BP-lowering interventions rather than BP lowering alone.

(10) Irrespective of the threshold BP above which BP-lowering treatment (lifestyle or pharmacological or other treatment) is recommended, the on-treatment BP target is 120–129/70–79 mmHg for all adults, provided this treatment is well tolerated. There are several important exceptions to these targets and individualized decision-making is always the most important priority.

13. Gaps in the evidence

(1) Drivers of worsening trajectories of BP control in women and men.

(2) Need for sex-specific data on epidemiology, risk factors, and pathophysiology of hypertension. Need for more prospective studies to assess women's and men's specific CVD risk factors pertinent to adults with elevated BP and hypertension, due to biological and socio-cultural conditions. This includes sex-specific weighting of traditional risk factors, as well as inclusion of sex-dependent, non-traditional, vascular risk factors such as stress, socio-economic conditions, and others.[1017,1018] We are also lacking data on sex-specific hormonal and genetic mechanisms and pathophysiology in the human.[1019] Another important area in need of investigation is a better understanding of the role of gender in the management of elevated BP and hypertension (including gender-driven barriers in accessing medical care and adherence).

(3) More widespread validation of home BP measuring devices. Validation protocols for cuffless BP measurement devices have just recently been proposed and need to be tested.

(4) Clinical effectiveness of HMOD in directing intensity of care and personalized approaches in managing elevated BP and hypertension.

(5) Best practice to screen and manage primary aldosteronism.

(6) Clinical benefits of treating low CVD-risk individuals with elevated BP and further data strengthening the use of BP-lowering medication among high-risk persons with baseline systolic BP of 120–129 mmHg.

(7) Need for more data on the sex-specific optimal dosing, effects, and adverse effects of BP-lowering drugs,[1020] in particular from specifically planned prospective randomized trials.

(8) More consideration for overall CVD outcomes of BP-lowering interventions.

(9) More European data (RCTs, real life) about the beneficial effect of treating patients with elevated BP and hypertension with polypills (inclusive of non-BP lowering medications).

(10) CVD outcomes-based data on MRAs as add-on therapy solely for resistant hypertension.

(11) Trials on the BP-lowering effects of newer antidiabetic drugs (such as SGLT2 inhibitors and GLP-1 receptor agonists) or drugs that now have indications for other conditions, such as finerenone or sacubitril-valsartan.

(12) Beneficial BP and CVD effects of increasing dietary potassium intake and other lifestyle interventions. Studies to disentangle the effect of sodium reduction vs. the effect of potassium supplementation on BP control and CVD outcomes.

(13) RCTs comparing single-pill combination therapy with fixed doses vs. multiple monotherapies and their effects on CVD outcomes.

(14) Cardiovascular outcomes trials of renal denervation.

(15) BP-lowering treatment RCTs on different ethnic and migrant groups established in Europe.

(16) Pharmacological BP management in young adults (aged <40 years) and better data on the efficacy of a life-course approach for the drug management of BP.[1021]

(17) CVD outcomes in moderately to severely frail and/or very elderly persons where BP medications have been deprescribed, and the impact of competing risks.

(18) Management of renal artery disease with haemodynamically stable but severe stenosis (i.e. without high-risk features).

(19) Need for clinical trials on managing hypertension in patients treated with anticancer drugs or anti-rejection drugs in recipients of an allograft transplant.

(20) Hypertension management in the setting of climate changes, global warming, air and other forms of pollution, pandemics, war zones, and in the context of drug restrictions experienced in some low-to-middle-income countries.

(21) Need to improve implementation of guidelines by healthcare providers.

(22) How to develop sustainable hypertension care at the intersection of growing numbers of patients and limited resources.

(23) Treat-to-target trials specifically testing BP-lowering drugs among drug-naïve persons with baseline BP of 120–129 mmHg and increased CVD risk.

14. 'What to do' and 'what not to do' messages from the guidelines

A selected sample of the main messages from these guidelines are provided in *Table 15*.

Table 15 What to do and what not to do

Recommendations	Class	Level
5. Measuring blood pressure		
It is recommended to measure BP using a validated and calibrated device, to enforce the correct measurement technique, and to apply a consistent approach to BP measurement for each patient.	I	B
All adult patients (≥18 years or older) are recommended to have their office and/or out-of-office BP measured on an opportunistic basis and recorded in their medical file, and be told what their current BP is.	I	C

Continued

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

Out-of-office BP measurement is recommended for diagnostic purposes, particularly because it can detect both white-coat hypertension and masked hypertension. Where out-of-office measurements are not logistically and/or economically feasible, then it is recommended that the diagnosis be confirmed with a repeat office BP measurement using the correct standardized measurement technique.	I	B
It is recommended that office BP should be measured in both arms at least at the first visit, because a between-arm systolic BP difference of >10 mmHg is associated with an increased CVD risk and may indicate arterial stenosis.	I	B
If a between-arm difference of >10 mmHg in systolic BP is recorded, then it is recommended that all subsequent BP readings use the arm with the higher BP reading.	I	B
Out-of-office BP measurement is recommended for ongoing management to quantify the effects of treatment and guide BP-lowering medication titration, and/or identify possible causes of side effects (e.g. symptomatic hypotension). Where out-of-office measurements are not logistically and/or economically feasible then ongoing management is recommended to be based on repeated office BP measurements using the correct standardized measurement technique.	I	B
It is recommended that all patients undergoing BP measurement also undergo pulse palpation at rest to determine heart rate and arrhythmias such as AF.	I	C
6. Definition and classification of elevated blood pressure and hypertension		
It is recommended that BP be categorized as non-elevated BP, elevated BP, and hypertension to aid treatment decisions.	I	B
It is recommended to use a risk-based approach in the treatment of elevated BP, and individuals with moderate or severe CKD, established CVD, HMOD, diabetes mellitus, or familial hypercholesterolaemia are considered at increased risk for CVD events.	I	B
SCORE2 is recommended for assessing 10-year risk of fatal and non-fatal CVD among individuals aged 40–69 years with elevated BP who are not already considered at increased risk due to moderate or severe CKD, established CVD, HMOD, diabetes mellitus, or familial hypercholesterolaemia.	I	B
SCORE2-OP is recommended for assessing the 10-year risk of fatal and non-fatal CVD among individuals aged ≥70 years with elevated BP who are not already considered at increased risk due to moderate or severe CKD, established CVD, HMOD, diabetes mellitus, or familial hypercholesterolaemia.	I	B
It is recommended that, irrespective of age, individuals with elevated BP and a SCORE2 or SCORE2-OP CVD risk of ≥10% be considered at increased risk for CVD for the purposes of risk-based management of their elevated BP.	I	B
7. Diagnosing hypertension and investigating underlying causes		
In individuals with increased CVD risk where their screening office BP is 120–139/70–89 mmHg, it is recommended to measure BP out-of-office, using ABPM and/or HBPM or, if not logistically feasible, make repeated office BP measurements on more than one visit.	I	B
Where screening office BP is 140–159/90–99 mmHg, it is recommended that the diagnosis of hypertension should be based on out-of-office BP measurement with ABPM and/or HBPM. If these measurements are not logistically or economically feasible, then diagnosis can be made on repeated office BP measurements on more than one visit.	I	B
Where screening BP is ≥160/100 mmHg: • It is recommended that BP 160–179/100–109 mmHg is confirmed as soon as possible (e.g. within 1 month) preferably by either home or ambulatory BP measurements. • It is recommended when BP is ≥180/110 mmHg that hypertensive emergency be excluded.	I	C
It is recommended to measure serum creatinine, eGFR, and urine ACR in all patients with hypertension.	I	A
If moderate-to-severe CKD is diagnosed, it is recommended to repeat measurements of serum creatinine, eGFR, and urine ACR at least annually.	I	C
A 12-lead ECG is recommended for all patients with hypertension.	I	B
Echocardiography is recommended in patients with hypertension and ECG abnormalities, or signs or symptoms of cardiac disease.	I	B
Fundoscopy is recommended if BP >180/110 mmHg in the work-up of hypertensive emergency and malignant hypertension, as well as in hypertensive patients with diabetes.	I	C
Routine genetic testing for patients with hypertension is not recommended.	III	C
It is recommended that patients with hypertension presenting with suggestive signs, symptoms, or medical history of secondary hypertension are appropriately screened for secondary hypertension.	I	B
8. Preventing and treating elevated blood pressure		
Restriction of sodium to approximately 2 g per day is recommended where possible in all adults with elevated BP and hypertension [this is equivalent to about 5 g of salt (sodium chloride) per day or about a teaspoon or less].	I	A
Moderate-intensity aerobic exercise of ≥150 min/week [moderate aerobic exercise (≥30 min, 5–7 days/week) or alternatively 75 min of vigorous exercise per week over 3 days] is recommended and should be complemented with low- or moderate-intensity dynamic or isometric resistance training (2–3 times/week) to reduce BP and CVD risk.	I	A
It is recommended to aim for a stable and healthy BMI (20–25 kg/m^2) and waist circumference values (<94 cm in men and <80 cm in women) to reduce BP and CVD risk.	I	A

Continued

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

Adopting a healthy and balanced diet such as the Mediterranean or DASH diets is recommended to help reduce BP and CVD risk.	I	A
Men and women are recommended to drink less alcohol than the upper limit, which is about 100 g/week of pure alcohol. How this translates into number of drinks depends on portion size (the standards of which differ per country), but most drinks contain 8–14 g of alcohol per drink. Preferably, it is recommended to avoid alcohol consumption to achieve best health outcomes.	I	B
It is recommended to restrict free sugar consumption, in particular sugar-sweetened beverages, to a maximum of 10% of energy intake. It is also recommended to discourage consumption of sugar-sweetened beverages, such as soft drinks and fruit juices, starting at young age.	I	B
It is recommended to stop tobacco smoking, initiate supportive care, and refer to smoking cessation programmes, as tobacco use strongly and independently causes CVD, CVD events, and all-cause mortality.	I	A
Among all BP-lowering drugs, ACE inhibitors, ARBs, dihydropyridine CCBs, and diuretics (thiazides and thiazide-like drugs such as chlorthalidone and indapamide) have demonstrated the most effective reduction of BP and CVD events, and are therefore recommended as first-line treatments to lower BP.	I	A
It is recommended that beta-blockers are combined with any of the other major BP-lowering drug classes when there are other compelling indications for their use, e.g. angina, post-myocardial infarction, HFrEF, or for heart rate control.	I	A
It is recommended to take medications at the most convenient time of day for the patient to establish a habitual pattern of medication taking to improve adherence.	I	B
Given trial evidence for more effective BP control vs. monotherapy, combination BP-lowering treatment is recommended for most patients with confirmed hypertension (BP ≥140/90 mmHg) as initial therapy. Preferred combinations are a RAS blocker (either an ACE inhibitor or an ARB) with a dihydropyridine CCB or diuretic. Exceptions to consider include patients aged ≥85 years, symptomatic orthostatic hypotension, moderate-to-severe frailty, and those with elevated BP (systolic BP 120–139 mmHg or diastolic BP 70–89 mmHg) with a concomitant indication for treatment.	I	B
In patients receiving combination BP-lowering treatment, fixed-dose single-pill combination treatment is recommended.	I	B
If BP is not controlled with a two-drug combination, increasing to a three-drug combination is recommended, usually a RAS blocker with a dihydropyridine CCB and a thiazide/thiazide-like diuretic, and preferably in a single-pill combination.	I	B
Combining two RAS blockers (ACE inhibitor and an ARB) is not recommended.	III	A
In adults with elevated BP and low/medium CVD risk (<10% over 10 years), BP lowering with lifestyle measures is recommended and can reduce the risk of CVD.	I	B
In adults with elevated BP and sufficiently high CVD risk, after 3 months of lifestyle intervention, BP lowering with pharmacological treatment is recommended for those with confirmed BP ≥130/80 mmHg to reduce CVD risk.	I	A
It is recommended that in hypertensive patients with confirmed BP ≥140/90 mmHg, irrespective of CVD risk, lifestyle measures and pharmacological BP-lowering treatment is initiated promptly to reduce CVD risk.	I	A
It is recommended to maintain BP-lowering drug treatment lifelong, even beyond the age of 85 years, if well tolerated.	I	A
8. Preventing and treating elevated blood pressure (blood pressure targets)		
To reduce CVD risk, it is recommended that treated systolic BP values in most adults be targeted to 120–129 mmHg, provided the treatment is well tolerated.	I	A
In cases where BP-lowering treatment is poorly tolerated and achieving a target systolic of 120–129 mmHg is not possible, it is recommended to target a systolic BP level that is 'as low as reasonably achievable' (ALARA principle).	I	A
8. Preventing and treating elevated blood pressure (renal denervation)		
Due to a lack of adequately powered outcomes trials demonstrating its safety and CVD benefits, renal denervation is not recommended as a first-line BP-lowering intervention for hypertension.	III	C
Renal denervation is not recommended for treating hypertension in patients with moderate-to-severely impaired renal function (eGFR <40 mL/min/1.73 m^2) or secondary causes of hypertension, until further evidence becomes available.	III	C
9. Managing specific patient groups or circumstances		
Young adults		
Comprehensive screening for the main causes of secondary hypertension is recommended in adults diagnosed with hypertension before the age of 40 years, except for obese young adults where it is recommended to start with an obstructive sleep apnoea evaluation.	I	B
Hypertension in pregnancy		
In women with gestational hypertension, starting drug treatment is recommended for those with confirmed office systolic BP ≥140 mmHg or diastolic BP ≥90 mmHg.	I	B
In pregnant women with chronic hypertension, starting drug treatment is recommended for those with confirmed office systolic BP ≥140 mmHg or diastolic BP ≥90 mmHg.	I	B
In women with chronic and gestational hypertension, it is recommended to lower BP below 140/90 mmHg but not below 80 mmHg for diastolic BP.	I	C

Continued

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

Dihydropyridine CCBs (preferably extended-release nifedipine), labetalol, and methyldopa are recommended first-line BP-lowering medications for treating hypertension in pregnancy.	I	C
In consultation with an obstetrician, low- to moderate-intensity exercise is recommended in all pregnant women without contraindications to reduce the risk of gestational hypertension and pre-eclampsia.	I	B
RAS blockers are not recommended during pregnancy.	III	B
Very old and frail patients; orthostatic hypotension		
It is recommended that treatment of elevated BP and hypertension among older patients aged <85 years who are not moderately to severely frail follows the same guidelines as for younger people, provided BP-lowering treatment is well tolerated.	I	A
It is recommended to maintain BP-lowering drug treatment lifelong, even beyond the age of 85 years, if well tolerated.	I	A
Before starting or intensifying BP-lowering medication, it is recommended to test for orthostatic hypotension, by first having the patient sit or lie for 5 min and then measuring BP 1 and/or 3 min after standing.	I	B
It is recommended to pursue non-pharmacological approaches as the first-line treatment of orthostatic hypotension among persons with supine hypertension. For such patients, it is also recommended to switch BP-lowering medications that worsen orthostatic hypotension to an alternative BP-lowering therapy and not to simply de-intensify therapy.	I	A
Diabetes		
In most adults with elevated BP and diabetes, after a maximum of 3 months of lifestyle intervention, BP lowering with pharmacological treatment is recommended for those with confirmed BP ≥130/80 mmHg to reduce CVD risk.	I	A
BP-lowering drug treatment is recommended for people with pre-diabetes or obesity when confirmed office BP is ≥140/90 mmHg or when office BP is 130–139/80–89 mmHg and the patient is at predicted 10-year risk of CVD ≥10% or with high-risk conditions, despite a maximum of 3 months of lifestyle therapy.	I	A
In persons with diabetes who are receiving BP-lowering drugs, it is recommended to target systolic BP to 120–129 mmHg, if tolerated.	I	A
Chronic kidney disease		
In patients with diabetic or non-diabetic moderate-to-severe CKD and confirmed BP ≥130/80 mmHg, lifestyle optimization and BP-lowering medication are recommended to reduce CVD risk, provided such treatment is well tolerated.	I	A
In adults with moderate-to-severe CKD who are receiving BP-lowering drugs and who have eGFR >30 mL/min/1.73 m^2, it is recommended to target systolic BP to 120–129 mmHg, if tolerated. Individualized BP targets are recommended for those with lower eGFR or renal transplantation.	I	A
In hypertensive patients with CKD and eGFR >20 mL/min/1.73 m^2, SGLT2 inhibitors are recommended to improve outcomes in the context of their modest BP-lowering properties.	I	A
Cardiac disease		
In patients with a history of myocardial infarction who require BP-lowering treatment, beta-blockers and RAS blockers are recommended as part of that treatment.	I	A
In patients with symptomatic angina who require BP-lowering treatment, beta-blockers and/or CCBs are recommended as part of that treatment.	I	A
In patients with symptomatic HFrEF/HFmrEF, the following treatments with BP-lowering effects are recommended to improve outcomes: ACE inhibitors (or ARBs if ACE inhibitors are not tolerated) or ARNi, beta-blockers, MRAs, and SGLT2 inhibitors.	I	A
In hypertensive patients with symptomatic HFpEF, SGLT2 inhibitors are recommended to improve outcomes in the context of their modest BP-lowering properties.	I	A
Other conditions		
It is recommended that the BP-lowering drug treatment strategy for preventing stroke should comprise a RAS blocker plus a CCB or a thiazide-like diuretic.	I	A
In patients with confirmed BP ≥130/80 mmHg with a history of TIA or stroke a systolic BP target 120–129 mmHg is recommended to reduce CVD outcomes, provided treatment is tolerated.	I	A
Renal artery angioplasty is not recommended in patients without confirmed haemodynamically significant renal artery stenosis.	III	A
10. Acute and short-term lowering of blood pressure		
Intracerebral haemorrhage or acute ischaemic stroke		
For patients with ischaemic stroke or TIA and an indication for BP lowering, it is recommended that BP lowering therapy should be commenced before hospital discharge.	I	B
In patients with intracerebral haemorrhage presenting with systolic BP ≥220 mmHg, acute reduction in systolic BP >70 mmHg from initial levels within 1 h of commencing treatment is not recommended.	III	B

Continued

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

Severe hypertension in pregnancy and pre-eclampsia		
In pre-eclampsia or eclampsia with hypertensive crisis, drug treatment with i.v. labetalol or nicardipine and magnesium is recommended.	I	C
In pre-eclampsia or eclampsia associated with pulmonary oedema, nitroglycerin given as an i.v. infusion is recommended.	I	C
In severe hypertension in pregnancy: • drug treatment with i.v. labetalol, oral methyldopa, or oral nifedipine is recommended. Intravenous hydralazine is a second-line option.	I	C
11. Patient-centred care in hypertension		
An informed discussion about CVD risk and treatment benefits tailored to the needs of a patient is recommended as part of hypertension management.	I	C
Home BP measurement for managing hypertension by using self-monitored BP is recommended to achieve better BP control.	I	B
Self-measurement, when properly performed, is recommended due to positive effects on the acceptance of a diagnosis of hypertension, patient empowerment, and adherence to treatment.	I	C
Multidisciplinary approaches in the management of patients with elevated BP and hypertension, including appropriate and safe task-shifting away from physicians are recommended to improve BP control.	I	A

ABPM, ambulatory blood pressure monitoring; ACE, angiotensin-converting enzyme; ACR, albumin:creatinine ratio; AF, atrial fibrillation; ALARA, as low as reasonably achievable; ARB, angiotensin receptor blocker; ARNi, angiotensin receptor–neprilysin inhibitor; BMI, body mass index; BP, blood pressure; CCB, calcium channel blocker; CKD, chronic kidney disease; CVD, cardiovascular disease; DASH, Dietary Approaches to Stop Hypertension; ECG, electrocardiogram; eGFR, estimated glomerular filtration rate; ESC, European Society of Cardiology; HBPM, home blood pressure monitoring; HFpEF, heart failure with preserved ejection fraction; HF(m)rEF, heart failure with (mildly) reduced ejection fraction; HMOD, hypertension-mediated organ damage; i.v., intravenous; MRA, mineralocorticoid receptor antagonist; RAS, renin–angiotensin system; SCORE2, Systematic COronary Risk Evaluation 2; SCORE2-OP, Systematic COronary Risk Evaluation 2-Older Persons; SGLT2, sodium–glucose co-transporter 2; TIA, transient ischaemic attack.

15. Evidence tables

Evidence tables are available at *European Heart Journal* online.

16. Data availability statement

No new data were generated or analysed in support of this research.

17. Author information

Author/task force Member Affiliations: Cian P. McCarthy, Cardiovascular Division Massachusetts General Hospital and Harvard Medical School Boston, MA, United States of America; **Rosa Maria Bruno**, PARCC, Université Paris Cité, Inserm, Paris, France, and Pharmacology & Hypertension, AP-HP, Hôpital Européen Georges Pompidou, Paris, France; **Sofie Brouwers**, Cardiovascular Center Aalst, Department of Cardiology, OLV Clinic Aalst, Aalst, Belgium, and Department of Experimental Pharmacology, Faculty of Medicine and Pharmacy, Vrije Universiteit Brussel, Brussels, Belgium; **Michelle D. Canavan**, School of Medicine, College of Medicine, Nursing and Health Sciences, University of Galway, Galway, Ireland, and Department of Geriatric Medicine, University Hospital Galway, Saolta Hospitals Group, Galway, Ireland; **Claudio Ceconi**, Motusmed Clinic, Brescia, Italy; **Ruxandra Maria Christodorescu**, Department V Internal Medicine, University of Medicine and Pharmacy V Babes, Timisoara, Romania, and Research Center, Institute of Cardiovascular Diseases, Timisoara, Romania; **Stella S. Daskalopoulou**, Medicine Research Institute of the McGill University Health Centre, McGill University, Montreal, Canada; **Charles J. Ferro**, Department of Renal Medicine, University Hospitals Birmingham, Birmingham, United Kingdom, and Institute of Cardiovascular Sciences University of Birmingham, Birmingham, United Kingdom; **Eva Gerdts**, Department of Clinical Science, University of Bergen, Bergen, Norway, and Department of Heart Disease, Haukeland University Hospital, Bergen, Norway; **Henner Hanssen**, Department Sport, Exercise and Health, Medical Faculty, University of Basel, Basel, Switzerland; **Julie Harris** (United Kingdom), ESC Patient Forum, Sophia Antipolis, France; **Lucas Lauder**, Department of cardiology, University hospital Basel, Basel, Switzerland, and Department of internal medicine III, Cardiology, Angiology, and Intensive Care Medicine, Saarland university medical center, Homburg, Germany; **Richard J. McManus**, Nuffield Department of Primary Care Health Sciences, University of Oxford, Oxford, United Kingdom; **Gerard J. Molloy**, School of Psychology, University of Galway, Galway, Ireland; **Kazem Rahimi**, Deep Medicine, Nuffield Department of Women's and Reproductive Health, University of Oxford, Oxford, United Kingdom; **Vera Regitz-Zagrosek**, Charite, University Medicine Berlin, Gender in Medicine, Berlin, Germany; **Gian Paolo Rossi**, Department of Medicine, DIMED, University of Padua, Padua, Italy; **Else Charlotte Sandset**, Department of Neurology, Oslo University Hospital, Oslo, Norway, The Norwegian Air Ambulance Foundation, Oslo, Norway, and Institute of Clinical Medicine University of Oslo, Oslo, Norway; **Bart Scheenaerts** (Belgium), ESC Patient Forum, Sophia Antipolis, France; **Jan A. Staessen**, Non-Profit Research Association Alliance for the Promotion of Preventive Medicine, Mechelen, Belgium, and Biomedical Research Group, Faculty of Medicine, University of Leuven, Leuven, Belgium; **Izabella Uchmanowicz**, Department of Nursing and Obstetrics, Faculty of Health Sciences, Wroclaw Medical University, Wroclaw, Poland; and **Maurizio Volterrani**, Exercise Science and Medicine San Raffaele Open University, Rome, Italy, and Cardiopulmonary Department, IRCCS San Raffaele, Rome, Italy.

18. Appendix

ESC Scientific Document Group

Includes Document Reviewers and ESC National Cardiac Societies.

Document Reviewers: Ana Abreu (CPG Review Co-ordinator) (Portugal), Michael Hecht Olsen (CPG Review Co-ordinator) (Denmark), Marco Ambrosetti (Italy), Emmanuel Androulakis (United Kingdom), Lia Evi Bang (Denmark), Jesper Nørgaard Bech (Denmark), Michael A. Borger (Germany), Pierre Boutouyrie (France), Luís Bronze (Portugal), Sergio Buccheri (Sweden), Regina Dalmau (Spain), Maria Carmen De Pablo Zarzosa (Spain), Christian Delles (United

© ESC 2024

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

Kingdom), Maria Manuela Fiuza (Portugal), Rahima Gabulova (Azerbaijan), Bjørn Olav Haugen (Norway), Christian Heiss (United Kingdom), Borja Ibanez (Spain), Stefan James (Sweden), Vikas Kapil (United Kingdom), Meral Kayikçioglu (Turkey), Lars Køber (Denmark), Konstantinos C. Koskinas (Switzerland), Emanuela Teresa Locati (Italy), Sharon MacDonald (United Kingdom), Anastasia S. Mihailidou (Australia), Borislava Mihaylova (United Kingdom), Richard Mindham (United Kingdom) Martin Bodtker Mortensen (Denmark), Sandor Nardai (Hungary), Lis Neubeck (United Kingdom), Jens Cosedis Nielsen (Denmark), Peter M. Nilsson (Sweden), Agnes A. Pasquet (Belgium), Mónica Mendes Pedro (Portugal), Eva Prescott (Denmark), Amina Rakisheva (Kazakhstan), Ernst Rietzschel (Belgium), Bianca Rocca (Italy), Xavier Rossello (Spain), Jean-Paul Schmid (Switzerland), Eduard Shantsila (United Kingdom), Isabella Sudano (Switzerland), Ana Teresa Timóteo (Portugal), Georgios Tsivgoulis (Greece), Andrea Ungar (Italy), Ilonca Vaartjes (Netherlands), Frank Visseren (Netherlands), Heinz Voeller (Germany), Christiaan Vrints (Belgium), Adam Witkowski (Poland), Maria-Christina Zennaro (France), and Katja Zeppenfeld (Netherlands).

ESC National Cardiac Societies actively involved in the review process of the 2024 ESC Guidelines for the management of elevated blood pressure and hypertension:

Albania: Albanian Society of Cardiology, Naltin Shuka; **Algeria:** Algerian Society of Cardiology, Nadia Laredj; **Austria:** Austrian Society of Cardiology, Noemi Pavo; **Azerbaijan:** Azerbaijan Society of Cardiology, Ulvi Mirzoyev; **Belgium:** Belgian Society of Cardiology, Philippe van de Borne; **Bosnia and Herzegovina:** Association of Cardiologists of Bosnia and Herzegovina, Šekib Sokolović; **Bulgaria:** Bulgarian Society of Cardiology, Arman Postadzhiyan; **Croatia:** Croatian Cardiac Society, Jure Samardzic; **Cyprus:** Cyprus Society of Cardiology, Petros Agathangelou; **Czechia:** Czech Society of Cardiology, Jiri Widimsky; **Denmark:** Danish Society of Cardiology, Michael Hecht Olsen; **Egypt:** Egyptian Society of Cardiology, Wael M. El-Kilany; **Estonia:** Estonian Society of Cardiology, Priit Pauklin; **Finland:** Finnish Cardiac Society, Jari A. Laukkanen; **France:** French Society of Cardiology, Romain Boulestreau; **Georgia:** Georgian Society of Cardiology, Bezhan Tsinamdzgvrishvili; **Germany:** German Cardiac Society, Ulrich Kintscher; **Greece:** Hellenic Society of Cardiology, Maria Marketou; **Hungary:** Hungarian Society of Cardiology, Dénes Páll; **Iceland:** Icelandic Society of Cardiology, Þórdís Jóna Hrafnkelsdóttir; **Ireland:** Irish Cardiac Society, Eamon Dolan; **Israel:** Israel Heart Society, Talya Wolak; **Italy:** Italian Federation of Cardiology, Grzegorz Bilo; **Kazakhstan:** Association of Cardiologists of Kazakhstan, Meiramgul Kapsimetovna Tundybayeva; **Kyrgyzstan:** Kyrgyz Society of Cardiology, Erkin Mirrakhimov, **Latvia:** Latvian Society of Cardiology, Karlis Trusinskis; **Lebanon:** Lebanese Society of Cardiology, Ghassan Kiwan; **Libya:** Libyan Cardiac Society, Omar Msalem; **Lithuania:** Lithuanian Society of Cardiology, Jolita Badarienė; **Luxembourg:** Luxembourg Society of Cardiology, Cristiana-Astra Banu; **Malta:** Maltese Cardiac Society, Matthew Mercieca Balbi; **Moldova (Republic of):** Moldavian Society of Cardiology, Alexandru Caraus; **Montenegro:** Montenegro Society of Cardiology, Aneta Boskovic; **Morocco:** Moroccan Society of Cardiology, Najat Mouine; **Netherlands:** Netherlands Society of Cardiology, Tom Vromen; **North Macedonia:** National Society of Cardiology of North Macedonia, Marijan Bosevski; **Norway:** Norwegian Society of Cardiology, Helga B. Midtbø; **Poland:** Polish Cardiac Society, Adrian Doroszko; **Portugal:** Portuguese Society of Cardiology, Hélder Dores; **Romania:** Romanian Society of Cardiology, Elisabeta Badila; **San Marino:** San Marino Society of Cardiology, Roberto Bini; **Serbia:** Cardiology Society of Serbia, Dragan Vojislav Simić; **Slovenia:** Slovenian Society of Cardiology, Zlatko Fras; **Spain:** Spanish Society of Cardiology, Pilar Mazón; **Sweden:** Swedish Society of Cardiology, Jonas Spaak; **Switzerland:** Swiss Society of Cardiology, Thilo Burkard; **Syrian Arab Republic:** Syrian Cardiovascular Association, Elias Barakat; **Tunisia:** Tunisian Society of Cardiology and Cardiovascular Surgery, Salem Abdessalem; **Türkiye:** Turkish Society of Cardiology, Yilmaz Gunes; **Ukraine:** Ukrainian Association of Cardiology, Yurij M. Sirenko; **United Kingdom of Great Britain and Northern Ireland:** British Cardiovascular Society, Adrian J. B. Brady; and **Uzbekistan:** Association of Cardiologists of Uzbekistan, Gulnoz Abdusattarovna Khamidullaeva.

ESC Clinical Practice Guidelines (CPG) Committee: Eva Prescott (Chairperson) (Denmark), Stefan James (Co-Chairperson) (Sweden), Elena Arbelo (Spain), Colin Baigent (United Kingdom), Michael A. Borger (Germany), Sergio Buccheri (Sweden), Borja Ibanez (Spain), Lars Køber (Denmark), Konstantinos C. Koskinas (Switzerland), John William McEvoy (Ireland), Borislava Mihaylova (United Kingdom), Richard Mindham (United Kingdom), Lis Neubeck (United Kingdom), Jens Cosedis Nielsen (Denmark), Agnes A. Pasquet (Belgium), Amina Rakisheva (Kazakhstan), Bianca Rocca (Italy), Xavier Rossello (Spain), Ilonca Vaartjes (Netherlands), Christiaan Vrints (Belgium), Adam Witkowski (Poland), and Katja Zeppenfeld (Netherlands).

19. References

1. Williams B, Mancia G, Spiering W, Rosei EA, Azizi M, Burnier M, *et al.* 2018 ESC/ESH Guidelines for the management of arterial hypertension. *Eur Heart J* 2018;**39**:3021–104. https://doi.org/10.1093/eurheartj/ehy339
2. Murray CJL, Aravkin AY, Zheng P, Abbafati C, Abbas KM, Abbasi-Kangevari M, *et al.* Global burden of 87 risk factors in 204 countries and territories, 1990–2019: a systematic analysis for the Global Burden of Disease Study 2019. *Lancet* 2020;**396**:1223–49. https://doi.org/10.1016/S0140-6736(20)30752-2
3. Mensah GA, Fuster V, Murray CJL, Roth GA, Mensah GA, Abate YH, *et al.* Global burden of cardiovascular diseases and risks, 1990–2022. *J Am Coll Cardiol* 2023;**82**:2350–473. https://doi.org/10.1016/j.jacc.2023.11.007
4. Johnson JL, Greaves L, Repta R. Better science with sex and gender: facilitating the use of a sex and gender-based analysis in health research. *Int J Equity Health* 2009;**8**:14. https://doi.org/10.1186/1475-9276-8-14
5. Mauvais-Jarvis F, Bairey Merz N, Barnes PJ, Brinton RD, Carrero J-J, DeMeo DL, *et al.* Sex and gender: modifiers of health, disease, and medicine. *Lancet* 2020;**396**:565–82. https://doi.org/10.1016/s0140-6736(20)31561-0
6. Camm AJ, Lüscher TF, Maurer G, Serruys PW (eds). *The ESC Textbook of Cardiovascular Medicine*. Oxford University Press, 2018.
7. Mayfield SK, Foti K, Moran AE, Blakeman DE, Frieden TR. Hypertension call to action: will we respond to the call with action? *Am J Hypertens* 2022;**35**:214–6. https://doi.org/10.1093/ajh/hpab191
8. Muntner P. The continuing challenge of low rates of blood pressure control among US adults. *Am J Hypertens* 2022;**35**:839–41. https://doi.org/10.1093/ajh/hpac075
9. Reuter H, Jordan J. Status of hypertension in Europe. *Curr Opin Cardiol* 2019;**34**:342–9. https://doi.org/10.1097/hco.0000000000000642
10. Brown JM, Siddiqui M, Calhoun DA, Carey RM, Hopkins PN, Williams GH, *et al.* The unrecognized prevalence of primary aldosteronism: a cross-sectional study. *Ann Intern Med* 2020;**173**:10–20. https://doi.org/10.7326/m20-0065
11. Oparil S, Acelajado MC, Bakris GL, Berlowitz DR, Cífková R, Dominiczak AF, *et al.* Hypertension. *Nat Rev Dis Primers* 2018;**4**:18014. https://doi.org/10.1038/nrdp.2018.14
12. Navaneethabalakrishnan S, Smith HL, Arenaz CM, Goodlett BL, McDermott JG, Mitchell BM. Update on immune mechanisms in hypertension. *Am J Hypertens* 2022;**35**:842–51. https://doi.org/10.1093/ajh/hpac077
13. Levy D, Garrison RJ, Savage DD, Kannel WB, Castelli WP. Prognostic implications of echocardiographically determined left ventricular mass in the Framingham heart study. *N Engl J Med* 1990;**322**:1561–6. https://doi.org/10.1056/nejm199005313222203
14. Devereux RB, Wachtell K, Gerdts E, Boman K, Nieminen MS, Papademetriou V, *et al.* Prognostic significance of left ventricular mass change during treatment of hypertension. *JAMA* 2004;**292**:2350–6. https://doi.org/10.1001/jama.292.19.2350

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

15. de Simone G, Gottdiener JS, Chinali M, Maurer MS. Left ventricular mass predicts heart failure not related to previous myocardial infarction: the cardiovascular health study. *Eur Heart J* 2008;**29**:741–7. https://doi.org/10.1093/eurheartj/ehm605
16. Verdecchia P, Porcellati C, Reboldi G, Gattobigio R, Borgioni C, Pearson TA, *et al.* Left ventricular hypertrophy as an independent predictor of acute cerebrovascular events in essential hypertension. *Circulation* 2001;**104**:2039–44. https://doi.org/10.1161/hc4201.097944
17. Hijazi Z, Verdecchia P, Oldgren J, Andersson U, Reboldi G, Di Pasquale G, *et al.* Cardiac biomarkers and left ventricular hypertrophy in relation to outcomes in patients with atrial fibrillation: experiences from the RE-LY trial. *J Am Heart Assoc* 2019;**8**:e010107. https://doi.org/10.1161/jaha.118.010107
18. Siedlinski M, Carnevale L, Xu X, Carnevale D, Evangelou E, Caulfield MJ, *et al.* Genetic analyses identify brain structures related to cognitive impairment associated with elevated blood pressure. *Eur Heart J* 2023;**44**:2114–25. https://doi.org/10.1093/eurheartj/ehad101
19. Cheung AK, Chang TI, Cushman WC, Furth SL, Hou FF, Ix JH, *et al.* KDIGO 2021 clinical practice guideline for the management of blood pressure in chronic kidney disease. *Kidney International* 2021;**99**:S1–87. https://doi.org/10.1016/j.kint.2020.11.003
20. Subbiah A, Bhowmik D. KDIGO recommendations on blood pressure management in chronic kidney disease. *Kidney Int* 2022;**101**:1299. https://doi.org/10.1016/j.kint.2022.02.036
21. Cheung CY, Biousse V, Keane PA, Schiffrin EL, Wong TY. Hypertensive eye disease. *Nat Rev Dis Primers* 2022;**8**:14. https://doi.org/10.1038/s41572-022-00342-0
22. Lewington S, Clarke R, Qizilbash N, Peto R, Collins R. Age-specific relevance of usual blood pressure to vascular mortality: a meta-analysis of individual data for one million adults in 61 prospective studies. *Lancet* 2002;**360**:1903–13. https://doi.org/10.1016/s0140-6736(02)11911-8
23. O'Rourke MF, Safar ME. Relationship between aortic stiffening and microvascular disease in brain and kidney: cause and logic of therapy. *Hypertension* 2005;**46**:200–4. https://doi.org/10.1161/01.Hyp.0000168052.00426.65
24. Gerdts E, Izzo R, Mancusi C, Losi MA, Manzi MV, Canciello G, *et al.* Left ventricular hypertrophy offsets the sex difference in cardiovascular risk (the Campania Salute Network). *Int J Cardiol* 2018;**258**:257–61. https://doi.org/10.1016/j.ijcard.2017.12.086
25. Gerdts E, Okin PM, de Simone G, Cramariuc D, Wachtell K, Boman K, *et al.* Gender differences in left ventricular structure and function during antihypertensive treatment: the Losartan intervention for endpoint reduction in hypertension study. *Hypertension* 2008;**51**:1109–14. https://doi.org/10.1161/hypertensionaha.107.107474
26. Gerdts E, Wachtell K, Omvik P, Otterstad JE, Oikarinen L, Boman K, *et al.* Left atrial size and risk of major cardiovascular events during antihypertensive treatment: Losartan Intervention for Endpoint Reduction in Hypertension Trial. *Hypertension* 2007;**49**:311–6. https://doi.org/10.1161/01.Hyp.0000254322.96189.85
27. Park JB, Schiffrin EL. Small artery remodeling is the most prevalent (earliest?) form of target organ damage in mild essential hypertension. *J Hypertens* 2001;**19**:921–30. https://doi.org/10.1097/00004872-200105000-00013
28. Vasan RS, Pan S, Xanthakis V, Beiser A, Larson MG, Seshadri S, *et al.* Arterial stiffness and long-term risk of health outcomes: the Framingham heart study. *Hypertension* 2022;**79**:1045–56. https://doi.org/10.1161/hypertensionaha.121.18776
29. Piskorz D. Hypertensive mediated organ damage and hypertension management. How to assess beneficial effects of antihypertensive treatments? *High Blood Press Cardiovasc Prev* 2020;**27**:9–17. https://doi.org/10.1007/s40292-020-00361-6
30. Vallelonga F, Cesareo M, Menon L, Airale L, Leone D, Astarita A, *et al.* Cardiovascular hypertension-mediated organ damage in hypertensive urgencies and hypertensive outpatients. *Front Cardiovasc Med* 2022;**9**:889554. https://doi.org/10.3389/fcvm.2022.889554
31. Vasan RS, Song RJ, Xanthakis V, Beiser A, DeCarli C, Mitchell GF, *et al.* Hypertension-mediated organ damage: prevalence, correlates, and prognosis in the community. *Hypertension* 2022;**79**:505–15. https://doi.org/10.1161/hypertensionaha.121.18502
32. Rapsomaniki E, Timmis A, George J, Pujades-Rodriguez M, Shah AD, Denaxas S, *et al.* Blood pressure and incidence of twelve cardiovascular diseases: lifetime risks, healthy life-years lost, and age-specific associations in 1·25 million people. *Lancet* 2014;**383**:1899–911. https://doi.org/10.1016/s0140-6736(14)60685-1
33. Malik R, Georgakis MK, Vujkovic M, Damrauer SM, Elliott P, Karhunen V, *et al.* Relationship between blood pressure and incident cardiovascular disease: linear and nonlinear Mendelian randomization analyses. *Hypertension* 2021;**77**:2004–13. https://doi.org/10.1161/hypertensionaha.120.16534
34. Ji H, Kim A, Ebinger JE, Niiranen TJ, Claggett BL, Bairey Merz CN, *et al.* Sex differences in blood pressure trajectories over the life course. *JAMA Cardiol* 2020;**5**:19–26. https://doi.org/10.1001/jamacardio.2019.5306
35. Salles GF, Reboldi G, Fagard RH, Cardoso CRL, Pierdomenico SD, Verdecchia P, *et al.* Prognostic effect of the nocturnal blood pressure fall in hypertensive patients: the ambulatory blood pressure collaboration in patients with hypertension (ABC-H) meta-analysis. *Hypertension* 2016;**67**:693–700. https://doi.org/10.1161/hypertensionaha.115.06981
36. Verdecchia P. Prognostic value of ambulatory blood pressure: current evidence and clinical implications. *Hypertension* 2000;**35**:844–51. https://doi.org/10.1161/01.hyp.35.3.844
37. Mauck GW, Smith CR, Geddes LA, Bourland JD. The meaning of the point of maximum oscillations in cuff pressure in the indirect measurement of blood pressure—part ii. *J Biomech Eng* 1980;**102**:28–33. https://doi.org/10.1115/1.3138195
38. Clark CE, Warren FC, Boddy K, McDonagh STJ, Moore SF, Goddard J, *et al.* Associations between systolic interarm differences in blood pressure and cardiovascular disease outcomes and mortality: individual participant data meta-analysis, development and validation of a prognostic algorithm: the INTERPRESS-IPD collaboration. *Hypertension* 2021;**77**:650–61. https://doi.org/10.1161/hypertensionaha.120.15997
39. Picone DS, Deshpande RA, Schultz MG, Fonseca R, Campbell NRC, Delles C, *et al.* Nonvalidated home blood pressure devices dominate the online marketplace in Australia: major implications for cardiovascular risk management. *Hypertension* 2020;**75**:1593–9. https://doi.org/10.1161/hypertensionaha.120.14719
40. Picone DS, Campbell NRC, Schutte AE, Olsen MH, Ordunez P, Whelton PK, *et al.* Validation status of blood pressure measuring devices sold globally. *JAMA* 2022;**327**:680–1. https://doi.org/10.1001/jama.2021.24464
41. Stergiou GS, O'Brien E, Myers M, Palatini P, Parati G, Kollias A, *et al.* STRIDE BP international initiative for accurate blood pressure measurement: systematic review of published validation studies of blood pressure measuring devices. *J Clin Hypertens (Greenwich)* 2019;**21**:1616–22. https://doi.org/10.1111/jch.13710
42. Stergiou GS, Alpert B, Mieke S, Asmar R, Atkins N, Eckert S, *et al.* A universal standard for the validation of blood pressure measuring devices: Association for the Advancement of Medical Instrumentation/European Society of Hypertension/International Organization for Standardization (AAMI/ESH/ISO) collaboration statement. *Hypertension* 2018;**71**:368–74. https://doi.org/10.1161/hypertensionaha.117.10237
43. Stergiou GS, Mukkamala R, Avolio A, Kyriakoulis KG, Mieke S, Murray A, *et al.* Cuffless blood pressure measuring devices: review and statement by the European Society of Hypertension working group on blood pressure monitoring and cardiovascular variability. *J Hypertens* 2022;**40**:1449–60. https://doi.org/10.1097/hjh.0000000000003224
44. Stergiou GS, Avolio AP, Palatini P, Kyriakoulis KG, Schutte AE, Mieke S, *et al.* European Society of Hypertension recommendations for the validation of cuffless blood pressure measuring devices: European Society of Hypertension Working Group on Blood Pressure Monitoring and Cardiovascular Variability. *J Hypertens* 2023;**41**:2074–87. https://doi.org/10.1097/hjh.0000000000003483
45. McAlister FA, Straus SE. Evidence based treatment of hypertension. Measurement of blood pressure: an evidence based review. *BMJ* 2001;**322**:908–11. https://doi.org/10.1136/bmj.322.7291.908
46. Fagius J, Karhuvaara S. Sympathetic activity and blood pressure increases with bladder distension in humans. *Hypertension* 1989;**14**:511–7. https://doi.org/10.1161/01.hyp.14.5.511
47. Stergiou GS, Kyriakoulis KG, Stambolliu E, Destounis A, Karpettas N, Kalogeropoulos P, *et al.* Blood pressure measurement in atrial fibrillation: review and meta-analysis of evidence on accuracy and clinical relevance. *J Hypertens* 2019;**37**:2430–41. https://doi.org/10.1097/hjh.0000000000002201
48. Clark CE, McDonagh STJ, McManus RJ. Accuracy of automated blood pressure measurements in the presence of atrial fibrillation: systematic review and meta-analysis. *J Hum Hypertens* 2019;**33**:352–64. https://doi.org/10.1038/s41371-018-0153-z
49. Verberk WJ, Omboni S, Kollias A, Stergiou GS. Screening for atrial fibrillation with automated blood pressure measurement: research evidence and practice recommendations. *Int J Cardiol* 2016;**203**:465–73. https://doi.org/10.1016/j.ijcard.2015.10.182
50. Manning DM, Kuchirka C, Kaminski J. Miscuffing: inappropriate blood pressure cuff application. *Circulation* 1983;**68**:763–6. https://doi.org/10.1161/01.cir.68.4.763
51. Irving G, Holden J, Stevens R, McManus RJ. Which cuff should I use? Indirect blood pressure measurement for the diagnosis of hypertension in patients with obesity: a diagnostic accuracy review. *BMJ Open* 2016;**6**:e012429. https://doi.org/10.1136/bmjopen-2016-012429
52. Keeley EC, Villanueva M, Chen YE, Gong Y, Handberg EM, Smith SM, *et al.* Attended vs unattended systolic blood pressure measurement: a randomized comparison in patients with cardiovascular disease. *J Clin Hypertens (Greenwich)* 2020;**22**:1987–92. https://doi.org/10.1111/jch.14037
53. Myers MG, Godwin M, Dawes M, Kiss A, Tobe SW, Grant FC, *et al.* Conventional versus automated measurement of blood pressure in primary care patients with systolic hypertension: randomised parallel design controlled trial. *BMJ* 2011;**342**:d286. https://doi.org/10.1136/bmj.d286
54. Clark CE. Inter-arm blood pressure difference, when is it a useful risk marker for cardiovascular events? *J Hum Hypertens* 2022;**36**:117–9. https://doi.org/10.1038/s41371-021-00629-x
55. Clark CE, Steele AM, Taylor RS, Shore AC, Ukoumunne OC, Campbell JL. Interarm blood pressure difference in people with diabetes: measurement and vascular and mortality implications: a cohort study. *Diabetes Care* 2014;**37**:1613–20. https://doi.org/10.2337/dc13-1576
56. Juraschek SP, Appel LJ, Mitchell CM, Mukamal KJ, Lipsitz LA, Blackford AL, *et al.* Comparison of supine and seated orthostatic hypotension assessments and their association with falls and orthostatic symptoms. *J Am Geriatr Soc* 2022;**70**:2310–9. https://doi.org/10.1111/jgs.17804

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

57. Stergiou GS, Palatini P, Parati G, O'Brien E, Januszewicz A, Lurbe E, *et al.* European Society of Hypertension practice guidelines for office and out-of-office blood pressure measurement. *J Hypertens* 2021;**39**:1293–302. https://doi.org/10.1097/hjh.0000000000002843
58. Parati G, Stergiou GS, Bilo G, Kollias A, Pengo M, Ochoa J, *et al.* Home blood pressure monitoring: methodology, clinical relevance and practical application: a 2021 position paper by the working group on blood pressure monitoring and cardiovascular variability of the European Society of Hypertension. *J Hypertens* 2021;**39**:1742–67. https://doi.org/10.1097/hjh.0000000000002922
59. Bradley CK, Choi E, Abdalla M, Mizuno H, Lam M, Cepeda M, *et al.* Use of different blood pressure thresholds to reduce the number of home blood pressure monitoring days needed for detecting hypertension. *Hypertension* 2023;**80**:2169–77. https://doi.org/10.1161/hypertensionaha.123.21118
60. Hodgkinson JA, Lee MM, Milner S, Bradburn P, Stevens R, Hobbs FDR, *et al.* Accuracy of blood-pressure monitors owned by patients with hypertension (ACCU-RATE study): a cross-sectional, observational study in central England. *Br J Gen Pract* 2020;**70**:e548–54. https://doi.org/10.3399/bjgp20X710381
61. Niiranen TJ, Asayama K, Thijs L, Johansson JK, Ohkubo T, Kikuya M, *et al.* Outcome-driven thresholds for home blood pressure measurement: international database of home blood pressure in relation to cardiovascular outcome. *Hypertension* 2013;**61**:27–34. https://doi.org/10.1161/hypertensionaha.111.00100
62. Parati G, Stergiou G, O'Brien E, Asmar R, Beilin L, Bilo G, *et al.* European Society of Hypertension practice guidelines for ambulatory blood pressure monitoring. *J Hypertens* 2014;**32**:1359–66. https://doi.org/10.1097/hjh.0000000000000221
63. Yang WY, Thijs L, Zhang ZY, Asayama K, Boggia J, Hansen TW, *et al.* Evidence-based proposal for the number of ambulatory readings required for assessing blood pressure level in research settings: an analysis of the IDACO database. *Blood Press* 2018;**27**:341–50. https://doi.org/10.1080/08037051.2018.1476057
64. Asayama K, Stolarz-Skrzypek K, Yang WY, Hansen TW, Brguljan-Hitij J, Odili AN, *et al.* What did we learn from the international databases on ambulatory and home blood pressure in relation to cardiovascular outcome? *Hypertens Res* 2023;**46**:934–49. https://doi.org/10.1038/s41440-023-01191-4
65. Drawz PE, Agarwal A, Dwyer JP, Horwitz E, Lash J, Lenoir K, *et al.* Concordance between blood pressure in the systolic blood pressure intervention trial and in routine clinical practice. *JAMA Intern Med* 2020;**180**:1655–63. https://doi.org/10.1001/jamainternmed.2020.5028
66. Jaeger BC, Bress AP, Bundy JD, Cheung AK, Cushman WC, Drawz PE, *et al.* Longer-term all-cause and cardiovascular mortality with intensive blood pressure control: a secondary analysis of a randomized clinical trial. *JAMA Cardiol* 2022;**7**:1138–46. https://doi.org/10.1001/jamacardio.2022.3345
67. Ntineri A, Niiranen TJ, McManus RJ, Lindroos A, Jula A, Schwartz C, *et al.* Ambulatory versus home blood pressure monitoring: frequency and determinants of blood pressure difference and diagnostic disagreement. *J Hypertens* 2019;**37**:1974–81. https://doi.org/10.1097/hjh.0000000000002148
68. Sheppard JP, Fletcher B, Gill P, Martin U, Roberts N, McManus RJ. Predictors of the home-clinic blood pressure difference: a systematic review and meta-analysis. *Am J Hypertens* 2016;**29**:614–25. https://doi.org/10.1093/ajh/hpv157
69. Franklin SS, Thijs L, Li Y, Hansen TW, Boggia J, Liu Y, *et al.* Masked hypertension in diabetes mellitus: treatment implications for clinical practice. *Hypertension* 2013;**61**:964–71. https://doi.org/10.1161/hypertensionaha.111.00289
70. Guirguis-Blake JM, Evans CV, Webber EM, Coppola EL, Perdue LA, Weyrich MS. Screening for hypertension in adults: updated evidence report and systematic review for the US preventive services task force. *JAMA* 2021;**325**:1657–69. https://doi.org/10.1001/jama.2020.21669
71. Xu W, Goldberg SI, Shubina M, Turchin A. Optimal systolic blood pressure target, time to intensification, and time to follow-up in treatment of hypertension: population based retrospective cohort study. *BMJ* 2015;**350**:h158. https://doi.org/10.1136/bmj.h158
72. Antza C, Farmakis I, Doundoulakis I, Akrivos E, Stalikas N, Zafeiropoulos S, *et al.* Reproducibility of masked hypertension and office-based hypertension: a systematic review and meta-analysis. *J Hypertens* 2022;**40**:1053–9. https://doi.org/10.1097/hjh.0000000000003111
73. Uhlig K, Patel K, Ip S, Kitsios GD, Balk EM. Self-measured blood pressure monitoring in the management of hypertension: a systematic review and meta-analysis. *Ann Intern Med* 2013;**159**:185–94. https://doi.org/10.7326/0003-4819-159-3-201308060-00008
74. Tucker KL, Sheppard JP, Stevens R, Bosworth HB, Bove A, Bray EP, *et al.* Self-monitoring of blood pressure in hypertension: a systematic review and individual patient data meta-analysis. *PLoS Med* 2017;**14**:e1002389. https://doi.org/10.1371/journal.pmed.1002389
75. McManus RJ, Mant J, Franssen M, Nickless A, Schwartz C, Hodgkinson J, *et al.* Efficacy of self-monitored blood pressure, with or without telemonitoring, for titration of antihypertensive medication (TASMINH4): an unmasked randomised controlled trial. *Lancet* 2018;**391**:949–59. https://doi.org/10.1016/s0140-6736(18)30309-x
76. McManus RJ, Little P, Stuart B, Morton K, Raftery J, Kelly J, *et al.* Home and Online Management and Evaluation of Blood Pressure (HOME BP) using a digital intervention in poorly controlled hypertension: randomised controlled trial. *BMJ* 2021;**372**:m4858. https://doi.org/10.1136/bmj.m4858
77. Li R, Liang N, Bu F, Hesketh T. The effectiveness of self-management of hypertension in adults using mobile health: systematic review and meta-analysis. *JMIR Mhealth Uhealth* 2020;**8**:e17776. https://doi.org/10.2196/17776
78. Acharya S, Neupane G, Seals A, Madhav KC, Giustini D, Sharma S, *et al.* Self-measured blood pressure-guided pharmacotherapy: a systematic review and meta-analysis of US-based telemedicine trials. *Hypertension* 2024;**81**:648–57. https://doi.org/10.1161/hypertensionaha.123.22109
79. Monahan M, Jowett S, Nickless A, Franssen M, Grant S, Greenfield S, *et al.* Cost-effectiveness of telemonitoring and self-monitoring of blood pressure for antihypertensive titration in primary care (TASMINH4). *Hypertension* 2019;**73**:1231–9. https://doi.org/10.1161/hypertensionaha.118.12415
80. Wood S, Greenfield SM, Sayeed Haque M, Martin U, Gill PS, Mant J, *et al.* Influence of ethnicity on acceptability of method of blood pressure monitoring: a cross-sectional study in primary care. *Br J Gen Pract* 2016;**66**:e577–586. https://doi.org/10.3399/bjgp16X685717
81. Stergiou GS, Karpettas N, Destounis A, Tzamouranis D, Nasothimiou E, Kollias A, *et al.* Home blood pressure monitoring alone vs. combined clinic and ambulatory measurements in following treatment-induced changes in blood pressure and organ damage. *Am J Hypertens* 2014;**27**:184–92. https://doi.org/10.1093/ajh/hpt206
82. Staessen JA, Byttebier G, Buntinx F, Celis H, O'Brien ET, Fagard R. Antihypertensive treatment based on conventional or ambulatory blood pressure measurement. A randomized controlled trial. Ambulatory blood pressure monitoring and treatment of hypertension investigators. *JAMA* 1997;**278**:1065–72. https://doi.org/10.1001/jama.1997.03550130039034
83. McEvoy JW, Leahy N, Parati G. The apples and oranges of blood pressure variability. *Hypertension* 2023;**80**:2556–8. https://doi.org/10.1161/hypertensionaha.123.21927
84. Ishikuro M, Obara T, Metoki H, Ohkubo T, Yamamoto M, Akutsu K, *et al.* Blood pressure measured in the clinic and at home during pregnancy among nulliparous and multiparous women: the BOSHI study. *Am J Hypertens* 2013;**26**:141–8. https://doi.org/10.1093/ajh/hps002
85. Bello NA, Woolley JJ, Cleary KL, Falzon L, Alpert BS, Oparil S, *et al.* Accuracy of blood pressure measurement devices in pregnancy: a systematic review of validation studies. *Hypertension* 2018;**71**:326–35. https://doi.org/10.1161/hypertensionaha.117.10295
86. Tucker KL, Mort S, Yu LM, Campbell H, Rivero-Arias O, Wilson HM, *et al.* Effect of self-monitoring of blood pressure on diagnosis of hypertension during higher-risk pregnancy: the BUMP 1 randomized clinical trial. *JAMA* 2022;**327**:1656–65. https://doi.org/10.1001/jama.2022.4712
87. Chappell LC, Tucker KL, Galal U, Yu L-M, Campbell H, Rivero-Arias O, *et al.* Effect of self-monitoring of blood pressure on blood pressure control in pregnant individuals with chronic or gestational hypertension: the BUMP 2 randomized clinical trial. *JAMA* 2022;**327**:1666–78. https://doi.org/10.1001/jama.2022.4726
88. Tita AT, Szychowski JM, Boggess K, Dugoff L, Sibai B, Lawrence K, *et al.* Treatment for mild chronic hypertension during pregnancy. *N Engl J Med* 2022;**386**:1781–92. https://doi.org/10.1056/NEJMoa2201295
89. Regitz-Zagrosek V, Roos-Hesselink JW, Bauersachs J, Blomström-Lundqvist C, Cífková R, De Bonis M, *et al.* 2018 ESC Guidelines for the management of cardiovascular diseases during pregnancy. *Eur Heart J* 2018;**39**:3165–241. https://doi.org/10.1093/eurheartj/ehy340
90. Tikhonoff V, Kuznetsova T, Thijs L, Cauwenberghs N, Stolarz-Skrzypek K, Seidlerová J, *et al.* Ambulatory blood pressure and long-term risk for atrial fibrillation. *Heart* 2018;**104**:1263–70. https://doi.org/10.1136/heartjnl-2017-312488
91. Lip GYH, Coca A, Kahan T, Boriani G, Manolis AS, Olsen MH, *et al.* Hypertension and cardiac arrhythmias: a consensus document from the European Heart Rhythm Association (EHRA) and ESC Council on Hypertension, endorsed by the Heart Rhythm Society (HRS), Asia-Pacific Heart Rhythm Society (APHRS) and Sociedad Latinoamericana de Estimulación Cardíaca y Electrofisiología (SOLEACE). *Europace* 2017;**19**:891–911. https://doi.org/10.1093/europace/eux091
92. Šelmytė-Besusparė A, Barysienė J, Petrikonytė D, Aidietis A, Marinskis G, Laucevičius A. Auscultatory versus oscillometric blood pressure measurement in patients with atrial fibrillation and arterial hypertension. *BMC Cardiovasc Disord* 2017;**17**:87. https://doi.org/10.1186/s12872-017-0521-6
93. Omboni S, Ballatore T, Rizzi F, Tomassini F, Campolo L, Panzeri E, *et al.* 24-Hour ambulatory blood pressure telemonitoring in patients at risk of atrial fibrillation: results from the TEMPLAR project. *Hypertens Res* 2022;**45**:1486–95. https://doi.org/10.1038/s41440-022-00932-1
94. Seccia TM, Letizia C, Muiesan ML, Lerco S, Cesari M, Bisogni V, *et al.* Atrial fibrillation as presenting sign of primary aldosteronism: results of the prospective appraisal on the prevalence of primary aldosteronism in hypertensive (PAPPHY) study. *J Hypertens* 2020;**38**:332–9. https://doi.org/10.1097/hjh.0000000000002250
95. McDonagh STJ, Mejzner N, Clark CE. Prevalence of postural hypotension in primary, community and institutional care: a systematic review and meta-analysis. *BMC Fam Pract* 2021;**22**:1. https://doi.org/10.1186/s12875-020-01313-8
96. Tran J, Hillebrand SL, Meskers CGM, Iseli RK, Maier AB. Prevalence of initial orthostatic hypotension in older adults: a systematic review and meta-analysis. *Age Ageing* 2021;**50**:1520–8. https://doi.org/10.1093/ageing/afab090
97. Juraschek SP, Hu JR, Cluett JL, Ishak AM, Mita C, Lipsitz LA, *et al.* Orthostatic hypotension, hypertension treatment, and cardiovascular disease: an individual participant meta-analysis. *JAMA* 2023;**330**:1459–71. https://doi.org/10.1001/jama.2023.18497

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

98. Juraschek SP, Cortez MM, Flack JM, Ghazi L, Kenny RA, Rahman M, *et al.* Orthostatic hypotension in adults with hypertension: a scientific statement from the American Heart Association. *Hypertension* 2024;**81**:e16–30. https://doi.org/10.1161/hyp.0000000000000236
99. Raber I, Belanger MJ, Farahmand R, Aggarwal R, Chiu N, Al Rifai M, *et al.* Orthostatic hypotension in hypertensive adults: Harry Goldblatt award for early career investigators 2021. *Hypertension* 2022;**79**:2388–96. https://doi.org/10.1161/hypertensionaha.122.18557
100. Wieling W, Kaufmann H, Claydon VE, van Wijnen VK, Harms MPM, Juraschek SP, *et al.* Diagnosis and treatment of orthostatic hypotension. *Lancet Neurol* 2022;**21**:735–46. https://doi.org/10.1016/s1474-4422(22)00169-7
101. Juraschek SP, Daya N, Rawlings AM, Appel LJ, Miller ER, Windham BG, *et al.* Association of history of dizziness and long-term adverse outcomes with early vs later orthostatic hypotension assessment times in middle-aged adults. *JAMA Intern Med* 2017;**177**:1316–23. https://doi.org/10.1001/jamainternmed.2017.2937
102. Brignole M, Moya A, de Lange FJ, Deharo J-C, Elliott PM, Fanciulli A, *et al.* Practical instructions for the 2018 ESC Guidelines for the diagnosis and management of syncope. *Eur Heart J* 2018;**39**:e43–80. https://doi.org/10.1093/eurheartj/ehy071
103. Rivasi G, Groppelli A, Brignole M, Soranna D, Zambon A, Bilo G, *et al.* Association between hypotension during 24 h ambulatory blood pressure monitoring and reflex syncope: the SynABPM 1 study. *Eur Heart J* 2022;**43**:3765–76. https://doi.org/10.1093/eurheartj/ehac347
104. Smith TO, Sillito JA, Goh CH, Abdel-Fattah A-R, Einarsson A, Soiza RL, *et al.* Association between different methods of assessing blood pressure variability and incident cardiovascular disease, cardiovascular mortality and all-cause mortality: a systematic review. *Age Ageing* 2020;**49**:184–92. https://doi.org/10.1093/ageing/afz178
105. Kim S, Xiao X, Chen J. Advances in photoplethysmography for personalized cardiovascular monitoring. *Biosensors (Basel)* 2022;**12**:863. https://doi.org/10.3390/bios12100863
106. Dagamseh A, Qananwah Q, Al Quran H, Shaker Ibrahim K. Towards a portable-noninvasive blood pressure monitoring system utilizing the photoplethysmogram signal. *Biomed Opt Express* 2021;**12**:7732–51. https://doi.org/10.1364/boe.444535
107. Sharman JE, Avolio AP, Baulmann J, Benetos A, Blacher J, Blizzard CL, *et al.* Validation of non-invasive central blood pressure devices: ARTERY Society task force consensus statement on protocol standardization. *Eur Heart J* 2017;**38**:2805–12. https://doi.org/10.1093/eurheartj/ehw632
108. Sharman JE, O'Brien E, Alpert B, Schutte AE, Delles C, Hecht Olsen M, *et al.* Lancet Commission on Hypertension group position statement on the global improvement of accuracy standards for devices that measure blood pressure. *J Hypertens* 2020;**38**:21–9. https://doi.org/10.1097/hjh.0000000000002246
109. Lunardi M, Muhammad F, Shahzad A, Nadeem A, Combe L, Simpkin AJ, *et al.* Performance of wearable watch-type home blood pressure measurement devices in a real-world clinical sample. *Clin Res Cardiol*. 2023. https://doi.org/10.1007/s00392-023-02353-7
110. Clark CE, Warren FC, Boddy K, McDonagh STJ, Moore SF, Teresa Alzamora M, *et al.* Higher arm versus lower arm systolic blood pressure and cardiovascular outcomes: a meta-analysis of individual participant data from the INTERPRESS-IPD collaboration. *Hypertension* 2022;**79**:2328–35. https://doi.org/10.1161/hypertensionaha.121.18921
111. Margolis KL, Dehmer SP, Sperl-Hillen J, O'Connor PJ, Asche SE, Bergdall AR, *et al.* Cardiovascular events and costs with home blood pressure telemonitoring and pharmacist management for uncontrolled hypertension. *Hypertension* 2020;**76**:1097–103. https://doi.org/10.1161/hypertensionaha.120.15492
112. Sheppard JP, Tucker KL, Davison WJ, Stevens R, Aekplakorn W, Bosworth H B, *et al.* Self-monitoring of blood pressure in patients with hypertension-related multimorbidity: systematic review and individual patient data meta-analysis. *Am J Hypertens* 2020;**33**:243–51. https://doi.org/10.1093/ajh/hpz182
113. Cooke G, Doust J, Sanders S. Is pulse palpation helpful in detecting atrial fibrillation? A systematic review. *J Fam Pract* 2006;**55**:130–4.
114. Arvanitis M, Qi G, Bhatt DL, Post WS, Chatterjee N, Battle A, *et al.* Linear and nonlinear Mendelian randomization analyses of the association between diastolic blood pressure and cardiovascular events: the J-curve revisited. *Circulation* 2021;**143**:895–906. https://doi.org/10.1161/circulationaha.120.049819
115. Whelton SP, McEvoy JW, Shaw L, Psaty BM, Lima JAC, Budoff M, *et al.* Association of normal systolic blood pressure level with cardiovascular disease in the absence of risk factors. *JAMA Cardiol* 2020;**5**:1011–8. https://doi.org/10.1001/jamacardio.2020.1731
116. Rahimi K, Bidel Z, Nazarzadeh M, Copland E, Canoy D, Ramakrishnan R, *et al.* Pharmacological blood pressure lowering for primary and secondary prevention of cardiovascular disease across different levels of blood pressure: an individual participant-level data meta-analysis. *Lancet* 2021;**397**:1625–36. https://doi.org/10.1016/S0140-6736(21)00590-0
117. Ji H, Niiranen TJ, Rader F, Henglin M, Kim A, Ebinger JE, *et al.* Sex differences in blood pressure associations with cardiovascular outcomes. *Circulation* 2021;**143**:761–3. https://doi.org/10.1161/circulationaha.120.049360
118. Kringeland E, Tell GS, Midtbø H, Igland J, Haugsgjerd TR, Gerdts E. Stage 1 hypertension, sex, and acute coronary syndromes during midlife: the Hordaland health study. *Eur J Prev Cardiol* 2022;**29**:147–54. https://doi.org/10.1093/eurjpc/zwab068
119. Appel LJ, Moore TJ, Obarzanek E, Vollmer WM, Svetkey LP, Sacks FM, *et al.* A clinical trial of the effects of dietary patterns on blood pressure. DASH collaborative research group. *N Engl J Med* 1997;**336**:1117–24. https://doi.org/10.1056/nejm199704173361601
120. Toledo E, Hu FB, Estruch R, Buil-Cosiales P, Corella D, Salas-Salvadó J, *et al.* Effect of the Mediterranean diet on blood pressure in the PREDIMED trial: results from a randomized controlled trial. *BMC Med* 2013;**11**:207. https://doi.org/10.1186/1741-7015-11-207
121. Sundström J, Arima H, Jackson R, Turnbull F, Rahimi K, Chalmers J, *et al.* Effects of blood pressure reduction in mild hypertension: a systematic review and meta-analysis. *Ann Intern Med* 2015;**162**:184–91. https://doi.org/10.7326/m14-0773
122. Ettehad D, Emdin CA, Kiran A, Anderson SG, Callender T, Emberson J, *et al.* Blood pressure lowering for prevention of cardiovascular disease and death: a systematic review and meta-analysis. *Lancet* 2016;**387**:957–67. https://doi.org/10.1016/s0140-6736(15)01225-8
123. Muntner P, Whelton PK. Using predicted cardiovascular disease risk in conjunction with blood pressure to guide antihypertensive medication treatment. *J Am Coll Cardiol* 2017;**69**:2446–56. https://doi.org/10.1016/j.jacc.2017.02.066
124. Herrett E, Strongman H, Gadd S, Tomlinson L, Nitsch D, Bhaskaran K, *et al.* The importance of blood pressure thresholds versus predicted cardiovascular risk on subsequent rates of cardiovascular disease: a cohort study in English primary care. *Lancet Healthy Longev* 2022;**3**:e22–30. https://doi.org/10.1016/s2666-7568(21)00281-6
125. Navar AM, Pencina MJ, Peterson ED. Assessing cardiovascular risk to guide hypertension diagnosis and treatment. *JAMA Cardiol* 2016;**1**:864–71. https://doi.org/10.1001/jamacardio.2016.2861
126. Kovell LC, Ahmed HM, Misra S, Whelton SP, Prokopowicz GP, Blumenthal RS, *et al.* US hypertension management guidelines: a review of the recent past and recommendations for the future. *J Am Heart Assoc* 2015;**4**:e002315. https://doi.org/10.1161/jaha.115.002315
127. McEvoy JW, Martin SS, Dardari ZA, Miedema MD, Sandfort V, Yeboah J, *et al.* Coronary artery calcium to guide a personalized risk-based approach to initiation and intensification of antihypertensive therapy. *Circulation* 2017;**135**:153–65. https://doi.org/10.1161/circulationaha.116.025471
128. Rossello X, Dorresteijn JA, Janssen A, Lambrinou E, Scherrenberg M, Bonnefoy-Cudraz E, *et al.* Risk prediction tools in cardiovascular disease prevention: a report from the ESC Prevention of CVD Programme led by the European Association of Preventive Cardiology (EAPC) in collaboration with the Acute Cardiovascular Care Association (ACCA) and the Association of Cardiovascular Nursing and Allied Professions (ACNAP). *Eur J Prev Cardiol* 2019;**26**:1534–44. https://doi.org/10.1177/2047487319846715
129. Ioannidis JPA. Diagnosis and treatment of hypertension in the 2017 ACC/AHA guidelines and in the real world. *JAMA* 2018;**319**:115–6. https://doi.org/10.1001/jama.2017.19672
130. Whelton PK, Carey RM, Aronow WS, Casey DE, Collins KJ, Dennison Himmelfarb C, *et al.* 2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA guideline for the prevention, detection, evaluation, and management of high blood pressure in adults: a report of the American College of Cardiology/American Heart Association task force on clinical practice guidelines. *Circulation* 2018;**138**:e484–594. https://doi.org/10.1161/cir.0000000000000596
131. Rahimi K, Bidel Z, Nazarzadeh M, Copland E, Canoy D, Wamil M, *et al.* Age-stratified and blood-pressure-stratified effects of blood-pressure-lowering pharmacotherapy for the prevention of cardiovascular disease and death: an individual participant-level data meta-analysis. *Lancet* 2021;**398**:1053–64. https://doi.org/10.1016/S0140-6736(21)01921-8
132. Brunström M, Carlberg B. Association of blood pressure lowering with mortality and cardiovascular disease across blood pressure levels: a systematic review and meta-analysis. *JAMA Intern Med* 2018;**178**:28–36. https://doi.org/10.1001/jamainternmed.2017.6015
133. Arguedas JA, Leiva V, Wright JM. Blood pressure targets in adults with hypertension. *Cochrane Database Syst Rev* 2020;**12**:CD004349. https://doi.org/10.1002/14651858.CD004349.pub3
134. Saiz LC, Gorricho J, Garjón J, Celaya MC, Erviti J, Leache L. Blood pressure targets for the treatment of people with hypertension and cardiovascular disease. *Cochrane Database Syst Rev* 2022;**11**:CD010315. https://doi.org/10.1002/14651858.CD010315.pub5
135. Wright JT Jr, Williamson JD, Whelton PK, Snyder JK, Sink KM, Rocco MV, *et al.* A randomized trial of intensive versus standard blood-pressure control. *N Engl J Med* 2015;**373**:2103–16. https://doi.org/10.1056/NEJMoa1511939
136. Zhang W, Zhang S, Deng Y, Wu S, Ren J, Sun G, *et al.* Trial of intensive blood-pressure control in older patients with hypertension. *N Engl J Med* 2021;**385**:1268–79. https://doi.org/10.1056/NEJMoa2111437
137. Cushman WC, Evans GW, Byington RP, Goff DC Jr, Grimm RH Jr, Cutler JA, *et al.* Effects of intensive blood-pressure control in type 2 diabetes mellitus. *N Engl J Med* 2010;**362**:1575–85. https://doi.org/10.1056/NEJMoa1001286
138. Ferrucci L, Furberg CD, Penninx BW, DiBari M, Williamson JD, Guralnik JM, *et al.* Treatment of isolated systolic hypertension is most effective in older patients with high-risk profile. *Circulation* 2001;**104**:1923–6. https://doi.org/10.1161/hc4101.097520

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

139. Leening MJ, Ferket BS, Steyerberg EW, Kavousi M, Deckers JW, Nieboer D, *et al.* Sex differences in lifetime risk and first manifestation of cardiovascular disease: prospective population based cohort study. *BMJ* 2014;**349**:g5992. https://doi.org/10.1136/bmj.g5992
140. Ference BA, Bhatt DL, Catapano AL, Packard CJ, Graham I, Kaptoge S, *et al.* Association of genetic variants related to combined exposure to lower low-density lipoproteins and lower systolic blood pressure with lifetime risk of cardiovascular disease. *JAMA* 2019;**322**:1381–91. https://doi.org/10.1001/jama.2019.14120
141. Kannel WB. Risk stratification in hypertension: new insights from the Framingham study. *Am J Hypertens* 2000;**13**:S3–10. https://doi.org/10.1016/s0895-7061(99)00252-6
142. Weycker D, Nichols GA, O'Keeffe-Rosetti M, Edelsberg J, Khan Z, Kaura S, *et al.* Risk-factor clustering and cardiovascular disease risk in hypertensive patients. *Am J Hypertens* 2007;**20**:599–607. https://doi.org/10.1016/j.amjhyper.2006.10.013
143. Constanti M, Floyd CN, Glover M, Boffa R, Wierzbicki AS, McManus RJ. Cost-effectiveness of initiating pharmacological treatment in stage one hypertension based on 10-year cardiovascular disease risk: a Markov modeling study. *Hypertension* 2021;**77**:682–91. https://doi.org/10.1161/hypertensionaha.120.14913
144. Forouzanfar MH, Liu P, Roth GA, Ng M, Biryukov S, Marczak L, *et al.* Global burden of hypertension and systolic blood pressure of at least 110 to 115 mm Hg, 1990–2015. *JAMA* 2017;**317**:165–82. https://doi.org/10.1001/jama.2016.19043
145. Karmali KN, Ning H, Goff DC, Lloyd-Jones DM. Identifying individuals at risk for cardiovascular events across the spectrum of blood pressure levels. *J Am Heart Assoc* 2015;**4**:e002126. https://doi.org/10.1161/jaha.115.002126
146. Liu J, Li Y, Ge J, Yan X, Zhang H, Zheng X, *et al.* Lowering systolic blood pressure to less than 120 mm Hg versus less than 140 mm Hg in patients with high cardiovascular risk with and without diabetes or previous stroke: an open-label, blinded-outcome, randomised trial. *Lancet* 2024;**404**:245–55. https://doi.org/10.1016/S0140-6736(24)01028-6
147. Wilson PW, D'Agostino RB, Levy D, Belanger AM, Silbershatz H, Kannel WB. Prediction of coronary heart disease using risk factor categories. *Circulation* 1998;**97**:1837–47. https://doi.org/10.1161/01.cir.97.18.1837
148. Sussman J, Vijan S, Hayward R. Using benefit-based tailored treatment to improve the use of antihypertensive medications. *Circulation* 2013;**128**:2309–17. https://doi.org/10.1161/circulationaha.113.002290
149. Karmali KN, Lloyd-Jones DM, van der Leeuw J, Goff DC, Yusuf S, Zanchetti A, *et al.* Blood pressure-lowering treatment strategies based on cardiovascular risk versus blood pressure: a meta-analysis of individual participant data. *PLoS Med* 2018;**15**:e1002538. https://doi.org/10.1371/journal.pmed.1002538
150. Gaziano TA, Steyn K, Cohen DJ, Weinstein MC, Opie LH. Cost-effectiveness analysis of hypertension guidelines in South Africa: absolute risk versus blood pressure level. *Circulation* 2005;**112**:3569–76. https://doi.org/10.1161/circulationaha.105.535922
151. Basu S, Sussman JB, Hayward RA. Black-white cardiovascular disease disparities after target-based versus personalized benefit-based lipid and blood pressure treatment. *MDM Policy Pract* 2017;**2**:2381468317725741. https://doi.org/10.1177/2381468317725741
152. Go AS, Chertow GM, Fan D, McCulloch CE, Hsu CY. Chronic kidney disease and the risks of death, cardiovascular events, and hospitalization. *N Engl J Med* 2004;**351**:1296–305. https://doi.org/10.1056/NEJMoa041031
153. Klooster CCV, Bhatt DL, Steg PG, Massaro JM, Dorresteijn JAN, Westerink J, *et al.* Predicting 10-year risk of recurrent cardiovascular events andcardiovascular interventions in patients with established cardiovascular disease: results from UCC-SMART and REACH. *Int J Cardiol* 2021;**325**:140–8. https://doi.org/10.1016/j.ijcard.2020.09.053
154. Kaasenbrood L, Boekholdt SM, van der Graaf Y, Ray KK, Peters RJG, Kastelein JJP, *et al.* Distribution of estimated 10-year risk of recurrent vascular events and residual risk in a secondary prevention population. *Circulation* 2016;**134**:1419–29. https://doi.org/10.1161/circulationaha.116.021314
155. Eisen A, Bhatt DL, Steg PG, Eagle KA, Goto S, Guo J, *et al.* Angina and future cardiovascular events in stable patients with coronary artery disease: insights from the Reduction of Atherothrombosis for Continued Health (REACH) registry. *J Am Heart Assoc* 2016;**5**:e004080. https://doi.org/10.1161/jaha.116.004080
156. Taylor CJ, Roalfe AK, Iles R, Hobbs FD. Ten-year prognosis of heart failure in the community: follow-up data from the Echocardiographic Heart of England Screening (ECHOES) study. *Eur J Heart Fail* 2012;**14**:176–84. https://doi.org/10.1093/eurjhf/hfr170
157. Shah KS, Xu H, Matsouaka RA, Bhatt DL, Heidenreich PA, Hernandez AF, *et al.* Heart failure with preserved, borderline, and reduced ejection fraction: 5–year outcomes. *J Am Coll Cardiol* 2017;**70**:2476–86. https://doi.org/10.1016/j.jacc.2017.08.074
158. Benjamin EJ, Wolf PA, D'Agostino RB, Silbershatz H, Kannel WB, Levy D. Impact of atrial fibrillation on the risk of death: the Framingham heart study. *Circulation* 1998;**98**:946–52. https://doi.org/10.1161/01.cir.98.10.946
159. Pandey A, Patel KV, Vongpatanasin W, Ayers C, Berry JD, Mentz RJ, *et al.* Incorporation of biomarkers into risk assessment for allocation of antihypertensive medication according to the 2017 ACC/AHA high blood pressure guideline: a pooled cohort analysis. *Circulation* 2019;**140**:2076–88. https://doi.org/10.1161/circulationaha.119.043337
160. Mulnier HE, Seaman HE, Raleigh VS, Soedamah-Muthu SS, Colhoun HM, Lawrenson RA, *et al.* Risk of myocardial infarction in men and women with type 2 diabetes in the UK: a cohort study using the general practice research database. *Diabetologia* 2008;**51**:1639–45. https://doi.org/10.1007/s00125-008-1076-y
161. Soedamah-Muthu SS, Fuller JH, Mulnier HE, Raleigh VS, Lawrenson RA, Colhoun HM. High risk of cardiovascular disease in patients with type 1 diabetes in the U.K.: a cohort study using the general practice research database. *Diabetes Care* 2006;**29**:798–804. https://doi.org/10.2337/diacare.29.04.06.dc05-1433
162. Wong ND, Glovaci D, Wong K, Malik S, Franklin SS, Wygant G, *et al.* Global cardiovascular disease risk assessment in United States adults with diabetes. *Diab Vasc Dis Res* 2012;**9**:146–52. https://doi.org/10.1177/1479164112436403
163. Paquette M, Bernard S, Cariou B, Hegele RA, Genest J, Trinder M, *et al.* Familial hypercholesterolemia-risk-score: a new score predicting cardiovascular events and cardiovascular mortality in familial hypercholesterolemia. *Arterioscler Thromb Vasc Biol* 2021;**41**:2632–40. https://doi.org/10.1161/atvbaha.121.316106
164. SCORE2-Diabetes Working Group and the ESC Cardiovascular Risk Collaboration. SCORE2-Diabetes: 10-year cardiovascular risk estimation in type 2 diabetes in Europe. *Eur Heart J* 2023;**44**:2544–56. https://doi.org/10.1093/eurheartj/ehad260
165. SCORE Working Group and the ESC Cardiovascular Risk Collaboration. SCORE2 risk prediction algorithms: new models to estimate 10-year risk of cardiovascular disease in Europe. *Eur Heart J* 2021;**42**:2439–54. https://doi.org/10.1093/eurheartj/ehab309
166. SCORE OP working group and ESC Cardiovascular Risk Collaboration. SCORE2-OP risk prediction algorithms: estimating incident cardiovascular event risk in older persons in four geographical risk regions. *Eur Heart J* 2021;**42**:2455–67. https://doi.org/10.1093/eurheartj/ehab312
167. Pignone M, Phillips CJ, Elasy TA, Fernandez A. Physicians' ability to predict the risk of coronary heart disease. *BMC Health Serv Res* 2003;**3**:13. https://doi.org/10.1186/1472-6963-3-13
168. Grover SA, Lowensteyn I, Esrey KL, Steinert Y, Joseph L, Abrahamowicz M. Do doctors accurately assess coronary risk in their patients? Preliminary results of the coronary health assessment study. *BMJ* 1995;**310**:975–8. https://doi.org/10.1136/bmj.310.6985.975
169. Friedmann PD, Brett AS, Mayo-Smith MF. Differences in generalists' and cardiologists' perceptions of cardiovascular risk and the outcomes of preventive therapy in cardiovascular disease. *Ann Intern Med* 1996;**124**:414–21. https://doi.org/10.7326/0003-4819-124-4-199602150-00005
170. Visseren FLJ, Mach F, Smulders YM, Carballo D, Koskinas KC, Bäck M, *et al.* ESC guidelines on cardiovascular disease prevention in clinical practice. *Eur Heart J* 2021;**42**:3227–337. https://doi.org/10.1093/eurheartj/ehab484
171. Xie X, Atkins E, Lv J, Bennett A, Neal B, Ninomiya T, *et al.* Effects of intensive blood pressure lowering on cardiovascular and renal outcomes: updated systematic review and meta-analysis. *Lancet* 2016;**387**:435–43. https://doi.org/10.1016/s0140-6736(15)00805-3
172. Herrett E, Gadd S, Jackson R, Bhaskaran K, Williamson E, van Staa T, *et al.* Eligibility and subsequent burden of cardiovascular disease of four strategies for blood pressure-lowering treatment: a retrospective cohort study. *Lancet* 2019;**394**:663–71. https://doi.org/10.1016/s0140-6736(19)31359-5
173. Matsushita K, Kaptoge S, Hageman SHJ, Sang Y, Ballew SH, Grams ME, *et al.* Including measures of chronic kidney disease to improve cardiovascular risk prediction by SCORE2 and SCORE2-OP. *Eur J Prev Cardiol* 2023;**30**:8–16. https://doi.org/10.1093/eurjpc/zwac176
174. Gerdts E, Sudano I, Brouwers S, Borghi C, Bruno RM, Ceconi C, *et al.* Sex differences in arterial hypertension. *Eur Heart J* 2022;**43**:4777–88. https://doi.org/10.1093/eurheartj/ehac470
175. Brown MC, Best KE, Pearce MS, Waugh J, Robson SC, Bell R. Cardiovascular disease risk in women with pre-eclampsia: systematic review and meta-analysis. *Eur J Epidemiol* 2013;**28**:1–19. https://doi.org/10.1007/s10654-013-9762-6
176. Honigberg MC, Zekavat SM, Aragam K, Klarin D, Bhatt DL, Scott NS, *et al.* Long-term cardiovascular risk in women with hypertension during pregnancy. *J Am Coll Cardiol* 2019;**74**:2743–54. https://doi.org/10.1016/j.jacc.2019.09.052
177. Leon LJ, McCarthy FP, Direk K, Gonzalez-Izquierdo A, Prieto-Merino D, Casas JP, *et al.* Preeclampsia and cardiovascular disease in a large UK pregnancy cohort of linked electronic health records: a CALIBER study. *Circulation* 2019;**140**:1050–60. https://doi.org/10.1161/circulationaha.118.038080
178. Haug EB, Horn J, Markovitz AR, Fraser A, Klykken B, Dalen H, *et al.* Association of conventional cardiovascular risk factors with cardiovascular disease after hypertensive disorders of pregnancy: analysis of the Nord-Trøndelag health study. *JAMA Cardiol* 2019;**4**:628–35. https://doi.org/10.1001/jamacardio.2019.1746
179. Wu P, Haththotuwa R, Kwok CS, Babu A, Kotronias RA, Rushton C, *et al.* Preeclampsia and future cardiovascular health: a systematic review and meta-analysis. *Circ Cardiovasc Qual Outcomes* 2017;**10**:e003497. https://doi.org/10.1161/circoutcomes.116.003497
180. Kramer CK, Campbell S, Retnakaran R. Gestational diabetes and the risk of cardiovascular disease in women: a systematic review and meta-analysis. *Diabetologia* 2019;**62**:905–14. https://doi.org/10.1007/s00125-019-4840-2
181. Kyriacou H, Al-Mohammad A, Muehlschlegel C, Foster-Davies L, Bruco MEF, Legard C, *et al.* The risk of cardiovascular diseases after miscarriage, stillbirth, and induced abortion: a systematic review and meta-analysis. *Eur Heart J Open* 2022;**2**:oeac065. https://doi.org/10.1093/ehjopen/oeac065

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

182. Kessous R, Shoham-Vardi I, Pariente G, Holcberg G, Sheiner E. An association between preterm delivery and long-term maternal cardiovascular morbidity. *Am J Obstet Gynecol* 2013;**209**:368.e1–8. https://doi.org/10.1016/j.ajog.2013.05.041
183. Markovitz AR, Stuart JJ, Horn J, Williams PL, Rimm EB, Missmer SA, *et al.* Does pregnancy complication history improve cardiovascular disease risk prediction? Findings from the HUNT study in Norway. *Eur Heart J* 2019;**40**:1113–20. https://doi.org/10.1093/eurheartj/ehy863
184. Saei Ghare Naz M, Sheidaei A, Aflatounian A, Azizi F, Ramezani Tehrani F. Does adding adverse pregnancy outcomes improve the Framingham cardiovascular risk score in women? Data from the Tehran Lipid and Glucose Study. *J Am Heart Assoc* 2022;**11**:e022349. https://doi.org/10.1161/jaha.121.022349
185. Gladstone RA, Pudwell J, Nerenberg KA, Grover SA, Smith GN. Cardiovascular risk assessment and follow-up of women after hypertensive disorders of pregnancy: a prospective cohort study. *J Obstet Gynaecol Can* 2019;**41**:1157–67.e1. https://doi.org/10.1016/j.jogc.2018.10.024
186. Patel AP, Wang M, Kartoun U, Ng K, Khera AV. Quantifying and understanding the higher risk of atherosclerotic cardiovascular disease among South Asian individuals: results from the UK Biobank prospective cohort study. *Circulation* 2021;**144**:410–22. https://doi.org/10.1161/circulationaha.120.052430
187. Tillin T, Hughes AD, Whincup P, Mayet J, Sattar N, McKeigue PM, *et al.* Ethnicity and prediction of cardiovascular disease: performance of QRISK2 and Framingham scores in a U.K. tri-ethnic prospective cohort study (SABRE—Southall And Brent REvisited). *Heart* 2014;**100**:60–7. https://doi.org/10.1136/heartjnl-2013-304474
188. Rabanal KS, Igland J, Tell GS, Jenum AK, Klemsdal TO, Ariansen I, *et al.* Validation of the cardiovascular risk model NORRISK 2 in South Asians and people with diabetes. *Scand Cardiovasc J* 2021;**55**:56–62. https://doi.org/10.1080/14017431.2020.1821909
189. Veronesi G, Gianfagna F, Giampaoli S, Chambless LE, Mancia G, Cesana G, *et al.* Improving long-term prediction of first cardiovascular event: the contribution of family history of coronary heart disease and social status. *Prev Med* 2014;**64**:75–80. https://doi.org/10.1016/j.ypmed.2014.04.007
190. Sivapalaratnam S, Boekholdt SM, Trip MD, Sandhu MS, Luben R, Kastelein JJP, *et al.* Family history of premature coronary heart disease and risk prediction in the EPIC-Norfolk prospective population study. *Heart* 2010;**96**:1985–9. https://doi.org/10.1136/hrt.2010.210740
191. Kimenai DM, Pirondini L, Gregson J, Prieto D, Pocock SJ, Perel P, *et al.* Socioeconomic deprivation: an important, largely unrecognized risk factor in primary prevention of cardiovascular disease. *Circulation* 2022;**146**:240–8. https://doi.org/10.1161/circulationaha.122.060042
192. Avina-Zubieta JA, Thomas J, Sadatsafavi M, Lehman AJ, Lacaille D. Risk of incident cardiovascular events in patients with rheumatoid arthritis: a meta-analysis of observational studies. *Ann Rheum Dis* 2012;**71**:1524–9. https://doi.org/10.1136/annrheumdis-2011-200726
193. Crowson CS, Matteson EL, Roger VL, Therneau TM, Gabriel SE. Usefulness of risk scores to estimate the risk of cardiovascular disease in patients with rheumatoid arthritis. *Am J Cardiol* 2012;**110**:420–4. https://doi.org/10.1016/j.amjcard.2012.03.044
194. Arts EE, Popa C, Den Broeder AA, Semb A G, Toms T, Kitas G D, *et al.* Performance of four current risk algorithms in predicting cardiovascular events in patients with early rheumatoid arthritis. *Ann Rheum Dis* 2015;**74**:668–74. https://doi.org/10.1136/annrheumdis-2013-204024
195. Arts EE, Popa CD, Den Broeder AA, Donders R, Sandoo A, Toms T, *et al.* Prediction of cardiovascular risk in rheumatoid arthritis: performance of original and adapted SCORE algorithms. *Ann Rheum Dis* 2016;**75**:674–80. https://doi.org/10.1136/annrheumdis-2014-206879
196. Gelfand JM, Neimann AL, Shin DB, Wang X, Margolis DJ, Troxel AB. Risk of myocardial infarction in patients with psoriasis. *JAMA* 2006;**296**:1735–41. https://doi.org/10.1001/jama.296.14.1735
197. Miller IM, Ellervik C, Yazdanyar S, Jemec GB. Meta-analysis of psoriasis, cardiovascular disease, and associated risk factors. *J Am Acad Dermatol* 2013;**69**:1014–24. https://doi.org/10.1016/j.jaad.2013.06.053
198. Eder L, Chandran V, Gladman DD. The Framingham risk score underestimates the extent of subclinical atherosclerosis in patients with psoriatic disease. *Ann Rheum Dis* 2014;**73**:1990–6. https://doi.org/10.1136/annrheumdis-2013-203433
199. Armstrong EJ, Harskamp CT, Armstrong AW. Psoriasis and major adverse cardiovascular events: a systematic review and meta-analysis of observational studies. *J Am Heart Assoc* 2013;**2**:e000062. https://doi.org/10.1161/jaha.113.000062
200. Elmets CA, Leonardi CL, Davis DMR, Gelfand JM, Lichten J, Mehta NN, *et al.* Joint AAD-NPF guidelines of care for the management and treatment of psoriasis with awareness and attention to comorbidities. *J Am Acad Dermatol* 2019;**80**:1073–113. https://doi.org/10.1016/j.jaad.2018.11.058
201. Lu X, Wang Y, Zhang J, Pu D, Hu N, Luo J, *et al.* Patients with systemic lupus erythematosus face a high risk of cardiovascular disease: a systematic review and meta-analysis. *Int Immunopharmacol* 2021;**94**:107466. https://doi.org/10.1016/j.intimp.2021.107466
202. Drosos GC, Konstantonis G, Sfikakis PP, Tektonidou MG. Underperformance of clinical risk scores in identifying vascular ultrasound-based high cardiovascular risk in systemic lupus erythematosus. *Eur J Prev Cardiol*. 2020. https://doi.org/10.1177/2047487320906650
203. Shah ASV, Stelzle D, Lee KK, Beck EJ, Alam S, Clifford S, *et al.* Global burden of atherosclerotic cardiovascular disease in people living with HIV: systematic review and meta-analysis. *Circulation* 2018;**138**:1100–12. https://doi.org/10.1161/circulationaha.117.033369
204. Triant VA, Perez J, Regan S, Massaro JM, Meigs JB, Grinspoon SK, *et al.* Cardiovascular risk prediction functions underestimate risk in HIV infection. *Circulation* 2018;**137**:2203–14. https://doi.org/10.1161/circulationaha.117.028975
205. Triant VA, Lyass A, Hurley LB, Borowsky LH, Ehrbar RQ, H W, *et al.* Cardiovascular Risk Estimation Is Suboptimal in People With HIV. *J Am Heart Assoc.* 2024;**13**:e029228. https://doi.org/10.1161/JAHA.123.029228
206. Correll CU, Solmi M, Veronese N, Bortolato B, Rosson S, Santonastaso P, *et al.* Prevalence, incidence and mortality from cardiovascular disease in patients with pooled and specific severe mental illness: a large-scale meta-analysis of 3,211,768 patients and 113,383,368 controls. *World Psychiatry* 2017;**16**:163–80. https://doi.org/10.1002/wps.20420
207. Lambert AM, Parretti HM, Pearce E, Price MJ, Riley M, Ryan R, *et al.* Temporal trends in associations between severe mental illness and risk of cardiovascular disease: a systematic review and meta-analysis. *PLoS Med* 2022;**19**:e1003960. https://doi.org/10.1371/journal.pmed.1003960
208. Cunningham R, Poppe K, Peterson D, Every-Palmer S, Soosay I, Jackson R. Prediction of cardiovascular disease risk among people with severe mental illness: a cohort study. *PLoS One* 2019;**14**:e0221521. https://doi.org/10.1371/journal.pone.0221521
209. Yeboah J, Young R, McClelland RL, Delaney JC, Polonsky TS, Dawood FZ, *et al.* Utility of nontraditional risk markers in atherosclerotic cardiovascular disease risk assessment. *J Am Coll Cardiol* 2016;**67**:139–47. https://doi.org/10.1016/j.jacc.2015.10.058
210. Akintoye E, Afonso L, Bengaluru Jayanna M, Bao W, Briasoulis A, Robinson J, *et al.* Prognostic utility of risk enhancers and coronary artery calcium score recommended in the 2018 ACC/AHA multisociety cholesterol treatment guidelines over the pooled cohort equation: insights from 3 large prospective cohorts. *J Am Heart Assoc* 2021;**10**:e019589. https://doi.org/10.1161/jaha.120.019589
211. Peters SA, den Ruijter HM, Bots ML, Moons KG. Improvements in risk stratification for the occurrence of cardiovascular disease by imaging subclinical atherosclerosis: a systematic review. *Heart* 2012;**98**:177–84. https://doi.org/10.1136/heartjnl-2011-300747
212. Nicolaides AN, Panayiotou AG, Griffin M, Tyllis T, Bond D, Georgiou N, *et al.* Arterial ultrasound testing to predict atherosclerotic cardiovascular events. *J Am Coll Cardiol* 2022;**79**:1969–82. https://doi.org/10.1016/j.jacc.2022.03.352
213. Laclaustra M, Casasnovas JA, Fernández-Ortiz A, Fuster V, León-Latre M, Jiménez-Borreguero LJ, *et al.* Femoral and carotid subclinical atherosclerosis association with risk factors and coronary calcium: the AWHS study. *J Am Coll Cardiol* 2016;**67**:1263–74. https://doi.org/10.1016/j.jacc.2015.12.056
214. López-Melgar B, Fernández-Friera L, Oliva B, García-Ruiz JM, Peñalvo JL, Gómez-Talavera S, *et al.* Subclinical atherosclerosis burden by 3D ultrasound in midlife: the PESA study. *J Am Coll Cardiol* 2017;**70**:301–13. https://doi.org/10.1016/j.jacc.2017.05.033
215. Ben-Shlomo Y, Spears M, Boustred C, May M, Anderson SG, Benjamin EJ, *et al.* Aortic pulse wave velocity improves cardiovascular event prediction: an individual participant meta-analysis of prospective observational data from 17,635 subjects. *J Am Coll Cardiol* 2014;**63**:636–46. https://doi.org/10.1016/j.jacc.2013.09.063
216. Ohkuma T, Ninomiya T, Tomiyama H, Kario K, Hoshide S, Kita Y, *et al.* Brachial-Ankle pulse wave velocity and the risk prediction of cardiovascular disease: an individual participant data meta-analysis. *Hypertension* 2017;**69**:1045–52. https://doi.org/10.1161/hypertensionaha.117.09097
217. Stone K, Veerasingam D, Meyer ML, Heffernan KS, Higgins S, Maria Bruno R, *et al.* Reimagining the value of brachial-ankle pulse wave velocity as a biomarker of cardiovascular disease risk-a call to action on behalf of VascAgeNet. *Hypertension* 2023;**80**:1980–92. https://doi.org/10.1161/hypertensionaha.123.21314
218. An DW, Hansen TW, Aparicio LS, Chori B, Huang Q-F, Wei F-F, *et al.* Derivation of an outcome-driven threshold for aortic pulse wave velocity: an individual-participant meta-analysis. *Hypertension* 2023;**80**:1949–59. https://doi.org/10.1161/hypertensionaha.123.21318
219. de Lemos JA, Ayers CR, Levine BD, deFilippi CR, Wang TJ, Hundley W, *et al.* Multimodality strategy for cardiovascular risk assessment: performance in 2 population-based cohorts. *Circulation* 2017;**135**:2119–32. https://doi.org/10.1161/circulationaha.117.027272
220. Hageman SHJ, Petitjean C, Pennells L, Kaptoge S, Pajouheshnia R, Tillmann T, *et al.* Improving 10-year cardiovascular risk prediction in apparently healthy people: flexible addition of risk modifiers on top of SCORE2. *Eur J Prev Cardiol* 2023;**30**:1705–14. https://doi.org/10.1093/eurjpc/zwad187
221. McEvoy JW, Chen Y, Nambi V, Ballantyne CM, Sharrett AR, Appel LJ, *et al.* High-sensitivity cardiac troponin T and risk of hypertension. *Circulation* 2015;**132**:825–33. https://doi.org/10.1161/circulationaha.114.014364
222. Hussain A, Sun W, Deswal A, de Lemos JA, McEvoy JW, Hoogeveen RC, *et al.* Association of NT-ProBNP, blood pressure, and cardiovascular events: the ARIC study. *J Am Coll Cardiol* 2021;**77**:559–71. https://doi.org/10.1016/j.jacc.2020.11.063
223. Stuart JJ, Tanz LJ, Cook NR, Spiegelman D, Missmer SA, Rimm EB, *et al.* Hypertensive disorders of pregnancy and 10-year cardiovascular risk prediction. *J Am Coll Cardiol* 2018;**72**:1252–63. https://doi.org/10.1016/j.jacc.2018.05.077

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

224. Timpka S, Fraser A, Schyman T, Stuart JJ, Åsvold BO, Mogren I, *et al.* The value of pregnancy complication history for 10-year cardiovascular disease risk prediction in middle-aged women. *Eur J Epidemiol* 2018;**33**:1003–10. https://doi.org/10.1007/s10654-018-0429-1
225. Blankenberg S, Salomaa V, Makarova N, Ojeda F, Wild P, Lackner KJ, *et al.* Troponin I and cardiovascular risk prediction in the general population: the BiomarCaRE consortium. *Eur Heart J* 2016;**37**:2428–37. https://doi.org/10.1093/eurheartj/ehw172
226. Zeller T, Tunstall-Pedoe H, Saarela O, Ojeda F, Schnabel RB, Tuovinen T, *et al.* High population prevalence of cardiac troponin I measured by a high-sensitivity assay and cardiovascular risk estimation: the MORGAM biomarker project Scottish cohort. *Eur Heart J* 2014;**35**:271–81. https://doi.org/10.1093/eurheartj/eht406
227. Schoenthaler AM, Lancaster KJ, Chaplin W, Butler M, Forsyth J, Ogedegbe G, *et al.* Cluster randomized clinical trial of FAITH (faith-based approaches in the treatment of hypertension) in blacks. *Circ Cardiovasc Qual Outcomes* 2018;**11**:e004691. https://doi.org/10.1161/circoutcomes.118.004691
228. Beaney T, Schutte AE, Stergiou GS, Borghi C, Burger D, Charchar F, *et al.* May Measurement Month 2019: the global blood pressure screening campaign of the international society of Hypertension. *Hypertension* 2020;**76**:333–41. https://doi.org/10.1161/hypertensionaha.120.14874
229. Victor RG, Lynch K, Li N, Blyler C, Muhammad E, Handler J, *et al.* A cluster-randomized trial of blood-pressure reduction in black barbershops. *N Engl J Med* 2018;**378**:1291–301. https://doi.org/10.1056/NEJMoa1717250
230. Sun Y, Mu J, Wang DW, Ouyang N, Xing L, Guo X, *et al.* A village doctor-led multifaceted intervention for blood pressure control in rural China: an open, cluster randomised trial. *Lancet* 2022;**399**:1964–75. https://doi.org/10.1016/s0140-6736(22)00325-7
231. Schmidt BM, Durao S, Toews I, Bavuma CM, Hohlfeld A, Nury E, *et al.* Screening strategies for hypertension. *Cochrane Database Syst Rev* 2020;**5**:CD013212. https://doi.org/10.1002/14651858.CD013212.pub2
232. Kaczorowski J, Chambers LW, Dolovich L, Paterson JM, Karwalajtys T, Gierman T, *et al.* Improving cardiovascular health at population level: 39 community cluster randomised trial of Cardiovascular Health Awareness Program (CHAP). *BMJ* 2011;**342**:d442. https://doi.org/10.1136/bmj.d442
233. Lindholt JS, Søgaard R, Rasmussen LM, Mejldal A, Lambrechtsen J, Steffensen FH, *et al.* Five-year outcomes of the Danish cardiovascular screening (DANCAVAS) trial. *N Engl J Med* 2022;**387**:1385–94. https://doi.org/10.1056/NEJMoa2208681
234. Sheppard JP, Schwartz CL, Tucker KL, McManus RJ. Modern management and diagnosis of hypertension in the United Kingdom: home care and self-care. *Ann Glob Health* 2016;**82**:274–87. https://doi.org/10.1016/j.aogh.2016.02.005
235. Andersson H, Hedström L, Bergman S, Bergh H. The outcome of two-step blood pressure screening in dental healthcare. *Scand J Public Health* 2018;**46**:623–9. https://doi.org/10.1177/1403494818759840
236. Zhang H, Thijs L, Kuznetsova T, Fagard RH, Li X, Staessen JA. Progression to hypertension in the non-hypertensive participants in the Flemish study on environment, genes and health outcomes. *J Hypertens* 2006;**24**:1719–27. https://doi.org/10.1097/01.hjh.0000242395.07473.92
237. Conen D, Aeschbacher S, Thijs L, Li Y, Boggia J, Asayama K, *et al.* Age-specific differences between conventional and ambulatory daytime blood pressure values. *Hypertension* 2014;**64**:1073–9. https://doi.org/10.1161/hypertensionaha.114.03957
238. Karnjanapiboonwong A, Anothaisintawee T, Chaikledkaew U, Dejthevaporn C, Attia J, Thakkinstian A. Diagnostic performance of clinic and home blood pressure measurements compared with ambulatory blood pressure: a systematic review and meta-analysis. *BMC Cardiovasc Disord* 2020;**20**:491. https://doi.org/10.1186/s12872-020-01736-2
239. Viera AJ, Yano Y, Lin FC, Simel DL, Yun J, Dave G, *et al.* Does this adult patient have hypertension?: The rational clinical examination systematic review. *JAMA* 2021;**326**:339–47. https://doi.org/10.1001/jama.2021.4533
240. Green BB, Anderson ML, Cook AJ, Ehrlich K, Hall YN, Hsu C, *et al.* Clinic, home, and kiosk blood pressure measurements for diagnosing hypertension: a randomized diagnostic study. *J Gen Intern Med* 2022;**37**:2948–56. https://doi.org/10.1007/s11606-022-07400-z
241. Kim JS, Rhee MY, Kim CH, Kim YR, Do U, Kim J-H, *et al.* Algorithm for diagnosing hypertension using out-of-office blood pressure measurements. *J Clin Hypertens (Greenwich)* 2021;**23**:1965–74. https://doi.org/10.1111/jch.14382
242. van den Born BH, Lip GYH, Brguljan-Hitij J, Cremer A, Segura J, Morales E, *et al.* ESC Council on Hypertension position document on the management of hypertensive emergencies. *Eur Heart J Cardiovasc Pharmacother* 2019;**5**:37–46. https://doi.org/10.1093/ehjcvp/pvy032
243. Ross S, Walker A, MacLeod MJ. Patient compliance in hypertension: role of illness perceptions and treatment beliefs. *J Hum Hypertens* 2004;**18**:607–13. https://doi.org/10.1038/sj.jhh.1001721
244. Hagger MS, Koch S, Chatzisarantis NLD, Orbell S. The common sense model of self-regulation: meta-analysis and test of a process model. *Psychol Bull* 2017;**143**:1117–54. https://doi.org/10.1037/bul0000118
245. Meyer D, Leventhal H, Gutmann M. Common-sense models of illness: the example of hypertension. *Health Psychol* 1985;**4**:115–35. https://doi.org/10.1037//0278-6133.4.2.115
246. O'Carroll RE, Chambers JA, Dennis M, Sudlow C, Johnston M. Improving adherence to medication in stroke survivors: a pilot randomised controlled trial. *Ann Behav Med* 2013;**46**:358–68. https://doi.org/10.1007/s12160-013-9515-5
247. Hollands GJ, Usher-Smith JA, Hasan R, Alexander F, Clarke N, Griffin SJ. Visualising health risks with medical imaging for changing recipients' health behaviours and risk factors: systematic review with meta-analysis. *PLoS Med* 2022;**19**:e1003920. https://doi.org/10.1371/journal.pmed.1003920
248. Burnier M. Medication adherence and persistence as the cornerstone of effective antihypertensive therapy. *Am J Hypertens* 2006;**19**:1190–6. https://doi.org/10.1016/j.amjhyper.2006.04.006
249. Parati G, Goncalves A, Soergel D, Bruno RM, Caiani EG, Gerdts E. New perspectives for hypertension management: progress in methodological and technological developments. *Eur J Prev Cardiol* 2023;**30**:48–60. https://doi.org/10.1093/eurjpc/zwac203
250. Naderi SH, Bestwick JP, Wald DS. Adherence to drugs that prevent cardiovascular disease: meta-analysis on 376,162 patients. *Am J Med* 2012;**125**:882–7.e1. https://doi.org/10.1016/j.amjmed.2011.12.013
251. Chang TE, Ritchey MD, Park S, Chang A, Odom EC, Durthaler J, *et al.* National rates of nonadherence to antihypertensive medications among insured adults with hypertension, 2015. *Hypertension* 2019;**74**:1324–32. https://doi.org/10.1161/hypertensionaha.119.13616
252. Qvarnström M, Kahan T, Kieler H, Brandt L, Hasselström J, Bengtsson Boström K, *et al.* Persistence to antihypertensive drug treatment in Swedish primary healthcare. *Eur J Clin Pharmacol* 2013;**69**:1955–64. https://doi.org/10.1007/s00228-013-1555-z
253. Burnier M, Egan BM. Adherence in hypertension. *Circ Res* 2019;**124**:1124–40. https://doi.org/10.1161/circresaha.118.313220
254. Curneen JMG, Rabbitt L, Browne D, O'Donoghue DF, Alansari Y, Harhen B, *et al.* Major disparities in patient-reported adherence compared to objective assessment of adherence using mass spectrometry: a prospective study in a tertiary-referral hypertension clinic. *Br J Clin Pharmacol* 2022;**89**:1948–55. https://doi.org/10.1111/bcp.15292
255. Corrao G, Parodi A, Nicotra F, Zambon A, Merlino L, Cesana G, *et al.* Better compliance to antihypertensive medications reduces cardiovascular risk. *J Hypertens* 2011;**29**:610–8. https://doi.org/10.1097/HJH.0b013e328342ca97
256. Kim S, Shin DW, Yun JM, Hwang Y, Park SK, Ko Y-J, *et al.* Medication adherence and the risk of cardiovascular mortality and hospitalization among patients with newly prescribed antihypertensive medications. *Hypertension* 2016;**67**:506–12. https://doi.org/10.1161/hypertensionaha.115.06731
257. Lane D, Lawson A, Burns A, Azizi M, Burnier M, Jones DJL, *et al.* Nonadherence in hypertension: how to develop and implement chemical adherence testing. *Hypertension* 2022;**79**:12–23. https://doi.org/10.1161/hypertensionaha.121.17596
258. Marshall IJ, Wolfe CD, McKevitt C. Lay perspectives on hypertension and drug adherence: systematic review of qualitative research. *BMJ* 2012;**345**:e3953. https://doi.org/10.1136/bmj.e3953
259. Schoenthaler A, Knafl GJ, Fiscella K, Ogedegbe G. Addressing the social needs of hypertensive patients: the role of patient–provider communication as a predictor of medication adherence. *Circ Cardiovasc Qual Outcomes* 2017;**10**:e003659. https://doi.org/10.1161/circoutcomes.117.003659
260. Schmieder RE, Wassmann S, Predel HG, Weisser B, Blettenberg J, Gillessen A, *et al.* Improved persistence to medication, decreased cardiovascular events and reduced all-cause mortality in hypertensive patients with use of single-pill combinations: results from the START-study. *Hypertension* 2023;**80**:1127–35. https://doi.org/10.1161/hypertensionaha.122.20810
261. Ruzicka M, Leenen FHH, Ramsay T, Bugeja A, Edwards C, McCormick B, *et al.* Use of directly observed therapy to assess treatment adherence in patients with apparent treatment-resistant hypertension. *JAMA Intern Med* 2019;**179**:1433–4. https://doi.org/10.1001/jamainternmed.2019.1455
262. Pio-Abreu A, Trani-Ferreira F, Silva GV, Bortolotto LA, Drager LF. Directly observed therapy for resistant/refractory hypertension diagnosis and blood pressure control. *Heart* 2022;**108**:1952–6. https://doi.org/10.1136/heartjnl-2022-320802
263. Tomaszewski M, White C, Patel P, Masca N, Damani R, Hepworth J, *et al.* High rates of non-adherence to antihypertensive treatment revealed by high-performance liquid chromatography-tandem mass spectrometry (HP LC-MS/MS) urine analysis. *Heart* 2014;**100**:855–61. https://doi.org/10.1136/heartjnl-2013-305063
264. Forette F, Seux ML, Staessen JA, *et al.* The prevention of dementia with antihypertensive treatment: new evidence from the systolic hypertension in Europe (Syst-Eur) study. *Arch Intern Med* 2002;**162**:2046–52. https://doi.org/10.1001/archinte.162.18.2046
265. Näslund U, Ng N, Lundgren A, Fhärm E, Grönlund C, Johansson H, *et al.* Visualization of asymptomatic atherosclerotic disease for optimum cardiovascular prevention (VIPVIZA): a pragmatic, open-label, randomised controlled trial. *Lancet* 2019;**393**:133–42. https://doi.org/10.1016/s0140-6736(18)32818-6
266. Bengtsson A, Norberg M, Ng N, Carlberg B, Grönlund C, Hultdin J, *et al.* The beneficial effect over 3 years by pictorial information to patients and their physician about subclinical atherosclerosis and cardiovascular risk: results from the VIPVIZA randomized clinical trial. *Am J Prev Cardiol* 2021;**7**:100199. https://doi.org/10.1016/j.ajpc.2021.100199
267. Sjölander M, Carlberg B, Norberg M, Näslund U, Ng N. Prescription of lipid-lowering and antihypertensive drugs following pictorial information about subclinical

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

atherosclerosis: a secondary outcome of a randomized clinical trial. *JAMA Netw Open* 2021;**4**:e2121683. https://doi.org/10.1001/jamanetworkopen.2021.21683
268. Muiesan ML, Salvetti M, Monteduro C, Bonzi B, Paini A, Viola S, *et al.* Left ventricular concentric geometry during treatment adversely affects cardiovascular prognosis in hypertensive patients. *Hypertension* 2004;**43**:731–8. https://doi.org/10.1161/01.HYP.0000121223.44837.de
269. Guerin AP, Blacher J, Pannier B, Marchais SJ, Safar ME, London GM, *et al.* Impact of aortic stiffness attenuation on survival of patients in end-stage renal failure. *Circulation* 2001;**103**:987–92. https://doi.org/10.1161/01.cir.103.7.987
270. Cardoso CRL, Salles GF. Prognostic value of changes in aortic stiffness for cardiovascular outcomes and mortality in resistant hypertension: a cohort study. *Hypertension* 2022;**79**:447–56. https://doi.org/10.1161/hypertensionaha.121.18498
271. Laurent S, Chatellier G, Azizi M, Calvet D, Choukroun G, Danchin N, *et al.* SPARTE study: normalization of arterial stiffness and cardiovascular events in patients with hypertension at medium to very high risk. *Hypertension* 2021;**78**:983–95. https://doi.org/10.1161/hypertensionaha.121.17579
272. Levin A, Stevens PE, Bilous RW, Coresh J, De Francisco ALM, De Jong PE, *et al.* Summary of recommendation statements. *Kidney Int Suppl (2011)* 2013;**3**:5–14. https://doi.org/10.1038/kisup.2012.77
273. Matsushita K, Coresh J, Sang Y, Chalmers J, Fox C, Guallar E, *et al.* Estimated glomerular filtration rate and albuminuria for prediction of cardiovascular outcomes: a collaborative meta-analysis of individual participant data. *Lancet Diabetes Endocrinol* 2015;**3**:514–25. https://doi.org/10.1016/s2213-8587(15)00040-6
274. Cheung AK, Rahman M, Reboussin DM, Craven TE, Greene T, Kimmel PL, *et al.* Effects of intensive BP control in CKD. *J Am Soc Nephrol* 2017;**28**:2812–23. https://doi.org/10.1681/asn.2017020148
275. Malhotra R, Nguyen HA, Benavente O, Mete M, Howard BV, Mant J, *et al.* Association between more intensive vs less intensive blood pressure lowering and risk of mortality in chronic kidney disease stages 3 to 5: a systematic review and meta-analysis. *JAMA Intern Med* 2017;**177**:1498–505. https://doi.org/10.1001/jamainternmed.2017.4377
276. Stevens PE, Levin A. Evaluation and management of chronic kidney disease: synopsis of the kidney disease: improving global outcomes 2012 clinical practice guideline. *Ann Intern Med* 2013;**158**:825–30. https://doi.org/10.7326/0003-4819-158-11-201306040-00007
277. Aboyans V, Ricco JB, Bartelink MEL, Björck M, Brodmann M, Cohnert T, *et al.* 2017 ESC Guidelines on the diagnosis and treatment of peripheral arterial diseases, in collaboration with the European Society for Vascular Surgery (ESVS): document covering atherosclerotic disease of extracranial carotid and vertebral, mesenteric, renal, upper and lower extremity arteriesEndorsed by: the European Stroke Organization (ESO) The Task Force for the Diagnosis and Treatment of Peripheral Arterial Diseases of the European Society of Cardiology (ESC) and of the European Society for Vascular Surgery (ESVS). *Eur Heart J* 2018;**39**:763–816. https://doi.org/10.1093/eurheartj/ehx095
278. Iribarren C, Round AD, Lu M, Okin PM, McNulty EJ. Cohort study of ECG left ventricular hypertrophy trajectories: ethnic disparities, associations with cardiovascular outcomes, and clinical utility. *J Am Heart Assoc* 2017;**6**:e004954. https://doi.org/10.1161/jaha.116.004954
279. Okin PM, Devereux RB, Jern S, Kjeldsen SE, Julius S, Dahlöf B. Baseline characteristics in relation to electrocardiographic left ventricular hypertrophy in hypertensive patients: the Losartan Intervention For Endpoint reduction (LIFE) in hypertension study. *Hypertension* 2000;**36**:766–73. https://doi.org/10.1161/01.hyp.36.5.766
280. Okin PM, Oikarinen L, Viitasalo M, Toivonen L, Kjeldsen SE, Nieminen MS, *et al.* Prognostic value of changes in the electrocardiographic strain pattern during antihypertensive treatment: the Losartan Intervention for End-point reduction in hypertension study (LIFE). *Circulation* 2009;**119**:1883–91. https://doi.org/10.1161/circulationaha.108.812313
281. Lehtonen AO, Puukka P, Varis J, Porthan K, Tikkanen JT, Nieminen MS, *et al.* Prevalence and prognosis of ECG abnormalities in normotensive and hypertensive individuals. *J Hypertens* 2016;**34**:959–66. https://doi.org/10.1097/hjh.0000000000000882
282. Pewsner D, Jüni P, Egger M, Battaglia M, Sundström J, Bachmann LM. Accuracy of electrocardiography in diagnosis of left ventricular hypertrophy in arterial hypertension: systematic review. *BMJ* 2007;**335**:711. https://doi.org/10.1136/bmj.39276.636354.AE
283. Kuznetsova T, Thijs L, Knez J, Cauwenberghs N, Petit T, Gu Y-M. Longitudinal changes in left ventricular diastolic function in a general population. *Circ Cardiovasc Imaging* 2015;**8**:e002882. https://doi.org/10.1161/circimaging.114.002882
284. Zhao L, Zierath R, Claggett B, Dorbala P, Matsushita K, Kitzman D. Longitudinal changes in left ventricular diastolic function in late life: the ARIC study. *JACC Cardiovasc Imaging* 2023;**16**:1133–45. https://doi.org/10.1016/j.jcmg.2023.02.022
285. Redfield MM, Jacobsen SJ, Burnett JC Jr, Mahoney DW, Bailey KR, Rodeheffer RJ. Burden of systolic and diastolic ventricular dysfunction in the community: appreciating the scope of the heart failure epidemic. *JAMA* 2003;**289**:194–202. https://doi.org/10.1001/jama.289.2.194
286. Sundström J, Lind L, Arnlöv J, Zethelius B, Andrén B, Lithell HO. Echocardiographic and electrocardiographic diagnoses of left ventricular hypertrophy predict mortality independently of each other in a population of elderly men. *Circulation* 2001;**103**:2346–51. https://doi.org/10.1161/01.cir.103.19.2346
287. Seko Y, Kato T, Yamaji Y, Haruna Y, Nakane E, Haruna T, *et al.* Discrepancy between left ventricular hypertrophy by echocardiography and electrocardiographic hypertrophy: clinical characteristics and outcomes. *Open Heart* 2021;**8**:e001765. https://doi.org/10.1136/openhrt-2021-001765
288. Modin D, Biering-Sørensen SR, Mogelvang R, Landler N, Jensen JS, Biering-Sørensen T. Prognostic value of echocardiography in hypertensive versus nonhypertensive participants from the general population. *Hypertension* 2018;**71**:742–51. https://doi.org/10.1161/hypertensionaha.117.10674
289. Armstrong AC, Jacobs DR Jr, Gidding SS, Colangelo LA, Gjesdal O, Lewis CE, *et al.* Framingham score and LV mass predict events in young adults: CARDIA study. *Int J Cardiol* 2014;**172**:350–5. https://doi.org/10.1016/j.ijcard.2014.01.003
290. Kuznetsova T, Thijs L, Knez J, Herbots L, Zhang Z, Staessen JA. Prognostic value of left ventricular diastolic dysfunction in a general population. *J Am Heart Assoc* 2014;**3**:e000789. https://doi.org/10.1161/jaha.114.000789
291. Marwick TH, Gillebert TC, Aurigemma G, *et al.* Recommendations on the use of echocardiography in adult hypertension: a report from the European Association of Cardiovascular Imaging (EACVI) and the American Society of Echocardiography (ASE). *Eur Heart J Cardiovasc Imaging* 2015;**16**:577–605. https://doi.org/10.1093/ehjci/jev076
292. Douglas PS, Garcia MJ, Haines DE, Lai WW, Manning WJ, Patel AR, *et al.* ACCF/ASE/AHA/ASNC/HFSA/HRS/SCAI/SCCM/SCCT/SCMR 2011 appropriate use criteria for echocardiography. A report of the American College of Cardiology Foundation appropriate use criteria task force, American Society of Echocardiography, American Heart Association, American Society of Nuclear Cardiology, Heart Failure Society of America, Heart Rhythm Society, Society for Cardiovascular Angiography and Interventions, Society of Critical Care Medicine, Society of Cardiovascular Computed Tomography, Society for Cardiovascular Magnetic Resonance American College of Chest Physicians. *J Am Soc Echocardiogr* 2011;**24**:229–67. https://doi.org/10.1016/j.echo.2010.12.008
293. van Rosendael AR, Bax AM, Smit JM, van den Hoogen IJ, Ma X, Al'Aref S, *et al.* Clinical risk factors and atherosclerotic plaque extent to define risk for major events in patients without obstructive coronary artery disease: the long-term coronary computed tomography angiography CONFIRM registry. *Eur Heart J Cardiovasc Imaging* 2020;**21**:479–88. https://doi.org/10.1093/ehjci/jez322
294. Mehta A, Pandey A, Ayers CR, Khera A, Sperling LS, Szklo M, *et al.* Predictive value of coronary artery calcium score categories for coronary events versus strokes: impact of sex and race: MESA and DHS. *Circ Cardiovasc Imaging* 2020;**13**:e010153. https://doi.org/10.1161/circimaging.119.010153
295. Bengtsson A, Lindvall K, Norberg M, Fhärm E. Increased knowledge makes a difference!—general practitioners' experiences of pictorial information about subclinical atherosclerosis for primary prevention: an interview study from the VIPVIZA trial. *Scand J Prim Health Care* 2021;**39**:77–84. https://doi.org/10.1080/02813432.2021.1882083
296. Ibanez B, Fernández-Ortiz A, Fernández-Friera L, García-Lunar I, Andrés V, Fuster V. Progression of early subclinical atherosclerosis (PESA) study: JACC focus seminar 7/8. *J Am Coll Cardiol* 2021;**78**:156–79. https://doi.org/10.1016/j.jacc.2021.05.011
297. Den Ruijter HM, Peters SA, Anderson TJ, Britton AR, Dekker JM, Eijkemans MJ. Common carotid intima-media thickness measurements in cardiovascular risk prediction: a meta-analysis. *JAMA* 2012;**308**:796–803. https://doi.org/10.1001/jama.2012.9630
298. Downie LE, Hodgson LA, DSylva C, McIntosh RL, Rogers SL, Connell P, *et al.* Hypertensive retinopathy: comparing the Keith-Wagener-Barker to a simplified classification. *J Hypertens* 2013;**31**:960–5. https://doi.org/10.1097/HJH.0b013e32835efea3
299. Liew G, Xie J, Nguyen H, Keay L, Kamran Ikram M, McGeechan K, *et al.* Hypertensive retinopathy and cardiovascular disease risk: 6 population-based cohorts meta-analysis. *Int J Cardiol Cardiovasc Risk Prev* 2023;**17**:200180. https://doi.org/10.1016/j.ijcrp.2023.200180
300. Padmanabhan S, Caulfield M, Dominiczak AF. Genetic and molecular aspects of hypertension. *Circ Res* 2015;**116**:937–59. https://doi.org/10.1161/circresaha.116.303647
301. Casey R, Neumann HPH, Maher ER. Genetic stratification of inherited and sporadic phaeochromocytoma and paraganglioma: implications for precision medicine. *Hum Mol Genet* 2020;**29**:R128–137. https://doi.org/10.1093/hmg/ddaa201
302. Rossi GP, Ceolotto G, Caroccia B, Lenzini L. Genetic screening in arterial hypertension. *Nat Rev Endocrinol* 2017;**13**:289–98. https://doi.org/10.1038/nrendo.2016.196
303. Cascón A, Calsina B, Monteagudo M, Mellid S, Díaz-Talavera A, Currás-Freixes M, *et al.* Genetic bases of pheochromocytoma and paraganglioma. *J Mol Endocrinol* 2023;**70**:e220167. https://doi.org/10.1530/jme-22-0167
304. Staessen JA, Wang J, Bianchi G, Birkenhäger WH. Essential hypertension. *Lancet* 2003;**361**:1629–41. https://doi.org/10.1016/s0140-6736(03)13302-8
305. Lenders JW, Duh QY, Eisenhofer G, Gimenez-Roqueplo A-P, Grebe SKG, Murad MH, *et al.* Pheochromocytoma and paraganglioma: an endocrine society clinical practice guideline. *J Clin Endocrinol Metab* 2014;**99**:1915–42. https://doi.org/10.1210/jc.2014-1498
306. Poulter NR, Prabhakaran D, Caulfield M. Hypertension. *Lancet* 2015;**386**:801–12. https://doi.org/10.1016/s0140-6736(14)61468-9
307. Mills KT, Bundy JD, Kelly TN, Reed JE, Kearney PM, Reynolds K, *et al.* Global disparities of hypertension prevalence and control: a systematic analysis of population-based

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

studies from 90 countries. *Circulation* 2016;**134**:441–50. https://doi.org/10.1161/circulationaha.115.018912
308. Wei FF, Zhang ZY, Huang QF, Staessen JA. Diagnosis and management of resistant hypertension: state of the art. *Nat Rev Nephrol* 2018;**14**:428–41. https://doi.org/10.1038/s41581-018-0006-6
309. Carey RM, Calhoun DA, Bakris GL, Brook RD, Daugherty SL, Dennison-Himmelfarb Cheryl R, *et al.* Resistant hypertension: detection, evaluation, and management: a scientific statement from the American Heart Association. *Hypertension* 2018;**72**:e53–90. https://doi.org/10.1161/hyp.0000000000000084
310. Carey RM, Sakhuja S, Calhoun DA, Whelton PK, Muntner P. Prevalence of apparent treatment-resistant hypertension in the United States. *Hypertension* 2019;**73**:424–31. https://doi.org/10.1161/hypertensionaha.118.12191
311. Vongpatanasin W. Resistant hypertension: a review of diagnosis and management. *JAMA* 2014;**311**:2216–24. https://doi.org/10.1001/jama.2014.5180
312. Azizi M, Sapoval M, Gosse P, Monge M, Bobrie G, Delsart P, *et al.* Optimum and stepped care standardised antihypertensive treatment with or without renal denervation for resistant hypertension (DENERHTN): a multicentre, open-label, randomised controlled trial. *Lancet* 2015;**385**:1957–65. https://doi.org/10.1016/s0140-6736(14)61942-5
313. Rossi GP, Bernini G, Caliumi C, Desideri G, Fabris B, Ferri C, *et al.* A prospective study of the prevalence of primary aldosteronism in 1,125 hypertensive patients. *J Am Coll Cardiol* 2006;**48**:2293–300. https://doi.org/10.1016/j.jacc.2006.07.059
314. Pedrosa RP, Drager LF, Gonzaga CC, Sousa MG, de Paula LKG, Amaro ACS, *et al.* Obstructive sleep apnea: the most common secondary cause of hypertension associated with resistant hypertension. *Hypertension* 2011;**58**:811–7. https://doi.org/10.1161/hypertensionaha.111.179788
315. Douma S, Petidis K, Doumas M, Papaefthimiou P, Triantafyllou A, Kartali N, *et al.* Prevalence of primary hyperaldosteronism in resistant hypertension: a retrospective observational study. *Lancet* 2008;**371**:1921–6. https://doi.org/10.1016/s0140-6736(08)60834-x
316. Monticone S, Burrello J, Tizzani D, Bertello C, Viola A, Buffolo F, *et al.* Prevalence and clinical manifestations of primary aldosteronism encountered in primary care practice. *J Am Coll Cardiol* 2017;**69**:1811–20. https://doi.org/10.1016/j.jacc.2017.01.052
317. Käyser SC, Deinum J, de Grauw WJ, Schalk BWM, Bor HJHJ, Lenders JWM, *et al.* Prevalence of primary aldosteronism in primary care: a cross-sectional study. *Br J Gen Pract* 2018;**68**:e114–22. https://doi.org/10.3399/bjgp18X694589
318. Rimoldi SF, Scherrer U, Messerli FH. Secondary arterial hypertension: when, who, and how to screen? *Eur Heart J* 2014;**35**:1245–54. https://doi.org/10.1093/eurheartj/eht534
319. Rossi GP, Bisogni V, Rossitto G, Maiolino G, Cesari M, Zhu R, *et al.* Practice recommendations for diagnosis and treatment of the most common forms of secondary hypertension. *High Blood Press Cardiovasc Prev* 2020;**27**:547–60. https://doi.org/10.1007/s40292-020-00415-9
320. Calhoun DA, Nishizaka MK, Zaman MA, Thakkar RB, Weissmann P. Hyperaldosteronism among black and white subjects with resistant hypertension. *Hypertension* 2002;**40**:892–6. https://doi.org/10.1161/01.hyp.0000040261.30455.b6
321. Jaffe G, Gray Z, Krishnan G, Stedman M, Zheng Y, Han J, *et al.* Screening rates for primary aldosteronism in resistant hypertension: a cohort study. *Hypertension* 2020;**75**:650–9. https://doi.org/10.1161/hypertensionaha.119.14359
322. Hundemer GL, Imsirovic H, Vaidya A, Yozamp N, Goupil R, Madore F, *et al.* Screening rates for primary aldosteronism among individuals with hypertension plus hypokalemia: a population-based retrospective cohort study. *Hypertension* 2022;**79**:178–86. https://doi.org/10.1161/hypertensionaha.121.18118
323. Monticone S, D'Ascenzo F, Moretti C, Williams TA, Veglio F, Gaita F, *et al.* Cardiovascular events and target organ damage in primary aldosteronism compared with essential hypertension: a systematic review and meta-analysis. *Lancet Diabetes Endocrinol* 2018;**6**:41–50. https://doi.org/10.1016/s2213-8587(17)30319-4
324. Savard S, Amar L, Plouin PF, Steichen O. Cardiovascular complications associated with primary aldosteronism: a controlled cross-sectional study. *Hypertension* 2013;**62**:331–6. https://doi.org/10.1161/hypertensionaha.113.01060
325. Hiramatsu K, Yamada T, Yukimura Y, Komiya I, Ichikawa K, Ishihara M, *et al.* A screening test to identify aldosterone-producing adenoma by measuring plasma renin activity. Results in hypertensive patients. *Arch Intern Med* 1981;**141**:1589–93. https://doi.org/10.1001/archinte.1981.00340130033011
326. Rossi GP. Primary aldosteronism: JACC state-of-the-art review. *J Am Coll Cardiol* 2019;**74**:2799–811. https://doi.org/10.1016/j.jacc.2019.09.057
327. Rossi GP, Ceolotto G, Rossitto G, Maiolino G, Cesari M, Seccia TM. Effects of mineralocorticoid and AT1 receptor antagonism on the aldosterone-renin ratio in primary aldosteronism—the EMIRA study. *J Clin Endocrinol Metab* 2020;**105**:2060–7. https://doi.org/10.1210/clinem/dgaa080
328. Rossi GP, Bisogni V, Bacca AV, Bacca AV, Belfiore A, Cesari M, *et al.* The 2020 Italian Society of Arterial Hypertension (SIIA) practical guidelines for the management of primary aldosteronism. *Int J Cardiol Hypertens* 2020;**5**:100029. https://doi.org/10.1016/j.ijchy.2020.100029
329. Faconti L, Kulkarni S, Delles C, Kapil V, Lewis P, Glover M, *et al.* Diagnosis and management of primary hyperaldosteronism in patients with hypertension: a practical approach endorsed by the British and Irish Hypertension Society. *J Hum Hypertens* 2024;**38**:8–18. https://doi.org/10.1038/s41371-023-00875-1
330. Olin JW, Gornik HL, Bacharach JM, Biller J, Fine LJ, Gray BH, *et al.* Fibromuscular dysplasia: state of the science and critical unanswered questions: a scientific statement from the American Heart Association. *Circulation* 2014;**129**:1048–78. https://doi.org/10.1161/01.cir.0000442577.96802.8c
331. Faconti L, Morselli F, Sinha M, Chrysochou C, Chowienczyk PJ. Fibromuscular dysplasia and hypertension—a statement on behalf of the British and Irish Hypertension Society. *J Hum Hypertens* 2021;**35**:1051–3. https://doi.org/10.1038/s41371-020-00456-6
332. Gornik HL, Persu A, Adlam D, Aparicio LS, Azizi M, Boulanger M, *et al.* First international consensus on the diagnosis and management of fibromuscular dysplasia. *J Hypertens* 2019;**37**:229–52. https://doi.org/10.1097/hjh.0000000000002019
333. Abrishami A, Khajehdehi A, Chung F. A systematic review of screening questionnaires for obstructive sleep apnea. *Can J Anaesth* 2010;**57**:423–38. https://doi.org/10.1007/s12630-010-9280-x
334. Chai-Coetzer CL, Antic NA, Hamilton GS, McArdle N, Wong K, Yee BJ, *et al.* Physician decision making and clinical outcomes with laboratory polysomnography or limited-channel sleep studies for obstructive sleep apnea: a randomized trial. *Ann Intern Med* 2017;**166**:332–40. https://doi.org/10.7326/m16-1301
335. Lenders JW, Eisenhofer G, Mannelli M, Pacak K. Phaeochromocytoma. *Lancet* 2005;**366**:665–75. https://doi.org/10.1016/s0140-6736(05)67139-5
336. Lenders JWM, Kerstens MN, Amar L, Prejbisz A, Robledo M, Taieb D, *et al.* Genetics, diagnosis, management and future directions of research of phaeochromocytoma and paraganglioma: a position statement and consensus of the Working Group on Endocrine Hypertension of the European Society of Hypertension. *J Hypertens* 2020;**38**:1443–56. https://doi.org/10.1097/hjh.0000000000002438
337. Mannelli M, Ianni L, Cilotti A, Conti A. Pheochromocytoma in Italy: a multicentric retrospective study. *Eur J Endocrinol* 1999;**141**:619–24. https://doi.org/10.1530/eje.0.1410619
338. Neumann HPH, Young WF Jr, Eng C. Pheochromocytoma and paraganglioma. *N Engl J Med* 2019;**381**:552–65. https://doi.org/10.1056/NEJMra1806651
339. Hundemer GL, Curhan GC, Yozamp N, Wang M, Vaidya A. Cardiometabolic outcomes and mortality in medically treated primary aldosteronism: a retrospective cohort study. *Lancet Diabetes Endocrinol* 2018;**6**:51–9. https://doi.org/10.1016/s2213-8587(17)30367-4
340. Jackson R, Lawes CM, Bennett DA, Milne RJ, Rodgers A. Treatment with drugs to lower blood pressure and blood cholesterol based on an individual's absolute cardiovascular risk. *Lancet* 2005;**365**:434–41.
341. Yang L, Sun J, Zhao M, Liang Y, Bovet P, Xi B. Elevated blood pressure in childhood and hypertension risk in adulthood: a systematic review and meta-analysis. *J Hypertens* 2020;**38**:2346–55. https://doi.org/10.1097/hjh.0000000000002550
342. Allen NB, Krefman AE, Labarthe D, Greenland P, Juonala M, Kähönen M, *et al.* Cardiovascular health trajectories from childhood through middle age and their association with subclinical atherosclerosis. *JAMA Cardiol* 2020;**5**:557–66. https://doi.org/10.1001/jamacardio.2020.0140
343. de Simone G, Mancusi C, Hanssen H, Genovesi S, Lurbe E, Parati G, *et al.* Hypertension in children and adolescents. *Eur Heart J* 2022;**43**:3290–301. https://doi.org/10.1093/eurheartj/ehac328
344. Falkstedt D, Koupil I, Hemmingsson T. Blood pressure in late adolescence and early incidence of coronary heart disease and stroke in the Swedish 1969 conscription cohort. *J Hypertens* 2008;**26**:1313–20. https://doi.org/10.1097/HJH.0b013e3282ffb17e
345. Jacobs DR Jr, Woo JG, Sinaiko AR, Daniels SR, Ikonen J, Juonala M, *et al.* Childhood cardiovascular risk factors and adult cardiovascular events. *N Engl J Med* 2022;**386**:1877–88. https://doi.org/10.1056/NEJMoa2109191
346. Oikonen M, Nuotio J, Magnussen CG, Viikari JSA, Taittonen L, Laitinen T, *et al.* Repeated blood pressure measurements in childhood in prediction of hypertension in adulthood. *Hypertension* 2016;**67**:41–7. https://doi.org/10.1161/hypertensionaha.115.06395
347. Garcia-Lunar I, van der Ploeg HP, Fernández Alvira JM, van Nassau F, Castellano Vázquez JM, van der Beek AJ, *et al.* Effects of a comprehensive lifestyle intervention on cardiovascular health: the TANSNIP-PESA trial. *Eur Heart J* 2022;**43**:3732–45. https://doi.org/10.1093/eurheartj/ehac378
348. Neal B, Wu Y, Feng X, Zhang R, Zhang Y, Shi J, *et al.* Effect of salt substitution on cardiovascular events and death. *N Engl J Med* 2021;**385**:1067–77. https://doi.org/10.1056/NEJMoa2105675
349. He FJ, Tan M, Ma Y, MacGregor GA. Salt reduction to prevent hypertension and cardiovascular disease: JACC state-of-the-art review. *J Am Coll Cardiol* 2020;**75**:632–47. https://doi.org/10.1016/j.jacc.2019.11.055
350. Ma Y, He FJ, Sun Q, Yuan C, Kieneker LM, Curhan GC, *et al.* 24-Hour urinary sodium and potassium excretion and cardiovascular risk. *N Engl J Med* 2022;**386**:252–63. https://doi.org/10.1056/NEJMoa2109794
351. Ma H, Xue Q, Wang X, Li X, Franco OH, Li Y, *et al.* Adding salt to foods and hazard of premature mortality. *Eur Heart J* 2022;**43**:2878–88. https://doi.org/10.1093/eurheartj/ehac208
352. Huang L, Trieu K, Yoshimura S, M Woodward, N Campbell, D Lackland, *et al.* Effect of dose and duration of reduction in dietary sodium on blood pressure levels: systematic

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

review and meta-analysis of randomised trials. *BMJ* 2020;**368**:m315. https://doi.org/10.1136/bmj.m315
353. Graudal N, Hubeck-Graudal T, Jürgens G, Taylor RS. Dose-response relation between dietary sodium and blood pressure: a meta-regression analysis of 133 randomized controlled trials. *Am J Clin Nutr* 2019;**109**:1273–8. https://doi.org/10.1093/ajcn/nqy384
354. Graudal NA, Hubeck-Graudal T, Jurgens G. Effects of low sodium diet versus high sodium diet on blood pressure, renin, aldosterone, catecholamines, cholesterol, and triglyceride. *Cochrane Database Syst Rev* 2020;**12**:CD004022. https://doi.org/10.1002/14651858.CD004022.pub5
355. Filippini T, Malavolti M, Whelton PK, Naska A, Orsini N, Vinceti M. Blood pressure effects of sodium reduction: dose–response meta-analysis of experimental studies. *Circulation* 2021;**143**:1542–67. https://doi.org/10.1161/circulationaha.120.050371
356. Gupta DK, Lewis CE, Varady KA, Su YR, Madhur MS, Lackland DT, *et al.* Effect of dietary sodium on blood pressure: a crossover trial. *JAMA* 2023;**330**:2258 . https://doi.org/10.1001/jama.2023.23651
357. He J, Gu D, Chen J, Jaquish CE, Rao DC, Hixson JE, *et al.* Gender difference in blood pressure responses to dietary sodium intervention in the GenSalt study. *J Hypertens* 2009;**27**:48–54. https://doi.org/10.1097/hjh.0b013e328316bb87
358. Sacks FM, Svetkey LP, Vollmer WM, Appel LJ, Bray GA, Harsha D, *et al.* Effects on blood pressure of reduced dietary sodium and the Dietary Approaches to Stop Hypertension (DASH) diet. *N Engl J Med* 2001;**344**:3–10. https://doi.org/10.1056/nejm200101043440101
359. MacGregor GA, Markandu ND, Sagnella GA, Singer DR, Cappuccio FP. Double-blind study of three sodium intakes and long-term effects of sodium restriction in essential hypertension. *Lancet* 1989;**2**:1244–7. https://doi.org/10.1016/s0140-6736(89)91852-7
360. Whelton PK, Appel LJ, Espeland MA, Applegate WB, Ettinger WH Jr, Kostis JB, *et al.* Sodium reduction and weight loss in the treatment of hypertension in older persons: a randomized controlled trial of nonpharmacologic interventions in the elderly (TONE). *JAMA* 1998;**279**:839–46. https://doi.org/10.1001/jama.279.11.839
361. He FJ, Li J, Macgregor GA. Effect of longer term modest salt reduction on blood pressure: Cochrane systematic review and meta-analysis of randomised trials. *BMJ* 2013;**346**:f1325. https://doi.org/10.1136/bmj.f1325
362. Stolarz-Skrzypek K, Kuznetsova T, Thijs L, Tikhonoff V, Seidlerová J, Richart T, *et al.* Fatal and nonfatal outcomes, incidence of hypertension, and blood pressure changes in relation to urinary sodium excretion. *JAMA* 2011;**305**:1777–85. https://doi.org/10.1001/jama.2011.574
363. Messerli FH, Hofstetter L, Syrogiannouli L, Rexhaj E, Siontis GCM, Seiler C, *et al.* Sodium intake, life expectancy, and all-cause mortality. *Eur Heart J* 2021;**42**:2103–12. https://doi.org/10.1093/eurheartj/ehaa947
364. Mente A, O'Donnell M, Rangarajan S, McQueen M, Dagenais G, Wielgosz A, *et al.* Urinary sodium excretion, blood pressure, cardiovascular disease, and mortality: a community-level prospective epidemiological cohort study. *Lancet* 2018;**392**:496–506. https://doi.org/10.1016/s0140-6736(18)31376-x
365. Au Yeung SL, Schooling CM. Impact of urinary sodium on cardiovascular disease and risk factors: a 2 sample Mendelian randomization study. *Clin Nutr* 2021;**40**:1990–6. https://doi.org/10.1016/j.clnu.2020.09.018
366. Zanetti D, Bergman H, Burgess S, Assimes TL, Bhalla V, Ingelsson E. Urinary albumin, sodium, and potassium and cardiovascular outcomes in the UK Biobank: observational and Mendelian randomization analyses. *Hypertension* 2020;**75**:714–22. https://doi.org/10.1161/hypertensionaha.119.14028
367. Nolan P, McEvoy JW. Salt restriction for treatment of hypertension—current state and future directions. *Curr Opin Cardiol* 2024;**39**:61–7. https://doi.org/10.1097/hco.0000000000001098
368. Yuan Y, Jin A, Neal B, Feng X, Qiao Q, Wang H, *et al.* Salt substitution and salt-supply restriction for lowering blood pressure in elderly care facilities: a cluster-randomized trial. *Nat Med* 2023;**29**:973–81. https://doi.org/10.1038/s41591-023-02286-8
369. Lechner K, Schunkert H. Recommendations on sodium intake for cardiovascular health: conviction or evidence? *Eur Heart J* 2020;**41**:3374–5. https://doi.org/10.1093/eurheartj/ehaa545
370. O'Donnell M, Mente A, Alderman MH, Brady AJB, Diaz R, Gupta R, *et al.* Salt and cardiovascular disease: insufficient evidence to recommend low sodium intake. *Eur Heart J* 2020;**41**:3363–73. https://doi.org/10.1093/eurheartj/ehaa586
371. Aburto NJ, Hanson S, Gutierrez H, Hooper L, Elliott P, Cappuccio FP. Effect of increased potassium intake on cardiovascular risk factors and disease: systematic review and meta-analyses. *BMJ* 2013;**346**:f1378. https://doi.org/10.1136/bmj.f1378
372. O'Donnell M, Mente A, Rangarajan S, McQueen MJ, Wang X, Liu L, *et al.* Urinary sodium and potassium excretion, mortality, and cardiovascular events. *N Engl J Med* 2014;**371**:612–23. https://doi.org/10.1056/NEJMoa1311889
373. Wouda RD, Boekholdt SM, Khaw KT, Wareham NJ, de Borst MH, Hoorn EJ, *et al.* Sex-specific associations between potassium intake, blood pressure, and cardiovascular outcomes: the EPIC-Norfolk study. *Eur Heart J* 2022;**43**:2867–75. https://doi.org/10.1093/eurheartj/ehac313
374. Filippini T, Naska A, Kasdagli MI, Torres D, Lopes C, Carvalho C, *et al.* Potassium intake and blood pressure: a dose-response meta-analysis of randomized controlled trials. *J Am Heart Assoc* 2020;**9**:e015719. https://doi.org/10.1161/jaha.119.015719
375. O'Donnell M, Yusuf S, Vogt L, Mente A, Messerli FH. Potassium intake: the Cinderella electrolyte. *Eur Heart J* 2023;**44**:4925–34. https://doi.org/10.1093/eurheartj/ehad628
376. Kidney Disease Outcomes Quality Initiative (K/DOQI). K/DOQI clinical practice guidelines on hypertension and antihypertensive agents in chronic kidney disease. *Am J Kidney Dis* 2004;**43**:S1–290. https://doi.org/10.1053/j.ajkd.2004.03.006
377. Ndanuko RN, Ibrahim R, Hapsari RA, Neale EP, Raubenheimer D, Charlton KE. Association between the urinary sodium to potassium ratio and blood pressure in adults: a systematic review and meta-analysis. *Adv Nutr* 2021;**12**:1751–67. https://doi.org/10.1093/advances/nmab036
378. Messerli FH, O'Donnell M, Mente A, Yusuf S. Settling the controversy of salt substitutes and stroke: sodium reduction or potassium increase? *Eur Heart J* 2022;**43**:3365–7. https://doi.org/10.1093/eurheartj/ehac160
379. Yin X, Paige E, Tian M, Li Q, Huang L, Yu J, *et al.* The proportion of dietary salt replaced with potassium-enriched salt in the SSaSS: implications for scale-up. *Hypertension* 2023;**80**:956–65. https://doi.org/10.1161/hypertensionaha.122.20115
380. Xu X, Zeng L, Jha V, Cobb LK, Shibuya K, Appel LJ, *et al.* Potassium-enriched salt substitutes: a review of recommendations in clinical management guidelines. *Hypertension* 2024;**81**:400–14. https://doi.org/10.1161/hypertensionaha.123.21343
381. Hanssen H, Boardman H, Deiseroth A, Moholdt T, Simonenko M, Kränkel N, *et al.* Personalized exercise prescription in the prevention and treatment of arterial hypertension: a consensus document from the European Association of Preventive Cardiology (EAPC) and the ESC Council on Hypertension. *Eur J Prev Cardiol* 2022;**29**:205–15. https://doi.org/10.1093/eurjpc/zwaa141
382. Van Hoof R, Hespel P, Fagard R, Lijnen P, Staessen J, Amery A. Effect of endurance training on blood pressure at rest, during exercise and during 24 hours in sedentary men. *Am J Cardiol* 1989;**63**:945–9. https://doi.org/10.1016/0002-9149(89)90145-8
383. MacDonald HV, Johnson BT, Huedo-Medina TB, Livingston J, Forsyth KC, Kraemer WJ, *et al.* Dynamic resistance training as stand-alone antihypertensive lifestyle therapy: a meta-analysis. *J Am Heart Assoc* 2016;**5**:e003231. https://doi.org/10.1161/jaha.116.003231
384. Hansford HJ, Parmenter BJ, McLeod KA, Wewege MA, Smart NA, Schutte AE, *et al.* The effectiveness and safety of isometric resistance training for adults with high blood pressure: a systematic review and meta-analysis. *Hypertens Res* 2021;**44**:1373–84. https://doi.org/10.1038/s41440-021-00720-3
385. Van Hoof R, Macor F, Lijnen P, Staessen J, Thijs L, Vanhees L, *et al.* Effect of strength training on blood pressure measured in various conditions in sedentary men. *Int J Sports Med* 1996;**17**:415–22. https://doi.org/10.1055/s-2007-972871
386. Leal JM, Galliano LM, Del Vecchio FB. Effectiveness of high-intensity interval training versus moderate-intensity continuous training in hypertensive patients: a systematic review and meta-analysis. *Curr Hypertens Rep* 2020;**22**:26. https://doi.org/10.1007/s11906-020-1030-z
387. Rossi A, Dikareva A, Bacon SL, Daskalopoulou SS. The impact of physical activity on mortality in patients with high blood pressure: a systematic review. *J Hypertens* 2012;**30**:1277–88. https://doi.org/10.1097/HJH.0b013e3283544669
388. Cuspidi C, Gherbesi E, Faggiano A, Sala C, Carugo S, Grassi G, *et al.* Masked hypertension and exaggerated blood pressure response to exercise: a review and meta-analysis. *Diagnostics (Basel)* 2023;**13**:1005. https://doi.org/10.3390/diagnostics13061005
389. Mariampillai JE, Liestøl K, Kjeldsen SE, Prestgaard EE, Engeseth K, Bodegard J, *et al.* Exercise systolic blood pressure at moderate workload is linearly associated with coronary disease risk in healthy men. *Hypertension* 2020;**75**:44–50. https://doi.org/10.1161/hypertensionaha.119.13528
390. Pelliccia A, Sharma S, Gati S, Bäck M, Börjesson M, Caselli S. 2020 ESC Guidelines on sports cardiology and exercise in patients with cardiovascular disease. *Eur Heart J* 2021;**42**:17–96. https://doi.org/10.1093/eurheartj/ehaa605
391. Bull FC, Al-Ansari SS, Biddle S, Borodulin K, Buman MP, Cardon G, *et al.* World Health Organization 2020 guidelines on physical activity and sedentary behaviour. *Br J Sports Med* 2020;**54**:1451–62. https://doi.org/10.1136/bjsports-2020-102955
392. Saco-Ledo G, Valenzuela PL, Ramírez-Jiménez M, Morales JS, Castillo-García A, Blumenthal JA, *et al.* Acute aerobic exercise induces short-term reductions in ambulatory blood pressure in patients with hypertension: a systematic review and meta-analysis. *Hypertension* 2021;**78**:1844–58. https://doi.org/10.1161/hypertensionaha.121.18099
393. Garber CE, Blissmer B, Deschenes MR, Franklin BA, Lamonte MJ, Lee I-M, *et al.* American College of Sports Medicine position stand. Quantity and quality of exercise for developing and maintaining cardiorespiratory, musculoskeletal, and neuromotor fitness in apparently healthy adults: guidance for prescribing exercise. *Med Sci Sports Exerc* 2011;**43**:1334–59. https://doi.org/10.1249/MSS.0b013e318213fefb
394. Edwards JJ, Deenmamode AHP, Griffiths M, Arnold O, Cooper NJ, Wiles JD, *et al.* Exercise training and resting blood pressure: a large-scale pairwise and network meta-analysis of randomised controlled trials. *Br J Sports Med* 2023;**57**:1317–26. https://doi.org/10.1136/bjsports-2022-106503
395. Fletcher GF, Ades PA, Kligfield P, Arena R, Balady GJ, Bittner VA, *et al.* Exercise standards for testing and training: a scientific statement from the American Heart Association. *Circulation* 2013;**128**:873–934. https://doi.org/10.1161/CIR.0b013e31829b5b44
396. Craft BB, Carroll HA, Lustyk MK. Gender differences in exercise habits and quality of life reports: assessing the moderating effects of reasons for exercise. *Int J Lib Arts Soc Sci* 2014;**2**:65–76.

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

397. Mathieu P, Poirier P, Pibarot P, Lemieux I, Després JP. Visceral obesity: the link among inflammation, hypertension, and cardiovascular disease. *Hypertension* 2009;**53**:577–84. https://doi.org/10.1161/hypertensionaha.108.110320
398. Chandra A, Neeland IJ, Berry JD, Ayers CR, Rohatgi A, Das SR, *et al.* The relationship of body mass and fat distribution with incident hypertension: observations from the Dallas heart study. *J Am Coll Cardiol* 2014;**64**:997–1002. https://doi.org/10.1016/j.jacc.2014.05.057
399. Neter JE, Stam BE, Kok FJ, Grobbee DE, Geleijnse JM. Influence of weight reduction on blood pressure: a meta-analysis of randomized controlled trials. *Hypertension* 2003;**42**:878–84. https://doi.org/10.1161/01.Hyp.0000094221.86888.Ae
400. Haase CL, Lopes S, Olsen AH, Satylganova A, Schnecke V, McEwan P. Weight loss and risk reduction of obesity-related outcomes in 0.5 million people: evidence from a UK primary care database. *Int J Obes (Lond)* 2021;**45**:1249–58. https://doi.org/10.1038/s41366-021-00788-4
401. Moore LL, Visioni AJ, Qureshi MM, Bradlee ML, Ellison RC, D'Agostino R. Weight loss in overweight adults and the long-term risk of hypertension: the Framingham study. *Arch Intern Med* 2005;**165**:1298–303. https://doi.org/10.1001/archinte.165.11.1298
402. Zomer E, Gurusamy K, Leach R, Trimmer C, Lobstein T, Morris S, *et al.* Interventions that cause weight loss and the impact on cardiovascular risk factors: a systematic review and meta-analysis. *Obes Rev* 2016;**17**:1001–11. https://doi.org/10.1111/obr.12433
403. Wing RR, Espeland MA, Clark JM, Hazuda HP, Knowler WC, Pownall HJ, *et al.* Association of weight loss maintenance and weight regain on 4-year changes in CVD risk factors: the action for health in diabetes (Look AHEAD) clinical trial. *Diabetes Care* 2016;**39**:1345–55. https://doi.org/10.2337/dc16-0509
404. Ma C, Avenell A, Bolland M, Hudson J, Stewart F, Robertson C, *et al.* Effects of weight loss interventions for adults who are obese on mortality, cardiovascular disease, and cancer: systematic review and meta-analysis. *BMJ* 2017;**359**:j4849. https://doi.org/10.1136/bmj.j4849
405. Semlitsch T, Krenn C, Jeitler K, Berghold A, Horvath K, Siebenhofer A. Long-term effects of weight-reducing diets in people with hypertension. *Cochrane Database Syst Rev* 2021;**2**:CD008274. https://doi.org/10.1002/14651858.CD008274.pub4
406. Wing RR, Bolin P, Brancati FL, Bray GA, Clark JM, Coday M, *et al.* Cardiovascular effects of intensive lifestyle intervention in type 2 diabetes. *N Engl J Med* 2013;**369**:145–54. https://doi.org/10.1056/NEJMoa1212914
407. Pagidipati NJ, Phelan M, Page C, Clowse M, Henao R, Peterson ED, *et al.* The importance of weight stabilization amongst those with overweight or obesity: results from a large health care system. *Prev Med Rep* 2021;**24**:101615. https://doi.org/10.1016/j.pmedr.2021.101615
408. Estruch R, Ros E, Salas-Salvadó J, Covas M-I, Corella D, Arós F, *et al.* Primary prevention of cardiovascular disease with a Mediterranean diet supplemented with extra-virgin olive oil or nuts. *N Engl J Med* 2018;**378**:e34. https://doi.org/10.1056/NEJMoa1800389
409. Delgado-Lista J, Alcala-Diaz JF, Torres-Peña JD, Quintana-Navarro GM, Fuentes F, Garcia-Rios A, *et al.* Long-term secondary prevention of cardiovascular disease with a Mediterranean diet and a low-fat diet (CORDIOPREV): a randomised controlled trial. *Lancet* 2022;**399**:1876–85. https://doi.org/10.1016/s0140-6736(22)00122-2
410. Blumenthal JA, Babyak MA, Hinderliter A, Watkins LL, Craighead L, Lin P-H, *et al.* Effects of the DASH diet alone and in combination with exercise and weight loss on blood pressure and cardiovascular biomarkers in men and women with high blood pressure: the ENCORE study. *Arch Intern Med* 2010;**170**:126–35. https://doi.org/10.1001/archinternmed.2009.470
411. Juraschek SP, Miller ER III, Weaver CM, Appel LJ. Effects of sodium reduction and the DASH diet in relation to baseline blood pressure. *J Am Coll Cardiol* 2017;**70**:2841–8. https://doi.org/10.1016/j.jacc.2017.10.011
412. Gay HC, Rao SG, Vaccarino V, Ali MK. Effects of different dietary interventions on blood pressure: systematic review and meta-analysis of randomized controlled trials. *Hypertension* 2016;**67**:733–9. https://doi.org/10.1161/hypertensionaha.115.06853
413. Siebenhofer A, Winterholer S, Jeitler K, Horvath K, Berghold A, Krenn C, *et al.* Long-term effects of weight-reducing drugs in people with hypertension. *Cochrane Database Syst Rev* 2021;**1**:CD007654. https://doi.org/10.1002/14651858.CD007654.pub5
414. Müller TD, Blüher M, Tschöp MH, DiMarchi RD. Anti-obesity drug discovery: advances and challenges. *Nat Rev Drug Discov* 2022;**21**:201–23. https://doi.org/10.1038/s41573-021-00337-8
415. Wilding JPH, Batterham RL, Calanna S, Davies M, Van Gaal LF, Lingvay I, *et al.* Once-weekly semaglutide in adults with overweight or obesity. *N Engl J Med* 2021;**384**:989–1002. https://doi.org/10.1056/NEJMoa2032183
416. Davies M, Færch L, Jeppesen OK, Pakseresht A, Pedersen SD, Perreault L, *et al.* Semaglutide 2·4 mg once a week in adults with overweight or obesity, and type 2 diabetes (STEP 2): a randomised, double-blind, double-dummy, placebo-controlled, phase 3 trial. *Lancet* 2021;**397**:971–84. https://doi.org/10.1016/s0140-6736(21)00213-0
417. Tasnim S, Tang C, Musini VM, Wright JM. Effect of alcohol on blood pressure. *Cochrane Database Syst Rev* 2020;**7**:CD012787. https://doi.org/10.1002/14651858.CD012787.pub2
418. Liu F, Liu Y, Sun X, Yin Z, Li H, Deng K, *et al.* Race- and sex-specific association between alcohol consumption and hypertension in 22 cohort studies: a systematic review and meta-analysis. *Nutr Metab Cardiovasc Dis* 2020;**30**:1249–59. https://doi.org/10.1016/j.numecd.2020.03.018
419. Griswold MG, Fullman N, Hawley C, Arian N, Zimsen SRM, Tymeson HD, *et al.* Alcohol use and burden for 195 countries and territories, 1990–2016: a systematic analysis for the Global Burden of Disease Study 2016. *Lancet* 2018;**392**:1015–35. https://doi.org/10.1016/s0140-6736(18)31310-2
420. D'Elia L, La Fata E, Galletti F, Scalfi L, Strazzullo P. Coffee consumption and risk of hypertension: a dose–response meta-analysis of prospective studies. *Eur J Nutr* 2019;**58**:271–80. https://doi.org/10.1007/s00394-017-1591-z
421. Hodgson JM, Puddey IB, Woodman RJ, Mulder TPJ, Fuchs D, Scott K, *et al.* Effects of black tea on blood pressure: a randomized controlled trial. *Arch Intern Med* 2012;**172**:186–8. https://doi.org/10.1001/archinte.172.2.186
422. Shah SA, Chu BW, Lacey CS, Riddock IC, Lee M, Dargush AE. Impact of acute energy drink consumption on blood pressure parameters: a meta-analysis. *Ann Pharmacother* 2016;**50**:808–15. https://doi.org/10.1177/1060028016656433
423. Shah SA, Szeto AH, Farewell R, Shek A, Fan D, Quach KN, *et al.* Impact of high volume energy drink consumption on electrocardiographic and blood pressure parameters: a randomized trial. *J Am Heart Assoc* 2019;**8**:e011318. https://doi.org/10.1161/jaha.118.011318
424. Basrai M, Schweinlin A, Menzel J, Mielke H, Weikert C, Dusemund B, *et al.* Energy drinks induce acute cardiovascular and metabolic changes pointing to potential risks for young adults: a randomized controlled trial. *J Nutr* 2019;**149**:441–50. https://doi.org/10.1093/jn/nxy303
425. Fung TT, Malik V, Rexrode KM, Manson JE, Willett WC, Hu FB. Sweetened beverage consumption and risk of coronary heart disease in women. *Am J Clin Nutr* 2009;**89**:1037–42. https://doi.org/10.3945/ajcn.2008.27140
426. Mullee A, Romaguera D, Pearson-Stuttard J, Viallon V, Stepien M, Freisling H, *et al.* Association between soft drink consumption and mortality in 10 European countries. *JAMA Intern Med* 2019;**179**:1479–90. https://doi.org/10.1001/jamainternmed.2019.2478
427. Farhangi MA, Nikniaz L, Khodarahmi M. Sugar-sweetened beverages increases the risk of hypertension among children and adolescence: a systematic review and dose–response meta-analysis. *J Transl Med* 2020;**18**:344. https://doi.org/10.1186/s12967-020-02511-9
428. Critchley JA, Capewell S. Mortality risk reduction associated with smoking cessation in patients with coronary heart disease: a systematic review. *JAMA* 2003;**290**:86–97. https://doi.org/10.1001/jama.290.1.86
429. Anthonisen NR, Skeans MA, Wise RA, Manfreda J, Kanner RE, Connett JE. The effects of a smoking cessation intervention on 14.5-year mortality: a randomized clinical trial. *Ann Intern Med* 2005;**142**:233–9. https://doi.org/10.7326/0003-4819-142-4-200502150-00005
430. Thomson B, Emberson J, Lacey B, Lewington S, Peto R, Jemal A. Association between smoking, smoking cessation, and mortality by race, ethnicity, and sex among US adults. *JAMA Netw Open* 2022;**5**:e2231480. https://doi.org/10.1001/jamanetworkopen.2022.31480
431. Yang JJ, Yu D, Shu XO, Wen W, Rahman S, Abe S. Reduction in total and major cause-specific mortality from tobacco smoking cessation: a pooled analysis of 16 population-based cohort studies in Asia. *Int J Epidemiol* 2022;**50**:2070–81. https://doi.org/10.1093/ije/dyab087
432. Martinez-Morata I, Sanchez TR, Shimbo D, Navas-Acien A. Electronic cigarette use and blood pressure endpoints: a systematic review. *Curr Hypertens Rep* 2020;**23**:2. https://doi.org/10.1007/s11906-020-01119-0
433. Kim SY, Jeong SH, Joo HJ, Park M, Park E-C, Kim JH, *et al.* High prevalence of hypertension among smokers of conventional and e-cigarette: using the nationally representative community dwelling survey. *Front Public Health* 2022;**10**:919585. https://doi.org/10.3389/fpubh.2022.919585
434. Groppelli A, Giorgi DM, Omboni S, Parati G, Mancia G. Persistent blood pressure increase induced by heavy smoking. *J Hypertens* 1992;**10**:495–9. https://doi.org/10.1097/00004872-199205000-00014
435. Primatesta P, Falaschetti E, Gupta S, Marmot MG, Poulter NR. Association between smoking and blood pressure: evidence from the health survey for England. *Hypertension* 2001;**37**:187–93. https://doi.org/10.1161/01.hyp.37.2.187
436. Stead LF, Buitrago D, Preciado N, Sanchez G, Hartmann-Boyce J, Lancaster T. Physician advice for smoking cessation. *Cochrane Database Syst Rev* 2013;**2013**:CD000165. https://doi.org/10.1002/14651858.CD000165.pub4
437. Rasmussen M, Lauridsen SV, Pedersen B, Backer V, Tønnesen H. Intensive versus short face-to-face smoking cessation interventions: a meta-analysis. *Eur Respir Rev* 2022;**31**:220063. https://doi.org/10.1183/16000617.0063-2022
438. Filippou CD, Thomopoulos CG, Kouremeti MM, Sotiropoulou LI, Nihoyannopoulos PI, Tousoulis DM, *et al.* Mediterranean diet and blood pressure reduction in adults with and without hypertension: a systematic review and meta-analysis of randomized controlled trials. *Clin Nutr* 2021;**40**:3191–200. https://doi.org/10.1016/j.clnu.2021.01.030
439. Cowell OR, Mistry N, Deighton K, Matu J, Griffiths A, Minihane AM, *et al.* Effects of a Mediterranean diet on blood pressure: a systematic review and meta-analysis of randomized controlled trials and observational studies. *J Hypertens* 2021;**39**:729–39. https://doi.org/10.1097/hjh.0000000000002667

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

440. Roerecke M, Tobe SW, Kaczorowski J, Bacon SL, Vafaei A, Hasan OSM, *et al.* Sex-specific associations between alcohol consumption and incidence of hypertension: a systematic review and meta-analysis of cohort studies. *J Am Heart Assoc* 2018;**7**: e008202. https://doi.org/10.1161/jaha.117.008202
441. Ding C, O'Neill D, Bell S, Stamatakis E, Britton A. Association of alcohol consumption with morbidity and mortality in patients with cardiovascular disease: original data and meta-analysis of 48,423 men and women. *BMC Med* 2021;**19**:167. https://doi.org/10.1186/s12916-021-02040-2
442. Marklund M, Singh G, Greer R, Cudhea F, Matsushita K, Micha R, *et al.* Estimated population wide benefits and risks in China of lowering sodium through potassium enriched salt substitution: modelling study. *BMJ* 2020;**369**:m824. https://doi.org/10.1136/bmj.m824
443. Pinho-Gomes AC, Azevedo L, Copland E, Canoy D, Nazarzadeh M, Ramakrishnan R, *et al.* Blood pressure-lowering treatment for the prevention of cardiovascular events in patients with atrial fibrillation: an individual participant data meta-analysis. *PLoS Med* 2021;**18**:e1003599. https://doi.org/10.1371/journal.pmed.1003599
444. Bidel Z, Nazarzadeh M, Canoy D, Copland E, Gerdts E, Woodward M, *et al.* Sex-specific effects of blood pressure lowering pharmacotherapy for the prevention of cardiovascular disease: an individual participant-level data meta-analysis. *Hypertension* 2023;**80**:2293–302. https://doi.org/10.1161/hypertensionaha.123.21496
445. Nazarzadeh M, Bidel Z, Canoy D, Copland E, Bennett DA, Dehghan A, *et al.* Blood pressure-lowering treatment for prevention of major cardiovascular diseases in people with and without type 2 diabetes: an individual participant-level data meta-analysis. *Lancet Diabetes Endocrinol* 2022;**10**:645–54. https://doi.org/10.1016/s2213-8587(22)00172-3
446. Thomopoulos C, Parati G, Zanchetti A. Effects of blood pressure lowering on outcome incidence in hypertension: 4. Effects of various classes of antihypertensive drugs—overview and meta-analyses. *J Hypertens* 2015;**33**:195–211. https://doi.org/10.1097/hjh.0000000000000447
447. Ishani A, Cushman WC, Leatherman SM, Lew RA, Woods P, Glassman PA, *et al.* Chlorthalidone vs. hydrochlorothiazide for hypertension-cardiovascular events. *N Engl J Med* 2022;**387**:2401–10. https://doi.org/10.1056/NEJMoa2212270
448. Ziff OJ, Samra M, Howard JP, Bromage DI, Ruschitzka F, Francis DP, *et al.* Beta-blocker efficacy across different cardiovascular indications: an umbrella review and meta-analytic assessment. *BMC Med* 2020;**18**:103. https://doi.org/10.1186/s12916-020-01564-3
449. Thomopoulos C, Bazoukis G, Tsioufis C, Mancia G. Beta-blockers in hypertension: overview and meta-analysis of randomized outcome trials. *J Hypertens* 2020;**38**: 1669–81. https://doi.org/10.1097/hjh.0000000000002523
450. Mancia G, Kjeldsen SE, Kreutz R, Pathak A, Grassi G, Esler M. Individualized beta-blocker treatment for high blood pressure dictated by medical comorbidities: indications beyond the 2018 European Society of Cardiology/European Society of Hypertension Guidelines. *Hypertension* 2022;**79**:1153–66. https://doi.org/10.1161/hypertensionaha.122.19020
451. Bradley HA, Wiysonge CS, Volmink JA, Mayosi BM, Opie LH. How strong is the evidence for use of beta-blockers as first-line therapy for hypertension? Systematic review and meta-analysis. *J Hypertens* 2006;**24**:2131–41. https://doi.org/10.1097/01.hjh.0000249685.58370.28
452. Wiysonge CS, Bradley HA, Volmink J, Mayosi BM, Opie LH. Beta-blockers for hypertension. *Cochrane Database Syst Rev* 2017;**1**:CD002003. https://doi.org/10.1002/14651858.CD002003.pub5
453. Elliott WJ, Meyer PM. Incident diabetes in clinical trials of antihypertensive drugs: a network meta-analysis. *Lancet* 2007;**369**:201–7. https://doi.org/10.1016/s0140-6736(07)60108-1
454. Bangalore S, Parkar S, Grossman E, Messerli FH. A meta-analysis of 94,492 patients with hypertension treated with beta blockers to determine the risk of new-onset diabetes mellitus. *Am J Cardiol* 2007;**100**:1254–62. https://doi.org/10.1016/j.amjcard.2007.05.057
455. Laurent S, Briet M, Boutouyrie P. Large and small artery cross-talk and recent morbidity-mortality trials in hypertension. *Hypertension* 2009;**54**:388–92. https://doi.org/10.1161/hypertensionaha.109.133116
456. Klingbeil AU, Schneider M, Martus P, Messerli FH, Schmieder RE. A meta-analysis of the effects of treatment on left ventricular mass in essential hypertension. *Am J Med* 2003;**115**:41–6. https://doi.org/10.1016/s0002-9343(03)00158-x
457. Zanchetti A, Bond MG, Hennig M, Neiss A, Mancia G, Dal Palù C, *et al.* Calcium antagonist lacidipine slows down progression of asymptomatic carotid atherosclerosis: principal results of the European Lacidipine Study on Atherosclerosis (ELSA), a randomized, double-blind, long-term trial. *Circulation* 2002;**106**:2422–7. https://doi.org/10.1161/01.cir.0000039288.86470.dd
458. Schiffrin EL, Deng LY. Comparison of effects of angiotensin I-converting enzyme inhibition and beta-blockade for 2 years on function of small arteries from hypertensive patients. *Hypertension* 1995;**25**:699–703. https://doi.org/10.1161/01.hyp.25.4.699
459. Williams B, MacDonald TM, Morant S, Webb DJ, Sever P, McInnes G, *et al.* Spironolactone versus placebo, bisoprolol, and doxazosin to determine the optimal treatment for drug-resistant hypertension (PATHWAY-2): a randomised, double-blind, crossover trial. *Lancet* 2015;**386**:2059–68. https://doi.org/10.1016/s0140-6736(15)00257-3
460. Agarwal R, Filippatos G, Pitt B, Anker SD, Rossing P, Joseph A, *et al.* Cardiovascular and kidney outcomes with finerenone in patients with type 2 diabetes and chronic kidney disease: the FIDELITY pooled analysis. *Eur Heart J* 2022;**43**:474–84. https://doi.org/10.1093/eurheartj/ehab777
461. Agarwal R, Ruilope LM, Ruiz-Hurtado G, Haller H, Schmieder RE, Anker SD, *et al.* Effect of finerenone on ambulatory blood pressure in chronic kidney disease in type 2 diabetes. *J Hypertens* 2023;**41**:295–302. https://doi.org/10.1097/hjh.0000000000003330
462. Bakris GL, Agarwal R, Anker SD, Pitt B, Ruilope LM, Rossing P, *et al.* Effect of finerenone on chronic kidney disease outcomes in type 2 diabetes. *N Engl J Med* 2020;**383**: 2219–29. https://doi.org/10.1056/NEJMoa2025845
463. The ALLHAT Officers Coordinators for the ALLHAT Collaborative Research Group. Major cardiovascular events in hypertensive patients randomized to doxazosin vs chlorthalidone: the antihypertensive and lipid-lowering treatment to prevent heart attack trial (ALLHAT). *JAMA* 2000;**283**:1967–75. https://doi.org/10.1001/jama.283.15.1967
464. Ruilope LM, Dukat A, Böhm M, Lacourcière Y, Gong J, Lefkowitz MP. Blood-pressure reduction with LCZ696, a novel dual-acting inhibitor of the angiotensin II receptor and neprilysin: a randomised, double-blind, placebo-controlled, active comparator study. *Lancet* 2010;**375**:1255–66. https://doi.org/10.1016/s0140-6736(09)61966-8
465. McMurray JJ, Packer M, Desai AS, Gong J, Lefkowitz MP, Rizkala AR, *et al.* Angiotensin–neprilysin inhibition versus enalapril in heart failure. *N Engl J Med* 2014;**371**:993–1004. https://doi.org/10.1056/NEJMoa1409077
466. Rakugi H, Kario K, Yamaguchi M, Sasajima T, Gotou H, Zhang J. Efficacy of sacubitril/valsartan versus olmesartan in Japanese patients with essential hypertension: a randomized, double-blind, multicenter study. *Hypertens Res* 2022;**45**:824–33. https://doi.org/10.1038/s41440-021-00819-7
467. Jackson AM, Jhund PS, Anand IS, Düngen H-D, Lam CSP, Lefkowitz MP, *et al.* Sacubitril-valsartan as a treatment for apparent resistant hypertension in patients with heart failure and preserved ejection fraction. *Eur Heart J* 2021;**42**:3741–52. https://doi.org/10.1093/eurheartj/ehab499
468. Kario K, Sun N, Chiang FT, Supasyndh O, Baek SH, Inubushi-Molessa A, *et al.* Efficacy and safety of LCZ696, a first-in-class angiotensin receptor neprilysin inhibitor, in Asian patients with hypertension: a randomized, double-blind, placebo-controlled study. *Hypertension* 2014;**63**:698–705. https://doi.org/10.1161/hypertensionaha.113.02002
469. Williams B, Cockcroft JR, Kario K, Zappe DH, Brunel PC, Wang Q, *et al.* Effects of sacubitril/valsartan versus olmesartan on central hemodynamics in the elderly with systolic hypertension: the PARAMETER study. *Hypertension* 2017;**69**:411–20. https://doi.org/10.1161/hypertensionaha.116.08556
470. Herrington WG, Savarese G, Haynes R, Marx N, Mellbin L, Lund LH, *et al.* Cardiac, renal, and metabolic effects of sodium-glucose co-transporter 2 inhibitors: a position paper from the European Society of Cardiology ad-hoc task force on sodium-glucose co-transporter 2 inhibitors. *Eur J Heart Fail* 2021;**23**:1260–75. https://doi.org/10.1002/ejhf.2286
471. Gupta R, Maitz T, Egeler D, Mehta A, Nyaeme M, Hajra A, *et al.* SGLT2 inhibitors in hypertension: role beyond diabetes and heart failure. *Trends Cardiovasc Med* 2022; **33**:479–86. https://doi.org/10.1016/j.tcm.2022.05.005
472. Kario K, Ferdinand KC, Vongpatanasin W. Are SGLT2 inhibitors new hypertension drugs? *Circulation* 2021;**143**:1750–3. https://doi.org/10.1161/circulationaha.121.053709
473. Yan C, Thijs L, Cao Y, Trenson S, Zhang Z-Y, Janssens S, *et al.* Opportunities of antidiabetic drugs in cardiovascular medicine: a meta-analysis and perspectives for trial design. *Hypertension* 2020;**76**:420–31. https://doi.org/10.1161/hypertensionaha.120.14791
474. Freeman MW, Halvorsen YD, Marshall W, Pater M, Isaacsohn J, Pearce C, *et al.* Phase 2 trial of baxdrostat for treatment-resistant hypertension. *N Engl J Med* 2023;**388**: 395–405. https://doi.org/10.1056/NEJMoa2213169
475. Laffin LJ, Rodman D, Luther JM, Vaidya A, Weir MR, Rajicic N, *et al.* Aldosterone synthase inhibition with lorundrostat for uncontrolled hypertension: the target-HTN randomized clinical trial. *JAMA* 2023;**330**:1140–50. https://doi.org/10.1001/jama.2023.16029
476. Schlaich MP, Bellet M, Weber MA, Danaietash P, Bakris GL, Flack JM, *et al.* Dual endothelin antagonist aprocitentan for resistant hypertension (PRECISION): a multicentre, blinded, randomised, parallel-group, phase 3 trial. *Lancet* 2022;**400**:1927–37. https://doi.org/10.1016/s0140-6736(22)02034-7
477. Desai AS, Webb DJ, Taubel J, Casey S, Cheng Y, Robbie GJ, *et al.* Zilebesiran, an RNA interference therapeutic agent for hypertension. *N Engl J Med* 2023;**389**:228–38. https://doi.org/10.1056/NEJMoa2208391
478. Law MR, Wald NJ, Morris JK, Jordan RE. Value of low dose combination treatment with blood pressure lowering drugs: analysis of 354 randomised trials. *BMJ* 2003; **326**:1427. https://doi.org/10.1136/bmj.326.7404.1427
479. Mahmud A, Feely J. Low-dose quadruple antihypertensive combination: more efficacious than individual agents—a preliminary report. *Hypertension* 2007;**49**:272–5. https://doi.org/10.1161/01.HYP.0000254479.66645.a3
480. Wald DS, Law M, Morris JK, Bestwick JP, Wald NJ. Combination therapy versus monotherapy in reducing blood pressure: meta-analysis on 11,000 participants from 42 trials. *Am J Med* 2009;**122**:290–300. https://doi.org/10.1016/j.amjmed.2008.09.038

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

481. Chow CK, Thakkar J, Bennett A, Hillis G, Burke M, Usherwood T, *et al.* Quarter-dose quadruple combination therapy for initial treatment of hypertension: placebo-controlled, crossover, randomised trial and systematic review. *Lancet* 2017;**389**:1035–42. https://doi.org/10.1016/s0140-6736(17)30260-x
482. Chow CK, Atkins ER, Hillis GS, Nelson MR, Reid CM, Schlaich MP, *et al.* Initial treatment with a single pill containing quadruple combination of quarter doses of blood pressure medicines versus standard dose monotherapy in patients with hypertension (QUARTET): a phase 3, randomised, double-blind, active-controlled trial. *Lancet* 2021;**398**:1043–52. https://doi.org/10.1016/s0140-6736(21)01922-x
483. Webster R, Salam A, de Silva HA, Selak V, Stepien S, Rajapakse S, *et al.* Fixed low-dose triple combination antihypertensive medication vs usual care for blood pressure control in patients with mild to moderate hypertension in Sri Lanka: a randomized clinical trial. *JAMA* 2018;**320**:566–79. https://doi.org/10.1001/jama.2018.10359
484. MacDonald TM, Williams B, Webb DJ, Morant S, Caulfield M, Cruickshank JK, *et al.* Combination therapy is superior to sequential monotherapy for the initial treatment of hypertension: a double-blind randomized controlled trial. *J Am Heart Assoc* 2017;**6**:e006986. https://doi.org/10.1161/jaha.117.006986
485. Salam A, Kanukula R, Atkins E, Wang X, Islam S, Kishore SP, *et al.* Efficacy and safety of dual combination therapy of blood pressure-lowering drugs as initial treatment for hypertension: a systematic review and meta-analysis of randomized controlled trials. *J Hypertens* 2019;**37**:1768–74. https://doi.org/10.1097/hjh.0000000000002096
486. Zhang ZY, Yu YL, Asayama K, Hansen TW, Maestre GE, Staessen JA. Starting antihypertensive drug treatment with combination therapy: controversies in hypertension—con side of the argument. *Hypertension* 2021;**77**:788–98. https://doi.org/10.1161/hypertensionaha.120.12858
487. Kahan T. Low-dose combination of blood pressure-lowering medicines. *Lancet* 2021;**398**:1022–3. https://doi.org/10.1016/s0140-6736(21)01964-4
488. Rea F, Corrao G, Merlino L, Mancia G. Initial antihypertensive treatment strategies and therapeutic inertia. *Hypertension* 2018;**72**:846–53. https://doi.org/10.1161/hypertensionaha.118.11308
489. Egan BM, Bandyopadhyay D, Shaftman SR, Wagner CS, Zhao Y, Yu-Isenberg KS. Initial monotherapy and combination therapy and hypertension control the first year. *Hypertension* 2012;**59**:1124–31. https://doi.org/10.1161/hypertensionaha.112.194167
490. Sundström J, Lind L, Nowrouzi S, Hagström E, Held C, Lytsy P, *et al.* Heterogeneity in blood pressure response to 4 antihypertensive drugs: a randomized clinical trial. *JAMA* 2023;**329**:1160–9. https://doi.org/10.1001/jama.2023.3322
491. Tsioufis K, Kreutz R, Sykara G, van Vugt J, Hassan T. Impact of single-pill combination therapy on adherence, blood pressure control, and clinical outcomes: a rapid evidence assessment of recent literature. *J Hypertens* 2020;**38**:1016–28. https://doi.org/10.1097/hjh.0000000000002381
492. Parati G, Kjeldsen S, Coca A, Cushman WC, Wang J. Adherence to single-pill versus free-equivalent combination therapy in hypertension: a systematic review and meta-analysis. *Hypertension* 2021;**77**:692–705. https://doi.org/10.1161/hypertensionaha.120.15781
493. Egan BM, Kjeldsen SE, Narkiewicz K, Kreutz R, Burnier M. Single-pill combinations, hypertension control and clinical outcomes: potential, pitfalls and solutions. *Blood Press* 2022;**31**:164–8. https://doi.org/10.1080/08037051.2022.2095254
494. Yusuf S, Joseph P, Dans A, Gao P, Teo K, Xavier D, *et al.* Polypill with or without aspirin in persons without cardiovascular disease. *N Engl J Med* 2021;**384**:216–28. https://doi.org/10.1056/NEJMoa2028220
495. Roshandel G, Khoshnia M, Poustchi H, Hemming K, Kamangar F, Gharavi A, *et al.* Effectiveness of polypill for primary and secondary prevention of cardiovascular diseases (Polylran): a pragmatic, cluster-randomised trial. *Lancet* 2019;**394**:672–83. https://doi.org/10.1016/s0140-6736(19)31791-x
496. Joseph P, Roshandel G, Gao P, Pais P, Lonn E, Xavier D, *et al.* Fixed-dose combination therapies with and without aspirin for primary prevention of cardiovascular disease: an individual participant data meta-analysis. *Lancet* 2021;**398**:1133–46. https://doi.org/10.1016/s0140-6736(21)01827-4
497. van Vark LC, Bertrand M, Akkerhuis KM, Brugts JJ, Fox K, Mourad J-J, *et al.* Angiotensin-converting enzyme inhibitors reduce mortality in hypertension: a meta-analysis of randomized clinical trials of renin–angiotensin–aldosterone system inhibitors involving 158 998 patients. *Eur Heart J* 2012;**33**:2088–97. https://doi.org/10.1093/eurheartj/ehs075
498. Parving HH, Brenner BM, McMurray JJ, de Zeeuw D, Haffner SM, Solomon SD, *et al.* Cardiorenal end points in a trial of aliskiren for type 2 diabetes. *N Engl J Med* 2012;**367**:2204–13. https://doi.org/10.1056/NEJMoa1208799
499. Fried LF, Emanuele N, Zhang JH, Brophy M, Conner TA, Duckworth W, *et al.* Combined angiotensin inhibition for the treatment of diabetic nephropathy. *N Engl J Med* 2013;**369**:1892–903. https://doi.org/10.1056/NEJMoa1303154
500. Yusuf S, Teo KK, Pogue J, Dyal L, Copland I, Schumacher H, *et al.* Telmisartan, ramipril, or both in patients at high risk for vascular events. *N Engl J Med* 2008;**358**:1547–59. https://doi.org/10.1056/NEJMoa0801317
501. Gupta AK, Arshad S, Poulter NR. Compliance, safety, and effectiveness of fixed-dose combinations of antihypertensive agents: a meta-analysis. *Hypertension* 2010;**55**:399–407. https://doi.org/10.1161/hypertensionaha.109.139816
502. Gnanenthiran SR, Wang N, Di Tanna GL, Salam A, Webster R, de Silva HA, *et al.* Association of low-dose triple combination therapy vs usual care with time at target blood pressure: a secondary analysis of the TRIUMPH randomized clinical trial. *JAMA Cardiol* 2022;**7**:645–50. https://doi.org/10.1001/jamacardio.2022.0471
503. Tam TS, Wu MH, Masson SC, Tsang MP, Stabler SN, Kinkade A, *et al.* Eplerenone for hypertension. *Cochrane Database Syst Rev* 2017;**2**:CD008996. https://doi.org/10.1002/14651858.CD008996.pub2
504. Pedersen ME, Cockcroft JR. The vasodilatory beta-blockers. *Curr Hypertens Rep* 2007;**9**:269–77. https://doi.org/10.1007/s11906-007-0050-2
505. Sica DA. Minoxidil: an underused vasodilator for resistant or severe hypertension. *J Clin Hypertens (Greenwich)* 2004;**6**:283–7. https://doi.org/10.1111/j.1524-6175.2004.03585.x
506. Muñoz D, Uzoije P, Reynolds C, Miller R, Walkley D, Pappalardo S, *et al.* Polypill for cardiovascular disease prevention in an underserved population. *N Engl J Med* 2019;**381**:1114–23. https://doi.org/10.1056/NEJMoa1815359
507. Castellano JM, Pocock SJ, Bhatt DL, Quesada AJ, Owen R, Fernandez-Ortiz A, *et al.* Polypill strategy in secondary cardiovascular prevention. *N Engl J Med* 2022;**387**:967–77. https://doi.org/10.1056/NEJMoa2208275
508. González-Juanatey JR, Cordero A, Castellano JM, Masana L, Dalmau R, Ruiz E, *et al.* The CNIC-polypill reduces recurrent major cardiovascular events in real-life secondary prevention patients in Spain: the NEPTUNO study. *Int J Cardiol* 2022;**361**:116–23. https://doi.org/10.1016/j.ijcard.2022.05.015
509. Jowett S, Barton P, Roalfe A, Fletcher K, Hobbs FDR, McManus RJ, *et al.* Cost-effectiveness analysis of use of a polypill versus usual care or best practice for primary prevention in people at high risk of cardiovascular disease. *PLoS One* 2017;**12**:e0182625. https://doi.org/10.1371/journal.pone.0182625
510. McManus RJ, Mant J, Bray EP, Holder R, Jones MI, Greenfield S, *et al.* Telemonitoring and self-management in the control of hypertension (TASMINH2): a randomised controlled trial. *Lancet* 2010;**376**:163–72. https://doi.org/10.1016/s0140-6736(10)60964-6
511. Clark CE, Smith LF, Taylor RS, Campbell JL. Nurse led interventions to improve control of blood pressure in people with hypertension: systematic review and meta-analysis. *BMJ* 2010;**341**:c3995. https://doi.org/10.1136/bmj.c3995
512. Mackenzie IS, Rogers A, Poulter NR, Williams B, Brown MJ, Webb DJ, *et al.* Cardiovascular outcomes in adults with hypertension with evening versus morning dosing of usual antihypertensives in the UK (TIME study): a prospective, randomised, open-label, blinded-endpoint clinical trial. *Lancet* 2022;**400**:1417–25. https://doi.org/10.1016/s0140-6736(22)01786-x
513. Conn VS, Ruppar TM, Chase JA, Enriquez M, Cooper PS. Interventions to improve medication adherence in hypertensive patients: systematic review and meta-analysis. *Curr Hypertens Rep* 2015;**17**:94. https://doi.org/10.1007/s11906-015-0606-5
514. Corrao G, Parodi A, Zambon A, Heiman F, Filippi A, Cricelli C, *et al.* Reduced discontinuation of antihypertensive treatment by two-drug combination as first step. Evidence from daily life practice. *J Hypertens* 2010;**28**:1584–90. https://doi.org/10.1097/HJH.0b013e328339f9fa
515. Krieger EM, Drager LF, Giorgi DMA, Pereira AC, Barreto-Filho JAS, Nogueira AR, *et al.* Spironolactone versus clonidine as a fourth-drug therapy for resistant hypertension: the ReHOT randomized study (resistant hypertension optimal treatment). *Hypertension* 2018;**71**:681–90. https://doi.org/10.1161/hypertensionaha.117.10662
516. Williams B, MacDonald TM, Morant SV, Webb DJ, Sever P, McInnes GT, *et al.* Endocrine and haemodynamic changes in resistant hypertension, and blood pressure responses to spironolactone or amiloride: the PATHWAY-2 mechanisms substudies. *Lancet Diabetes Endocrinol* 2018;**6**:464–75. https://doi.org/10.1016/s2213-8587(18)30071-8
517. Mann JF, Schmieder RE, McQueen M, Dyal L, Schumacher H, Pogue J, *et al.* Renal outcomes with telmisartan, ramipril, or both, in people at high vascular risk (the ONTARGET study): a multicentre, randomised, double-blind, controlled trial. *Lancet* 2008;**372**:547–53. https://doi.org/10.1016/s0140-6736(08)61236-2
518. Jones DW, Whelton PK, Allen N, Clark D III, Gidding SS, Muntner P, *et al.* Management of stage 1 hypertension in adults with a low 10-year risk for cardiovascular disease: filling a guidance gap: a scientific statement from the American Heart Association. *Hypertension* 2021;**77**:e58–67. https://doi.org/10.1161/hyp.0000000000000195
519. Ali DH, Kiliç B, Hart HE, Bots ML, Biermans MCJ, Spiering W, *et al.* Therapeutic inertia in the management of hypertension in primary care. *J Hypertens* 2021;**39**:1238–45. https://doi.org/10.1097/hjh.0000000000002783
520. Appel LJ, Champagne CM, Harsha DW, Cooper LS, Obarzanek E, Elmer PJ, *et al.* Effects of comprehensive lifestyle modification on blood pressure control: main results of the PREMIER clinical trial. *JAMA* 2003;**289**:2083–93. https://doi.org/10.1001/jama.289.16.2083
521. Appel LJ, Sacks FM, Carey VJ, Obarzanek E, Swain JF, Miller ER, *et al.* Effects of protein, monounsaturated fat, and carbohydrate intake on blood pressure and serum lipids: results of the OmniHeart randomized trial. *JAMA* 2005;**294**:2455–64. https://doi.org/10.1001/jama.294.19.2455
522. Thomopoulos C, Parati G, Zanchetti A. Effects of blood-pressure-lowering treatment on outcome incidence. 12. Effects in individuals with high-normal and normal blood pressure: overview and meta-analyses of randomized trials. *J Hypertens* 2017;**35**:2150–60. https://doi.org/10.1097/hjh.0000000000001547
523. Williamson JD, Supiano MA, Applegate WB, Berlowitz DR, Campbell RC, Chertow GM, *et al.* Intensive vs standard blood pressure control and cardiovascular disease

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

outcomes in adults aged ≥75 years: a randomized clinical trial. *JAMA* 2016;**315**:2673–82. https://doi.org/10.1001/jama.2016.7050
524. Beckett N, Peters R, Leonetti G, Duggan J, Fagard R, Thijs L, *et al.* Subgroup and per-protocol analyses from the hypertension in the very elderly trial. *J Hypertens* 2014;**32**:1478–87; discussion 1487. https://doi.org/10.1097/hjh.0000000000000195
525. Jowett S, Kodabuckus S, Ford GA, Hobbs FDR, Lown M, Mant J, *et al.* Cost-effectiveness of antihypertensive deprescribing in primary care: a Markov modelling study using data from the OPTiMISE trial. *Hypertension* 2022;**79**:1122–31. https://doi.org/10.1161/hypertensionaha.121.18726
526. Odden MC, McClure LA, Sawaya BP, White CL, Peralta CA, Field TS, *et al.* Achieved blood pressure and outcomes in the secondary prevention of small subcortical strokes trial. *Hypertension* 2016;**67**:63–9. https://doi.org/10.1161/hypertensionaha.115.06480
527. Chen T, Shao F, Chen K, Wang Y, Wu Z, Wang Y, *et al.* Time to clinical benefit of intensive blood pressure lowering in patients 60 years and older with hypertension: a secondary analysis of randomized clinical trials. *JAMA Intern Med* 2022;**182**:660–7. https://doi.org/10.1001/jamainternmed.2022.1657
528. Neaton JD, Grimm RH Jr, Prineas RJ, Stamler J, Grandits GA, Elmer PJ, *et al.* Treatment of mild hypertension study. Final results. Treatment of mild hypertension study research group. *JAMA* 1993;**270**:713–24. https://doi.org/10.1001/jama.1993.03510060059034
529. Wu J, Kraja AT, Oberman A, Lewis C, Ellison R, Arnett D, *et al.* A summary of the effects of antihypertensive medications on measured blood pressure. *Am J Hypertens* 2005;**18**:935–42. https://doi.org/10.1016/j.amjhyper.2005.01.011
530. Canoy D, Copland E, Nazarzadeh M, Ramakrishnan R, Pinho-Gomes A-C, Salam A, *et al.* Antihypertensive drug effects on long-term blood pressure: an individual-level data meta-analysis of randomised clinical trials. *Heart* 2022;**108**:1281–9. https://doi.org/10.1136/heartjnl-2021-320171
531. Paz MA, de-La-Sierra A, Sáez M, Barceló MA, Rodríguez JJ, Castro S, *et al.* Treatment efficacy of anti-hypertensive drugs in monotherapy or combination: ATOM systematic review and meta-analysis of randomized clinical trials according to PRISMA statement. *Medicine (Baltimore)* 2016;**95**:e4071. https://doi.org/10.1097/md.0000000000004071
532. Morales-Salinas A, Kones R. Concerning the degradation of β-blocker use in the 2018 ESC/ESH hypertension guidelines. *Eur Heart J* 2019;**40**:2091. https://doi.org/10.1093/eurheartj/ehz125
533. Lindholm LH, Carlberg B, Samuelsson O. Should beta blockers remain first choice in the treatment of primary hypertension? A meta-analysis. *Lancet* 2005;**366**:1545–53. https://doi.org/10.1016/s0140-6736(05)67573-3
534. Larochelle P, Tobe SW, Lacourcière Y. β-Blockers in hypertension: studies and meta-analyses over the years. *Can J Cardiol* 2014;**30**:S16–22. https://doi.org/10.1016/j.cjca.2014.02.012
535. Reboussin DM, Allen NB, Griswold ME, Guallar E, Hong Y, Lackland DT, *et al.* Systematic review for the 2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA guideline for the prevention, detection, evaluation, and management of high blood pressure in adults: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *Hypertension* 2018;**71**:e116–35. https://doi.org/10.1161/hyp.0000000000000067
536. Wright JM, Musini VM, Gill R. First-line drugs for hypertension. *Cochrane Database Syst Rev* 2018;**4**:CD001841. https://doi.org/10.1002/14651858.CD001841.pub3
537. Johnston GD. Dose-response relationships with antihypertensive drugs. *Pharmacol Ther* 1992;**55**:53–93. https://doi.org/10.1016/0163-7258(92)90029-y
538. Law MR, Morris JK, Wald NJ. Use of blood pressure lowering drugs in the prevention of cardiovascular disease: meta-analysis of 147 randomised trials in the context of expectations from prospective epidemiological studies. *BMJ* 2009;**338**:b1665. https://doi.org/10.1136/bmj.b1665
539. Messerli FH, Bangalore S, Schmieder RE. Wilder's principle: pre-treatment value determines post-treatment response. *Eur Heart J* 2015;**36**:576–9. https://doi.org/10.1093/eurheartj/ehu467
540. Lasserson DS, Buclin T, Glasziou P. How quickly should we titrate antihypertensive medication? Systematic review modelling blood pressure response from trial data. *Heart* 2011;**97**:1771–5. https://doi.org/10.1136/hrt.2010.221473
541. Thomopoulos C, Parati G, Zanchetti A. Effects of blood pressure lowering on outcome incidence in hypertension. 1. Overview, meta-analyses, and meta-regression analyses of randomized trials. *J Hypertens* 2014;**32**:2285–95. https://doi.org/10.1097/hjh.0000000000000378
542. Verdecchia P, Staessen JA, Angeli F, de Simone G, Achilli A, Ganau A, *et al.* Usual versus tight control of systolic blood pressure in non-diabetic patients with hypertension (Cardio-Sis): an open-label randomised trial. *Lancet* 2009;**374**:525–33. https://doi.org/10.1016/s0140-6736(09)61340-4
543. Benavente OR, Coffey CS, Conwit R, Hart RG, McClure LA, Pearce LA, *et al.* Blood-pressure targets in patients with recent lacunar stroke: the SPS3 randomised trial. *Lancet* 2013;**382**:507–15. https://doi.org/10.1016/s0140-6736(13)60852-1
544. Vaduganathan M, Claggett BL, Juraschek SP, Solomon SD. Assessment of long-term benefit of intensive blood pressure control on residual life span: secondary analysis of the Systolic Blood Pressure Intervention Trial (SPRINT). *JAMA Cardiol* 2020;**5**:576–81. https://doi.org/10.1001/jamacardio.2019.6192
545. Lewis CE, Fine LJ, Beddhu S, Cheung AK, Cushman WC, Cutler JA, *et al.* Final report of a trial of intensive versus standard blood-pressure control. *N Engl J Med* 2021;**384**:1921–30. https://doi.org/10.1056/NEJMoa1901281
546. Beddhu S, Chertow GM, Cheung AK, Cushman WC, Rahman M, Greene T, *et al.* Influence of baseline diastolic blood pressure on effects of intensive compared with standard blood pressure control. *Circulation* 2018;**137**:134–43. https://doi.org/10.1161/circulationaha.117.030848
547. McEvoy JW, Daya N, Rahman F, Hoogeveen RC, Blumenthal RS, Shah AM, *et al.* Association of isolated diastolic hypertension as defined by the 2017 ACC/AHA blood pressure guideline with incident cardiovascular outcomes. *JAMA* 2020;**323**:329–38. https://doi.org/10.1001/jama.2019.21402
548. Jacobsen AP, Al Rifai M, Arps K, Whelton SP, Budoff MJ, Nasir K, *et al.* A cohort study and meta-analysis of isolated diastolic hypertension: searching for a threshold to guide treatment. *Eur Heart J* 2021;**42**:2119–29. https://doi.org/10.1093/eurheartj/ehab111
549. McEvoy JW, Yang WY, Thijs L, Zhang Z-Y, Melgarejo JD, Boggia J, *et al.* Isolated diastolic hypertension in the IDACO study: an age-stratified analysis using 24-hour ambulatory blood pressure measurements. *Hypertension* 2021;**78**:1222–31. https://doi.org/10.1161/hypertensionaha.121.17766
550. Albasri A, Hattle M, Koshiaris C, Dunnigan A, Paxton B, Fox SE, *et al.* Association between antihypertensive treatment and adverse events: systematic review and meta-analysis. *BMJ* 2021;**372**:n189. https://doi.org/10.1136/bmj.n189
551. Savoia C, Volpe M, Grassi G, Borghi C, Agabiti Rosei E, Touyz RM, *et al.* Personalized medicine-a modern approach for the diagnosis and management of hypertension. *Clin Sci (Lond)* 2017;**131**:2671–85. https://doi.org/10.1042/cs20160407
552. Egan BM, Basile JN, Rehman SU, Davis PB, Grob CH 3rd, Riehle JF, *et al.* Plasma renin test–guided drug treatment algorithm for correcting patients with treated but uncontrolled hypertension: a randomized controlled trial. *Am J Hypertens* 2009;**22**:792–801. https://doi.org/10.1038/ajh.2009.63
553. Huang KY, Tseng PT, Wu YC, Tu Y-K, Stubbs B, Su K-P, *et al.* Do beta-adrenergic blocking agents increase asthma exacerbation? A network meta-analysis of randomized controlled trials. *Sci Rep* 2021;**11**:452. https://doi.org/10.1038/s41598-020-79837-3
554. Bennett M, Chang CL, Tatley M, Savage R, Hancox RJ. The safety of cardioselective β_1-blockers in asthma: literature review and search of global pharmacovigilance safety reports. *ERJ Open Res* 2021;**7**:00801-2020. https://doi.org/10.1183/23120541.00801-2020
555. Assimes TL, Elstein E, Langleben A, Suissa S. Long-term use of antihypertensive drugs and risk of cancer. *Pharmacoepidemiol Drug Saf* 2008;**17**:1039–49. https://doi.org/10.1002/pds.1656
556. Zhang Y, Song M, Chan AT, Meyerhardt JA, Willett WC, Giovannucci EL. Long-term use of antihypertensive medications, hypertension and colorectal cancer risk and mortality: a prospective cohort study. *Br J Cancer* 2022;**127**:1974–82. https://doi.org/10.1038/s41416-022-01975-4
557. Grossman E, Messerli FH. Long-term safety of antihypertensive therapy. *Prog Cardiovasc Dis* 2006;**49**:16–25. https://doi.org/10.1016/j.pcad.2006.06.002
558. Parati G, Stergiou GS, Dolan E, Bilo G. Blood pressure variability: clinical relevance and application. *J Clin Hypertens (Greenwich)* 2018;**20**:1133–7. https://doi.org/10.1111/jch.13304
559. Schutte AE, Kollias A, Stergiou GS. Blood pressure and its variability: classic and novel measurement techniques. *Nat Rev Cardiol* 2022;**19**:643–54. https://doi.org/10.1038/s41569-022-00690-0
560. Lauder L, Azizi M, Kirtane AJ, Böhm M, Mahfoud F. Device-based therapies for arterial hypertension. *Nat Rev Cardiol* 2020;**17**:614–28. https://doi.org/10.1038/s41569-020-0364-1
561. Mahfoud F, Schlaich MP, Lobo MD. Device therapy of hypertension. *Circ Res* 2021;**128**:1080–99. https://doi.org/10.1161/circresaha.121.318091
562. DiBona GF. Sympathetic nervous system and hypertension. *Hypertension* 2013;**61**:556–60. https://doi.org/10.1161/hypertensionaha.111.00633
563. DiBona GF, Esler M. Translational medicine: the antihypertensive effect of renal denervation. *Am J Physiol Regul Integr Comp Physiol* 2010;**298**:R245–253. https://doi.org/10.1152/ajpregu.00647.2009
564. Böhm M, Kario K, Kandzari DE, Mahfoud F, Weber MA, Schmieder RE, *et al.* Efficacy of catheter-based renal denervation in the absence of antihypertensive medications (SPYRAL HTN-OFF MED Pivotal): a multicentre, randomised, sham-controlled trial. *Lancet* 2020;**395**:1444–51. https://doi.org/10.1016/s0140-6736(20)30554-7
565. Weber MA, Kirtane AJ, Weir MR, Radhakrishnan J, Das T, Berk M, *et al.* The REDUCE HTN: REINFORCE: randomized, sham-controlled trial of bipolar radiofrequency renal denervation for the treatment of hypertension. *JACC Cardiovasc Interv* 2020;**13**:461–70. https://doi.org/10.1016/j.jcin.2019.10.061
566. Kandzari DE, Böhm M, Mahfoud F, Townsend RR, Weber MA, Pocock S, *et al.* Effect of renal denervation on blood pressure in the presence of antihypertensive drugs: 6-month efficacy and safety results from the SPYRAL HTN-ON MED proof-of-concept randomised trial. *Lancet* 2018;**391**:2346–55. https://doi.org/10.1016/s0140-6736(18)30951-6
567. Azizi M, Schmieder RE, Mahfoud F, Weber MA, Daemen J, Davies J, *et al.* Endovascular ultrasound renal denervation to treat hypertension (RADIANCE-HTN SOLO): a multicentre, international, single-blind, randomised, sham-controlled trial. *Lancet* 2018;**391**:2335–45. https://doi.org/10.1016/s0140-6736(18)31082-1

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

568. Azizi M, Sanghvi K, Saxena M, Gosse P, Reilly JP, Levy T, *et al.* Ultrasound renal denervation for hypertension resistant to a triple medication pill (RADIANCE-HTN TRIO): a randomised, multicentre, single-blind, sham-controlled trial. *Lancet* 2021;**397**:2476–86. https://doi.org/10.1016/s0140-6736(21)00788-1
569. Mahfoud F, Böhm M, Schmieder R, Narkiewicz K, Ewen S, Ruilope L, *et al.* Effects of renal denervation on kidney function and long-term outcomes: 3-year follow-up from the global SYMPLICITY registry. *Eur Heart J* 2019;**40**:3474–82. https://doi.org/10.1093/eurheartj/ehz118
570. Bhatt DL, Vaduganathan M, Kandzari DE, Leon MB, Rocha-Singh K, Townsend RR, *et al.* Long-term outcomes after catheter-based renal artery denervation for resistant hypertension: final follow-up of the randomised SYMPLICITY HTN-3 trial. *Lancet* 2022;**400**:1405–16. https://doi.org/10.1016/s0140-6736(22)01787-1
571. Mahfoud F, Kandzari DE, Kario K, Townsend RR, Weber MA, Schmieder RE, *et al.* Long-term efficacy and safety of renal denervation in the presence of antihypertensive drugs (SPYRAL HTN-ON MED): a randomised, sham-controlled trial. *Lancet* 2022;**399**:1401–10. https://doi.org/10.1016/s0140-6736(22)00455-x
572. Rader F, Kirtane AJ, Wang Y, Daemen J, Lurz P, Sayer J, *et al.* Durability of blood pressure reduction after ultrasound renal denervation: three-year follow-up of the treatment arm of the randomised RADIANCE-HTN SOLO trial. *EuroIntervention* 2022;**18**:e677–85. https://doi.org/10.4244/eij-d-22-00305
573. Al Ghorani H, Kulenthiran S, Recktenwald MJM, Lauder L, Kunz M, Götzinger F, *et al.* 10-year outcomes of catheter-based renal denervation in patients with resistant hypertension. *J Am Coll Cardiol* 2023;**81**:517–9. https://doi.org/10.1016/j.jacc.2022.11.038
574. Stone P, Campbell J, Thompson S, Walker J. A prospective, randomized study comparing ultrasound versus fluoroscopic guided femoral arterial access in noncardiac vascular patients. *J Vasc Surg* 2020;**72**:259–67. https://doi.org/10.1016/j.jvs.2019.09.051
575. Jolly SS, Yusuf S, Cairns J, Niemelä K, Xavier D, Widimsky P, *et al.* Radial versus femoral access for coronary angiography and intervention in patients with acute coronary syndromes (RIVAL): a randomised, parallel group, multicentre trial. *Lancet* 2011;**377**:1409–20. https://doi.org/10.1016/s0140-6736(11)60404-2
576. Townsend RR, Walton A, Hettrick DA, Hickey GL, Weil J, Sharp ASP, *et al.* Review and meta-analysis of renal artery damage following percutaneous renal denervation with radiofrequency renal artery ablation. *EuroIntervention* 2020;**16**:89–96. https://doi.org/10.4244/eij-d-19-00902
577. Sanders MF, Reitsma JB, Morpey M, Gremmels H, Bots ML, Pisano A, *et al.* Renal safety of catheter-based renal denervation: systematic review and meta-analysis. *Nephrol Dial Transplant* 2017;**32**:1440–7. https://doi.org/10.1093/ndt/gfx088
578. Ahmad Y, Francis DP, Bhatt DL, Howard JP. Renal denervation for hypertension: a systematic review and meta-analysis of randomized, blinded, placebo-controlled trials. *JACC Cardiovasc Interv* 2021;**14**:2614–24. https://doi.org/10.1016/j.jcin.2021.09.020
579. Chowdhury EK, Reid CM, Zomer E, Kelly DJ, Liew D. Cost-effectiveness of renal denervation therapy for treatment-resistant hypertension: a best case scenario. *Am J Hypertens* 2018;**31**:1156–63. https://doi.org/10.1093/ajh/hpy108
580. Sharp ASP, Cao KN, Esler MD, Kandzari DE, Lobo MD, Schmieder RE, *et al.* Cost-effectiveness of catheter-based radiofrequency renal denervation for the treatment of uncontrolled hypertension: an analysis for the UK based on recent clinical evidence. *Eur Heart J Qual Care Clin Outcomes* 2024:qcae001. https://doi.org/10.1093/ehjqcco/qcae001
581. Fengler K, Reimann P, Rommel KP, Kresoja K-P, Blazek S, Unterhuber M, *et al.* Comparison of long-term outcomes for responders versus non-responders following renal denervation in resistant hypertension. *J Am Heart Assoc* 2021;**10**:e022429. https://doi.org/10.1161/jaha.121.022429
582. Mahfoud F, Mancia G, Schmieder RE, Ruilope L, Narkiewicz K, Schlaich M, *et al.* Cardiovascular risk reduction after renal denervation according to time in therapeutic systolic blood pressure range. *J Am Coll Cardiol* 2022;**80**:1871–80. https://doi.org/10.1016/j.jacc.2022.08.802
583. Byrne RA, Rossello X, Coughlan JJ, Barbato E, Berry C, Chieffo A, *et al.* ESC guidelines for the management of acute coronary syndromes. *Eur Heart J* 2023;**44**:3720–826. https://doi.org/10.1093/eurheartj/ehad191
584. Singh RR, Denton KM. Renal denervation. *Hypertension* 2018;**72**:528–36. https://doi.org/10.1161/hypertensionaha.118.10265
585. Barbato E, Azizi M, Schmieder RE, Lauder L, Böhm M, Brouwers S, *et al.* Renal denervation in the management of hypertension in adults. A clinical consensus statement of the ESC Council on Hypertension and the European Association of Percutaneous Cardiovascular Interventions (EAPCI). *Eur Heart J* 2023;**44**:1313–30. https://doi.org/10.1093/eurheartj/ehad054
586. Bhatt DL, Kandzari DE, O'Neill WW, D'Agostino R, Flack JM, Katzen BT, *et al.* A controlled trial of renal denervation for resistant hypertension. *N Engl J Med* 2014;**370**:1393–401. https://doi.org/10.1056/NEJMoa1402670
587. Desch S, Okon T, Heinemann D, Kulle K, Röhnert K, Sonnabend M, *et al.* Randomized sham-controlled trial of renal sympathetic denervation in mild resistant hypertension. *Hypertension* 2015;**65**:1202–8. https://doi.org/10.1161/hypertensionaha.115.05283
588. Mathiassen ON, Vase H, Bech JN, Christensen KL, Buus NH, Schroeder AP, *et al.* Renal denervation in treatment-resistant essential hypertension. A randomized, SHAM-controlled, double-blinded 24–h blood pressure-based trial. *J Hypertens* 2016;**34**:1639–47. https://doi.org/10.1097/hjh.0000000000000977
589. Townsend RR, Mahfoud F, Kandzari DE, Kario K, Pocock S, Weber MA, *et al.* Catheter-based renal denervation in patients with uncontrolled hypertension in the absence of antihypertensive medications (SPYRAL HTN-OFF MED): a randomised, sham-controlled, proof-of-concept trial. *Lancet* 2017;**390**:2160–70. https://doi.org/10.1016/s0140-6736(17)32281-x
590. Kario K, Yokoi Y, Okamura K, Fujihara M, Ogoyama Y, Yamamoto E, *et al.* Catheter-based ultrasound renal denervation in patients with resistant hypertension: the randomized, controlled REQUIRE trial. *Hypertens Res* 2022;**45**:221–31. https://doi.org/10.1038/s41440-021-00754-7
591. Neuzil P, Merkely B, Erglis A, Marinskis G, de Groot JR, Schmidinger H, *et al.* Pacemaker-mediated programmable hypertension control therapy. *J Am Heart Assoc* 2017;**6**:e006974. https://doi.org/10.1161/jaha.117.006974
592. Kalarus Z, Merkely B, Neužil P, Grabowski M, Mitkowski P, Marinskis G, *et al.* Pacemaker-based cardiac neuromodulation therapy in patients with hypertension: a pilot study. *J Am Heart Assoc* 2021;**10**:e020492. https://doi.org/10.1161/jaha.120.020492
593. Musini VM, Tejani AM, Bassett K, Puil L, Wright JM. Pharmacotherapy for hypertension in adults 60 years or older. *Cochrane Database Syst Rev* 2019;**6**:CD000028. https://doi.org/10.1002/14651858.CD000028.pub3
594. Musini VM, Gueyffier F, Puil L, Salzwedel DM, Wright JM. Pharmacotherapy for hypertension in adults aged 18 to 59 years. *Cochrane Database Syst Rev* 2017;**8**:CD008276. https://doi.org/10.1002/14651858.CD008276.pub2
595. Thomopoulos C, Parati G, Zanchetti A. Effects of blood-pressure-lowering treatment in hypertension: 9. Discontinuations for adverse events attributed to different classes of antihypertensive drugs: meta-analyses of randomized trials. *J Hypertens* 2016;**34**:1921–32. https://doi.org/10.1097/hjh.0000000000001052
596. Benetos A, Petrovic M, Strandberg T. Hypertension management in older and frail older patients. *Circ Res* 2019;**124**:1045–60. https://doi.org/10.1161/circresaha.118.313236
597. Frey L, Gravestock I, Pichierri G, Steurer J, Burgstaller JM. Serious adverse events in patients with target-oriented blood pressure management: a systematic review. *J Hypertens* 2019;**37**:2135–44. https://doi.org/10.1097/hjh.0000000000002176
598. Palmer SC, Sciancalepore M, Strippoli GF. Trial quality in nephrology: how are we measuring up? *Am J Kidney Dis* 2011;**58**:335–7. https://doi.org/10.1053/j.ajkd.2011.06.006
599. Carriazo S, Sarafidis P, Ferro CJ, Ortiz A. Blood pressure targets in CKD 2021: the never-ending guidelines debacle. *Clin Kidney J* 2022;**15**:845–51. https://doi.org/10.1093/ckj/sfac014
600. Kessler A, Sollie S, Challacombe B, Briggs K, Van Hemelrijck M. The global prevalence of erectile dysfunction: a review. *BJU Int* 2019;**124**:587–99. https://doi.org/10.1111/bju.14813
601. Doumas M, Douma S. The effect of antihypertensive drugs on erectile function: a proposed management algorithm. *J Clin Hypertens (Greenwich)* 2006;**8**:359–64. https://doi.org/10.1111/j.1524-6175.2005.05285.x
602. Ismail SB, Noor NM, Hussain NHN, Sulaiman Z, Shamsudin MA, Irfan M. Angiotensin receptor blockers for erectile dysfunction in hypertensive men: a brief meta-analysis of randomized control trials. *Am J Mens Health* 2019;**13**:1557988319892735. https://doi.org/10.1177/1557988319892735
603. Lawson AJ, Hameed MA, Brown R, Cappuccio FP, George S, Hinton T, *et al.* Nonadherence to antihypertensive medications is related to pill burden in apparent treatment-resistant hypertensive individuals. *J Hypertens* 2020;**38**:1165–73. https://doi.org/10.1097/hjh.0000000000002398
604. Choudhry NK, Kronish IM, Vongpatanasin W, Ferdinand KC, Pavlik VN, Egan BM, *et al.* Medication adherence and blood pressure control: a scientific statement from the American heart association. *Hypertension* 2022;**79**:e1–14. https://doi.org/10.1161/hyp.0000000000000203
605. Liao CT, Toh HS, Sun L, Yang C-T, Hu A, Wei D, *et al.* Cost-effectiveness of intensive vs standard blood pressure control among older patients with hypertension. *JAMA Netw Open* 2023;**6**:e230708. https://doi.org/10.1001/jamanetworkopen.2023.0708
606. O'Donoghue P, O'Halloran AM, Kenny RA, Romero-Ortuno R. Do the frail experience more adverse events from intensive blood pressure control? A 2-year prospective study in the Irish Longitudinal Study on Ageing (TILDA). *EClinicalMedicine* 2022;**45**:101304. https://doi.org/10.1016/j.eclinm.2022.101304
607. Sexton DJ, Canney M, O'Connell MDL, Moore P, Little MA, O'Seaghdha CM, *et al.* Injurious falls and syncope in older community-dwelling adults meeting inclusion criteria for SPRINT. *JAMA Intern Med* 2017;**177**:1385–7. https://doi.org/10.1001/jamainternmed.2017.2924
608. Masoli JAH, Sheppard JP, Rajkumar C. Hypertension management in older patients-are the guideline blood pressure targets appropriate? *Age Ageing* 2022;**51**:afab226. https://doi.org/10.1093/ageing/afab226
609. Richter D, Guasti L, Walker D, Lambrinou E, Lionis C, Abreu A, *et al.* Frailty in cardiology: definition, assessment and clinical implications for general cardiology. A consensus document of the Council for Cardiology Practice (CCP), Association for Acute Cardio Vascular Care (ACVC), Association of Cardiovascular Nursing and Allied Professions (ACNAP), European Association of Preventive Cardiology (EAPC), European Heart Rhythm Association (EHRA), Council on Valvular Heart Diseases (VHD), Council on Hypertension (CHT), Council of Cardio-Oncology (CCO),

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

Working Group (WG) Aorta and Peripheral Vascular Diseases, WG e-Cardiology, WG Thrombosis, of the European Society of Cardiology, European Primary Care Cardiology Society (EPCCS). *Eur J Prev Cardiol* 2022;**29**:216–27. https://doi.org/10.1093/eurjpc/zwaa167
610. Hughes D, Judge C, Murphy R, Loughlin E, Costello M, Whiteley W, *et al.* Association of blood pressure lowering with incident dementia or cognitive impairment: a systematic review and meta-analysis. *JAMA* 2020;**323**:1934–44. https://doi.org/10.1001/jama.2020.4249
611. Peters R, Xu Y, Fitzgerald O, Aung HL, Beckett N, Bulpitt C, *et al.* Blood pressure lowering and prevention of dementia: an individual patient data meta-analysis. *Eur Heart J* 2022;**43**:4980–90. https://doi.org/10.1093/eurheartj/ehac584
612. Pathak A, Poulter NR, Kavanagh M, Kreutz R, Burnier M. Improving the management of hypertension by tackling awareness, adherence, and clinical inertia: a symposium report. *Am J Cardiovasc Drugs* 2022;**22**:251–61. https://doi.org/10.1007/s40256-021-00505-6
613. Sheppard JP, Lown M, Burt J, Temple E, Lowe R, Ashby H, *et al.* Generalizability of blood pressure lowering trials to older patients: cross-sectional analysis. *J Am Geriatr Soc* 2020;**68**:2508–15. https://doi.org/10.1111/jgs.16749
614. Thomopoulos C, Parati G, Zanchetti A. Effects of blood pressure lowering treatment in hypertension: 8. Outcome reductions vs. discontinuations because of adverse drug events—meta-analyses of randomized trials. *J Hypertens* 2016;**34**:1451–63. https://doi.org/10.1097/hjh.0000000000000972
615. Jordan J, Tank J, Reuter H. Risk-benefit assessment of intense blood pressure lowering. *Hypertension* 2019;**74**:1302–4. https://doi.org/10.1161/hypertensionaha.119.13835
616. Rietz H, Pennlert J, Nordström P, Brunström M. Prevalence, time-trends and clinical characteristics of hypertension in young adults: nationwide cross-sectional study of 1.7 million Swedish 18-year-olds, 1969–2010. *J Hypertens* 2022;**40**:1231–8. https://doi.org/10.1097/hjh.0000000000003141
617. Ostchega Y, Fryar CD, Nwankwo T, Nguyen DT. Hypertension prevalence among adults aged 18 and over: United States, 2017–2018. *NCHS Data Brief* 2020;**364**:1–8.
618. O'Neil A, Scovelle AJ, Milner AJ, Kavanagh A. Gender/sex as a social determinant of cardiovascular risk. *Circulation* 2018;**137**:854–64. https://doi.org/10.1161/circulationaha.117.028595
619. Bruno RM, Pucci G, Rosticci M, Guarino L, Guglielmo C, Agabiti Rosei C, *et al.* Association between lifestyle and systemic arterial hypertension in young adults: a national, survey-based, cross-sectional study. *High Blood Press Cardiovasc Prev* 2016;**23**:31–40. https://doi.org/10.1007/s40292-016-0135-6
620. Suzuki Y, Kaneko H, Yano Y, Okada A, Itoh H, Matsuoka S, *et al.* Association of cardiovascular health metrics with risk of transition to hypertension in non-hypertensive young adults. *Am J Hypertens* 2022;**35**:858–66. https://doi.org/10.1093/ajh/hpac057
621. Haggart RC, Bartels CM, Smith MA, Johnson HM. Sociodemographics and hypertension control among young adults with incident hypertension: a multidisciplinary group practice observational study. *J Hypertens* 2018;**36**:2425–33. https://doi.org/10.1097/hjh.0000000000001872
622. Liu J, Bu X, Wei L, Wang X, Lai L, Dong C, *et al.* Global burden of cardiovascular diseases attributable to hypertension in young adults from 1990 to 2019. *J Hypertens* 2021;**39**:2488–96. https://doi.org/10.1097/hjh.0000000000002958
623. Zhang Y, Moran AE. Trends in the prevalence, awareness, treatment, and control of hypertension among young adults in the United States, 1999 to 2014. *Hypertension* 2017;**70**:736–42. https://doi.org/10.1161/hypertensionaha.117.09801
624. Yano Y, Stamler J, Garside DB, Daviglus ML, Franklin SS, Carnethon MR, *et al.* Isolated systolic hypertension in young and middle-aged adults and 31-year risk for cardiovascular mortality: the Chicago Heart Association Detection Project in Industry study. *J Am Coll Cardiol* 2015;**65**:327–35. https://doi.org/10.1016/j.jacc.2014.10.060
625. Noilhan C, Barigou M, Bieler L, Amar J, Chamontin B, Bouhanick B, *et al.* Causes of secondary hypertension in the young population: a monocentric study. *Ann Cardiol Angeiol (Paris)* 2016;**65**:159–64. https://doi.org/10.1016/j.ancard.2016.04.016
626. Jones ESW, Esack I, Mangena P, Rayner BL. Hypertension in adolescents and young adults referred to a tertiary hypertension clinic in Cape Town, South Africa. *Medicine (Baltimore)* 2020;**99**:e23137. https://doi.org/10.1097/md.0000000000023137
627. Chasan-Taber L, Willett WC, Manson JE, Spiegelman D, Hunter DJ, Curhan G, *et al.* Prospective study of oral contraceptives and hypertension among women in the United States. *Circulation* 1996;**94**:483–9. https://doi.org/10.1161/01.cir.94.3.483
628. Lubianca JN, Faccin CS, Fuchs FD. Oral contraceptives: a risk factor for uncontrolled blood pressure among hypertensive women. *Contraception* 2003;**67**:19–24. https://doi.org/10.1016/s0010-7824(02)00429-8
629. The American College of Obstetricians and Gynecologists. ACOG practice bulletin No. 206: use of hormonal contraception in women with coexisting medical conditions. *Obstet Gynecol* 2019;**133**:e128–50. https://doi.org/10.1097/aog.0000000000003072
630. Glisic M, Shahzad S, Tsoli S, Chadni M, Asllanaj E, Rojas LZ, *et al.* Association between progestin-only contraceptive use and cardiometabolic outcomes: a systematic review and meta-analysis. *Eur J Prev Cardiol* 2018;**25**:1042–52. https://doi.org/10.1177/2047487318774847
631. Archer DF, Ahrendt HJ, Drouin D. Drospirenone-only oral contraceptive: results from a multicenter noncomparative trial of efficacy, safety and tolerability. *Contraception* 2015;**92**:439–44. https://doi.org/10.1016/j.contraception.2015.07.014
632. Pappaccogli M, Di Monaco S, Warchoł-Celińska E, Lorthioir A, Amar L, Aparicio LS, *et al.* The European/International Fibromuscular Dysplasia Registry and Initiative (FEIRI)—clinical phenotypes and their predictors based on a cohort of 1000 patients. *Cardiovasc Res* 2021;**117**:950–9. https://doi.org/10.1093/cvr/cvaa102
633. Thompson P, Logan I, Tomson C, Sheerin N, Ellam T. Obesity, sex, race, and early onset hypertension: implications for a refined investigation strategy. *Hypertension* 2020;**76**:859–65. https://doi.org/10.1161/hypertensionaha.120.15557
634. Saladini F, Fania C, Mos L, Mazzer A, Casiglia E, Palatini P. Office pulse pressure is a predictor of favorable outcome in young- to middle-aged subjects with stage 1 hypertension. *Hypertension* 2017;**70**:537–42. https://doi.org/10.1161/hypertensionaha.117.09516
635. Hoeltzenbein M, Tissen-Diabaté T, Fietz AK, Zinke S, Kayser A, Meister R, *et al.* Increased rate of birth defects after first trimester use of angiotensin converting enzyme inhibitors—treatment or hypertension related? An observational cohort study. *Pregnancy Hypertens* 2018;**13**:65–71. https://doi.org/10.1016/j.preghy.2018.04.022
636. Ahmed B, Tran DT, Zoega H, Kennedy SE, Jorm LR, Havard A. Maternal and perinatal outcomes associated with the use of renin-angiotensin system (RAS) blockers for chronic hypertension in early pregnancy. *Pregnancy Hypertens* 2018;**14**:156–61. https://doi.org/10.1016/j.preghy.2018.09.010
637. Cífková R, Johnson MR, Kahan T, Brguljan J, Williams B, Coca A, *et al.* Peripartum management of hypertension: a position paper of the ESC Council on Hypertension and the European Society of Hypertension. *Eur Heart J Cardiovasc Pharmacother* 2020;**6**:384–93. https://doi.org/10.1093/ehjcvp/pvz082
638. Magee LA, Brown MA, Hall DR, Gupte S, Hennessy A, Karumanchi SA, *et al.* The 2021 International Society for the Study of Hypertension in Pregnancy classification, diagnosis & management recommendations for international practice. *Pregnancy Hypertens* 2022;**27**:148–69. https://doi.org/10.1016/j.preghy.2021.09.008
639. Kassebaum NJ, Barber RM, Bhutta ZA, Dandona L, Gething PW, Hay SI, *et al.* Global, regional, and national levels of maternal mortality, 1990–2015: a systematic analysis for the Global Burden of Disease Study 2015. *The Lancet* 2016;**388**:1775–812. https://doi.org/10.1016/S0140-6736(16)31470-2
640. Garovic VD, White WM, Vaughan L, Saiki M, Parashuram S, Garcia-Valencia O, *et al.* Incidence and long-term outcomes of hypertensive disorders of pregnancy. *J Am Coll Cardiol* 2020;**75**:2323–34. https://doi.org/10.1016/j.jacc.2020.03.028
641. Heida KY, Franx A, van Rijn BB, Eijkemans MJC, Boer JMA, Verschuren MWM, *et al.* Earlier age of onset of chronic hypertension and type 2 diabetes mellitus after a hypertensive disorder of pregnancy or gestational diabetes mellitus. *Hypertension* 2015;**66**:1116–22. https://doi.org/10.1161/hypertensionaha.115.06005
642. Garovic VD, Bailey KR, Boerwinkle E, Hunt SC, Weder AB, Curb D, *et al.* Hypertension in pregnancy as a risk factor for cardiovascular disease later in life. *J Hypertens* 2010;**28**:826–33. https://doi.org/10.1097/HJH.0b013e328335c29a
643. Brown MA, Magee LA, Kenny LC, Karumanchi SA, McCarthy FP, Saito S, *et al.* Hypertensive disorders of pregnancy: ISSHP classification, diagnosis, and management recommendations for international practice. *Hypertension* 2018;**72**:24–43. https://doi.org/10.1161/hypertensionaha.117.10803
644. Homer CS, Brown MA, Mangos G, Davis GK. Non-proteinuric pre-eclampsia: a novel risk indicator in women with gestational hypertension. *J Hypertens* 2008;**26**:295–302. https://doi.org/10.1097/HJH.0b013e3282f1a953
645. Davis GK, Roberts LM, Mangos GJ, Brown MA. Comparisons of auscultatory hybrid and automated sphygmomanometers with mercury sphygmomanometry in hypertensive and normotensive pregnant women: parallel validation studies. *J Hypertens* 2015;**33**:499–505; discussion 505–496. https://doi.org/10.1097/hjh.0000000000000420
646. Schmella MJ, Clifton RG, Althouse AD, Roberts JM. Uric acid determination in gestational hypertension: is it as effective a delineator of risk as proteinuria in high-risk women? *Reprod Sci* 2015;**22**:1212–9. https://doi.org/10.1177/1933719115572477
647. Rolnik DL, Wright D, Poon LC, O'Gorman N, Syngelaki A, de Paco Matallana C, *et al.* Aspirin versus placebo in pregnancies at high risk for preterm preeclampsia. *N Engl J Med* 2017;**377**:613–22. https://doi.org/10.1056/NEJMoa1704559
648. Chappell LC, Shennan AH. Assessment of proteinuria in pregnancy. *BMJ* 2008;**336**:968–9. https://doi.org/10.1136/bmj.39540.657928.BE
649. Jeon HR, Jeong DH, Lee JY, Woo EY, Shin GT, Kim S-Y. sFlt-1/PlGF ratio as a predictive and prognostic marker for preeclampsia. *J Obstet Gynaecol Res* 2021;**47**:2318–23. https://doi.org/10.1111/jog.14815
650. Bateman BT, Bansil P, Hernandez-Diaz S, Mhyre JM, Callaghan WM, Kuklina EV. Prevalence, trends, and outcomes of chronic hypertension: a nationwide sample of delivery admissions. *Am J Obstet Gynecol* 2012;**206**:134.e1–8. https://doi.org/10.1016/j.ajog.2011.10.878
651. Zhou J, Azizan EAB, Cabrera CP, Fernandes-Rosa FL, Boulkroun S, Argentesi G, *et al.* Somatic mutations of GNA11 and GNAQ in CTNNB1-mutant aldosterone-producing adenomas presenting in puberty, pregnancy or menopause. *Nat Genet* 2021;**53**:1360–72. https://doi.org/10.1038/s41588-021-00906-y
652. Lenders JWM, Langton K, Langenhuijsen JF, Eisenhofer G. Pheochromocytoma and pregnancy. *Endocrinol Metab Clin North Am* 2019;**48**:605–17. https://doi.org/10.1016/j.ecl.2019.05.006
653. Bancos I, Atkinson E, Eng C, Young WF Jr, Neumann HPH, Yukina M, *et al.* Maternal and fetal outcomes in phaeochromocytoma and pregnancy: a multicentre retrospective

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

cohort study and systematic review of literature. *Lancet Diabetes Endocrinol* 2021;**9**:13–21. https://doi.org/10.1016/s2213-8587(20)30363-6
654. Martínez-Vizcaíno V, Sanabria-Martínez G, Fernández-Rodríguez R, Cavero-Redondo I, Pascual-Morena C, Álvarez-Bueno C, *et al.* Exercise during pregnancy for preventing gestational diabetes mellitus and hypertensive disorders: an umbrella review of randomised controlled trials and an updated meta-analysis. *Bjog* 2023;**130**:264–75. https://doi.org/10.1111/1471-0528.17304
655. Mottola MF, Davenport MH, Ruchat SM, Davies GA, Poitras VJ, Gray CE, *et al.* 2019 Canadian guideline for physical activity throughout pregnancy. *Br J Sports Med* 2018;**52**:1339–46. https://doi.org/10.1136/bjsports-2018-100056
656. Askie LM, Duley L, Henderson-Smart DJ, Stewart LA. Antiplatelet agents for prevention of pre-eclampsia: a meta-analysis of individual patient data. *Lancet* 2007;**369**:1791–8. https://doi.org/10.1016/s0140-6736(07)60712-0
657. Duley L, Meher S, Hunter KE, Seidler AL, Askie LM. Antiplatelet agents for preventing pre-eclampsia and its complications. *Cochrane Database Syst Rev* 2019;**2019**:CD004659. https://doi.org/10.1002/14651858.CD004659.pub3
658. Hofmeyr GJ, Lawrie TA, Atallah ÁN, Torloni MR. Calcium supplementation during pregnancy for preventing hypertensive disorders and related problems. *Cochrane Database Syst Rev* 2018;**10**:CD001059. https://doi.org/10.1002/14651858.CD001059.pub5
659. Dwarkanath P, Muhihi A, Sudfeld CR, Wylie BJ, Wang M, Perumal N, *et al.* Two randomized trials of low-dose calcium supplementation in pregnancy. *N Engl J Med* 2024;**390**:143–53. https://doi.org/10.1056/NEJMoa2307212
660. Abalos E, Duley L, Steyn DW, Gialdini C. Antihypertensive drug therapy for mild to moderate hypertension during pregnancy. *Cochrane Database Syst Rev* 2018;**10**:CD002252. https://doi.org/10.1002/14651858.CD002252.pub4
661. Magee LA, von Dadelszen P, Rey E, Ross S, Asztalos E, Murphy KE, *et al.* Less-tight versus tight control of hypertension in pregnancy. *N Engl J Med* 2015;**372**:407–17. https://doi.org/10.1056/NEJMoa1404595
662. Redman CW. Fetal outcome in trial of antihypertensive treatment in pregnancy. *Lancet* 1976;**2**:753–6. https://doi.org/10.1016/s0140-6736(76)90597-3
663. Cockburn J, Moar VA, Ounsted M, Redman CW. Final report of study on hypertension during pregnancy: the effects of specific treatment on the growth and development of the children. *Lancet* 1982;**1**:647–9. https://doi.org/10.1016/s0140-6736(82)92202-4
664. Bellos I, Pergialiotis V, Papapanagiotou A, Loutradis D, Daskalakis G. Comparative efficacy and safety of oral antihypertensive agents in pregnant women with chronic hypertension: a network metaanalysis. *Am J Obstet Gynecol* 2020;**223**:525–37. https://doi.org/10.1016/j.ajog.2020.03.016
665. Lydakis C, Lip GY, Beevers M, Beevers DG. Atenolol and fetal growth in pregnancies complicated by hypertension. *Am J Hypertens* 1999;**12**:541–7. https://doi.org/10.1016/s0895-7061(99)00031-x
666. Sridharan K, Sequeira RP. Drugs for treating severe hypertension in pregnancy: a network meta-analysis and trial sequential analysis of randomized clinical trials. *Br J Clin Pharmacol* 2018;**84**:1906–16. https://doi.org/10.1111/bcp.13649
667. Bhat AD, Keasler PM, Kolluru L, Dombrowski MM, Palanisamy A, Singh PM, *et al.* Treatment of acute-onset hypertension in pregnancy: a network meta-analysis of randomized controlled trials comparing anti-hypertensives and route of administration. *Pregnancy Hypertens* 2023;**34**:74–82. https://doi.org/10.1016/j.preghy.2023.10.005
668. Ehikioya E, Okobi OE, Beeko MAE, Abanga R, Abah NNI, Briggs L, *et al.* Comparing intravenous labetalol and intravenous hydralazine for managing severe gestational hypertension. *Cureus* 2023;**15**:e42332. https://doi.org/10.7759/cureus.42332
669. Halpern DG, Weinberg CR, Pinnelas R, Mehta-Lee S, Economy KE, Valente AM, *et al.* Use of medication for cardiovascular disease during pregnancy: JACC state-of-the-art review. *J Am Coll Cardiol* 2019;**73**:457–76. https://doi.org/10.1016/j.jacc.2018.10.075
670. Goel A, Maski MR, Bajracharya S, Wenger JB, Zhang D, Salahuddin S, *et al.* Epidemiology and mechanisms of de novo and persistent hypertension in the postpartum period. *Circulation* 2015;**132**:1726–33. https://doi.org/10.1161/circulationaha.115.015721
671. Behrens I, Basit S, Melbye M, Lykke JA, Wohlfahrt J, Bundgaard H, *et al.* Risk of post-pregnancy hypertension in women with a history of hypertensive disorders of pregnancy: nationwide cohort study. *BMJ* 2017;**358**:j3078. https://doi.org/10.1136/bmj.j3078
672. Barrett PM, McCarthy FP, Evans M, Kublickas M, Perry IJ, Stenvinkel P, *et al.* Hypertensive disorders of pregnancy and the risk of chronic kidney disease: a Swedish registry-based cohort study. *PLoS Med* 2020;**17**:e1003255. https://doi.org/10.1371/journal.pmed.1003255
673. Giorgione V, Ridder A, Kalafat E, Khalil A, Thilaganathan B. Incidence of postpartum hypertension within 2 years of a pregnancy complicated by pre-eclampsia: a systematic review and meta-analysis. *BJOG* 2021;**128**:495–503. https://doi.org/10.1111/1471-0528.16545
674. Benschop L, Duvekot JJ, Versmissen J, van Broekhoven V, Steegers EAP, Roeters van Lennep JE, *et al.* Blood pressure profile 1 year after severe preeclampsia. *Hypertension* 2018;**71**:491–8. https://doi.org/10.1161/hypertensionaha.117.10338
675. Kitt J, Fox R, Frost A, Shanyinde M, Tucker K, Bateman PA, *et al.* Long-term blood pressure control after hypertensive pregnancy following physician-optimized self-management: the POP-HT randomized clinical trial. *JAMA* 2023;**330**:1991–9. https://doi.org/10.1001/jama.2023.21523
676. van Oostwaard MF, Langenveld J, Schuit E, Papatsonis DNM, Brown MA, Byaruhanga RN, *et al.* Recurrence of hypertensive disorders of pregnancy: an individual patient data metaanalysis. *Am J Obstet Gynecol* 2015;**212**:624.e1–17. https://doi.org/10.1016/j.ajog.2015.01.009
677. Brown MA, Mackenzie C, Dunsmuir W, Roberts L, Ikin K, Matthews J, *et al.* Can we predict recurrence of pre-eclampsia or gestational hypertension? *BJOG* 2007;**114**:984–93. https://doi.org/10.1111/j.1471-0528.2007.01376.x
678. Heimberger S, Perdigao JL, Mueller A, Shahul S, Naseem H, Minhas R, *et al.* Effect of blood pressure control in early pregnancy and clinical outcomes in African American women with chronic hypertension. *Pregnancy Hypertens* 2020;**20**:102–7. https://doi.org/10.1016/j.preghy.2020.03.008
679. Salazar MR, Espeche WG, Balbín E, Leiva Sisnieguez CE, Leiva Sisnieguez BC, Stavile RN, *et al.* Office blood pressure values and the necessity of out-of-office measurements in high-risk pregnancies. *J Hypertens* 2019;**37**:1838–44. https://doi.org/10.1097/hjh.0000000000002140
680. Buawangpong N, Teekachunhatean S, Koonrungsesomboon N. Adverse pregnancy outcomes associated with first-trimester exposure to angiotensin-converting enzyme inhibitors or angiotensin II receptor blockers: a systematic review and meta-analysis. *Pharmacol Res Perspect* 2020;**8**:e00644. https://doi.org/10.1002/prp2.644
681. Weber-Schoendorfer C, Kayser A, Tissen-Diabaté T, Winterfeld U, Eleftheriou G, Te Winkel B, *et al.* Fetotoxic risk of AT1 blockers exceeds that of angiotensin-converting enzyme inhibitors: an observational study. *J Hypertens* 2020;**38**:133–41. https://doi.org/10.1097/hjh.0000000000002233
682. Chen X, Mao G, Leng SX. Frailty syndrome: an overview. *Clin Interv Aging* 2014;**9**:433–41. https://doi.org/10.2147/cia.S45300
683. Rodriguez-Mañas L, Fried LP. Frailty in the clinical scenario. *Lancet* 2015;**385**:e7–9. https://doi.org/10.1016/s0140-6736(14)61595-6
684. Fried LP, Tangen CM, Walston J, Newman AB, Hirsch C, Gottdiener J, *et al.* Frailty in older adults: evidence for a phenotype. *J Gerontol A Biol Sci Med Sci* 2001;**56**:M146–156. https://doi.org/10.1093/gerona/56.3.m146
685. Fried LP, Kronmal RA, Newman AB, Bild DE, Mittelmark MB, Polak JF, *et al.* Risk factors for 5-year mortality in older adults: the cardiovascular health study. *JAMA* 1998;**279**:585–92. https://doi.org/10.1001/jama.279.8.585
686. Martens DS, Thijs L, Latosinska A, Trenson S, Siwy J, Zhang Z-Y, *et al.* Urinary peptidomic profiles to address age-related disabilities: a prospective population study. *Lancet Healthy Longev* 2021;**2**:e690–703. https://doi.org/10.1016/s2666-7568(21)00226-9
687. Monahan KD. Effect of aging on baroreflex function in humans. *Am J Physiol Regul Integr Comp Physiol* 2007;**293**:R3–r12. https://doi.org/10.1152/ajpregu.00031.2007
688. Paleczny B, Niewiński P, Rydlewska A, Piepoli MF, Borodulin-Nadzieja L, Jankowska EA, *et al.* Age-related reflex responses from peripheral and central chemoreceptors in healthy men. *Clin Auton Res* 2014;**24**:285–96. https://doi.org/10.1007/s10286-014-0263-9
689. Anker D, Santos-Eggimann B, Zwahlen M, Santschi V, Rodondi N, Wolfson C, *et al.* Blood pressure in relation to frailty in older adults: a population-based study. *J Clin Hypertens (Greenwich)* 2019;**21**:1895–904. https://doi.org/10.1111/jch.13722
690. Ravindrarajah R, Hazra NC, Hamada S, Charlton J, Jackson SHD, Dregan A, *et al.* Systolic blood pressure trajectory, frailty, and all-cause mortality >80 years of age: cohort study using electronic health records. *Circulation* 2017;**135**:2357–68. https://doi.org/10.1161/circulationaha.116.026687
691. Searle SD, Mitnitski A, Gahbauer EA, Gill TM, Rockwood K. A standard procedure for creating a frailty index. *BMC Geriatr* 2008;**8**:24. https://doi.org/10.1186/1471-2318-8-24
692. Clegg A, Bates C, Young J, Ryan R, Nichols L, Ann Teale E, *et al.* Development and validation of an electronic frailty index using routine primary care electronic health record data. *Age Ageing* 2016;**45**:353–60. https://doi.org/10.1093/ageing/afw039
693. Rockwood K, Song X, MacKnight C, Bergman H, Hogan DB, McDowell I, *et al.* A global clinical measure of fitness and frailty in elderly people. *CMAJ* 2005;**173**:489–95. https://doi.org/10.1503/cmaj.050051
694. Nguyen TN, Harris K, Woodward M, Chalmers J, Cooper M, Hamet P, *et al.* The impact of frailty on the effectiveness and safety of intensive glucose control and blood pressure-lowering therapy for people with type 2 diabetes: results from the ADVANCE trial. *Diabetes Care* 2021;**44**:1622–9. https://doi.org/10.2337/dc20-2664
695. Masoli JAH, Delgado J, Pilling L, Strain D, Melzer D. Blood pressure in frail older adults: associations with cardiovascular outcomes and all-cause mortality. *Age Ageing* 2020;**49**:807–13. https://doi.org/10.1093/ageing/afaa028
696. Aparicio LS, Thijs L, Boggia J, Jacobs L, Barochiner J, Odili AN, *et al.* Defining thresholds for home blood pressure monitoring in octogenarians. *Hypertension* 2015;**66**:865–73. https://doi.org/10.1161/hypertensionaha.115.05800
697. Gaffney B, Jacobsen AP, Pallippattu AW, Leahy N, McEvoy JW. The diastolic blood pressure J-curve in hypertension management: links and risk for cardiovascular disease. *Integr Blood Press Control* 2021;**14**:179–87. https://doi.org/10.2147/ibpc.S286957
698. Warwick J, Falaschetti E, Rockwood K, Mitnitski A, Thijs L, Beckett N, *et al.* No evidence that frailty modifies the positive impact of antihypertensive treatment in very elderly people: an investigation of the impact of frailty upon treatment effect in the HYpertension in the Very Elderly Trial (HYVET) study, a double-blind,

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

placebo-controlled study of antihypertensives in people with hypertension aged 80 and over. *BMC Med* 2015;**13**:78. https://doi.org/10.1186/s12916-015-0328-1
699. Todd OM, Wilkinson C, Hale M, Wong NL, Hall M, Sheppard JP, *et al.* Is the association between blood pressure and mortality in older adults different with frailty? A systematic review and meta-analysis. *Age Ageing* 2019;**48**:627–35. https://doi.org/10.1093/ageing/afz072
700. Li Y, Thijs L, Zhang ZY, Asayama K, Hansen TW, Boggia J, *et al.* Opposing age-related trends in absolute and relative risk of adverse health outcomes associated with out-of-office blood pressure. *Hypertension* 2019;**74**:1333–42. https://doi.org/10.1161/hypertensionaha.119.12958
701. Beckett NS, Peters R, Fletcher AE, Staessen JA, Liu L, Dumitrascu D, *et al.* Treatment of hypertension in patients 80 years of age or older. *N Engl J Med* 2008;**358**:1887–98. https://doi.org/10.1056/NEJMoa0801369
702. Peters R, Beckett N, McCormack T, Fagard R, Fletcher A, Bulpitt C. Treating hypertension in the very elderly-benefits, risks, and future directions, a focus on the hypertension in the very elderly trial. *Eur Heart J* 2014;**35**:1712–8. https://doi.org/10.1093/eurheartj/eht464
703. Travers J, Romero-Ortuno R, Bailey J, Cooney MT. Delaying and reversing frailty: a systematic review of primary care interventions. *Br J Gen Pract* 2019;**69**:e61–9. https://doi.org/10.3399/bjgp18X700241
704. Bogaerts JMK, von Ballmoos LM, Achterberg WP, Gussekloo J, Streit S, van der Ploeg MA, *et al.* Do we AGREE on the targets of antihypertensive drug treatment in older adults: a systematic review of guidelines on primary prevention of cardiovascular diseases. *Age Ageing* 2022;**51**:afab192. https://doi.org/10.1093/ageing/afab192
705. Sheppard JP, Benetos A, McManus RJ. Antihypertensive deprescribing in older adults: a practical guide. *Curr Hypertens Rep* 2022;**24**:571–80. https://doi.org/10.1007/s11906-022-01215-3
706. McDonnell CC, Rogers KC, Regen SM, Finks SW. The Fall Risk with Alpha blockers Given InitiaL dose or Elderly status (FRAGILE) study. *Ann Pharmacother* 2020;**54**:226–31. https://doi.org/10.1177/1060028019880305
707. Hiremath S, Ruzicka M, Petrcich W, McCallum MK, Hundemer GL, Tanuseputro P, *et al.* Alpha-blocker use and the risk of hypotension and hypotension-related clinical events in women of advanced age. *Hypertension* 2019;**74**:645–51. https://doi.org/10.1161/hypertensionaha.119.13289
708. Li H, Xu TY, Li Y, Chia Y-C, Buranakitjaroen P, Cheng H-M, *et al.* Role of α1-blockers in the current management of hypertension. *J Clin Hypertens (Greenwich)* 2022;**24**:1180–6. https://doi.org/10.1111/jch.14556
709. Ravindrarajah R, Dregan A, Hazra NC, Hamada S, Jackson SHD, Gulliford MC, *et al.* Declining blood pressure and intensification of blood pressure management among people over 80 years: cohort study using electronic health records. *J Hypertens* 2017;**35**:1276–82. https://doi.org/10.1097/hjh.0000000000001291
710. Chun S, Han K, Lee S, Cho M-H, Jeong S-M, Jung H-W, *et al.* Impact of frailty on the relationship between blood pressure and cardiovascular diseases and mortality in young-old adults. *J Pers Med* 2022;**12**:418. https://doi.org/10.3390/jpm12030418
711. Khan N, McAlister FA. Re-examining the efficacy of beta-blockers for the treatment of hypertension: a meta-analysis. *CMAJ* 2006;**174**:1737–42. https://doi.org/10.1503/cmaj.060110
712. Lavan AH, Gallagher P, Parsons C, O'Mahony D. STOPPFrail (Screening Tool of Older Persons Prescriptions in Frail adults with limited life expectancy): consensus validation. *Age Ageing* 2017;**46**:600–7. https://doi.org/10.1093/ageing/afx005
713. Eeftinck Schattenkerk DW, van Gorp J, Vogt L, Peters RJ, van den Born BH. Isolated systolic hypertension of the young and its association with central blood pressure in a large multi-ethnic population. The HELIUS study. *Eur J Prev Cardiol* 2018;**25**:1351–9. https://doi.org/10.1177/2047487318777430
714. Franklin SS, Jacobs MJ, Wong ND, L'Italien GJ, Lapuerta P. Predominance of isolated systolic hypertension among middle-aged and elderly US hypertensives: analysis based on National Health and Nutrition Examination Survey (NHANES) III. *Hypertension* 2001;**37**:869–74. https://doi.org/10.1161/01.hyp.37.3.869
715. Van Bortel LM, Laurent S, Boutouyrie P, Chowienczyk P, Cruickshank JK, De Backer T, *et al.* Expert consensus document on the measurement of aortic stiffness in daily practice using carotid-femoral pulse wave velocity. *J Hypertens* 2012;**30**:445–8. https://doi.org/10.1097/HJH.0b013e32834fa8b0
716. The Reference Values for Arterial Stiffness Collaboration. Determinants of pulse wave velocity in healthy people and in the presence of cardiovascular risk factors: 'establishing normal and reference values'. *Eur Heart J* 2010;**31**:2338–50. https://doi.org/10.1093/eurheartj/ehq165
717. Li Y, Wei FF, Thijs L, Boggia J, Asayama K, Hansen TW, *et al.* Ambulatory hypertension subtypes and 24-hour systolic and diastolic blood pressure as distinct outcome predictors in 8341 untreated people recruited from 12 populations. *Circulation* 2014;**130**:466–74. https://doi.org/10.1161/circulationaha.113.004876
718. Staessen JA, Gasowski J, Wang JG, Thijs L, Hond ED, Boissel J-P, *et al.* Risks of untreated and treated isolated systolic hypertension in the elderly: meta-analysis of outcome trials. *Lancet* 2000;**355**:865–72. https://doi.org/10.1016/s0140-6736(99)07330-4
719. McEniery CM, Yasmin, Wallace S, Maki-Petaja K, McDonnell B, Sharman JE, *et al.* Increased stroke volume and aortic stiffness contribute to isolated systolic hypertension in young adults. *Hypertension* 2005;**46**:221–6. https://doi.org/10.1161/01.HYP.0000165310.84801.e0
720. Palatini P, Rosei EA, Avolio A, Bilo G, Casiglia E, Ghiadoni L, *et al.* Isolated systolic hypertension in the young: a position paper endorsed by the European Society of Hypertension. *J Hypertens* 2018;**36**:1222–36. https://doi.org/10.1097/hjh.0000000000001726
721. Unger T, Borghi C, Charchar F, Khan NA, Poulter NR, Prabhakaran D, *et al.* 2020 International Society of Hypertension global hypertension practice guidelines. *Hypertension* 2020;**75**:1334–57. https://doi.org/10.1161/hypertensionaha.120.15026
722. Yano Y, Chang Kim H, Lee H, Azahar N, Ahmed S, Kitaoka K, *et al.* Response to isolated diastolic hypertension and risk of cardiovascular disease: controversies in hypertension—con side of the argument. *Hypertension* 2022;**79**:1579. https://doi.org/10.1161/hypertensionaha.122.19493
723. Jacobsen AP, McKittrick M, Daya N, Al Rifai M, McEvoy JW. Isolated diastolic hypertension and risk of cardiovascular disease: controversies in hypertension—con side of the argument. *Hypertension* 2022;**79**:1571–8. https://doi.org/10.1161/hypertensionaha.122.18458
724. Blank SG, Mann SJ, James GD, West JE, Pickering TG. Isolated elevation of diastolic blood pressure. Real or artifactual? *Hypertension* 1995;**26**:383–9. https://doi.org/10.1161/01.hyp.26.3.383
725. Cho SMJ, Lee H, Koyama S, Zou RS, Schuermans A, Ganesh S, *et al.* Cumulative diastolic blood pressure burden in normal systolic blood pressure and cardiovascular disease. *Hypertension* 2024;**81**:273–81. https://doi.org/10.1161/hypertensionaha.123.22160
726. Grobman B, Turkson-Ocran RN, Staessen JA, Yu Y-L, Lipsitz LA, Mukamal KJ, *et al.* Body position and orthostatic hypotension in hypertensive adults: results from the Syst-Eur trial. *Hypertension* 2023;**80**:820–7. https://doi.org/10.1161/hypertensionaha.122.20602
727. Juraschek SP, Taylor AA, Wright JT Jr, Evans GW, Miller ER, Plante TB, *et al.* Orthostatic hypotension, cardiovascular outcomes, and adverse events: results from SPRINT. *Hypertension* 2020;**75**:660–7. https://doi.org/10.1161/hypertensionaha.119.14309
728. Juraschek SP, Daya N, Appel LJ, Miller ER, McEvoy JW, Matsushita K, *et al.* Orthostatic hypotension and risk of clinical and subclinical cardiovascular disease in middle-aged adults. *J Am Heart Assoc* 2018;**7**:e008884. https://doi.org/10.1161/jaha.118.008884
729. Fleg JL, Evans GW, Margolis KL, Barzilay J, Basile JN, Bigger JT, *et al.* Orthostatic hypotension in the ACCORD (Action to Control Cardiovascular Risk in Diabetes) blood pressure trial: prevalence, incidence, and prognostic significance. *Hypertension* 2016;**68**:888–95. https://doi.org/10.1161/hypertensionaha.116.07474
730. Juraschek SP, Hu JR, Cluett JL, Ishak A, Mita C, Lipsitz LA, *et al.* Effects of intensive blood pressure treatment on orthostatic hypotension : a systematic review and individual participant-based meta-analysis. *Ann Intern Med* 2021;**174**:58–68. https://doi.org/10.7326/m20-4298
731. Ylitalo A, Airaksinen KE, Sellin L, Huikuri HV. Effects of combination antihypertensive therapy on baroreflex sensitivity and heart rate variability in systemic hypertension. *Am J Cardiol* 1999;**83**:885–9. https://doi.org/10.1016/s0002-9149(98)01067-4
732. Ganjehei L, Massumi A, Razavi M, Wilson JM. Orthostatic hypotension as a manifestation of vitamin B12 deficiency. *Tex Heart Inst J* 2012;**39**:722–3.
733. Shimbo D, Barrett Bowling C, Levitan EB, Deng L, Sim JJ, Huang L, *et al.* Short-term risk of serious fall injuries in older adults initiating and intensifying treatment with antihypertensive medication. *Circ Cardiovasc Qual Outcomes* 2016;**9**:222–9. https://doi.org/10.1161/circoutcomes.115.002524
734. Krediet CT, van Lieshout JJ, Bogert LW, Immink RV, Kim Y-S, Wieling W, *et al.* Leg crossing improves orthostatic tolerance in healthy subjects: a placebo-controlled crossover study. *Am J Physiol Heart Circ Physiol* 2006;**291**:H1768–72. https://doi.org/10.1152/ajpheart.00287.2006
735. Okamoto LE, Diedrich A, Baudenbacher FJ, Harder R, Whitfield JS, Iqbal F, *et al.* Efficacy of servo-controlled splanchnic venous compression in the treatment of orthostatic hypotension: a randomized comparison with midodrine. *Hypertension* 2016;**68**:418–26. https://doi.org/10.1161/hypertensionaha.116.07199
736. Okamoto LE, Celedonio JE, Smith EC, Gamboa A, Shibao CA, Diedrich A, *et al.* Local passive heat for the treatment of hypertension in autonomic failure. *J Am Heart Assoc* 2021;**10**:e018979. https://doi.org/10.1161/jaha.120.018979
737. Buyken AE, von Eckardstein A, Schulte H, Cullen P, Assmann G. Type 2 diabetes mellitus and risk of coronary heart disease: results of the 10-year follow-up of the PROCAM study. *Eur J Cardiovasc Prev Rehabil* 2007;**14**:230–6. https://doi.org/10.1097/HJR.0b013e3280142037
738. Nathan DM. Long-term complications of diabetes mellitus. *N Engl J Med* 1993;**328**:1676–85. https://doi.org/10.1056/nejm199306103282306
739. Marx N, Federici M, Schütt K, Müller-Wieland D, Ajjan RA, Antunes MJ, *et al.* 2023 ESC Guidelines for the management of cardiovascular disease in patients with diabetes. *Eur Heart J* 2023;**44**:4043–140. https://doi.org/10.1093/eurheartj/ehad192
740. Pylypchuk R, Wells S, Kerr A, Poppe K, Harwood M, Mehta S, *et al.* Cardiovascular risk prediction in type 2 diabetes before and after widespread screening: a derivation and validation study. *Lancet* 2021;**397**:2264–74. https://doi.org/10.1016/s0140-6736(21)00572-9
741. Danaei G, Fahimi S, Lu Y, Zhou B, Hajifathalian K, Di Cesare M, *et al.* Effects of diabetes definition on global surveillance of diabetes prevalence and diagnosis: a pooled analysis

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

of 96 population-based studies with 331288 participants. *Lancet Diabetes Endocrinol* 2015;**3**:624–37. https://doi.org/10.1016/S2213-8587(15)00129-1
742. Wright AK, Suarez-Ortegon MF, Read SH, Kontopantelis E, Buchan I, Emsley R, *et al.* Risk factor control and cardiovascular event risk in people with type 2 diabetes in primary and secondary prevention settings. *Circulation* 2020;**142**:1925–36. https://doi.org/10.1161/circulationaha.120.046783
743. Adamsson Eryd S, Gudbjörnsdottir S, Manhem K, Rosengren A, Svensson A-M, Miftaraj M, *et al.* Blood pressure and complications in individuals with type 2 diabetes and no previous cardiovascular disease: national population based cohort study. *BMJ* 2016;**354**:i4070. https://doi.org/10.1136/bmj.i4070
744. Brunström M, Carlberg B. Effect of antihypertensive treatment at different blood pressure levels in patients with diabetes mellitus: systematic review and meta-analyses. *BMJ* 2016;**352**:i717. https://doi.org/10.1136/bmj.i717
745. Emdin CA, Rahimi K, Neal B, Callender T, Perkovic V, Patel A. Blood pressure lowering in type 2 diabetes: a systematic review and meta-analysis. *JAMA* 2015;**313**:603–15. https://doi.org/10.1001/jama.2014.18574
746. Thomopoulos C, Parati G, Zanchetti A. Effects of blood-pressure-lowering treatment on outcome incidence in hypertension: 10 – Should blood pressure management differ in hypertensive patients with and without diabetes mellitus? Overview and meta-analyses of randomized trials. *J Hypertens* 2017;**35**:922–44. https://doi.org/10.1097/hjh.0000000000001276
747. Rahman F, McEvoy JW, Ohkuma T, Marre M, Hamet P, Harrap S, *et al.* Effects of blood pressure lowering on clinical outcomes according to baseline blood pressure and cardiovascular risk in patients with type 2 diabetes mellitus. *Hypertension* 2019;**73**:1291–9. https://doi.org/10.1161/hypertensionaha.118.12414
748. Agashe S, Petak S. Cardiac autonomic neuropathy in diabetes mellitus. *Methodist Debakey Cardiovasc J* 2018;**14**:251–6. https://doi.org/10.14797/mdcj-14-4-251
749. Beddhu S, Chertow GM, Greene T, Whelton PK, Ambrosius WT, Cheung AK, *et al.* Effects of intensive systolic blood pressure lowering on cardiovascular events and mortality in patients with type 2 diabetes mellitus on standard glycemic control and in those without diabetes mellitus: reconciling results from ACCORD BP and SPRINT. *J Am Heart Assoc* 2018;**7**:e009326. https://doi.org/10.1161/jaha.118.009326
750. Buckley LF, Dixon DL, Wohlford GF, Wijesinghe DS, Baker WL, Van Tassell BW. Intensive versus standard blood pressure control in SPRINT-eligible participants of ACCORD-BP. *Diabetes Care* 2017;**40**:1733–8. https://doi.org/10.2337/dc17-1366
751. Brouwer TF, Vehmeijer JT, Kalkman DN, Berger WR, van den Born B-JH, Peters RJ, *et al.* Intensive blood pressure lowering in patients with and patients without type 2 diabetes: a pooled analysis from two randomized trials. *Diabetes Care* 2018;**41**:1142–8. https://doi.org/10.2337/dc17-1722
752. Shi S, Gouskova N, Najafzadeh M, Wei LJ, Kim DH. Intensive versus standard blood pressure control in type 2 diabetes: a restricted mean survival time analysis of a randomised controlled trial. *BMJ Open* 2021;**11**:e050335. https://doi.org/10.1136/bmjopen-2021-050335
753. Palmer SC, Mavridis D, Navarese E, Craig JC, Tonelli M, Salanti G, *et al.* Comparative efficacy and safety of blood pressure-lowering agents in adults with diabetes and kidney disease: a network meta-analysis. *Lancet* 2015;**385**:2047–56. https://doi.org/10.1016/s0140-6736(14)62459-4
754. Ying A, Arima H, Czernichow S, Woodward M, Huxley R, Turnbull F, *et al.* Effects of blood pressure lowering on cardiovascular risk according to baseline body-mass index: a meta-analysis of randomised trials. *Lancet* 2015;**385**:867–74. https://doi.org/10.1016/s0140-6736(14)61171-5
755. Nazarzadeh M, Bidel Z, Canoy D, Copland E, Wamil M, Majert J, *et al.* Blood pressure lowering and risk of new-onset type 2 diabetes: an individual participant data meta-analysis. *Lancet* 2021;**398**:1803–10. https://doi.org/10.1016/s0140-6736(21)01920-6
756. Kearney PM, Whelton M, Reynolds K, Muntner P, Whelton PK, He J, *et al.* Global burden of hypertension: analysis of worldwide data. *Lancet* 2005;**365**:217–23. https://doi.org/10.1016/s0140-6736(05)17741-1
757. Jager KJ, Kovesdy C, Langham R, Rosenberg M, Jha V, Zoccali C, *et al.* A single number for advocacy and communication—worldwide more than 850 million individuals have kidney diseases. *Kidney Int* 2019;**96**:1048–50. https://doi.org/10.1016/j.kint.2019.07.012
758. Foreman KJ, Marquez N, Dolgert A, Fukutaki K, Fullman N, McGaughey M, *et al.* Forecasting life expectancy, years of life lost, and all-cause and cause-specific mortality for 250 causes of death: reference and alternative scenarios for 2016–40 for 195 countries and territories. *Lancet* 2018;**392**:2052–90. https://doi.org/10.1016/s0140-6736(18)31694-5
759. Ortiz A, Sanchez-Niño MD, Crespo-Barrio M, De-Sequera-Ortiz P, Fernández-Giráldez E, García-Maset R, *et al.* The Spanish Society of Nephrology (SENEFRO) commentary to the Spain GBD 2016 report: keeping chronic kidney disease out of sight of health authorities will only magnify the problem. *Nefrologia (Engl Ed)* 2019;**39**:29–34. https://doi.org/10.1016/j.nefro.2018.09.002
760. Law JP, Pickup L, Pavlovic D, Townend JN, Ferro CJ. Hypertension and cardiomyopathy associated with chronic kidney disease: epidemiology, pathogenesis and treatment considerations. *J Hum Hypertens* 2023;**37**:1–19. https://doi.org/10.1038/s41371-022-00751-4
761. Fay KS, Cohen DL. Resistant hypertension in people with CKD: a review. *Am J Kidney Dis* 2021;**77**:110–21. https://doi.org/10.1053/j.ajkd.2020.04.017
762. Webster AC, Nagler EV, Morton RL, Masson P. Chronic kidney disease. *Lancet* 2017;**389**:1238–52. https://doi.org/10.1016/s0140-6736(16)32064-5
763. Ninomiya T, Perkovic V, Turnbull F, Neal B, Barzi F, Cass A, *et al.* Blood pressure lowering and major cardiovascular events in people with and without chronic kidney disease: meta-analysis of randomised controlled trials. *BMJ* 2013;**347**:f5680. https://doi.org/10.1136/bmj.f5680
764. Aggarwal R, Petrie B, Bala W, Chiu N. Mortality outcomes with intensive blood pressure targets in chronic kidney disease patients. *Hypertension* 2019;**73**:1275–82. https://doi.org/10.1161/hypertensionaha.119.12697
765. Bangalore S, Toklu B, Gianos E, Schwartzbard A, Weintraub H, Ogedegbe G, *et al.* Optimal systolic blood pressure target after SPRINT: insights from a network meta-analysis of randomized trials. *Am J Med* 2017;**130**:707–19.e8. https://doi.org/10.1016/j.amjmed.2017.01.004
766. Lv J, Ehteshami P, Sarnak MJ, Tighiouart H, Jun M, Ninomiya T, *et al.* Effects of intensive blood pressure lowering on the progression of chronic kidney disease: a systematic review and meta-analysis. *CMAJ* 2013;**185**:949–57. https://doi.org/10.1503/cmaj.121468
767. Tsai WC, Wu HY, Peng YS, Yang J-Y, Chen H-Y, Chiu Y-L, *et al.* Association of intensive blood pressure control and kidney disease progression in nondiabetic patients with chronic kidney disease: a systematic review and meta-analysis. *JAMA Intern Med* 2017;**177**:792–9. https://doi.org/10.1001/jamainternmed.2017.0197
768. Thompson S, Wiebe N, Padwal RS, Gyenes G, Headley SAE, Radhakrishnan J, *et al.* The effect of exercise on blood pressure in chronic kidney disease: a systematic review and meta-analysis of randomized controlled trials. *PLoS One* 2019;**14**:e0211032. https://doi.org/10.1371/journal.pone.0211032
769. Huang M, Lv A, Wang J, Xu N, Ma G, Zhai Z, *et al.* Exercise training and outcomes in hemodialysis patients: systematic review and meta-analysis. *Am J Nephrol* 2019;**50**:240–54. https://doi.org/10.1159/000502447
770. Zhang Y, He D, Zhang W, Xing Y, Guo Y, Wang F, *et al.* ACE inhibitor benefit to kidney and cardiovascular outcomes for patients with non-dialysis chronic kidney disease stages 3–5: a network meta-analysis of randomised clinical trials. *Drugs* 2020;**80**:797–811. https://doi.org/10.1007/s40265-020-01290-3
771. Jafar TH, Stark PC, Schmid CH, Landa M, Maschio G, de Jong PE, *et al.* Progression of chronic kidney disease: the role of blood pressure control, proteinuria, and angiotensin-converting enzyme inhibition: a patient-level meta-analysis. *Ann Intern Med* 2003;**139**:244–52. https://doi.org/10.7326/0003-4819-139-4-200308190-00006
772. Xie X, Liu Y, Perkovic V, Li X, Ninomiya T, Hou W, *et al.* Renin-angiotensin system inhibitors and kidney and cardiovascular outcomes in patients with CKD: a Bayesian network meta-analysis of randomized clinical trials. *Am J Kidney Dis* 2016;**67**:728–41. https://doi.org/10.1053/j.ajkd.2015.10.011
773. Wu HY, Huang JW, Lin HJ, Liao W-C, Peng Y-S, Hung K-Y, *et al.* Comparative effectiveness of renin-angiotensin system blockers and other antihypertensive drugs in patients with diabetes: systematic review and Bayesian network meta-analysis. *BMJ* 2013;**347**:f6008. https://doi.org/10.1136/bmj.f6008
774. Agarwal R, Sinha AD, Cramer AE, Balmes-Fenwick M, Dickinson JH, Ouyang F, *et al.* Chlorthalidone for hypertension in advanced chronic kidney disease. *N Engl J Med* 2021;**385**:2507–19. https://doi.org/10.1056/NEJMoa2110730
775. Beddhu S, Shen J, Cheung AK, Kimmel PL, Chertow GM, Wei G, *et al.* Implications of early decline in eGFR due to intensive BP control for cardiovascular outcomes in SPRINT. *J Am Soc Nephrol* 2019;**30**:1523–33. https://doi.org/10.1681/asn.2018121261
776. Herrington WG, Staplin N, Wanner C, Green JB, Hauske SJ, Emberson JR, *et al.* Empagliflozin in patients with chronic kidney disease. *N Engl J Med* 2023;**388**:117–27. https://doi.org/10.1056/NEJMoa2204233
777. Heerspink HJL, Stefánsson BV, Correa-Rotter R, Chertow GM, Greene T, Hou F-F, *et al.* Dapagliflozin in patients with chronic kidney disease. *N Engl J Med* 2020;**383**:1436–46. https://doi.org/10.1056/NEJMoa2024816
778. Pitt B, Filippatos G, Agarwal R, Anker SD, Bakris GL, Rossing P, *et al.* Cardiovascular events with finerenone in kidney disease and type 2 diabetes. *N Engl J Med* 2021;**385**:2252–63. https://doi.org/10.1056/NEJMoa2110956
779. Ku E, McCulloch CE, Inker LA, Tighiouart H, Schaefer F, Wühl E, *et al.* Intensive BP control in patients with CKD and risk for adverse outcomes. *J Am Soc Nephrol* 2023;**34**:385–93. https://doi.org/10.1681/asn.0000000000000072
780. Lewis EJ, Hunsicker LG, Clarke WR, Berl T, Pohl MA, Lewis JB, *et al.* Renoprotective effect of the angiotensin-receptor antagonist irbesartan in patients with nephropathy due to type 2 diabetes. *N Engl J Med* 2001;**345**:851–60. https://doi.org/10.1056/NEJMoa011303
781. Brenner BM, Cooper ME, de Zeeuw D, Keane WF, Mitch WE, Parving H-H, *et al.* Effects of losartan on renal and cardiovascular outcomes in patients with type 2 diabetes and nephropathy. *N Engl J Med* 2001;**345**:861–9. https://doi.org/10.1056/NEJMoa011161
782. The GISEN Group (Gruppo Italiano di Studi Epidemiologici in Nefrologia). Randomised placebo-controlled trial of effect of ramipril on decline in glomerular filtration rate and risk of terminal renal failure in proteinuric, non-diabetic nephropathy. *Lancet* 1997;**349**:1857–63. https://doi.org/10.1016/S0140-6736(96)11445-8

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

783. Cruickshank JM, Thorp JM, Zacharias FJ. Benefits and potential harm of lowering high blood pressure. *Lancet* 1987;**1**:581–4. https://doi.org/10.1016/s0140-6736(87)90231-5
784. Staessen J, Bulpitt C, Clement D, De Leeuw P, Fagard R, Fletcher A, *et al.* Relation between mortality and treated blood pressure in elderly patients with hypertension: report of the European working party on high blood pressure in the elderly. *BMJ* 1989;**298**:1552–6. https://doi.org/10.1136/bmj.298.6687.1552
785. Böhm M, Schumacher H, Teo KK, Lonn EM, Mahfoud F, Mann JFE, *et al.* Achieved blood pressure and cardiovascular outcomes in high-risk patients: results from ONTARGET and TRANSCEND trials. *Lancet* 2017;**389**:2226–37. https://doi.org/10.1016/s0140-6736(17)30754-7
786. Böhm M, Ferreira JP, Mahfoud F, Duarte K, Pitt B, Zannad F, *et al.* Myocardial reperfusion reverses the J-curve association of cardiovascular risk and diastolic blood pressure in patients with left ventricular dysfunction and heart failure after myocardial infarction: insights from the EPHESUS trial. *Eur Heart J* 2020;**41**:1673–83. https://doi.org/10.1093/eurheartj/ehaa132
787. McEvoy JW, Chen Y, Rawlings A, Hoogeveen RC, Ballantyne CM, Blumenthal RS, *et al.* Diastolic blood pressure, subclinical myocardial damage, and cardiac events: implications for blood pressure control. *J Am Coll Cardiol* 2016;**68**:1713–22. https://doi.org/10.1016/j.jacc.2016.07.754
788. Mancia G, Kreutz R, Brunström M, Burnier M, Grassi G, Januszewicz A, *et al.* 2023 ESH Guidelines for the management of arterial hypertension The Task Force for the management of arterial hypertension of the European Society of Hypertension: Endorsed by the International Society of Hypertension (ISH) and the European Renal Association (ERA). *J Hypertens* 2023;**41**:1874–2071. https://doi.org/10.1097/hjh.0000000000003480
789. Carvalho JJ, Baruzzi RG, Howard PF, Poulter N, Alpers M P, Franco L J, *et al.* Blood pressure in four remote populations in the INTERSALT study. *Hypertension* 1989;**14**:238–46. https://doi.org/10.1161/01.hyp.14.3.238
790. Birrane JP, Foschi M, Sacco S, McEvoy JW. Another nail in the coffin of causality for the diastolic blood pressure J curve. *Hypertension* 2022;**79**:794–7. https://doi.org/10.1161/hypertensionaha.122.18997
791. Mueller NT, Noya-Alarcon O, Contreras M, Appel LJ, Dominguez-Bello MG. Association of age with blood pressure across the lifespan in isolated Yanomami and Yekwana villages. *JAMA Cardiol* 2018;**3**:1247–9. https://doi.org/10.1001/jamacardio.2018.3676
792. Ilkun OL, Greene T, Cheung AK, Whelton PK, Wei G, Boucher RE, *et al.* The influence of baseline diastolic blood pressure on the effects of intensive blood pressure lowering on cardiovascular outcomes and all-cause mortality in type 2 diabetes. *Diabetes Care* 2020;**43**:1878–84. https://doi.org/10.2337/dc19-2047
793. Shihab S, Boucher RE, Abraham N, Wei G, Beddhu S. Influence of baseline diastolic blood pressure on the effects of intensive systolic blood pressure lowering on the risk of stroke. *Hypertension* 2022;**79**:785–93. https://doi.org/10.1161/hypertensionaha.121.18172
794. Vahanian A, Beyersdorf F, Praz F, Milojevic M, Baldus S, Bauersachs J, *et al.* 2021 ESC/EACTS Guidelines for the management of valvular heart disease. *Eur Heart J* 2022;**43**:561–632. https://doi.org/10.1093/eurheartj/ehab395
795. McDonagh TA, Metra M, Adamo M, Gardner RS, Baumbach A, Böhm M, *et al.* 2021 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure. *Eur Heart J* 2021;**42**:3599–726. https://doi.org/10.1093/eurheartj/ehab368
796. Mancusi C, de Simone G, Brguljan Hitij J, Sudano I, Mahfoud F, Parati G, *et al.* Management of patients with combined arterial hypertension and aortic valve stenosis: a consensus document from the Council on Hypertension and Council on Valvular Heart Disease of the European Society of Cardiology, the European Association of Cardiovascular Imaging (EACVI), and the European Association of Percutaneous Cardiovascular Interventions (EAPCI). *Eur Heart J Cardiovasc Pharmacother* 2021;**7**:242–50. https://doi.org/10.1093/ehjcvp/pvaa040
797. Nielsen OW, Sajadieh A, Sabbah M, Greve AM, Olsen MH, Boman K, *et al.* Assessing optimal blood pressure in patients with asymptomatic aortic valve stenosis: the simvastatin ezetimibe in aortic stenosis study (SEAS). *Circulation* 2016;**134**:455–68. https://doi.org/10.1161/circulationaha.115.021213
798. McDonagh TA, Metra M, Adamo M, Gardner RS, Baumbach A, Böhm M, *et al.* 2023 focused update of the 2021 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure. *Eur Heart J* 2023;**44**:3627–39. https://doi.org/10.1093/eurheartj/ehad195
799. Yusuf S, Pfeffer MA, Swedberg K, Granger CB, Held P, McMurray JJV, *et al.* Effects of candesartan in patients with chronic heart failure and preserved left-ventricular ejection fraction: the CHARM-preserved trial. *Lancet* 2003;**362**:777–81. https://doi.org/10.1016/s0140-6736(03)14285-7
800. Pitt B, Pfeffer MA, Assmann SF, Boineau R, Anand IS, Claggett B, *et al.* Spironolactone for heart failure with preserved ejection fraction. *N Engl J Med* 2014;**370**:1383–92. https://doi.org/10.1056/NEJMoa1313731
801. Nazarzadeh M, Pinho-Gomes AC, Bidel Z, Canoy D, Dehghan A, Byrne KS, *et al.* Genetic susceptibility, elevated blood pressure, and risk of atrial fibrillation: a Mendelian randomization study. *Genome Med* 2021;**13**:38. https://doi.org/10.1186/s13073-021-00849-3
802. Emdin CA, Anderson SG, Salimi-Khorshidi G, Woodward M, MacMahon S, Dwyer T, *et al.* Usual blood pressure, atrial fibrillation and vascular risk: evidence from 4.3 million adults. *Int J Epidemiol* 2017;**46**:162–72. https://doi.org/10.1093/ije/dyw053
803. Hindricks G, Potpara T, Dagres N, Arbelo E, Bax JJ, Blomström-Lundqvist C, *et al.* 2020 ESC Guidelines for the diagnosis and management of atrial fibrillation developed in collaboration with the European Association for Cardio-Thoracic Surgery (EACTS): The Task Force for the diagnosis and management of atrial fibrillation of the European Society of Cardiology (ESC) Developed with the special contribution of the European Heart Rhythm Association (EHRA) of the ESC. *Eur Heart J* 2021;**42**:373–498. https://doi.org/10.1093/eurheartj/ehaa612
804. Alber J, Alladi S, Bae HJ, Barton DA, Beckett LA, Bell JM, *et al.* White matter hyperintensities in vascular contributions to cognitive impairment and dementia (VCID): knowledge gaps and opportunities. *Alzheimers Dement (N Y)* 2019;**5**:107–17. https://doi.org/10.1016/j.trci.2019.02.001
805. Georgakis MK, Duering M, Wardlaw JM, Dichgans M. WMH and long-term outcomes in ischemic stroke: a systematic review and meta-analysis. *Neurology* 2019;**92**:e1298–308. https://doi.org/10.1212/wnl.0000000000007142
806. Jokinen H, Koikkalainen J, Laakso HM, Melkas S, Nieminen T, Brander A, *et al.* Global burden of small vessel disease-related brain changes on MRI predicts cognitive and functional decline. *Stroke* 2020;**51**:170–8. https://doi.org/10.1161/strokeaha.119.026170
807. Potter T, Lioutas VA, Tano M, Pan A, Meeks J, Woo D, *et al.* Cognitive impairment after intracerebral hemorrhage: a systematic review of current evidence and knowledge gaps. *Front Neurol* 2021;**12**:716632. https://doi.org/10.3389/fneur.2021.716632
808. Wang F, Hua S, Zhang Y, Zhu J, Liu R, Jiang Z. Association between small vessel disease markers, medial temporal lobe atrophy and cognitive impairment after stroke: a systematic review and meta-analysis. *J Stroke Cerebrovasc Dis* 2021;**30**:105460. https://doi.org/10.1016/j.jstrokecerebrovasdis.2020.105460
809. Ungvari Z, Toth P, Tarantini S, Prodan CI, Sorond F, Merkely B, *et al.* Hypertension-induced cognitive impairment: from pathophysiology to public health. *Nat Rev Nephrol* 2021;**17**:639–54. https://doi.org/10.1038/s41581-021-00430-6
810. Kelly D, Rothwell PM. Disentangling the multiple links between renal dysfunction and cerebrovascular disease. *J Neurol Neurosurg Psychiatry* 2020;**91**:88–97. https://doi.org/10.1136/jnnp-2019-320526
811. Papanastasiou CA, Theochari CA, Zareifopoulos N, Arfaras-Melainis A, Giannakoulas G, Karamitsos TD, *et al.* Atrial fibrillation is associated with cognitive impairment, all-cause dementia, vascular dementia, and Alzheimer's disease: a systematic review and meta-analysis. *J Gen Intern Med* 2021;**36**:3122–35. https://doi.org/10.1007/s11606-021-06954-8
812. Li J, Wu Y, Zhang D, Nie J. Associations between heart failure and risk of dementia: a PRISMA-compliant meta-analysis. *Medicine (Baltimore)* 2020;**99**:e18492. https://doi.org/10.1097/md.0000000000018492
813. Canavan M, O'Donnell MJ. Hypertension and cognitive impairment: a review of mechanisms and key concepts. *Front Neurol* 2022;**13**:821135. https://doi.org/10.3389/fneur.2022.821135
814. Dawson J, Béjot Y, Christensen LM, De Marchis GM, Dichgans M, Hagberg G, *et al.* European Stroke Organisation (ESO) guideline on pharmacological interventions for long-term secondary prevention after ischaemic stroke or transient ischaemic attack. *Eur Stroke J* 2022;**7**:I–XLI. https://doi.org/10.1177/23969873221100032
815. Kleindorfer DO, Towfighi A, Chaturvedi S, Cockroft KM, Gutierrez J, Lombardi-Hill D, *et al.* 2021 Guideline for the prevention of stroke in patients with stroke and transient ischemic attack: a guideline from the American Heart Association/American Stroke Association. *Stroke* 2021;**52**:e364–467. https://doi.org/10.1161/str.0000000000000375
816. Zonneveld TP, Richard E, Vergouwen MD, Nederkoorn PJ, de Haan RJ, Roos YB, *et al.* Blood pressure-lowering treatment for preventing recurrent stroke, major vascular events, and dementia in patients with a history of stroke or transient ischaemic attack. *Cochrane Database Syst Rev* 2018;**7**:CD007858. https://doi.org/10.1002/14651858.CD007858.pub2
817. Boncoraglio GB, Del Giovane C, Tramacere I. Antihypertensive drugs for secondary prevention after ischemic stroke or transient ischemic attack: a systematic review and meta-analysis. *Stroke* 2021;**52**:1974–82. https://doi.org/10.1161/strokeaha.120.031945
818. Fischer U, Cooney MT, Bull LM, Silver LE, Chalmers J, Anderson CS, *et al.* Acute post-stroke blood pressure relative to premorbid levels in intracerebral haemorrhage versus major ischaemic stroke: a population-based study. *Lancet Neurol* 2014;**13**:374–84. https://doi.org/10.1016/s1474-4422(14)70031-6
819. Yusuf S, Diener HC, Sacco RL, Cotton D, Ôunpuu S, Lawton WA, *et al.* Telmisartan to prevent recurrent stroke and cardiovascular events. *N Engl J Med* 2008;**359**:1225–37. https://doi.org/10.1056/NEJMoa0804593
820. PROGRESS Collaborative Group. Randomised trial of a perindopril-based blood-pressure-lowering regimen among 6,105 individuals with previous stroke or transient ischaemic attack. *Lancet* 2001;**358**:1033–41. https://doi.org/10.1016/s0140-6736(01)06178-5
821. McGurgan IJ, Kelly PJ, Turan TN, Rothwell PM. Long-term secondary prevention: management of blood pressure after a transient ischemic attack or stroke. *Stroke* 2022;**53**:1085–103. https://doi.org/10.1161/strokeaha.121.035851
822. Rudd AG, Bowen A, Young GR, James MA. The latest national clinical guideline for stroke. *Clin Med (Lond)* 2017;**17**:154–5. https://doi.org/10.7861/clinmedicine.17-2-154

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

823. Arima H, Chalmers J, Woodward M, Anderson C, Rodgers A, Davis S, *et al.* Lower target blood pressures are safe and effective for the prevention of recurrent stroke: the PROGRESS trial. *J Hypertens* 2006;**24**:1201–8. https://doi.org/10.1097/01.hjh.0000226212.34055.86
824. Kitagawa K, Yamamoto Y, Arima H, Maeda T, Sunami N, Kanzawa T, *et al.* Effect of standard vs intensive blood pressure control on the risk of recurrent stroke: a randomized clinical trial and meta-analysis. *JAMA Neurol* 2019;**76**:1309–18. https://doi.org/10.1001/jamaneurol.2019.2167
825. Suchard MA, Schuemie MJ, Krumholz HM, You SC, Chen R, Pratt N, *et al.* Comprehensive comparative effectiveness and safety of first-line antihypertensive drug classes: a systematic, multinational, large-scale analysis. *Lancet* 2019;**394**:1816–26. https://doi.org/10.1016/s0140-6736(19)32317-7
826. Wang WT, You LK, Chiang CE, Sung S-H, Chuang S-Y, Cheng H-M, *et al.* Comparative effectiveness of blood pressure-lowering drugs in patients who have already suffered from stroke: traditional and Bayesian network meta-analysis of randomized trials. *Medicine (Baltimore)* 2016;**95**:e3302. https://doi.org/10.1097/md.0000000000003302
827. Bath PM, Scutt P, Blackburn DJ, Ankolekar S, Krishnan K, Ballard C, *et al.* Intensive versus guideline blood pressure and lipid lowering in patients with previous stroke: main results from the pilot 'Prevention of Decline in Cognition after Stroke Trial' (PODCAST) randomised controlled trial. *PLoS One* 2017;**12**:e0164608. https://doi.org/10.1371/journal.pone.0164608
828. Mant J, McManus RJ, Roalfe A, Fletcher K, Taylor CJ, Martin U, *et al.* Different systolic blood pressure targets for people with history of stroke or transient ischaemic attack: PAST-BP (Prevention After Stroke—Blood Pressure) randomised controlled trial. *BMJ* 2016;**352**:i708. https://doi.org/10.1136/bmj.i708
829. Kolmos M, Christoffersen L, Kruuse C. Recurrent ischemic stroke—a systematic review and meta-analysis. *J Stroke Cerebrovasc Dis* 2021;**30**:105935. https://doi.org/10.1016/j.jstrokecerebrovasdis.2021.105935
830. Livingston G, Huntley J, Sommerlad A, Ames D, Ballard C, Banerjee S, *et al.* Dementia prevention, intervention, and care: 2020 report of the Lancet Commission. *Lancet* 2020;**396**:413–46. https://doi.org/10.1016/s0140-6736(20)30367-6
831. Ou YN, Tan CC, Shen XN, Xu W, Hou X-H, Dong Q, *et al.* Blood pressure and risks of cognitive impairment and dementia: a systematic review and meta-analysis of 209 prospective studies. *Hypertension* 2020;**76**:217–25. https://doi.org/10.1161/hypertensionaha.120.14993
832. Abell JG, Kivimäki M, Dugravot A, Tabak AG, Fayosse A, Shipley M, *et al.* Association between systolic blood pressure and dementia in the Whitehall II cohort study: role of age, duration, and threshold used to define hypertension. *Eur Heart J* 2018;**39**:3119–25. https://doi.org/10.1093/eurheartj/ehy288
833. Alpérovitch A, Blachier M, Soumaré A, Ritchie K, Dartigues J-F, Richard-Harston S, *et al.* Blood pressure variability and risk of dementia in an elderly cohort, the Three-City study. *Alzheimers Dement* 2014;**10**:S330–337. https://doi.org/10.1016/j.jalz.2013.05.1777
834. McGrath ER, Beiser AS, DeCarli C, Plourde KL, Vasan RS, Greenberg S, *et al.* Blood pressure from mid- to late life and risk of incident dementia. *Neurology* 2017;**89**:2447–54. https://doi.org/10.1212/wnl.0000000000004741
835. Skoog I, Nilsson L, Persson G, Lernfelt B, Landahl S, Palmertz B, *et al.* 15-year longitudinal study of blood pressure and dementia. *Lancet* 1996;**347**:1141–5. https://doi.org/10.1016/s0140-6736(96)90608-x
836. Launer LJ, Masaki K, Petrovitch H, Foley D, Havlik RJ. The association between midlife blood pressure levels and late-life cognitive function. The Honolulu-Asia aging study. *JAMA* 1995;**274**:1846–51. https://doi.org/10.1001/jama.1995.03530230032026
837. Gottesman RF, Albert MS, Alonso A, Coker LH, Coresh J, Davis SM, *et al.* Associations between midlife vascular risk factors and 25-year incident dementia in the atherosclerosis risk in Communities (ARIC) cohort. *JAMA Neurol* 2017;**74**:1246–54. https://doi.org/10.1001/jamaneurol.2017.1658
838. Levine DA, Springer MV, Brodtmann A. Blood pressure and vascular cognitive impairment. *Stroke* 2022;**53**:1104–13. https://doi.org/10.1161/strokeaha.121.036140
839. Bosch J, O'Donnell M, Swaminathan B, Lonn EM, Sharma M, Dagenais G, *et al.* Effects of blood pressure and lipid lowering on cognition: results from the HOPE-3 study. *Neurology* 2019;**92**:e1435–46. https://doi.org/10.1212/wnl.0000000000007174
840. Applegate WB, Pressel S, Wittes J, Luhr J, Shekelle RB, Camel GH, *et al.* Impact of the treatment of isolated systolic hypertension on behavioral variables. Results from the systolic hypertension in the elderly program. *Arch Intern Med* 1994;**154**:2154–60. https://doi.org/10.1001/archinte.1994.00420190047006
841. Williamson JD, Pajewski NM, Auchus AP, Bryan RN, Chelune G, Cheung AK, *et al.* Effect of intensive vs standard blood pressure control on probable dementia: a randomized clinical trial. *JAMA* 2019;**321**:553–61. https://doi.org/10.1001/jama.2018.21442
842. Peters R, Beckett N, Forette F, Tuomilehto J, Clarke R, Ritchie C, *et al.* Incident dementia and blood pressure lowering in the Hypertension in the Very Elderly Trial cognitive function assessment (HYVET-COG): a double-blind, placebo controlled trial. *Lancet Neurol* 2008;**7**:683–9. https://doi.org/10.1016/s1474-4422(08)70143-1
843. Lithell H, Hansson L, Skoog I, Elmfeldt D, Hofman A, Olofsson B, *et al.* The Study on Cognition and Prognosis in the Elderly (SCOPE): principal results of a randomized double-blind intervention trial. *J Hypertens* 2003;**21**:875–86. https://doi.org/10.1097/00004872-200305000-00011
844. White WB, Wakefield DB, Moscufo N, Guttmann CRG, Kaplan RF, Bohannon RW, *et al.* Effects of intensive versus standard ambulatory blood pressure control on cerebrovascular outcomes in older people (INFINITY). *Circulation* 2019;**140**:1626–35. https://doi.org/10.1161/circulationaha.119.041603
845. Tzourio C, Anderson C, Chapman N, Woodward M, Neal B, MacMahon S, *et al.* Effects of blood pressure lowering with perindopril and indapamide therapy on dementia and cognitive decline in patients with cerebrovascular disease. *Arch Intern Med* 2003;**163**:1069–75. https://doi.org/10.1001/archinte.163.9.1069
846. Peters R, Yasar S, Anderson CS, Andrews S, Antikainen R, Arima H, *et al.* Investigation of antihypertensive class, dementia, and cognitive decline: a meta-analysis. *Neurology* 2020;**94**:e267–81. https://doi.org/10.1212/wnl.0000000000008732
847. Yang W, Luo H, Ma Y, Si S, Zhao H. Effects of antihypertensive drugs on cognitive function in elderly patients with hypertension: a review. *Aging Dis* 2021;**12**:841–51. https://doi.org/10.14336/ad.2020.1111
848. Iseli R, Nguyen VTV, Sharmin S, Reijnierse EM, Lim WK, Maier AB, *et al.* Orthostatic hypotension and cognition in older adults: a systematic review and meta-analysis. *Exp Gerontol* 2019;**120**:40–9. https://doi.org/10.1016/j.exger.2019.02.017
849. Ernst ME, Ryan J, Chowdhury EK, Margolis KL, Beilin LJ, Reid CM, *et al.* Long-term blood pressure variability and risk of cognitive decline and dementia among older adults. *J Am Heart Assoc* 2021;**10**:e019613. https://doi.org/10.1161/jaha.120.019613
850. Nakamura K, Stefanescu Schmidt A. Treatment of hypertension in coarctation of the aorta. *Curr Treat Options Cardiovasc Med* 2016;**18**:40. https://doi.org/10.1007/s11936-016-0462-x
851. Dijkema EJ, Leiner T, Grotenhuis HB. Diagnosis, imaging and clinical management of aortic coarctation. *Heart* 2017;**103**:1148–55. https://doi.org/10.1136/heartjnl-2017-311173
852. Panzer J, Bové T, Vandekerckhove K, De Wolf D. Hypertension after coarctation repair-a systematic review. *Transl Pediatr* 2022;**11**:270–9. https://doi.org/10.21037/tp-21-418
853. Schaefer BM, Lewin MB, Stout KK, Gill E, Prueitt A, Byers P H, *et al.* The bicuspid aortic valve: an integrated phenotypic classification of leaflet morphology and aortic root shape. *Heart* 2008;**94**:1634–8. https://doi.org/10.1136/hrt.2007.132092
854. Wang J, Deng W, Lv Q, Li Y, Liu T, Xie M, *et al.* Aortic dilatation in patients with bicuspid aortic valve. *Front Physiol* 2021;**12**:615175. https://doi.org/10.3389/fphys.2021.615175
855. Davies RR, Kaple RK, Mandapati D, Gallo A, Botta DM, Elefteriades JA, *et al.* Natural history of ascending aortic aneurysms in the setting of an unreplaced bicuspid aortic valve. *Ann Thorac Surg* 2007;**83**:1338–44. https://doi.org/10.1016/j.athoracsur.2006.10.074
856. Lindman BR, Otto CM. Time to treat hypertension in patients with aortic stenosis. *Circulation* 2013;**128**:1281–3. https://doi.org/10.1161/circulationaha.113.005275
857. Erbel R, Aboyans V, Boileau C, Bossone E, Di Bartolomeo R, Eggebrecht H, *et al.* 2014 ESC Guidelines on the diagnosis and treatment of aortic diseases: document covering acute and chronic aortic diseases of the thoracic and abdominal aorta of the adult. The Task Force for the Diagnosis and Treatment of Aortic Diseases of the European Society of Cardiology (ESC). *Eur Heart J* 2014;**35**:2873–926. https://doi.org/10.1093/eurheartj/ehu281
858. Shores J, Berger KR, Murphy EA, Pyeritz RE. Progression of aortic dilatation and the benefit of long-term beta-adrenergic blockade in Marfan's syndrome. *N Engl J Med* 1994;**330**:1335–41. https://doi.org/10.1056/nejm199405123301902
859. Groenink M, den Hartog AW, Franken R, Radonic T, de Waard V, Timmermans J, *et al.* Losartan reduces aortic dilatation rate in adults with Marfan syndrome: a randomized controlled trial. *Eur Heart J* 2013;**34**:3491–500. https://doi.org/10.1093/eurheartj/eht334
860. Pitcher A, Spata E, Emberson J, Davies K, Halls H, Holland L, *et al.* Angiotensin receptor blockers and β blockers in Marfan syndrome: an individual patient data meta-analysis of randomised trials. *Lancet* 2022;**400**:822–31. https://doi.org/10.1016/s0140-6736(22)01534-3
861. Mazzolai L, Rodriguez-Palomares JF, Teixido-Tura G, Lanzi S, Boc V, Bossone E, *et al.* 2024 ESC Guidelines for the management peripheral arterial and aortic diseases. *Eur Heart J* 2024;**45**:3538–700. https://doi.org/10.1093/eurheartj/ehae179
862. Rechel B, Mladovsky P, Ingleby D, Mackenbach JP, McKee M. Migration and health in an increasingly diverse Europe. *Lancet* 2013;**381**:1235–45. https://doi.org/10.1016/s0140-6736(12)62086-8
863. Modesti PA, Reboldi G, Cappuccio FP, Agyemang C, Remuzzi G, Rapi S, *et al.* Panethnic differences in blood pressure in Europe: a systematic review and meta-analysis. *PLoS One* 2016;**11**:e0147601. https://doi.org/10.1371/journal.pone.0147601
864. Whelton PK, Einhorn PT, Muntner P, Appel LJ, Cushman WC, Diez Roux AV, *et al.* Research needs to improve hypertension treatment and control in African Americans. *Hypertension* 2016;**68**:1066–72. https://doi.org/10.1161/hypertensionaha.116.07905
865. Kaufman JS, Cooper RS, McGee DL. Socioeconomic status and health in blacks and whites: the problem of residual confounding and the resiliency of race. *Epidemiology* 1997;**8**:621–8.
866. Agyemang C, van Oeffelen AA, Norredam M, Kappelle LJ, Klijn CJM, Bots ML, *et al.* Socioeconomic inequalities in stroke incidence among migrant groups: analysis of nationwide data. *Stroke* 2014;**45**:2397–403. https://doi.org/10.1161/strokeaha.114.005505

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

867. Mehanna M, Gong Y, McDonough CW, Beitelshees AL, Gums JG, Chapman AB, *et al.* Blood pressure response to metoprolol and chlorthalidone in European and African Americans with hypertension. *J Clin Hypertens (Greenwich)* 2017;**19**:1301–8. https://doi.org/10.1111/jch.13094
868. Faconti L, McNally RJ, Farukh B, Adeyemi O, Cruickshank JK, Wilkinson IB, *et al.* Differences in hypertension phenotypes between Africans and Europeans: role of environment. *J Hypertens* 2020;**38**:1278–85. https://doi.org/10.1097/hjh.0000000000002403
869. Schutte AE, Kruger R, Gafane-Matemane LF, Breet Y, Strauss-Kruger M, Cruickshank JK, *et al.* Ethnicity and arterial stiffness. *Arterioscler Thromb Vasc Biol* 2020;**40**:1044–54. https://doi.org/10.1161/atvbaha.120.313133
870. Erlinger TP, Vollmer WM, Svetkey LP, Appel LJ. The potential impact of nonpharmacologic population-wide blood pressure reduction on coronary heart disease events: pronounced benefits in African-Americans and hypertensives. *Prev Med* 2003;**37**:327–33. https://doi.org/10.1016/s0091-7435(03)00140-3
871. The ALLHAT Officers Coordinators for the ALLHAT Collaborative Research Group. Major outcomes in high-risk hypertensive patients randomized to angiotensin-converting enzyme inhibitor or calcium channel blocker vs diuretic: the Antihypertensive and Lipid-Lowering Treatment to Prevent Heart Attack Trial (ALLHAT). *JAMA* 2002;**288**:2981–97. https://doi.org/10.1001/jama.288.23.2981
872. Wright JT Jr, Dunn JK, Cutler JA, Davis BR, Cushman WC, Ford CE, *et al.* Outcomes in hypertensive black and nonblack patients treated with chlorthalidone, amlodipine, and lisinopril. *JAMA* 2005;**293**:1595–608. https://doi.org/10.1001/jama.293.13.1595
873. Wright JT Jr, Harris-Haywood S, Pressel S, Barzilay J, Baimbridge C, Bareis CJ, *et al.* Clinical outcomes by race in hypertensive patients with and without the metabolic syndrome: antihypertensive and lipid-lowering treatment to prevent heart attack trial (ALLHAT). *Arch Intern Med* 2008;**168**:207–17. https://doi.org/10.1001/archinternmed.2007.66
874. Ojji DB, Mayosi B, Francis V, Badri M, Cornelius V, Smythe W, *et al.* Comparison of dual therapies for lowering blood pressure in Black Africans. *N Engl J Med* 2019;**380**:2429–39. https://doi.org/10.1056/NEJMoa1901113
875. van der Linden EL, Couwenhoven BN, Beune E, Daams JG, van den Born B-JH, Agyemang C. Hypertension awareness, treatment and control among ethnic minority populations in Europe: a systematic review and meta-analysis. *J Hypertens* 2021;**39**:202–13. https://doi.org/10.1097/hjh.0000000000002651
876. Agyemang C, Nyaaba G, Beune E, Meeks K, Owusu-Dabo E, Addo J, *et al.* Variations in hypertension awareness, treatment, and control among Ghanaian migrants living in Amsterdam, Berlin, London, and nonmigrant Ghanaians living in rural and urban Ghana—the RODAM study. *J Hypertens* 2018;**36**:169–77. https://doi.org/10.1097/hjh.0000000000001520
877. van Laer SD, Snijder MB, Agyemang C, Peters RJ, van den Born BH. Ethnic differences in hypertension prevalence and contributing determinants – the HELIUS study. *Eur J Prev Cardiol* 2018;**25**:1914–22. https://doi.org/10.1177/2047487318803241
878. Costello HM, Gumz ML. Circadian rhythm, clock genes, and hypertension: recent advances in hypertension. *Hypertension* 2021;**78**:1185–96. https://doi.org/10.1161/hypertensionaha.121.14519
879. O'Brien E, Sheridan J, O'Malley K. Dippers and non-dippers. *Lancet* 1988;**2**:397. https://doi.org/10.1016/s0140-6736(88)92867-x
880. Pickering TG, Shimbo D, Haas D. Ambulatory blood-pressure monitoring. *N Engl J Med* 2006;**354**:2368–74. https://doi.org/10.1056/NEJMra060433
881. Abdalla M, Goldsmith J, Muntner P, Diaz KM, Reynolds K, Schwartz JE, *et al.* Is isolated nocturnal hypertension a reproducible phenotype? *Am J Hypertens* 2016;**29**:33–8. https://doi.org/10.1093/ajh/hpv058
882. Mancia G, Facchetti R, Bombelli M, Quarti-Trevano F, Cuspidi C, Grassi G. Short- and long-term reproducibility of nighttime blood pressure phenotypes and nocturnal blood pressure reduction. *Hypertension* 2021;**77**:1745–55. https://doi.org/10.1161/hypertensionaha.120.16827
883. Nolde JM, Kiuchi MG, Lugo-Gavidia LM, Ho JK, Chan J, Matthews VB, *et al.* Nocturnal hypertension: a common phenotype in a tertiary clinical setting associated with increased arterial stiffness and central blood pressure. *J Hypertens* 2021;**39**:250–8. https://doi.org/10.1097/hjh.0000000000002620
884. Thomas SJ, Booth JN III, Jaeger BC, Hubbard D, Sakhuja S, Abdalla M, *et al.* Association of sleep characteristics with nocturnal hypertension and nondipping blood pressure in the CARDIA study. *J Am Heart Assoc* 2020;**9**:e015062. https://doi.org/10.1161/jaha.119.015062
885. Kario K, Hoshide S, Haimoto H, Yamagiwa K, Uchiba K, Nagasaka S, *et al.* Sleep blood pressure self-measured at home as a novel determinant of organ damage: Japan morning surge home blood pressure (J-HOP) study. *J Clin Hypertens (Greenwich)* 2015;**17**:340–8. https://doi.org/10.1111/jch.12500
886. de la Sierra A, Gorostidi M, Banegas JR, Segura J, de la Cruz JJ, Ruilope LM, *et al.* Nocturnal hypertension or nondipping: which is better associated with the cardiovascular risk profile? *Am J Hypertens* 2014;**27**:680–7. https://doi.org/10.1093/ajh/hpt175
887. Wijkman M, Länne T, Engvall J, Lindström T, Östgren CJ, Nystrom FH, *et al.* Masked nocturnal hypertension—a novel marker of risk in type 2 diabetes. *Diabetologia* 2009;**52**:1258–64. https://doi.org/10.1007/s00125-009-1369-9
888. Muntner P, Lewis CE, Diaz KM, Carson AP, Kim Y, Calhoun D, *et al.* Racial differences in abnormal ambulatory blood pressure monitoring measures: results from the coronary artery risk development in young adults (CARDIA) study. *Am J Hypertens* 2015;**28**:640–8. https://doi.org/10.1093/ajh/hpu193
889. Thomas SJ, Booth JN III, Bromfield SG, Seals SR, Spruill TM, Ogedegbe G, *et al.* Clinic and ambulatory blood pressure in a population-based sample of African Americans: the Jackson heart study. *J Am Soc Hypertens* 2017;**11**:204–212.e205. https://doi.org/10.1016/j.jash.2017.02.001
890. Husain A, Lin FC, Tuttle LA, Olsson E, Viera AJ. The reproducibility of racial differences in ambulatory blood pressure phenotypes and measurements. *Am J Hypertens* 2017;**30**:961–7. https://doi.org/10.1093/ajh/hpx079
891. Li Y, Wang JG. Isolated nocturnal hypertension: a disease masked in the dark. *Hypertension* 2013;**61**:278–83. https://doi.org/10.1161/hypertensionaha.111.00217
892. Hoshide S, Kario K, de la Sierra A, Bilo G, Schillaci G, Banegas JR, *et al.* Ethnic differences in the degree of morning blood pressure surge and in its determinants between Japanese and European hypertensive subjects: data from the ARTEMIS study. *Hypertension* 2015;**66**:750–6. https://doi.org/10.1161/hypertensionaha.115.05958
893. Banegas JR, Ruilope LM, de la Sierra A, Gorostidi M, Segura J, Martell N, *et al.* High prevalence of masked uncontrolled hypertension in people with treated hypertension. *Eur Heart J* 2014;**35**:3304–12. https://doi.org/10.1093/eurheartj/ehu016
894. Huang JF, Zhang DY, Sheng CS, An D-W, Li M, Cheng Y-B, *et al.* Isolated nocturnal hypertension in relation to host and environmental factors and clock genes. *J Clin Hypertens (Greenwich)* 2022;**24**:1255–62. https://doi.org/10.1111/jch.14532
895. Matsumoto T, Tabara Y, Murase K, Setoh K, Kawaguchi T, Nagashima S, *et al.* Nocturia and increase in nocturnal blood pressure: the Nagahama study. *J Hypertens* 2018;**36**:2185–92. https://doi.org/10.1097/hjh.0000000000001802
896. Seif F, Patel SR, Walia HK, Rueschman M, Bhatt DL, Blumenthal RS, *et al.* Obstructive sleep apnea and diurnal nondipping hemodynamic indices in patients at increased cardiovascular risk. *J Hypertens* 2014;**32**:267–75. https://doi.org/10.1097/hjh.0000000000000011
897. Kimura G, Dohi Y, Fukuda M. Salt sensitivity and circadian rhythm of blood pressure: the keys to connect CKD with cardiovascular events. *Hypertens Res* 2010;**33**:515–20. https://doi.org/10.1038/hr.2010.47
898. Drawz PE, Alper AB, Anderson AH, Brecklin CS, Charleston J, Chen J, *et al.* Masked hypertension and elevated nighttime blood pressure in CKD: prevalence and association with target organ damage. *Clin J Am Soc Nephrol* 2016;**11**:642–52. https://doi.org/10.2215/cjn.08530815
899. Kanno A, Metoki H, Kikuya M, Terawaki H, Hara A, Hashimoto T, *et al.* Usefulness of assessing masked and white-coat hypertension by ambulatory blood pressure monitoring for determining prevalent risk of chronic kidney disease: the Ohasama study. *Hypertens Res* 2010;**33**:1192–8. https://doi.org/10.1038/hr.2010.139
900. Kushiro T, Kario K, Saito I, Teramukai S, Sato Y, Okuda Y, *et al.* Increased cardiovascular risk of treated white coat and masked hypertension in patients with diabetes and chronic kidney disease: the HONEST study. *Hypertens Res* 2017;**40**:87–95. https://doi.org/10.1038/hr.2016.87
901. Lurbe E, Redon J, Kesani A, Pascual JM, Tacons J, Alvarez V, *et al.* Increase in nocturnal blood pressure and progression to microalbuminuria in type 1 diabetes. *N Engl J Med* 2002;**347**:797–805. https://doi.org/10.1056/NEJMoa013410
902. Cuspidi C, Sala C, Tadic M, Gherbesi E, De Giorgi A, Grassi G, *et al.* Clinical and prognostic significance of a reverse dipping pattern on ambulatory monitoring: an updated review. *J Clin Hypertens (Greenwich)* 2017;**19**:713–21. https://doi.org/10.1111/jch.13023
903. Wu Q, Hong M, Xu J, Tang X, Zhu L, Gao P, *et al.* Diurnal blood pressure pattern and cardiac damage in hypertensive patients with primary aldosteronism. *Endocrine* 2021;**72**:835–43. https://doi.org/10.1007/s12020-021-02606-3
904. Kario K, Hoshide S, Mizuno H, Kabutoya T, Nishizawa M, Yoshida T, *et al.* Nighttime blood pressure phenotype and cardiovascular prognosis: practitioner-based nationwide JAMP study. *Circulation* 2020;**142**:1810–20. https://doi.org/10.1161/circulationaha.120.049730
905. Kario K, Hoshide S, Nagai M, Okawara Y, Kanegae H. Sleep and cardiovascular outcomes in relation to nocturnal hypertension: the J-HOP nocturnal blood pressure study. *Hypertens Res* 2021;**44**:1589–96. https://doi.org/10.1038/s41440-021-00709-y
906. Wang Q, Wang Y, Wang J, Zhang L, Zhao MH. Nocturnal systolic hypertension and adverse prognosis in patients with CKD. *Clin J Am Soc Nephrol* 2021;**16**:356–64. https://doi.org/10.2215/cjn.14420920
907. Hansen TW, Li Y, Boggia J, Thijs L, Richart T, Staessen JA. Predictive role of the nighttime blood pressure. *Hypertension* 2011;**57**:3–10. https://doi.org/10.1161/hypertensionaha.109.133900
908. Staessen JA, Thijs L, Fagard R, O'Brien ET, Clement D, de Leeuw PW, *et al.* Predicting cardiovascular risk using conventional vs ambulatory blood pressure in older patients with systolic hypertension. *JAMA* 1999;**282**:539–46. https://doi.org/10.1001/jama.282.6.539
909. Verdecchia P, Porcellati C, Schillaci G, Borgioni C, Ciucci A, Battistelli M, *et al.* Ambulatory blood pressure. An independent predictor of prognosis in essential hypertension. *Hypertension* 1994;**24**:793–801. https://doi.org/10.1161/01.hyp.24.6.793
910. Ohkubo T, Hozawa A, Yamaguchi J, Kikuya M, Ohmori K, Michimata M, *et al.* Prognostic significance of the nocturnal decline in blood pressure in individuals with and without high 24-h blood pressure: the Ohasama study. *J Hypertens* 2002;**20**:2183–9. https://doi.org/10.1097/00004872-200211000-00017

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

911. Nagai M, Hoshide S, Ishikawa J, Shimada K, Kario K. Ambulatory blood pressure as an independent determinant of brain atrophy and cognitive function in elderly hypertension. *J Hypertens* 2008;**26**:1636–41. https://doi.org/10.1097/HJH.0b013e3283018333
912. Staessen JA, Thijs L, Ohkubo T, Kikuya M, Richart T, Boggia J, *et al.* Thirty years of research on diagnostic and therapeutic thresholds for the self-measured blood pressure at home. *Blood Press Monit* 2008;**13**:352–65. https://doi.org/10.1097/MBP.0b013e3283108f93
913. Kario K, Shimada K. Risers and extreme-dippers of nocturnal blood pressure in hypertension: antihypertensive strategy for nocturnal blood pressure. *Clin Exp Hypertens* 2004;**26**:177–89. https://doi.org/10.1081/ceh-120028556
914. Tan X, Sundström J, Lind L, Franzon K, Kilander L, Benedict C, *et al.* Reverse dipping of systolic blood pressure is associated with increased dementia risk in older men: a longitudinal study over 24 years. *Hypertension* 2021;**77**:1383–90. https://doi.org/10.1161/hypertensionaha.120.16711
915. Judd E, Calhoun DA. Apparent and true resistant hypertension: definition, prevalence and outcomes. *J Hum Hypertens* 2014;**28**:463–8. https://doi.org/10.1038/jhh.2013.140
916. Denker MG, Haddad DB, Townsend RR, Cohen DL. Blood pressure control 1 year after referral to a hypertension specialist. *J Clin Hypertens (Greenwich)* 2013;**15**:624–9. https://doi.org/10.1111/jch.12146
917. Blumenthal JA, Hinderliter AL, Smith PJ, Mabe S, Watkins LL, Craighead L, *et al.* Effects of lifestyle modification on patients with resistant hypertension: results of the TRIUMPH randomized clinical trial. *Circulation* 2021;**144**:1212–26. https://doi.org/10.1161/circulationaha.121.055329
918. Hung SC, Kuo KL, Peng CH, Wu C-H, Lien Y-C, Wang Y-C, *et al.* Volume overload correlates with cardiovascular risk factors in patients with chronic kidney disease. *Kidney Int* 2014;**85**:703–9. https://doi.org/10.1038/ki.2013.336
919. Taler SJ, Textor SC, Augustine JE. Resistant hypertension: comparing hemodynamic management to specialist care. *Hypertension* 2002;**39**:982–8. https://doi.org/10.1161/01.hyp.0000016176.16042.2f
920. Gaddam KK, Nishizaka MK, Pratt-Ubunama MN, Pimenta E, Aban I, Oparil S, *et al.* Characterization of resistant hypertension: association between resistant hypertension, aldosterone, and persistent intravascular volume expansion. *Arch Intern Med* 2008;**168**:1159–64. https://doi.org/10.1001/archinte.168.11.1159
921. Roush GC, Ernst ME, Kostis JB, Tandon S, Sica DA. Head-to-head comparisons of hydrochlorothiazide with indapamide and chlorthalidone: antihypertensive and metabolic effects. *Hypertension* 2015;**65**:1041–6. https://doi.org/10.1161/hypertensionaha.114.05021
922. Peterzan MA, Hardy R, Chaturvedi N, Hughes AD. Meta-analysis of dose-response relationships for hydrochlorothiazide, chlorthalidone, and bendroflumethiazide on blood pressure, serum potassium, and urate. *Hypertension* 2012;**59**:1104–9. https://doi.org/10.1161/hypertensionaha.111.190637
923. Chen C, Zhu XY, Li D, Lin Q, Zhou K. Clinical efficacy and safety of spironolactone in patients with resistant hypertension: a systematic review and meta-analysis. *Medicine (Baltimore)* 2020;**99**:e21694. https://doi.org/10.1097/md.0000000000021694
924. Zhao D, Liu H, Dong P, Zhao J. A meta-analysis of add-on use of spironolactone in patients with resistant hypertension. *Int J Cardiol* 2017;**233**:113–7. https://doi.org/10.1016/j.ijcard.2016.12.158
925. Bazoukis G, Thomopoulos C, Tsioufis C. Effect of mineralocorticoid antagonists on blood pressure lowering: overview and meta-analysis of randomized controlled trials in hypertension. *J Hypertens* 2018;**36**:987–94. https://doi.org/10.1097/hjh.0000000000001671
926. Oxlund CS, Henriksen JE, Tarnow L, Schousboe K, Gram J, Jacobsen IA. Low dose spironolactone reduces blood pressure in patients with resistant hypertension and type 2 diabetes mellitus: a double blind randomized clinical trial. *J Hypertens* 2013;**31**:2094–102. https://doi.org/10.1097/HJH.0b013e3283638b1a
927. Desai R, Park H, Brown JD, Mohandas R, Pepine CJ, Smith SM. Comparative safety and effectiveness of aldosterone antagonists versus beta-blockers as fourth agents in patients with apparent resistant hypertension. *Hypertension* 2022;**79**:2305–15. https://doi.org/10.1161/hypertensionaha.122.19280
928. Juurlink DN, Mamdani MM, Lee DS, Kopp A, Austin PC, Laupacis A, *et al.* Rates of hyperkalemia after publication of the randomized aldactone evaluation study. *N Engl J Med* 2004;**351**:543–51. https://doi.org/10.1056/NEJMoa040135
929. Agarwal R, Rossignol P, Romero A, Garza D, Mayo MR, Warren S, *et al.* Patiromer versus placebo to enable spironolactone use in patients with resistant hypertension and chronic kidney disease (AMBER): a phase 2, randomised, double-blind, placebo-controlled trial. *Lancet* 2019;**394**:1540–50. https://doi.org/10.1016/s0140-6736(19)32135-x
930. Burton TJ, Mackenzie IS, Balan K, Koo B, Bird N, Soloviev DV, *et al.* Evaluation of the sensitivity and specificity of (11)C-metomidate positron emission tomography (PET)-CT for lateralizing aldosterone secretion by Conn's adenomas. *J Clin Endocrinol Metab* 2012;**97**:100–9. https://doi.org/10.1210/jc.2011-1537
931. Rossi GP, Maiolino G, Seccia TM. Adrenal venous sampling: where do we stand? *Endocrinol Metab Clin North Am* 2019;**48**:843–58. https://doi.org/10.1016/j.ecl.2019.08.012
932. Wu X, Senanayake R, Goodchild E, Bashari WA, Salsbury J, Cabrera CP, *et al.* [(11)C] metomidate PET-CT versus adrenal vein sampling for diagnosing surgically curable primary aldosteronism: a prospective, within-patient trial. *Nat Med* 2023;**29**:190–202. https://doi.org/10.1038/s41591-022-02114-5
933. Lenzini L, Prisco S, Caroccia B, Rossi GP. Saga of familial hyperaldosteronism: yet a new channel. *Hypertension* 2018;**71**:1010–4. https://doi.org/10.1161/hypertensionaha.118.11150
934. Parthasarathy HK, Ménard J, White WB, Young WF, Williams GH, Williams B, *et al.* A double-blind, randomized study comparing the antihypertensive effect of eplerenone and spironolactone in patients with hypertension and evidence of primary aldosteronism. *J Hypertens* 2011;**29**:980–90. https://doi.org/10.1097/HJH.0b013e3283455ca5
935. Shagjaa T, Sanga V, Rossi GP. Skin hyperpigmentation due to post-surgical adrenal insufficiency regressed with the dexamethasone treatment. *J Clin Med* 2022;**11**:5379. https://doi.org/10.3390/jcm11185379
936. Sanga V, Lenzini L, Seccia TM, Rossi GP. Familial hyperaldosteronism type 1 and pregnancy: successful treatment with low dose dexamethasone. *Blood Press* 2021;**30**:133–7. https://doi.org/10.1080/08037051.2020.1863771
937. Persu A, Touzé E, Mousseaux E, Barral X, Joffre F, Plouin P-F. Diagnosis and management of fibromuscular dysplasia: an expert consensus. *Eur J Clin Invest* 2012;**42**:338–47. https://doi.org/10.1111/j.1365-2362.2011.02577.x
938. Wheatley K, Ives N, Gray R, Kalra PA, Moss JG, Baigent C, *et al.* Revascularization versus medical therapy for renal-artery stenosis. *N Engl J Med* 2009;**361**:1953–62. https://doi.org/10.1056/NEJMoa0905368
939. Cooper CJ, Murphy TP, Cutlip DE, Jamerson K, Henrich W, Reid DM, *et al.* Stenting and medical therapy for atherosclerotic renal-artery stenosis. *N Engl J Med* 2014;**370**:13–22. https://doi.org/10.1056/NEJMoa1310753
940. Bailey SR, Beckman JA, Dao TD, Misra S, Sobieszczyk PS, White CJ, *et al.* ACC/AHA/SCAI/SIR/SVM 2018 appropriate use criteria for peripheral artery intervention: a report of the American College of Cardiology Appropriate Use Criteria Task Force, American Heart Association, Society for Cardiovascular Angiography and Interventions, Society of Interventional Radiology, and Society for Vascular Medicine. *J Am Coll Cardiol* 2019;**73**:214–37. https://doi.org/10.1016/j.jacc.2018.10.002
941. Trinquart L, Mounier-Vehier C, Sapoval M, Gagnon N, Plouin PF. Efficacy of revascularization for renal artery stenosis caused by fibromuscular dysplasia: a systematic review and meta-analysis. *Hypertension* 2010;**56**:525–32. https://doi.org/10.1161/hypertensionaha.110.152918
942. Sanga V, Bertoli E, Crimì F, Barbiero G, Battistel M, Teresa Seccia M, *et al.* Pickering syndrome: an overlooked renovascular cause of recurrent heart failure. *J Am Heart Assoc* 2023;**12**:e030474. https://doi.org/10.1161/jaha.123.030474
943. Bhalla V, Textor SC, Beckman JA, Casanegra AI, Cooper CJ, Kim ESH, *et al.* Revascularization for renovascular disease: a scientific statement from the American Heart Association. *Hypertension* 2022;**79**:e128–43. https://doi.org/10.1161/hyp.0000000000000217
944. Bravo E, Fouad-Tarazi F, Rossi G, Imamura M, Lin WW, Madkour MA, *et al.* A re-evaluation of the hemodynamics of pheochromocytoma. *Hypertension* 1990;**15**:I128–131. https://doi.org/10.1161/01.hyp.15.2_suppl.i128
945. Cohen JB, Brown NJ, Brown SA, Dent S, van Dorst DCH, Herrmann SM, *et al.* Cancer therapy-related hypertension: a scientific statement from the American Heart Association. *Hypertension* 2023;**80**:e46–57. https://doi.org/10.1161/hyp.0000000000000224
946. Lyon AR, López-Fernández T, Couch LS, Asteggiano R, Aznar MC, Bergler-Klein J, *et al.* ESC guidelines on cardio-oncology developed in collaboration with the European Hematology Association (EHA), the European Society for Therapeutic Radiology and Oncology (ESTRO) and the International Cardio-Oncology Society (IC-OS). *Eur Heart J* 2022;**43**:4229–361. https://doi.org/10.1093/eurheartj/ehac244
947. Boulestreau R, van den Born BH, Lip GYH, Gupta A. Malignant hypertension: current perspectives and challenges. *J Am Heart Assoc* 2022;**11**:e023397. https://doi.org/10.1161/jaha.121.023397
948. Ma L, Hu X, Song L, Chen X, Ouyang M, Billot L, *et al.* The third intensive care bundle with blood pressure reduction in acute cerebral haemorrhage trial (INTERACT3): an international, stepped wedge cluster randomised controlled trial. *Lancet* 2023;**402**:27–40. https://doi.org/10.1016/s0140-6736(23)00806-1
949. Moullaali TJ, Wang X, Sandset EC, Woodhouse LJ, Law ZK, Arima H, *et al.* Early lowering of blood pressure after acute intracerebral haemorrhage: a systematic review and meta-analysis of individual patient data. *J Neurol Neurosurg Psychiatry* 2022;**93**:6–13. https://doi.org/10.1136/jnnp-2021-327195
950. Wang X, Di Tanna GL, Moullaali TJ, Martin R' H, Shipes VB, Robinson TG, *et al.* J-shape relation of blood pressure reduction and outcome in acute intracerebral hemorrhage: a pooled analysis of INTERACT2 and ATACH-II individual participant data. *Int J Stroke* 2022;**17**:1129–36. https://doi.org/10.1177/17474930211064076
951. Qureshi AI, Huang W, Lobanova I, Barsan WG, Hanley DF, Hsu CY, *et al.* Outcomes of intensive systolic blood pressure reduction in patients with intracerebral hemorrhage and excessively high initial systolic blood pressure: post hoc analysis of a randomized clinical trial. *JAMA Neurol* 2020;**77**:1355–65. https://doi.org/10.1001/jamaneurol.2020.3075
952. Bath PM, Krishnan K. Interventions for deliberately altering blood pressure in acute stroke. *Cochrane Database Syst Rev* 2014;**2014**:CD000039. https://doi.org/10.1002/14651858.CD000039.pub3

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

953. Berge E, Whiteley W, Audebert H, De Marchis GM, Fonseca AC, Padiglioni C, *et al.* European Stroke Organisation (ESO) guidelines on intravenous thrombolysis for acute ischaemic stroke. *Eur Stroke J* 2021;**6**:I–LXII. https://doi.org/10.1177/2396987321989865
954. LeCouffe NE, Kappelhof M, Treurniet KM, Lingsma HF, Zhang G, van den Wijngaard IR, *et al.* 2B, 2C, or 3: what should be the angiographic target for endovascular treatment in ischemic stroke? *Stroke* 2020;**51**:1790–6. https://doi.org/10.1161/strokeaha.119.028891
955. Mistry EA, Hart KW, Davis LT, Gao Y, Prestigiacomo CJ, Mittal S, *et al.* Blood pressure management after endovascular therapy for acute ischemic stroke: the BEST-II randomized clinical trial. *JAMA* 2023;**330**:821–31. https://doi.org/10.1001/jama.2023.14330
956. Lee M, Ovbiagele B, Hong KS, Wu Y-L, Lee J-E, Rao NM, *et al.* Effect of blood pressure lowering in early ischemic stroke: meta-analysis. *Stroke* 2015;**46**:1883–9. https://doi.org/10.1161/strokeaha.115.009552
957. Anderson CS, Huang Y, Lindley RI, Chen X, Arima H, Chen G, *et al.* Intensive blood pressure reduction with intravenous thrombolysis therapy for acute ischaemic stroke (ENCHANTED): an international, randomised, open-label, blinded-endpoint, phase 3 trial. *Lancet* 2019;**393**:877–88. https://doi.org/10.1016/s0140-6736(19)30038-8
958. Mazighi M, Richard S, Lapergue B, Sibon I, Gory B, Berge J, *et al.* Safety and efficacy of intensive blood pressure lowering after successful endovascular therapy in acute ischaemic stroke (BP-TARGET): a multicentre, open-label, randomised controlled trial. *Lancet Neurol* 2021;**20**:265–74. https://doi.org/10.1016/s1474-4422(20)30483-x
959. Yang P, Song L, Zhang Y, Zhang X, Chen X, Li Y, *et al.* Intensive blood pressure control after endovascular thrombectomy for acute ischaemic stroke (ENCHANTED2/MT): a multicentre, open-label, blinded-endpoint, randomised controlled trial. *Lancet* 2022;**400**:1585–96. https://doi.org/10.1016/s0140-6736(22)01882-7
960. Sandset EC, Anderson CS, Bath PM, Christensen H, Fischer U, Gąsecki D, *et al.* European Stroke Organisation (ESO) guidelines on blood pressure management in acute ischaemic stroke and intracerebral haemorrhage. *Eur Stroke J* 2021;**6**:XIviii–lxxxix. https://doi.org/10.1177/23969873211012133
961. Anderson CS, Heeley E, Huang Y, Wang J, Stapf C, Delcourt C, *et al.* Rapid blood-pressure lowering in patients with acute intracerebral hemorrhage. *N Engl J Med* 2013;**368**:2355–65. https://doi.org/10.1056/NEJMoa1214609
962. Qureshi AI, Palesch YY, Barsan WG, Hanley DF, Hsu CY, Martin RL, *et al.* Intensive blood-pressure lowering in patients with acute cerebral hemorrhage. *N Engl J Med* 2016;**375**:1033–43. https://doi.org/10.1056/NEJMoa1603460
963. Moullaali TJ, Wang X, Woodhouse LJ, Law ZK, Delcourt C, Sprigg N, *et al.* Lowering blood pressure after acute intracerebral haemorrhage: protocol for a systematic review and meta-analysis using individual patient data from randomised controlled trials participating in the blood pressure in acute stroke collaboration (BASC). *BMJ Open* 2019;**9**:e030121. https://doi.org/10.1136/bmjopen-2019-030121
964. American College of Obstetricians and Gynecologists. Gestational hypertension and preeclampsia: ACOG practice bulletin, number 222. *Obstet Gynecol* 2020;**135**:e237–60. https://doi.org/10.1097/aog.0000000000003891
965. Garovic VD, Dechend R, Easterling T, Karumanchi SA, Baird SM, Magee LA, *et al.* Hypertension in pregnancy: diagnosis, blood pressure goals, and pharmacotherapy: a scientific statement from the American Heart Association. *Hypertension* 2022;**79**:e21–41. https://doi.org/10.1161/hyp.0000000000000208
966. Magee LA, Nicolaides KH, von Dadelszen P. Preeclampsia. *N Engl J Med* 2022;**386**:1817–32. https://doi.org/10.1056/NEJMra2109523
967. Altman D, Carroli G, Duley L, Farrell B, Moodley J, Neilson J, *et al.* Do women with pre-eclampsia, and their babies, benefit from magnesium sulphate? The Magpie trial: a randomised placebo-controlled trial. *Lancet* 2002;**359**:1877–90. https://doi.org/10.1016/s0140-6736(02)08778-0
968. Poon LC, Magee LA, Verlohren S, Shennan A, von Dadelszen P, Sheiner E, *et al.* A literature review and best practice advice for second and third trimester risk stratification, monitoring, and management of pre-eclampsia: compiled by the pregnancy and non-communicable diseases committee of FIGO (the International Federation of Gynecology and Obstetrics). *Int J Gynaecol Obstet* 2021;**154 Suppl 1**:3–31. https://doi.org/10.1002/ijgo.13763
969. Muhammad S, Usman H, Dawha YM, Yahya A, Yekeen A, Bako B. Comparison of intravenous labetalol and hydralazine for severe hypertension in pregnancy in Northeastern Nigeria: a randomized controlled trial. *Pregnancy Hypertens* 2022;**29**:1–6. https://doi.org/10.1016/j.preghy.2022.05.001
970. Magee LA, Cham C, Waterman EJ, Ohlsson A, von Dadelszen P. Hydralazine for treatment of severe hypertension in pregnancy: meta-analysis. *BMJ* 2003;**327**:955–60. https://doi.org/10.1136/bmj.327.7421.955
971. Wu HZ, Cheng Y, Yu D, Li J-B, Jiang Y-F, Zhu Z-N. Different dosage regimens of nifedipine, labetalol, and hydralazine for the treatment of severe hypertension during pregnancy: a network meta-analysis of randomized controlled trials. *Hypertens Pregnancy* 2022;**41**:126–38. https://doi.org/10.1080/10641955.2022.2056196
972. Halvorsen S, Mehilli J, Cassese S, Hall TS, Abdelhamid M, Barbato E, *et al.* 2022 ESC Guidelines on cardiovascular assessment and management of patients undergoing non-cardiac surgery. *Eur Heart J* 2022;**43**:3826–924. https://doi.org/10.1093/eurheartj/ehac270
973. Lizano-Díez I, Poteet S, Burniol-Garcia A, Cerezales M. The burden of perioperative hypertension/hypotension: a systematic review. *PLoS One* 2022;**17**:e0263737. https://doi.org/10.1371/journal.pone.0263737
974. Futier E, Lefrant JY, Guinot PG, Godet T, Lorne E, Cuvillon P, *et al.* Effect of individualized vs standard blood pressure management strategies on postoperative organ dysfunction among high-risk patients undergoing major surgery: a randomized clinical trial. *JAMA* 2017;**318**:1346–57. https://doi.org/10.1001/jama.2017.14172
975. Sanders RD, Hughes F, Shaw A, Thompson A, Bader A, Hoeft A, *et al.* Perioperative quality initiative consensus statement on preoperative blood pressure, risk and outcomes for elective surgery. *Br J Anaesth* 2019;**122**:552–62. https://doi.org/10.1016/j.bja.2019.01.018
976. Messina A, Robba C, Calabrò L, Zambelli D, Iannuzzi F, Molinari E, *et al.* Association between perioperative fluid administration and postoperative outcomes: a 20-year systematic review and a meta-analysis of randomized goal-directed trials in major visceral/noncardiac surgery. *Crit Care* 2021;**25**:43. https://doi.org/10.1186/s13054-021-03464-1
977. Meng L, Yu W, Wang T, Zhang L, Heerdt PM, Gelb AW. Blood pressure targets in perioperative care. *Hypertension* 2018;**72**:806–17. https://doi.org/10.1161/hypertensionaha.118.11688
978. Wanner PM, Wulff DU, Djurdjevic M, Korte W, Schnider TW, Filipovic M. Targeting higher intraoperative blood pressures does not reduce adverse cardiovascular events following noncardiac surgery. *J Am Coll Cardiol* 2021;**78**:1753–64. https://doi.org/10.1016/j.jacc.2021.08.048
979. Blessberger H, Lewis SR, Pritchard MW, Fawcett LJ, Domanovits H, Schlager O, *et al.* Perioperative beta-blockers for preventing surgery-related mortality and morbidity in adults undergoing non-cardiac surgery. *Cochrane Database Syst Rev* 2019;**9**:CD013438. https://doi.org/10.1002/14651858.CD013438
980. McGory ML, Maggard MA, Ko CY. A meta-analysis of perioperative beta blockade: what is the actual risk reduction? *Surgery* 2005;**138**:171–9. https://doi.org/10.1016/j.surg.2005.03.022
981. Kertai MD, Cooter M, Pollard RJ, Buhrman W, Aronson S, Mathew JP, *et al.* Is compliance with surgical care improvement project cardiac (SCIP-Card-2) measures for perioperative β-blockers associated with reduced incidence of mortality and cardiovascular-related critical quality indicators after noncardiac surgery? *Anesth Analg* 2018;**126**:1829–38. https://doi.org/10.1213/ane.0000000000002577
982. Roshanov PS, Rochwerg B, Patel A, Salehian O, Duceppe E, Belley-Côté EP, *et al.* Withholding versus continuing angiotensin-converting enzyme inhibitors or angiotensin II receptor blockers before noncardiac surgery: an analysis of the vascular events in noncardiac surgery patlents cOhort evaluatioN prospective cohort. *Anesthesiology* 2017;**126**:16–27. https://doi.org/10.1097/aln.0000000000001404
983. Shiffermiller JF, Monson BJ, Vokoun CW, Beachy MW, Smith MP, Sullivan JN, *et al.* Prospective randomized evaluation of preoperative angiotensin-converting enzyme inhibition (PREOP-ACEI). *J Hosp Med* 2018;**13**:661–7. https://doi.org/10.12788/jhm.3036
984. Hollmann C, Fernandes NL, Biccard BM. A systematic review of outcomes associated with withholding or continuing angiotensin-converting enzyme inhibitors and angiotensin receptor blockers before noncardiac surgery. *Anesth Analg* 2018;**127**:678–87. https://doi.org/10.1213/ane.0000000000002837
985. Ackland GL, Patel A, Abbott TEF, Begum S, Dias P, Crane DR, *et al.* Discontinuation vs. continuation of renin–angiotensin system inhibition before non-cardiac surgery: the SPACE trial. *Eur Heart J* 2024;**45**:1146–55. https://doi.org/10.1093/eurheartj/ehad716
986. McEvoy MD, Gupta R, Koepke EJ, Feldheiser A, Michard F, Levett D, *et al.* Perioperative quality initiative consensus statement on postoperative blood pressure, risk and outcomes for elective surgery. *Br J Anaesth* 2019;**122**:575–86. https://doi.org/10.1016/j.bja.2019.01.019
987. Ekman I, Swedberg K, Taft C, Lindseth A, Norberg A, Brink E, *et al.* Person-centered care—ready for prime time. *Eur J Cardiovasc Nurs* 2011;**10**:248–51. https://doi.org/10.1016/j.ejcnurse.2011.06.008
988. Krist AH, Tong ST, Aycock RA, Longo DR. Engaging patients in decision-making and behavior change to promote prevention. *Stud Health Technol Inform* 2017;**240**:284–302. https://doi.org/10.3233/ISU-170826
989. Johnson RA, Huntley A, Hughes RA, Hughes RA, Cramer H, Turner KM, *et al.* Interventions to support shared decision making for hypertension: a systematic review of controlled studies. *Health Expect* 2018;**21**:1191–207. https://doi.org/10.1111/hex.12826
990. Johnson RL, Roter D, Powe NR, Cooper LA. Patient race/ethnicity and quality of patient–physician communication during medical visits. *Am J Public Health* 2004;**94**:2084–90. https://doi.org/10.2105/ajph.94.12.2084
991. Williams MV, Davis T, Parker RM, Weiss BD. The role of health literacy in patient-physician communication. *Fam Med* 2002;**34**:383–9.
992. Karmali KN, Persell SD, Perel P, Lloyd-Jones DM, Berendsen MA, Huffman MD. Risk scoring for the primary prevention of cardiovascular disease. *Cochrane Database Syst Rev* 2017;**3**:CD006887. https://doi.org/10.1002/14651858.CD006887.pub4
993. Ma C, Zhou Y, Zhou W, Huang C. Evaluation of the effect of motivational interviewing counselling on hypertension care. *Patient Educ Couns* 2014;**95**:231–7. https://doi.org/10.1016/j.pec.2014.01.011
994. Huang X, Xu N, Wang Y, Sun Y, Guo A. The effects of motivational interviewing on hypertension management: a systematic review and meta-analysis. *Patient Educ Couns* 2023;**112**:107760. https://doi.org/10.1016/j.pec.2023.107760

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026

995. Glaser E, Richard C, Lussier MT. The impact of a patient web communication intervention on reaching treatment suggested guidelines for chronic diseases: a randomized controlled trial. *Patient Educ Couns* 2017;**100**:2062–70. https://doi.org/10.1016/j.pec.2017.05.022
996. Richard AA, Shea K. Delineation of self-care and associated concepts. *J Nurs Scholarsh* 2011;**43**:255–64. https://doi.org/10.1111/j.1547-5069.2011.01404.x
997. Wilkinson A, Whitehead L. Evolution of the concept of self-care and implications for nurses: a literature review. *Int J Nurs Stud* 2009;**46**:1143–7. https://doi.org/10.1016/j.ijnurstu.2008.12.011
998. Bodenheimer T, Lorig K, Holman H, Grumbach K. Patient self-management of chronic disease in primary care. *JAMA* 2002;**288**:2469–75. https://doi.org/10.1001/jama.288.19.2469
999. Barlow J, Wright C, Sheasby J, Turner A, Hainsworth J. Self-management approaches for people with chronic conditions: a review. *Patient Educ Couns* 2002;**48**:177–87. https://doi.org/10.1016/s0738-3991(02)00032-0
1000. Hallberg I, Ranerup A, Kjellgren K. Supporting the self-management of hypertension: patients' experiences of using a mobile phone-based system. *J Hum Hypertens* 2016;**30**:141–6. https://doi.org/10.1038/jhh.2015.37
1001. Almeida GO, Aidar FJ, Matos DG, Almeida-Neto PF, Melo EV, Barreto Filho JAS, *et al.* Non-targeted self-measurement of blood pressure: association with self-medication, unscheduled emergency visits and anxiety. *Medicina (Kaunas)* 2021;**57**:75. https://doi.org/10.3390/medicina57010075
1002. Greaves F, Joshi I, Campbell M, Roberts S, Patel N, Powell J, *et al.* What is an appropriate level of evidence for a digital health intervention? *Lancet* 2019;**392**:2665–7. https://doi.org/10.1016/s0140-6736(18)33129-5
1003. Morton K, Dennison L, May C, Murray E, Little P, McManus RJ, *et al.* Using digital interventions for self-management of chronic physical health conditions: a meta-ethnography review of published studies. *Patient Educ Couns* 2017;**100**:616–35. https://doi.org/10.1016/j.pec.2016.10.019
1004. McKinstry B, Hanley J, Wild S, Pagliari C, Paterson M, Lewis S, *et al.* Telemonitoring based service redesign for the management of uncontrolled hypertension: multicentre randomised controlled trial. *BMJ* 2013;**346**:f3030. https://doi.org/10.1136/bmj.f3030
1005. Persell SD, Peprah YA, Lipiszko D, Lee JY, Li JJ, Ciolino JD, *et al.* Effect of home blood pressure monitoring via a smartphone hypertension coaching application or tracking application on adults with uncontrolled hypertension: a randomized clinical trial. *JAMA Netw Open* 2020;**3**:e200255. https://doi.org/10.1001/jamanetworkopen.2020.0255
1006. Andersson U, Nilsson PM, Kjellgren K, Hoffmann M, Wennersten A, Midlöv P, *et al.* PERson-centredness in Hypertension management using Information Technology: a randomized controlled trial in primary care. *J Hypertens* 2023;**41**:246–53. https://doi.org/10.1097/hjh.0000000000003322
1007. Bergland OU, Halvorsen LV, Søraas CL, Hjørnholm U, Kjær VN, Rognstad S, *et al.* Detection of nonadherence to antihypertensive treatment by measurements of serum drug concentrations. *Hypertension* 2021;**78**:617–28. https://doi.org/10.1161/hypertensionaha.121.17514
1008. Durand H, Hayes P, Morrissey EC, Newell J, Casey M, Murphy AW, *et al.* Medication adherence among patients with apparent treatment-resistant hypertension: systematic review and meta-analysis. *J Hypertens* 2017;**35**:2346–57. https://doi.org/10.1097/hjh.0000000000001502
1009. Kronish IM, Thorpe CT, Voils CI. Measuring the multiple domains of medication nonadherence: findings from a Delphi survey of adherence experts. *Transl Behav Med* 2021;**11**:104–13. https://doi.org/10.1093/tbm/ibz133
1010. DiMatteo MR. Social support and patient adherence to medical treatment: a meta-analysis. *Health Psychol* 2004;**23**:207–18. https://doi.org/10.1037/0278-6133.23.2.207
1011. Houle SK, Chatterley T, Tsuyuki RT. Multidisciplinary approaches to the management of high blood pressure. *Curr Opin Cardiol* 2014;**29**:344–53. https://doi.org/10.1097/hco.0000000000000071
1012. Woodham NS, Taneepanichskul S, Somrongthong R, Kitsanapun A, Sompakdee B. Effectiveness of a multidisciplinary approach intervention to improve blood pressure control among elderly hypertensive patients in rural Thailand: a quasi-experimental study. *J Multidiscip Healthc* 2020;**13**:571–80. https://doi.org/10.2147/jmdh.S254286
1013. Mattei da Silva ÂT, de Fátima Mantovani M, Castanho Moreira R, Perez Arthur J, Molina de Souza R. Nursing case management for people with hypertension in primary health care: a randomized controlled trial. *Res Nurs Health* 2020;**43**:68–78. https://doi.org/10.1002/nur.21994
1014. He J, Ouyang N, Guo X, Sun G, Li Z, Mu J, *et al.* Effectiveness of a non-physician community health-care provider-led intensive blood pressure intervention versus usual care on cardiovascular disease (CRHCP): an open-label, blinded-endpoint, cluster-randomised trial. *Lancet* 2023;**401**:928–38. https://doi.org/10.1016/s0140-6736(22)02603-4
1015. Creegan D, McEvoy JW. Selected highlights in the updated treatment of hypertension. *Trends Cardiovasc Med* https://doi.org/10.1016/j.tcm.2023.11.001
1016. Jaffe MG, Lee GA, Young JD, Sidney S, Go AS. Improved blood pressure control associated with a large-scale hypertension program. *JAMA* 2013;**310**:699–705. https://doi.org/10.1001/jama.2013.108769
1017. Hänsel M, Steigmiller K, Luft AR, Gebhard C, Held U, Wegener S, *et al.* Neurovascular disease in Switzerland: 10-year trends show non-traditional risk factors on the rise and higher exposure in women. *Eur J Neurol* 2022;**29**:2851–60. https://doi.org/10.1111/ene.15434
1018. de Ruiter SC, Schmidt AF, Grobbee DE, den Ruijter HM, Peters SAE. Sex-specific Mendelian randomisation to assess the causality of sex differences in the effects of risk factors and treatment: spotlight on hypertension. *J Hum Hypertens* 2023;**37**:602–8. https://doi.org/10.1038/s41371-023-00821-1
1019. Chapman N, Ching SM, Konradi AO, Nuyt AM, Khan T, Twumasi-Ankrah B, *et al.* Arterial hypertension in women: state of the art and knowledge gaps. *Hypertension* 2023;**80**:1140–9. https://doi.org/10.1161/hypertensionaha.122.20448
1020. Tamargo J, Caballero R, Mosquera ED. Sex and gender differences in the treatment of arterial hypertension. *Expert Rev Clin Pharmacol* 2023;**16**:329–47. https://doi.org/10.1080/17512433.2023.2189585
1021. Olsen MH, Angell SY, Asma S, Boutouyrie P, Burger D, Chirinos JA, *et al.* A call to action and a lifecourse strategy to address the global burden of raised blood pressure on current and future generations: the Lancet Commission on hypertension. *Lancet* 2016;**388**:2665–712. https://doi.org/10.1016/s0140-6736(16)31134-5

Downloaded from https://academic.oup.com/eurheartj/article/45/38/3912/7741010 by guest on 14 February 2026